Part II

Relative Poincaré duality for differentiable stacks

Abstract

Following ideas from motivic homotopy theory, we introduce for every separated differentiable stack $\mathcal{X}$ an ∞-category $\mathrm{SH}(\mathcal{X})$ of genuine sheaves of spectra on $\mathcal{X}$, which for manifolds is given by ordinary sheaves of spectra while on the classifying stack of a compact Lie group is given by genuine G-spectra. We prove a form of relative Poincaré duality in this setting: for a proper representable submersion f of separated differentiable stacks, there is an equivalence $f_\sharp \simeq f_*(- \otimes S^{T_f})$ between its relative homology and a twist of its relative cohomology by the relative tangent sphere bundle of f. When specialized to quotient stacks of equivariant smooth manifolds, this recovers both equivariant Atiyah duality and the Wirthmüller isomorphism in stable equivariant homotopy theory.

II.1 Introduction

The goal of this article is to establish a version of relative Poincaré duality for separated differentiable stacks, following ideas from Hoyois [Hoy17] in the setting of equivariant motivic homotopy theory.

Relative Poincaré duality for smooth manifolds

Poincaré duality is an important result relating the homology and cohomology of manifolds: if M is a compact orientable n-dimensional manifold, then forming the cap product with the fundamental class $[M] \in H_n(M)$ induces an isomorphism of graded abelian groups

$$- \cap [M] \colon H^{n-*}(M;\mathbb{Z}) \xrightarrow{\cong} H_*(M;\mathbb{Z}).$$

The orientability condition on M can be dropped if one replaces the left-hand side by the cohomology of the *orientation sheaf* $\mathcal{O}_M \in \mathrm{Shv}(M;\mathrm{Ab})$; the resulting isomorphism $H^{n-*}(M;\mathcal{O}_M) \xrightarrow{\cong} H_*(M)$ is known as *twisted Poincaré duality*.

There is a more general version of Poincaré duality for *families* of manifolds, known as *relative Poincaré duality*. Consider a proper smooth submersion $f \colon M \to N$, thought of as an N-indexed family of compact smooth manifolds $M_x = f^{-1}(x)$ for $x \in N$. The pullback functor on sheaves of spectra $f^* \colon \mathrm{Shv}(N;\mathrm{Sp}) \to \mathrm{Shv}(M;\mathrm{Sp})$ admits both a left adjoint $f_\sharp$ and a right adjoint f_*, which we think of as the *relative sheaf homology/cohomology* of f, respectively. Indeed, given a sheaf of spectra $\mathcal{F}$ on M, one can show that the stalks of the sheaves $f_\sharp(\mathcal{F})$ and $f_*(\mathcal{F})$ on N are given by the sheaf homology/cohomology of the restriction of $\mathcal{F}$ to each of the fibers of f. Relative Poincaré duality states that these two functors $f_\sharp$ and f_* agree up to a 'twist' by the tangent sphere bundle $S^{T_f} \in \mathrm{Shv}(M;\mathrm{Sp})$, the suspension spectrum of the one-point compactification of the relative tangent bundle of f:

Proposition (Relative Poincaré duality, Volpe [Vol21, Proposition 6.18, Theorem 7.11])**.** *Let $f \colon M \to N$ be a proper smooth submersion between smooth manifolds. Then there is for every sheaf $\mathcal{F} \in \mathrm{Shv}(M;\mathrm{Sp})$ a natural equivalence $f_\sharp(\mathcal{F}) \simeq f_*(\mathcal{F} \otimes S^{T_f})$ of sheaves of spectra on N.*

Relative Poincaré duality for differentiable stacks

The goal of this article is to generalize the above result to the setting of (separated[1]) differentiable stacks. We may think of a differentiable stack as a generalization of a smooth manifold which is allowed to have *singularities*: instead of being locally isomorphic to a Euclidean space $\mathbb{R}^n$, it will locally be isomorphic to the quotient of $\mathbb{R}^n$ by a linear action of a compact Lie group G. While the geometric behavior of differentiable stacks closely parallels that of smooth manifolds, the allowed singularities make it possible to capture equivariant phenomena.

For every differentiable stack $\mathcal{X}$, we will define a stable ∞-category $\mathrm{SH}(\mathcal{X})$ of *genuine sheaves of spectra on* $\mathcal{X}$. When $\mathcal{X} = M$ is a smooth manifold, this reduces to the ∞-category $\mathrm{Shv}(M;\mathrm{Sp})$ of ordinary sheaves of spectra on M, while for a classifying stack $\mathbb{B}G$ of a compact Lie group G it recovers the ∞-category Sp_G of genuine G-spectra. The construction of $\mathrm{SH}(\mathcal{X})$ follows the definition of the stable motivic homotopy category in motivic homotopy theory: First one forms the ∞-category $\mathrm{H}(\mathcal{X}) = \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Sub}_{/\mathcal{X}})$ of homotopy invariant sheaves on the site of representable submersions over $\mathcal{X}$ (equipped with the open cover topology). Then, at least locally in $\mathcal{X}$, one defines $\mathrm{SH}(\mathcal{X})$ by inverting all sphere bundles of vector bundles over $\mathcal{X}$, which determines $\mathrm{SH}(\mathcal{X})$ for an arbitrary differentiable stack by asking the assignment $\mathcal{X} \mapsto \mathrm{SH}(\mathcal{X})$ to satisfy descent with respect to open covers.

Consider a proper representable submersion $f\colon \mathcal{Y} \to \mathcal{X}$ of differentiable stacks, which we want to think of as an $\mathcal{X}$-indexed family of compact smooth manifolds. This map admits a relative tangent bundle T_f over $\mathcal{Y}$, producing a tangent sphere bundle $S^{T_f} \in \mathrm{SH}(\mathcal{Y})$. The map f defines a symmetric monoidal pullback functor $f^*\colon \mathrm{SH}(\mathcal{X}) \to \mathrm{SH}(\mathcal{Y})$, which admits both a left adjoint $f_\sharp$ as well as a right adjoint f_*, thought of as the relative homology and cohomology of f. The following is our main result:

Theorem A (Relative Poincaré duality for genuine sheaves, Theorem 6.1.7)**.** *Let $f\colon \mathcal{Y} \to \mathcal{X}$ be a proper representable submersion between separated differentiable stacks. Then there is for every $\mathcal{F} \in \mathrm{SH}(\mathcal{Y})$ a natural equivalence $f_\sharp(\mathcal{F}) \simeq f_*(\mathcal{F} \otimes S^{T_f})$ in $\mathrm{SH}(\mathcal{X})$.*

As a consequence we will deduce relative Atiyah duality for differentiable stacks, and we will prove proper base change, smooth-proper base change and the proper projection formula for the pushforward functors $p_*\colon \mathrm{SH}(\mathcal{Y}) \to \mathrm{SH}(\mathcal{X})$ of proper maps $p\colon \mathcal{Y} \to \mathcal{X}$.

[1]In this introduction, all differentiable stacks are assumed separated.

Organization

In Chapter II.2 we recall standard material on differentiable stacks, including their relation to Lie groupoids, local properties of maps of stacks and the notion of vector bundles over stacks. In Chapter II.3 we discuss various geometrical aspects of differentiable stacks, including the coarse moduli space of a stack (Section 3.1), open complements of closed substacks (Section 3.2), the isotropy groups of a stack (Section 3.4) and the notions of relative tangent bundles and normal bundles (Section 3.5). A geometrically well-behaved class of differentiable stacks, discussed in Section 3.3, are the *separated* differentiable stacks: those whose diagonal is proper. In Section 3.6 we show that every embedding of separated differentiable stacks admits a *tubular neighborhood* and in Section 3.7 we prove that every separated differentiable stack is locally isomorphic to a quotient stack $\mathbb{R}^n//G$ for some smooth linear action of a compact Lie group G on a Euclidean space $\mathbb{R}^n$.

In Chapter II.4 we introduce for every separated differentiable stack $\mathcal{X}$ the ∞-categories $\mathrm{H}(\mathcal{X})$ and $\mathrm{SH}(\mathcal{X})$ of *genuine sheaves of animae/spectra on* $\mathcal{X}$, show that ordinary sheaves are both a localization and a colocalization of genuine sheaves, and prove basic properties for genuine sheaves like smooth base change and the smooth projection formula. In Section 4.4, we show that genuine sheaves over classifying stacks of compact Lie groups give back classical equivariant homotopy theory:

Theorem B (Theorem 4.4.16, Proposition 4.4.17)**.** *For a compact Lie group G, there are equivalences of* ∞*-categories* $\mathrm{H}(\mathbb{B}G) \simeq \mathrm{An}_G$ *and* $\mathrm{SH}(\mathbb{B}G) \simeq \mathrm{Sp}_G$.

In Section 4.5 we show that the assignments $\mathcal{X} \mapsto \mathrm{H}(\mathcal{X})$ and $\mathcal{X} \mapsto \mathrm{SH}(\mathcal{X})$ admit universal characterizations, phrased in the language of *pullback formalisms* introduced by [DG22].

In Chapter II.5, we prove the localization theorem for pointed genuine sheaves:

Theorem C (Theorem II.5.2.16)**.** *Let* $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ *be a closed embedding of differentiable stacks and let* $j\colon \mathcal{U} \hookrightarrow \mathcal{X}$ *be its open complement. Then the functor* $i_*\colon \mathrm{H}(\mathcal{Z})_* \to \mathrm{H}(\mathcal{X})_*$ *is fully faithful, and there is a preferred cofiber sequence* $j_\sharp j^* \xrightarrow{\text{counit}} \mathrm{id} \xrightarrow{\text{unit}} i_* i^*$.

In Chapter II.6 we give a proof of our main result, Theorem A stated above. We give a precise formulation in Section 6.1. The proof, which will be given in Section 6.4, is close in spirit to the proof of Atiyah duality for a compact smooth manifold M, using a version of *Pontryagin-Thom collapse map* constructed in Section 6.2. In Section 6.3 we introduce the auxiliary notion of a *kernel operator*. In Section 6.5 we discuss various important consequences of relative Poincaré duality, like relative Atiyah duality, proper base change and smooth-proper base change.

Conventions

In contrast to most sources on differentiable stacks, we will fully work in the homotopy-theoretic setting of ∞-categories, and adopt the standard notations and terminology from this setting. One notable exception, following [CS23, Section 5.1.4], is that we use the word 'anima' rather than 'space' to refer to the notion of an ∞-groupoid, and accordingly write An for the ∞-topos of animae/∞-groupoids. Similarly we write An_G for the ∞-category of G-spaces for a compact Lie group G and refer to its objects as 'genuine G-animae'.

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A major source of inspiration for this work was the article [Hoy17], whose proof strategy we will follow closely. Additionally, important references that provided significant insights include [Kha19], [CD19], [Vol21], and [KR21].

II.2 Foundations on differentiable stacks

In this chapter, we will recall some of the foundations on differentiable stacks: sheaves of groupoids on the site of differentiable manifolds and open coverings which admit a representable atlas. All of the material in this chapter is well-known and sources on this topic are plentiful; a selection is [Pro96; BX11; Met03; Ler10; Hoy13]. Our treatment differs somewhat from these sources in that we will work in the homotopy-theoretic setting of sheaves of animae/∞-groupoids, sometimes called ∞-*sheaves*, rather than sheaves of groupoids; a recollection of the theory of sheaf ∞-topoi is provided in Appendix E.4. We will see in Corollary 2.3.24 below that the objects of interest, the differentiable stacks, are in fact sheaves of groupoids, making our approach equivalent to the classical approach.

2.1 Representable morphisms of stacks

We start by introducing stacks on the site of smooth manifolds and discussing the notion of representable morphisms between stacks.

Definition 2.1.1. Let Diff denote the ordinary category of smooth manifolds[1] and smooth maps. We turn it into a site by equipping it with the open cover topology and let $\mathrm{Shv}(\mathrm{Diff})$ denote the associated ∞-topos of sheaves of animae on Diff. We refer to an object $\mathcal{X} \in \mathrm{Shv}(\mathrm{Diff})$ as a *stack on* Diff, or as a *stack* for short.

Note that the site $(\mathrm{Diff}, \mathrm{open})$ is subcanonical: for every smooth manifold $M \in \mathrm{Diff}$, the representable functor $\mathrm{Hom}_{\mathrm{Diff}}(-, M)\colon \mathrm{Diff} \to \mathrm{Set} \hookrightarrow \mathrm{An}$ is a sheaf. In other words, the Yoneda embedding $\mathrm{Diff} \hookrightarrow \mathrm{PSh}(\mathrm{Diff})$ factors through the subcategory of sheaves:

$$y\colon \mathrm{Diff} \hookrightarrow \mathrm{Shv}(\mathrm{Diff}).$$

[1]All smooth manifolds are assumed to be Hausdorff. They are allowed to be *impure*: different path components may have different dimensions.

We will often abuse notation by identifying smooth manifolds M with their associated sheaves $y(M)$; accordingly, we say that a stack $\mathcal{X}$ on Diff *is a smooth manifold* if it lies in the essential image of the Yoneda embedding. The Yoneda embedding y preserves all limits, and furthermore preserves arbitrary coproducts due to the sheaf condition.

Recall from Appendix E.2 the notion of an *effective epimorphism* in an ∞-topos. In the case of Shv(Diff), the effective epimorphisms can be characterized as those maps admitting *local sections*:

Definition 2.1.2. Let M be a smooth manifold and let $f : \mathcal{X} \to M$ be a map of sheaves. We say that f *admits local sections* if there exists an open cover $\{U_i\}_{i \in I}$ of M such that the map $U_i \hookrightarrow M$ factors through f for each $i \in I$.

Proposition 2.1.3. *If M is a smooth manifold, then a map $f : \mathcal{X} \to M$ in* Shv(Diff) *is an effective epimorphism if and only if it admits local sections.*

Proof. This is a special case of Lemma E.38. ◻

Since the site Diff does not admit all pullbacks, there is a subtlety in the definition of representable morphisms between stacks on Diff. However, pullbacks along *smooth submersions* always exist in Diff, see for example Proposition C.9. This leads to the following two-step definition of representability, which guarantees that every smooth map between smooth manifolds is representable when considered as a morphism of stacks:

Definition 2.1.4 (Representable morphisms, [EG11, Definition 2.1])**.** Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of stacks on Diff.

(1) We call f a *representable submersion* if for any smooth manifold M and any morphism $M \to \mathcal{Y}$, the fiber product $M \times_{\mathcal{Y}} \mathcal{X}$ is a smooth manifold and the induced map $M \times_{\mathcal{Y}} \mathcal{X} \to M$ is a submersion. If the map $M \times_{\mathcal{Y}} \mathcal{X} \to M$ is surjective for every M, we say that f is a *representable surjective submersion*.

(2) We say f *representable* if for any representable submersion $M \to \mathcal{Y}$ from a smooth manifold M, the pullback $M \times_{\mathcal{Y}} \mathcal{X}$ is a smooth manifold.

It is clear from the definition that every representable submersion is a representable morphism. Furthermore, every smooth map $f : M \to N$ of smooth manifolds is representable when considered as a morphism of stacks, and it is a representable submersion of stacks if and only if it is a smooth submersion of smooth manifolds.

Warning 2.1.5. We warn the reader that representability behaves reasonably only when there is a sufficiently large supply of representable submersions into $\mathcal{Y}$. For instance, if there are no representable submersions from smooth manifolds to $\mathcal{Y}$, then every morphism $\mathcal{X} \to \mathcal{Y}$ is representable according to the above definition.

Remark 2.1.6. A smooth submersion between smooth manifolds has local sections if and only if it is surjective, see e.g. Lemma C.10. It follows from Proposition 2.1.3 that a representable submersion $f\colon \mathcal{X} \to \mathcal{Y}$ of stacks is in fact a representable *surjective* submersion if and only if it is an effective epimorphism.

Definition 2.1.7 (Open substack)**.** Let $j\colon \mathcal{U} \to \mathcal{X}$ be a representable submersion of stacks on Diff. We say that j exhibits $\mathcal{U}$ as an *open substack* of $\mathcal{X}$ if for every smooth manifold M and any morphism $M \to \mathcal{X}$, the induced map of smooth manifolds $M \times_{\mathcal{X}} \mathcal{U} \to M$ is an open embedding.

As we will establish next, representability of a stack is a local condition:

Lemma 2.1.8. *Let $\mathcal{X}$ be a stack on* Diff *and assume that there exists a collection of open substacks $\{j_\alpha\colon U_\alpha \hookrightarrow \mathcal{X}\}_{\alpha\in I}$ of $\mathcal{X}$ such that the map $\coprod_{\alpha\in I} U_\alpha \to \mathcal{X}$ is an effective epimorphism.*[2] *If each U_α is a smooth manifold, then so is $\mathcal{X}$.*

Proof. By the definition of effective epimorphisms, recalled in Definition E.12, $\mathcal{X}$ is the colimit of the Čech nerve of the map $\coprod_{\alpha\in I} U_\alpha \to \mathcal{X}$. Since each of the maps $f_\alpha\colon U_\alpha \hookrightarrow \mathcal{X}$ is assumed to be a representable submersion, the iterated pullbacks $U_{\alpha_1} \times_{\mathcal{X}} \cdots \times_{\mathcal{X}} U_{\alpha_n}$ are all smooth manifolds again, hence the Čech nerve defines a simplicial diagram in Diff. We define a topological space M as the colimit of this simplicial diagram, regarded as a diagram in the ordinary category Top of topological spaces. More explicitly:

$$M = \operatorname{coeq}\left(\bigsqcup_{\alpha\neq\beta} U_\alpha \times_{\mathcal{X}} U_\beta \rightrightarrows \bigsqcup_{\alpha} U_\alpha\right).$$

Each of the maps $U_\alpha \to M$ is an open embedding, so M admits an open cover by the smooth manifolds U_α. As the smooth structures are compatible on their intersections, M itself obtains the structure of a smooth manifold. It follows that M is the colimit in Diff of the simplicial diagram, and consequently also the colimit in Shv(Diff) due to descent with respect to open covers. It follows that M is equivalent to $\mathcal{X}$, proving the claim. □

[2]Such a collection of open substacks is called an ‘open cover’, see Definition 4.1.2.

Corollary 2.1.9. *Let $\mathcal{X} \to M$ be a morphism of stacks on* Diff *and assume that M admits an open cover $\{V_\alpha\}$ such that each of the pullback stacks $V_\alpha \times_M \mathcal{X}$ is a smooth manifold. Then $\mathcal{X}$ is a smooth manifold.*

Proof. The maps $V_\alpha \times_M \mathcal{X} \to \mathcal{X}$ are pullbacks of the open embeddings $V_\alpha \hookrightarrow \mathcal{X}$ and thus are open substacks of $\mathcal{X}$. As the map $\bigsqcup_{\alpha \in I} V_\alpha \to M$ is an effective epimorphism, so is their base chane $\bigsqcup_{\alpha\in I} V_\alpha \times_M \mathcal{X} \to \mathcal{X}$ along the map $\mathcal{X} \to M$. The claim thus follows from Lemma 2.1.8. □

Proposition 2.1.10. *Consider a pullback diagram*

$$\begin{tikzcd} \mathcal{X}' \arrow[r] \arrow[d, "f'"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{X} \arrow[d, "f"] \\ \mathcal{Y}' \arrow[r, "g"] & \mathcal{Y} \end{tikzcd}$$

of stacks on Diff.

(1) If f is a representable submersion, then f' is a representable submersion.

(2) If g is a representable submersion and f is representable, then f' is representable.

(3) If g is an effective epimorphism and f' is a representable submersion, then f is a representable submersion.

(4) If g is a representable surjective submersion and f' is representable, then f is representable.

Proof. Parts (1) and (2) are immediate from the definitions. For part (3), assume that g is an effective epimorphism and let $M \to \mathcal{Y}$ be a morphism from a smooth manifold. It follows from Proposition 2.1.3 that the map $M \to \mathcal{Y}$ locally factors through $g\colon \mathcal{Y}' \twoheadrightarrow \mathcal{Y}$. Since the condition we are proving may be checked locally in M by Corollary 2.1.9, we may assume that already the map $M \to \mathcal{Y}$ factors through $\mathcal{Y}'$ via some map $s\colon M \to \mathcal{Y}'$. Now consider the following pullback diagram:

$$\begin{tikzcd} N \arrow[r] \arrow[d, "f''"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{X}' \arrow[r] \arrow[d, "f'"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{X} \arrow[d, "f"] \\ M \arrow[r, "s"] & \mathcal{Y}' \arrow[r, two heads, "g"] & \mathcal{Y}. \end{tikzcd}$$

By assumption, f' is a representable submersion, and it follows that the pullback $N = M \times_{\mathcal{Y}'} \mathcal{X}' \cong M \times_{\mathcal{Y}} \mathcal{X}$ is a smooth manifold and that the map $f''\colon N \to M$ is a submersion. This proves that f is a representable submersion, finishing the proof of (3).

For part (4), let $M \to \mathcal{Y}$ be a representable submersion from some smooth manifold M. We have to show that the pullback $M \times_{\mathcal{Y}} \mathcal{X}$ is again a smooth manifold. By pulling back the whole situation along $M \to \mathcal{Y}$, we may assume that $\mathcal{Y} = M$ is a smooth manifold. We are then in the situation of a pullback square

$$\begin{tikzcd} N \arrow[r, "g'"] \arrow[d, "f'"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{X} \arrow[d, "f"] \\ M' \arrow[r, "g", two heads] & M \end{tikzcd}$$

in $\mathrm{Shv}(\mathrm{Diff})$, where the stacks M, M' and N are all smooth manifolds and g is a surjective submersion. We have to show that $\mathcal{X}$ is a smooth manifold as well.

As g is a surjective smooth submersion, it follows from Lemma C.10 that g admits local sections: there exists an open cover $\{U_i\}_{i\in I}$ of M such that for every $i \in I$ there is a smooth map $s_i \colon U_i \to M'$ such that the composite $g \circ s_i \colon U_i \to M$ is equal to the inclusion of U_i into M. By Corollary 2.1.9, it suffices to show that each of the pullback stacks $\mathcal{X}_i := \mathcal{X} \times_M U_i$ is a smooth manifold. By base changing the whole situation along the map $U_i \hookrightarrow M$, we may thus assume that the map $g \colon M' \twoheadrightarrow M$ admits a section $s \colon M \to M'$.

We may now define a map $t = (s \circ f, \mathrm{id}) \colon \mathcal{X} \to N = M' \times_M \mathcal{X}$. Since it is a section of the map $g' \colon N \to \mathcal{X}$, it follows that the stack $\mathcal{X}$ is a retract in $\mathrm{Shv}(\mathrm{Diff})$ of the smooth manifold N. It follows from idempotent completeness of Diff, Corollary C.5, that $\mathcal{X}$ is itself a smooth manifold, finishing the proof. □

Corollary 2.1.11. *Representable submersions and representable morphisms of stacks are closed under compositions.*

Proof. The statement about representable submersions is clear from the definition, since smooth submersions of smooth manifolds are closed under composition. The statement about representable morphisms follows the definition, using part (1) of Proposition 2.1.10. □

Lemma 2.1.12. *Let $f_1 \colon \mathcal{X}_1 \to \mathcal{Y}_1$ and $f_2 \colon \mathcal{X}_2 \to \mathcal{Y}_2$ be two representable submersions of stacks on* Diff. *Then their cartesian product $f_1 \times f_2 \colon \mathcal{X}_1 \times \mathcal{X}_2 \to \mathcal{Y}_1 \times \mathcal{Y}_2$ is again a representable submersion.*

Proof. Let $g = (g_1, g_2) \colon M \to \mathcal{Y}_1 \times \mathcal{Y}_2$ be a map from a smooth manifold. By assumption, the pullbacks $N_1 := M \times_{\mathcal{Y}_1} \mathcal{X}_1$ and $N_2 := M \times_{\mathcal{Y}_2} \mathcal{X}_2$ are smooth manifolds, and the maps $N_1 \to M$ and $N_2 \to M$ are smooth submersions. Now consider the following pullback diagram:

$$\begin{tikzcd} N \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & N_1 \times N_2 \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{X}_1 \times \mathcal{X}_2 \arrow[d, "f_1 \times f_2"] \\ M \arrow[r, "\Delta"] & M \times M \arrow[r, "g_1 \times g_2"] & \mathcal{Y}_1 \times \mathcal{Y}_2. \end{tikzcd}$$

Since the vertical map $N_1 \times N_2 \to M \times M$ is a submersion, the left pullback exists in the category Diff, and it follows that N is a smooth manifold. Since the bottom composite is the map $g = (g_1, g_2)$, this finishes the proof. $\square$

Lemma 2.1.13. *Let $\mathcal{X}$ and $\mathcal{Y}$ be stacks on* Diff*. Then any representable morphism $f\colon \mathcal{X} \to \mathcal{Y}$ is 0-truncated.*

Proof. A map f in a sheaf topos is 0-truncated if and only if its base change along any map $M \to \mathcal{Y}$ from a representable object M is 0-truncated. As f is representable, the pullback $N := M \times_{\mathcal{Y}} \mathcal{X}$ is a smooth manifold. It follows that the base change $N \to M$ of f is a smooth map between smooth manifolds, and in particular 0-truncated, finishing the proof. $\square$

2.2 Differentiable stacks

In this section, we introduce a special class of stacks on Diff called *differentiable stacks*, whose geometric behavior closely parallels that of smooth manifolds. These objects will play a central role throughout the entire remainder of this article.

Definition 2.2.1 (Differentiable stack)**.** Let $\mathcal{X}$ be a stack on Diff. A *representable atlas* for $\mathcal{X}$ is a smooth manifold M together with a representable surjective submersion $p\colon M \twoheadrightarrow \mathcal{X}$. A stack which admits a representable atlas is called a *differentiable stack*. We let

$$\mathrm{DiffStk} \subseteq \mathrm{Shv}(\mathrm{Diff})$$

denote the full subcategory spanned by the differentiable stacks.

Differentiable stacks are closed under a variety of constructions.

Lemma 2.2.2. *If $\mathcal{X}$ and $\mathcal{Y}$ are differentiable stacks with atlases $M \twoheadrightarrow \mathcal{X}$ and $N \twoheadrightarrow \mathcal{Y}$, then the cartesian product $\mathcal{X} \times \mathcal{Y}$ is a differentiable stack with atlas $M \times N \twoheadrightarrow \mathcal{X} \times \mathcal{Y}$.*

Proof. The product map is an effective epimorphism whose source is a smooth manifold. By Lemma 2.1.12, it is also a representable submersion, finishing the proof. $\square$

Lemma 2.2.3. *Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a representable morphism, and assume that $M \twoheadrightarrow \mathcal{Y}$ is a representable atlas for $\mathcal{Y}$. Then the pullback $N := M \times_Y \mathcal{X} \to \mathcal{X}$ is a representable atlas for $\mathcal{X}$.*

Proof. The map $N \to \mathcal{X}$ is a representable submersion and an effective epimorphism since it is a base change of the map $M \twoheadrightarrow \mathcal{Y}$. Furthermore, the assumption that f is a representable submersion guarantees that N is a smooth manifold. □

Lemma 2.2.4. *Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a representable morphism of stacks.*

(1) If $\mathcal{Y}$ is a differentiable stack, then also $\mathcal{X}$ is a differentiable stack;

(2) If f is a representable surjective submersion and $\mathcal{X}$ is a differentiable stack, then also $\mathcal{Y}$ is a differentiable stack.

Proof. Part (1) is immediate from Lemma 2.2.3. For part (2), note that any representable atlas $M \twoheadrightarrow \mathcal{X}$ of $\mathcal{X}$ gives a representable atlas for $\mathcal{Y}$ by postcomposing with f. □

As a consequence, we see that differentiable stacks are closed under pullbacks along representable submersions on Shv(Diff):

Corollary 2.2.5. *Consider a pullback square*

$$\begin{tikzcd} \mathcal{X}' \arrow[r, "g'"] \arrow[d, "f'"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{X} \arrow[d, "f"] \\ \mathcal{Y}' \arrow[r, "g"] & \mathcal{Y} \end{tikzcd}$$

of stacks on Diff. *Assume that $\mathcal{X}$, $\mathcal{Y}$ and $\mathcal{Y}'$ are differentiable stacks and that f is a representable submersion. Then $\mathcal{X}'$ is also a differentiable stack.*

Proof. By Proposition 2.1.10, the morphism f' is a representable submersion, hence the claim follows from Lemma 2.2.4. □

Lemma 2.2.6. *Let $\mathcal{X}$ be a differentiable stack with representable atlas $M \twoheadrightarrow \mathcal{X}$. Then a morphism $f\colon \mathcal{Y} \to \mathcal{X}$ is representable if and only if the pullback $M \times_{\mathcal{X}} \mathcal{Y}$ is a smooth manifold. Furthermore, f is a representable submersion if and only if its base change $M \times_{\mathcal{X}} \mathcal{Y} \to M$ of f is a smooth submersion of smooth manifolds.*

Proof. This is immediate from Proposition 2.1.10, using that any morphism between smooth manifolds is representable. □

Corollary 2.2.7. *Let $\mathcal{Y}$ be a differentiable stack. Then the diagonal $\Delta\colon \mathcal{Y} \to \mathcal{Y} \times \mathcal{Y}$ is representable.*

Proof. Let $M \twoheadrightarrow \mathcal{Y}$ be a representable atlas for $\mathcal{Y}$. By Lemma 2.2.2, the product map $M \times M \twoheadrightarrow \mathcal{Y} \times \mathcal{Y}$ is again a representable atlas. By Lemma 2.2.6, it thus suffices to show that the differentiable stack $(M \times M) \times_{\mathcal{Y}\times\mathcal{Y}} \mathcal{Y} \cong M \times_{\mathcal{Y}} M$ is a smooth manifold. This is immediate since the map $M \twoheadrightarrow \mathcal{Y}$ is a representable submersion. □

Corollary 2.2.8. *Let $\mathcal{X}$ be a differentiable stack. Then any morphism $N \to \mathcal{X}$ from a smooth manifold N is representable.*

Proof. Let $M \twoheadrightarrow \mathcal{Y}$ be a representable atlas for $\mathcal{X}$. By Lemma 2.2.6 it suffices to show that the pullback $M \times_{\mathcal{X}} N$ is a smooth manifold. This is immediate from the fact that the map $M \twoheadrightarrow \mathcal{X}$ is a representable submersion. □

2.3 Classifying stacks of Lie groupoids

A rich supply of differentiable stacks is provided by the *classifying stacks* of Lie groupoids. In fact, we will show in Corollary 2.3.8 below that every differentiable stack is (non-canonically) equivalent to the classifying stack $\mathbb{B}\mathcal{G}$ of some Lie groupoid $\mathcal{G}$.

We refer to Appendix D for background material on Lie groupoids. We will also make use of some foundational results on the relation between groupoid objects and effective epimorphisms in an ∞-topos, recalled in Appendix E.2.

Definition 2.3.1 (Nerve of a Lie groupoid)**.** The *nerve* $N\mathcal{G}$ of a Lie groupoid $\mathcal{G} = (\mathcal{G}_1 \rightrightarrows \mathcal{G}_0)$ is the simplicial manifold $N\mathcal{G}\colon \Delta^{\mathrm{op}} \to \mathrm{Diff}$ whose n-simplices are given by the manifold

$$\mathcal{G}_n := \mathcal{G}_1 \times_{s,\mathcal{G}_0,t} \cdots \times_{s,\mathcal{G}_0,t} \mathcal{G}_1,$$

given as the n-fold iterated fiber product of $\mathcal{G}_1$ over $\mathcal{G}_0$ in Diff. The face maps of $N\mathcal{G}$ are obtained from source, target and multiplication maps of $\mathcal{G}$, while the degeneracy maps of $N\mathcal{G}$ are obtained from the unit map of $\mathcal{G}$.

Definition 2.3.2 (Classifying stack)**.** Let $\mathcal{G} = (\mathcal{G}_1 \rightrightarrows \mathcal{G}_0)$ be a Lie groupoid. We define its *classifying stack* $\mathbb{B}\mathcal{G} \in \mathrm{Shv}(\mathrm{Diff})$ as the geometric realization of its nerve in Shv(Diff):

$$\mathbb{B}\mathcal{G} := |N\mathcal{G}| = \mathrm{colim}\left(\cdots \; \mathcal{G}_2 \;\substack{\longrightarrow\\\longrightarrow\\\longrightarrow}\; \mathcal{G}_1 \;\rightrightarrows\; \mathcal{G}_0 \right) \quad \in \quad \mathrm{Shv}(\mathrm{Diff}).$$

The classifying stack $\mathbb{B}\mathcal{G}$ of a Lie groupoid is indeed a differentiable stack:

Lemma 2.3.3. *Let $\mathcal{G}$ be a Lie groupoid. Then the canonical map $\mathcal{G}_0 \twoheadrightarrow \mathbb{B}\mathcal{G}$ is a representable atlas for $\mathbb{B}\mathcal{G}$. In particular, $\mathbb{B}\mathcal{G}$ is a differentiable stack.*

Proof. As groupoid objects in ∞-topoi are effective, the map $\mathcal{G}_0 \twoheadrightarrow \mathbb{B}\mathcal{G}$ is an effective epimorphism, see e.g. Corollary E.15. Since $\mathcal{G}_0$ is a smooth manifold, it remains to show that the map $\mathcal{G}_0 \twoheadrightarrow \mathbb{B}\mathcal{G}$ is a representable submersion. By Proposition 2.1.10, it suffices to show that this is the case after pulling it back along an effective epimorphism, for example along $\mathcal{G}_0 \twoheadrightarrow \mathbb{B}\mathcal{G}$ itself. But this base change $\mathrm{pr}_1 \colon \mathcal{G}_0 \times_{\mathbb{B}\mathcal{G}} \mathcal{G}_0 \to \mathcal{G}_0$ is equivalent to the map source map $s \colon \mathcal{G}_1 \to \mathcal{G}_0$, which by assumption on $\mathcal{G}$ is a submersion of smooth manifolds and hence a representable submersion in $\mathrm{Shv}(\mathrm{Diff})$. $\square$

Example 2.3.4 (Classifying stack of Lie group)**.** Every Lie group G can be regarded as a one-object Lie groupoid $G \rightrightarrows \mathrm{pt}$ and thus gives rise to a differentiable stack $\mathbb{B}G$ called the *classifying stack of G*.

Example 2.3.5 (Global quotient stack)**.** Let M be a smooth manifold and let G be a Lie group acting smoothly on M. We obtain a Lie groupoid $G \ltimes M = (G \times M \rightrightarrows M)$, called the *action groupoid*. The associated differentiable stack $\mathbb{B}(G \ltimes M)$ is denoted by $M//G$ and is called the *quotient stack* of M by G. Since the action groupoid $G \ltimes M$ is functorial in M, we obtain a functor

$$-//G \colon \mathrm{Diff}_G \to \mathrm{DiffStk},$$

where Diff_G denotes the category of smooth G-manifolds. Note that the quotient stack $\mathrm{pt}//G$ of the point is precisely the classifying stack $\mathbb{B}G$ of G.

A differentiable stack $\mathcal{X}$ is called a *global quotient stack* if it is equivalent to a quotient stack $M//G$ for some *compact* Lie group G and some smooth G-manifold. We let $\mathrm{QtStk} \subseteq \mathrm{DiffStk}$ denote the full subcategory of global quotient stacks.

By Lemma 2.3.3, every Lie groupoid gives rise to a differentiable stack. We will now show that, conversely, every differentiable stack $\mathcal{X}$ can be presented by a Lie groupoid $\mathcal{G}$, in the sense that there exists an equivalence $\mathcal{X} \simeq \mathbb{B}\mathcal{G}$ of stacks. In fact, we will see that there is a one-to-one correspondence between choices of an atlas for $\mathcal{X}$ and choices of a presentation of $\mathcal{X}$ by a Lie groupoid.

Definition 2.3.6. We denote by $\mathrm{Atl}^{\mathrm{rep}}(\mathrm{Shv}(\mathrm{Diff})) \subseteq \mathrm{Fun}([1], \mathrm{Shv}(\mathrm{Diff}))$ the full subcategory of the arrow category of $\mathrm{Shv}(\mathrm{Diff})$ spanned by the representable atlases $M \twoheadrightarrow \mathcal{X}$ in the sense of Definition 2.2.1.

Proposition 2.3.7. *Sending a Lie groupoid $\mathcal{G}$ to the representable atlas $\mathcal{G}_0 \twoheadrightarrow \mathbb{B}\mathcal{G}$ defines an equivalence of ∞-categories* $\mathrm{LieGrpd} \xrightarrow{\sim} \mathrm{Atl}^{\mathrm{rep}}(\mathrm{Shv}(\mathrm{Diff}))$.

Proof. Recall from Proposition E.13 the equivalence

$$\mathrm{Grpd}(\mathrm{Shv}(\mathrm{Diff})) \xrightarrow{\sim} \mathrm{Atl}(\mathrm{Shv}(\mathrm{Diff}))$$

between the ∞-category of groupoid objects in Shv(Diff) and the ∞-category of atlases (i.e. effective epimorphisms) in Shv(Diff). This equivalence is given by sending a groupoid object $\mathcal{G}$ to the effective epimorphism $\mathcal{G}_0 \twoheadrightarrow \mathbb{B}\mathcal{G}$, and its inverse sends an effective epimorphism to its Čech nerve. We claim that this equivalence restricts to the desired equivalence $\mathrm{LieGrpd} \xrightarrow{\sim} \mathrm{Atl}^{\mathrm{rep}}(\mathrm{Shv}(\mathrm{Diff}))$ of full subcategories. By Lemma 2.3.3, the map $\mathcal{G}_0 \twoheadrightarrow \mathbb{B}\mathcal{G}$ is indeed a representable atlas if $\mathcal{G}$ is a Lie groupoid. Conversely, if $p\colon M \twoheadrightarrow \mathcal{X}$ is a representable atlas, its Čech nerve $\check{C}(p)$ is given at level $[n] \in \Delta$ by the n-fold fiber product

$$\check{C}(p)_n \simeq M \times_{\mathcal{X}} M \times_{\mathcal{X}} \cdots \times_{\mathcal{X}} M,$$

which is a smooth manifold by the assumptions that M is a manifold and that the map p is a representable submersion. It follows that $\check{C}(p)$ is the nerve of a Lie groupoid $\mathcal{G} = (M \times_{\mathcal{X}} M \rightrightarrows M)$, finishing the proof. □

Corollary 2.3.8. *Every differentiable stack $\mathcal{X}$ is equivalent to the classifying stack $\mathbb{B}\mathcal{G}$ of some Lie groupoid $\mathcal{G}$.*

Proof. Let $M \twoheadrightarrow \mathcal{X}$ be a representable atlas for $\mathcal{X}$. By Proposition 2.3.7, there exists a Lie groupoid $\mathcal{G}$ with $M \cong \mathcal{G}_0$ and $\mathcal{X} \simeq \mathbb{B}\mathcal{G}$. This finishes the proof. □

Warning 2.3.9. Although we may think of Lie groupoids as *presentations* of differentiable stacks, it is not true that a morphism of Lie groupoids $\mathcal{H} \to \mathcal{G}$ contains the same data as a morphism $\mathbb{B}\mathcal{H} \to \mathbb{B}\mathcal{G}$ of stacks. Indeed, as made precise by Proposition 2.3.7, morphisms of stacks $\mathbb{B}\mathcal{H} \to \mathbb{B}\mathcal{G}$ comes from a morphism of Lie groupoids $\mathcal{H} \to \mathcal{G}$ precisely if it is compatible with the preferred atlases $\mathcal{H}_0 \twoheadrightarrow \mathbb{B}\mathcal{H}$ and $\mathcal{G}_0 \twoheadrightarrow \mathbb{B}\mathcal{G}$.

2.3.1 Differentiable stacks are Lie groupoids up to Morita equivalence

We will now recall the well-known fact that two Lie groupoids $\mathcal{H}$ and $\mathcal{G}$ have equivalent classifying stacks if and only if they are Morita equivalent, in the sense of Definition D.18.

We start by characterizing when a morphism of Lie groupoids $f\colon \mathcal{H} \to \mathcal{G}$ induces an equivalence of differentiable stacks $\mathbb{B}f\colon \mathbb{B}\mathcal{H} \xrightarrow{\sim} \mathbb{B}\mathcal{G}$. For this, we need to recall the notions of fully faithfulness and essential surjectivity of morphisms of Lie groupoids.

Definition 2.3.10 ([HF19, Section 6.1])**.** Let $f\colon \mathcal{H} \to \mathcal{G}$ be a morphism of Lie groupoids.

(1) The morphism f is called *fully faithful* if the square

$$\begin{tikzcd} \mathcal{H}_1 \arrow[r, "f_1"] \arrow[d, "{(s,t)}"'] & \mathcal{G}_1 \arrow[d, "{(s,t)}"] \\ \mathcal{H}_0 \times \mathcal{H}_0 \arrow[r, "f_0 \times f_0"] & \mathcal{G}_0 \times \mathcal{G}_0 \end{tikzcd}$$

is a pullback square.

(2) The morphism f is called *essentially surjective* if the composite map

$$\mathcal{G}_1 \times_{\mathcal{G}_0} \mathcal{H}_0 \xrightarrow{\mathrm{pr}_1} \mathcal{G}_1 \xrightarrow{t} \mathcal{G}_0$$

is a surjective submersion;

(3) The morphism f is called a *Morita map* if it is both fully faithful and essentially surjective.

(4) The morphism f is called *strongly surjective* if the smooth map $f_0 \colon \mathcal{H}_0 \to \mathcal{G}_0$ is a surjective submersion.

(5) The morphism f is called a *Morita fibration* if it is both fully faithful and strongly surjective.

Warning 2.3.11. The conventions on the terminology of Morita maps are not very consistent in the literature: the notion we call 'Morita map' appears in the literature under the names 'equivalence' [Moe02, Definition 2.4], 'weak equivalence' [Pro96, Definition 12], 'essential equivalence' [Met03, Definition 58] and 'Morita equivalence' [Car11, Definition I.2.20]. Morita fibrations also appear under the names 'Morita morphisms' [BX11, Definition 2.24] and 'elementary Morita equivalences' [Noo05, Section 8]. Our convention follows [HF19, Section 6.1].

Remark 2.3.12 (Morita fibrations are Morita maps)**.** Every strongly surjective morphism f of Lie groupoids is essentially surjective, and thus every Morita fibration is a Morita map. Indeed, the condition in (4) that the map f_0 is a surjective submersion implies that the projection map $\mathrm{pr}_1 \colon \mathcal{G}_1 \times_{\mathcal{G}_0} \mathcal{H}_0 \to \mathcal{G}_1$ is a surjective submersion, as it is a base change of f_0. Since the target map $t \colon \mathcal{G}_1 \to \mathcal{G}_0$ is always a surjective submersion, it follows that $t \circ \mathrm{pr}_1$ is also a surjective submersion, which is condition (2).

The following result characterizes the properties of fully faithfulness and essential surjectivity of a morphism of Lie groupoids in terms of its underlying map of differentiable stacks.

Lemma 2.3.13 ([Met03, Proposition 60], [Car11, Proposition I.2.5])**.** *Let $f\colon \mathcal{H} \to \mathcal{G}$ be a morphism of Lie groupoids, and consider the induced map $\mathbb{B}f\colon \mathbb{B}\mathcal{H} \to \mathbb{B}\mathcal{G}$ on differentiable stacks.*

(1) The map $\mathbb{B}f$ is a monomorphism if and only if f is fully faithful;

(2) The map $\mathbb{B}f$ is an representable surjective submersion if and only if f is essentially surjective;

(3) The map $\mathbb{B}f$ is an equivalence if and only if f is a Morita map.

Proof. For part (1), consider the following diagram:

$$\begin{tikzcd}
\mathcal{H}_1 \arrow[rr, "f_1"] \arrow[dr] \arrow[dd, "{(s,t)}"'] & & \mathcal{G}_1 \arrow[dr] \arrow[dd, "{(s,t)}" near start] & \\
& \mathbb{B}\mathcal{H} \arrow[rr, crossing over, "\mathbb{B}f" near start] & & \mathbb{B}\mathcal{G} \arrow[dd, "\Delta"] \\
\mathcal{H}_0 \times \mathcal{H}_0 \arrow[rr, "f_0 \times f_0" near end] \arrow[dr, two heads] & & \mathcal{G}_0 \times \mathcal{G}_0 \arrow[dr, two heads] & \\
& \mathbb{B}\mathcal{H} \times \mathbb{B}\mathcal{H} \arrow[rr, "\mathbb{B}f \times \mathbb{B}f"] \arrow[from=uu, crossing over, "\Delta" near start] & & \mathbb{B}\mathcal{G} \times \mathbb{B}\mathcal{G}.
\end{tikzcd}$$

The map $\mathbb{B}f\colon \mathbb{B}\mathcal{H} \to \mathbb{B}\mathcal{G}$ is a monomorphism if and only if the front square of the diagram is a pullback square. Since the map $\mathcal{H}_0 \twoheadrightarrow \mathbb{B}\mathcal{H}$ is an effective epimorphism and the left and right faces of the cube are pullback squares, it follows from the pasting lemma for pullback squares that the front square is a pullback square if and only if the back square is. As the back square defines fully faithfulness of f, this finishes the proof of (1)

For part (2), consider the following pullback diagram:

$$\begin{tikzcd}
\mathcal{G}_1 \times_{\mathcal{G}_0} \mathcal{H}_0 \arrow[r, "\mathrm{pr}_1"] \arrow[d, "\mathrm{pr}_2"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{G}_1 \arrow[r, "t"] \arrow[d, "s"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{G}_0 \arrow[d, two heads] \\
\mathcal{H}_0 \arrow[r, "f_0"] & \mathcal{G}_0 \arrow[r] & \mathbb{B}\mathcal{G}.
\end{tikzcd}$$

The bottom composite $\mathcal{H}_0 \to \mathcal{G}_0 \to \mathbb{B}\mathcal{G}$ is equivalent to the composite $\mathcal{H}_0 \twoheadrightarrow \mathbb{B}\mathcal{H} \xrightarrow{\mathbb{B}f} \mathbb{B}\mathcal{G}$. As the map $\mathcal{H}_0 \twoheadrightarrow \mathbb{B}\mathcal{H}$ is a representable surjective submersion, it follows from part (4) of Lemma C.7 that the map $\mathbb{B}f$ is a representable surjective submersion if and only if the bottom composite $\mathcal{H}_0 \to \mathbb{B}\mathcal{G}$ is. Since this may be checked after pulling back along the atlas $\mathcal{G}_0 \twoheadrightarrow \mathbb{B}\mathcal{G}$, this is equivalent to the condition that the top composite $t \circ \mathrm{pr}_1\colon \mathcal{G}_1 \times_{\mathcal{G}_0} \mathcal{H}_0 \to \mathcal{G}_0$

is a surjective submersion. Since this is the definition of fully faithfulness of f, this finishes the proof of (2).

Part (3) is a direct consequence of (1) and (2) since a morphism of stacks is an equivalence if and only if it is both a monomorphism and a representable surjective submersion. $\square$

We deduce that two Lie groupoids have equivalent classifying stacks if and only if they are Morita equivalent:

Proposition 2.3.14 (cf. [Pro96], [BX11, Theorem 2.26], [Met03])**.** *Given two Lie groupoids $\mathcal{H}$ and $\mathcal{G}$, the following conditions are equivalent:*

(1) The classifying stacks $\mathbb{B}\mathcal{H}$ and $\mathbb{B}\mathcal{G}$ are equivalent;

(2) The Lie groupoids $\mathcal{H}$ and $\mathcal{G}$ are Morita equivalent, in the sense of Definition D.18;

(3) There exists a third Lie groupoid $\mathcal{K}$ along with Morita fibrations $\mathcal{K} \to \mathcal{H}$ and $\mathcal{K} \to \mathcal{G}$;

(4) There exists a third Lie groupoid $\mathcal{K}$ along with Morita maps $\mathcal{K} \to \mathcal{H}$ and $\mathcal{K} \to \mathcal{G}$.

Proof. To prove that (1) implies (2), assume that an equivalence of stacks $\mathbb{B}\mathcal{H} \simeq \mathbb{B}\mathcal{G}$ has been given. It follows that the composite $\mathcal{H}_0 \twoheadrightarrow \mathbb{B}\mathcal{H} \simeq \mathbb{B}\mathcal{G}$ is a representable atlas for $\mathbb{B}\mathcal{G}$. We define a smooth manifold P as the following pullback:

$$\begin{tikzcd} P \arrow[r, two heads, "\beta"] \arrow[d, two heads, "\alpha"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{H}_0 \arrow[d, two heads] \\ \mathcal{G}_0 \arrow[r, two heads] & \mathbb{B}\mathcal{G}. \end{tikzcd}$$

Observe that the map $\alpha\colon P \twoheadrightarrow \mathcal{G}_0$ is a principal $\mathcal{H}$-bundle, in the sense of Definition D.15, as it is a base change of the principal $\mathcal{H}$-bundle $\mathcal{H}_0 \twoheadrightarrow \mathbb{B}\mathcal{G}$. Similarly the map $\beta\colon P \twoheadrightarrow \mathcal{H}_0$ is a principal $\mathcal{G}$-bundle. It is immediate that the actions of $\mathcal{G}$ and $\mathcal{H}$ commute as they act on two separate components of $P = \mathcal{G}_0 \times_{\mathbb{B}\mathcal{G}} \mathcal{H}_0$, and thus the triple (P, α, β) forms a Morita equivalence between $\mathcal{G}$ and $\mathcal{H}$.

To prove that (2) implies (3), assume given a Morita equivalence (P, α, β) between $\mathcal{G}$ and $\mathcal{H}$, in the sense of Definition D.18. We start by defining a Lie groupoid $\mathcal{K}$. Set $\mathcal{K}_0 = P$ and define $\mathcal{K}_1$ via the following pullback diagram:

$$\begin{tikzcd} \mathcal{K}_1 \arrow[r, "(\alpha_1, \beta_1)"] \arrow[d, "s"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{G}_1 \times \mathcal{H}_1 \arrow[d, "s \times s"] \\ P \arrow[r, "(\alpha, \beta)"] & \mathcal{G}_0 \times \mathcal{H}_0. \end{tikzcd}$$

The commuting actions of $\mathcal{G}$ and $\mathcal{H}$ on P define an action map $t\colon \mathcal{K}_1 \to P$ and one checks that the maps $s,t\colon \mathcal{K}_1 \to P = \mathcal{K}_0$ are part of a Lie groupoid structure $\mathcal{K} = (\mathcal{K}_1 \overset{s}{\underset{t}{\rightrightarrows}} \mathcal{K}_0)$. The Lie groupoid $\mathcal{K}$ comes equipped with morphisms of Lie groupoids $(\alpha_1,\alpha)\colon \mathcal{K} \to \mathcal{G}$ and $(\beta_1,\beta)\colon \mathcal{K} \to \mathcal{H}$. We claim that these are Morita fibrations, thus finishing the proof of (3). By symmetry, it suffices to check this for (α_1,α). Strong surjectivity is clear, as the map $\alpha\colon \mathcal{K}_0 = P \to \mathcal{G}_0$ is a surjective submersion by assumption. For fully faithfulness, we must show that the map

$$\mathcal{K}_1 \to \mathcal{G}_1 \times_{\mathcal{G}_0\times\mathcal{G}_0} (P\times P)$$

is a diffeomorphism. To provide an inverse, consider the shear map $\mathrm{shear}\colon \mathcal{H}_1 \times_{\mathcal{H}_0} P \to P \times_{\mathcal{G}_0} P$, which is a diffeomorphism as $\alpha\colon P \to \mathcal{G}_0$ is a principal $\mathcal{H}$-bundle. An inverse to the above map is then given by

$$\mathcal{G}_1 \times_{\mathcal{G}_0\times\mathcal{G}_0} (P\times P) \to \mathcal{K}_1 = (\mathcal{G}_1 \times \mathcal{H}_1) \times_{\mathcal{G}_0\times\mathcal{H}_0} P, \quad (g,p,p') \mapsto (g, \mathrm{shear}^{-1}(p,p')),$$

giving the desired claim.

It is immediate from Remark 2.3.12 that (3) implies (4).

Finally, to see that (4) implies (1), observe that the Morita maps $\mathcal{K} \to \mathcal{G}$ and $\mathcal{K} \to \mathcal{H}$ induce equivalences of stacks

$$\mathbb{B}\mathcal{H} \xleftarrow{\sim} \mathbb{B}\mathcal{K} \xrightarrow{\sim} \mathbb{B}\mathcal{G}$$

by Lemma 2.3.13. It follows that $\mathbb{B}\mathcal{H}$ and $\mathbb{B}\mathcal{G}$ are equivalent, as desired. □

As shown above, the classifying stack functor $\mathbb{B}\colon \mathrm{LieGrpd} \to \mathrm{DiffStk}$ sends Morita maps to equivalences. A well-known result in the theory of differentiable stacks says that this functor is in fact *universal* with this property. For completeness we will state the precise result, although we shall not make use of it.

Theorem 2.3.15 ([Pro96, Section 3.4, Section 4])**.** *The functor* $\mathbb{B}\colon \mathrm{LieGrpd} \to \mathrm{DiffStk}$ *exhibits the* $(2,1)$*-category*[3] DiffStk *as the* $(2,1)$*-categorical Dwyer-Kan localization of the category* LieGrpd *at the Morita maps: for any other* $(2,1)$*-category* $\mathcal{D}$*, precomposition with* $\mathbb{B}$ *induces a fully faithful functor*

$$\mathrm{Fun}(\mathrm{DiffStk},\mathcal{D}) \hookrightarrow \mathrm{Fun}(\mathrm{LieGrpd},\mathcal{D})$$

whose essential image consists of those functors $\mathrm{LieGrpd} \to \mathcal{D}$ *which send Morita maps to equivalences.*

Remark 2.3.16. One can prove that the functor $\mathbb{B}\colon \mathrm{LieGrpd} \to \mathrm{DiffStk}$ is in fact an ∞-categorical Dwyer-Kan localization, see [Nui16].

[3] See Corollary 2.3.24 below.

2.3.2 Classification of Lie groupoid actions on smooth manifolds

Given a Lie groupoid $\mathcal{G}$, recall from Definition D.14 the notion of a *left action* of $\mathcal{G}$ on a smooth manifold: given a smooth submersion $M \to \mathcal{G}_0$, a left $\mathcal{G}$-action on M consists of an associative and unital action $a\colon \mathcal{G}_1 \times_{s,\mathcal{G}_0} M \to M$ over $\mathcal{G}_0$. A smooth manifold equipped with a left $\mathcal{G}$-action is called a *smooth $\mathcal{G}$-manifold*. In this subsection, we show that such left actions are classified by representable maps of stacks to the classifying stack $\mathbb{B}\mathcal{G}$. More precisely, we show that the category $\mathrm{Diff}_{\mathcal{G}}$ of smooth $\mathcal{G}$-manifolds is equivalent to the subcategory of $\mathrm{DiffStk}_{/\mathbb{B}\mathcal{G}}$ spanned by the representable submersions into $\mathbb{B}\mathcal{G}$, where the equivalence sends M to the map $M//\mathcal{G} \to \mathrm{pt}//\mathcal{G} = \mathbb{B}\mathcal{G}$.

Definition 2.3.17. Given a smooth $\mathcal{G}$-manifold M, we obtain a Lie groupoid

$$\mathcal{G} \ltimes M = \left(\mathcal{G}_1 \times_{\mathcal{G}_0} M \overset{\mathrm{pr}_2}{\underset{a}{\rightrightarrows}} M \right)$$

called the *action groupoid* of M. We define the *quotient stack* $M//\mathcal{G} \in \mathrm{Shv}(\mathrm{Diff})$ as the classifying stack $\mathbb{B}(\mathcal{G} \ltimes M)$. The Lie groupoid $\mathcal{G} \ltimes M$ comes equipped with a morphism of Lie groupoids to $\mathcal{G}$:

$$\begin{tikzcd}
\mathcal{G}_1 \times_{\mathcal{G}_0} M \arrow[r, shift left, "\mathrm{pr}_2"] \arrow[r, shift right, "a"'] \arrow[d, "\mathrm{pr}_1"'] & M \arrow[d] \\
\mathcal{G}_1 \arrow[r, shift left, "s"] \arrow[r, shift right, "t"'] & \mathcal{G}_0.
\end{tikzcd}$$

The notion of left actions of Lie groupoids on smooth manifolds may be phrased in terms of groupoid actions in the ∞-topos $\mathrm{Shv}(\mathrm{Diff})$ of stacks on Diff. Recall from Definition E.17 that a left action of a groupoid object $\mathcal{G}$ in an ∞-topos $\mathcal{B}$ consists of another groupoid object $\mathcal{H}$ in $\mathcal{B}$ equipped with a *cartesian* morphism $c\colon \mathcal{H} \to \mathcal{G}$ of simplicial objects in $\mathcal{B}$, meaning that all the naturality squares of c are pullback squares. In the case at hand, observe that the map $\mathcal{G} \ltimes M \to \mathcal{G}$ of Lie groupoids from Definition 2.3.17 is a cartesian morphism of groupoid objects in $\mathrm{Shv}(\mathrm{Diff})$, thus producing a functor

$$\mathcal{G} \ltimes -\colon \mathrm{Diff}_{\mathcal{G}} \to \mathrm{Act}_{\mathcal{G}}(\mathrm{Shv}(\mathrm{Diff}))$$

from the category of smooth $\mathcal{G}$-manifolds to the ∞-category of stacks on Diff equipped with a $\mathcal{G}$-action.

Lemma 2.3.18. *For a Lie groupoid $\mathcal{G}$, the functor $\mathcal{G} \ltimes -\colon \mathrm{Diff}_{\mathcal{G}} \to \mathrm{Act}_{\mathcal{G}}(\mathrm{Shv}(\mathrm{Diff}))$ is fully faithful. An object $c\colon \mathcal{H} \to \mathcal{G}$ of $\mathrm{Act}_{\mathcal{G}}(\mathrm{Shv}(\mathrm{Diff}))$ lies inside the essential image of this functor if and only if $\mathcal{H}_0 \in \mathrm{Shv}(\mathrm{Diff})$ is a smooth manifold and the map $\mathcal{H}_0 \to \mathcal{G}_0$ is a smooth submersion.*

Proof. Notice that a morphism $\varphi\colon M \to M'$ of smooth $\mathcal{G}$-manifolds determines and is determined by the induced morphism $\mathcal{G} \ltimes M \to \mathcal{G} \ltimes M'$ of Lie groupoids over $\mathcal{G}$, so that this functor is fully faithful. Furthermore, if $c\colon \mathcal{H} \to \mathcal{G}$ is a cartesian morphism of groupoid objects of $\mathrm{Shv}(\mathrm{Diff})$ such that $\mathcal{H}_0$ is a smooth manifold, then it follows that $\mathcal{H}_n \simeq \mathcal{H}_0 \times_{\mathcal{G}_0} \mathcal{G}_n$ is a smooth manifold for all $n \geq 0$, so that $\mathcal{H}$ is in fact a Lie groupoid. Defining $M := \mathcal{H}_0$, the map $d_1\colon \mathcal{G}_1 \times_{\mathcal{G}_0} M \simeq \mathcal{H}_1 \to \mathcal{H}_0 = M$ determines an action of $\mathcal{G}$ on M and one observes that $\mathcal{H}$ is equivalent to the action groupoid $\mathcal{G} \ltimes M$, showing that $\mathcal{H}$ lies in the essential image. □

Corollary 2.3.19. *Let $\mathcal{G}$ be a Lie groupoid. Sending a smooth $\mathcal{G}$-manifold M to the map $M//\mathcal{G} \to \mathbb{B}\mathcal{G}$ defines a fully faithful functor*

$$\mathrm{Diff}_{\mathcal{G}} \hookrightarrow \mathrm{DiffStk}_{/\mathbb{B}\mathcal{G}}$$

whose essential image consists of the representable submersions $\mathcal{X} \to \mathbb{B}\mathcal{G}$.

Proof. By Proposition E.20, there is an equivalence of ∞-categories

$$\mathrm{Act}_{\mathcal{G}}(\mathrm{Shv}(\mathrm{Diff})) \xrightarrow{\sim} \mathrm{Shv}(\mathrm{Diff})_{/\mathbb{B}\mathcal{G}}$$

given by sending a cartesian morphism $\varphi\colon \mathcal{H} \to \mathcal{G}$ of groupoid objects in $\mathrm{Shv}(\mathrm{Diff})$ to the map $\mathbb{B}\varphi\colon \mathbb{B}\mathcal{H} \to \mathbb{B}\mathcal{G}$. Since there is a cartesian square

$$\begin{tikzcd}
\mathcal{H}_0 \arrow[r, two heads] \arrow[d, "\varphi_0"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathbb{B}\mathcal{G} \arrow[d, "\mathbb{B}\varphi"] \\
\mathcal{G}_0 \arrow[r, two heads] & \mathbb{B}\mathcal{G},
\end{tikzcd}$$

it follows from Proposition 2.1.10 that the map φ_0 is a representable submersion if and only if $\mathbb{B}\varphi$ is a representable submersion. It follows from Lemma 2.3.18 that the subcategory $\mathrm{Diff}_{\mathcal{G}}$ of $\mathrm{Act}_{\mathcal{G}}(\mathrm{Shv}(\mathrm{Diff}))$ corresponds to the full subcategory of $\mathrm{Shv}(\mathrm{Diff})_{/\mathbb{B}\mathcal{G}}$ spanned by the representable submersions. As the source of every representable submersion $\mathcal{X} \to \mathbb{B}\mathcal{G}$ is a differentiable stack by Lemma 2.2.4, the claim follows. □

Corollary 2.3.20. *Let G be a Lie group. The assignment $M \mapsto (M//G \to \mathbb{B}G)$ defines a fully faithful functor*

$$-//G\colon \mathrm{Diff}_G \hookrightarrow \mathrm{DiffStk}_{/\mathbb{B}G},$$

whose essential image consists of the representable morphisms $\mathcal{X} \to \mathbb{B}G$.

Proof. Given Corollary 2.3.19, it remains to show that a morphism $f\colon \mathcal{X} \to \mathbb{B}G$ of differentiable stacks is representable if and only if it is a representable submersion. One direction is immediate, so assume that $\mathcal{X} \to \mathbb{B}G$ is representable. Consider the following pullback square:

$$\begin{tikzcd} M \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{X} \arrow[d, "f"] \\ \mathrm{pt} \arrow[r, two heads] & \mathbb{B}G. \end{tikzcd}$$

As the map $\mathrm{pt} \twoheadrightarrow \mathbb{B}G$ is a representable submersion, it follows that the pullback M is a smooth manifold. In particular, the map $M \to \mathrm{pt}$ is a smooth submersion. By Lemma 2.2.6, it follows that also $f: \mathcal{X} \to \mathbb{B}G$ is a representable submersion, finishing the proof. $\square$

Corollary 2.3.21. *A differentiable stack $\mathcal{X}$ is a global quotient stack in the sense of Example 2.3.5 if and only if it admits a representable submersion $\mathcal{X} \to \mathbb{B}G$ for some compact Lie group G.* $\square$

2.3.3 Classification of principal bundles on smooth manifolds

Let $\mathcal{G}$ be a Lie groupoid and let N be a smooth manifold. Recall from Definition D.15 the notion of a *smooth principal $\mathcal{G}$-bundle on N*: a surjective smooth submersion $p: P \twoheadrightarrow N$ equipped with a left $\mathcal{G}$-action $a: \mathcal{G}_1 \times_{\mathcal{G}_0} P \to P$ living over N for which the shear map $(a, \mathrm{pr}_2): \mathcal{G}_1 \times_{\mathcal{G}_0} P \to P \times_N P$ is a diffeomorphism. The goal of this subsection is to give a classification of smooth principal $\mathcal{G}$-bundles on N in terms of morphisms $N \to \mathbb{B}\mathcal{G}$ of differentiable stacks: there is an equivalence of ∞-groupoids

$$\mathrm{Hom}_{\mathrm{DiffStk}}(N, \mathbb{B}\mathcal{G}) \simeq \{\text{Smooth principal } \mathcal{G}\text{-bundles on } N\}.$$

Since the right-hand side is simply a 1-groupoid, it follows in particular that the stack $\mathbb{B}\mathcal{G}$ is a 1-stack, making our definition of differentiable stacks compatible with the standard literature.

We start by comparing the explicit definition of smooth principal bundles for Lie groupoids to the general topos-theoretic notion of principal bundles, recalled in Appendix E.3.

Proposition 2.3.22. *Let $\mathcal{G}$ be a Lie groupoid en let N be a smooth manifold. Then there is an equivalence*

$$\mathrm{PrnBdl}_{\mathcal{G}}(N) \xrightarrow{\sim} \mathrm{PrnBdl}_{\mathcal{G}}(\mathrm{Shv}(\mathrm{Diff}))_N$$

between the category of smooth principal $\mathcal{G}$-bundles over N, as defined in Definition D.15, and the ∞-category of principal $\mathcal{G}$-bundles over N internal to the ∞-topos Shv(Diff)*, as defined in Definition E.24.*

Proof. For a smooth principal $\mathcal{G}$-bundle $p\colon P \to N$, the action groupoid $\mathcal{G} \ltimes P$ is by assumption (2) of Definition D.15 a groupoid object in the slice $\mathrm{Diff}_{/N}$. The groupoid map $\mathcal{G} \ltimes P \to \mathcal{G}$ lifts by adjunction to a groupoid map $\mathcal{G} \ltimes P \to \mathcal{G} \times N$ in the slice $\mathrm{Diff}_{/N}$. This map is still a cartesian natural transformation, so forms an object of $\mathrm{Act}_{\mathcal{G}}(\mathrm{Diff}_{/N}) \subseteq \mathrm{Act}_{\mathcal{G}}(\mathrm{Shv}(\mathrm{Diff})_{/N})$. This produces a functor

$$\mathrm{PrnBdl}_{\mathcal{G}}(N) \to \mathrm{Act}_{\mathcal{G}}(\mathrm{Shv}(\mathrm{Diff})_{/N}).$$

Since a map $P \to P'$ of smooth principal $\mathcal{G}$-bundles over N is equivalently specified by a map $\mathcal{G} \ltimes P \to \mathcal{G} \ltimes P'$ commuting with the maps to N and to $\mathcal{G}$, this is a fully faithful functor. We claim that this inclusion factors through the full subcategory $\mathrm{PrnBdl}_{\mathcal{G}}(\mathrm{Shv}(\mathrm{Diff}))_N$ of principal $\mathcal{G}$-bundles over N internal to $\mathrm{Shv}(\mathrm{Diff})$. Notice that the map $(a, \mathrm{pr}_2)\colon \mathcal{G}_1 \times_{\mathcal{G}_0} P \to P \times_N P$ in condition (3) of Definition D.15 is precisely the shear map (internal to $\mathrm{Diff}_{/N}$) from Definition E.21, so that the map $P \to N$ is a formally principal $\mathcal{G}$-bundle in the sense of Definition E.24. Furthermore, the condition (1) from Definition D.15 that the map $P \to N$ is a surjective smooth submersion is by Remark 2.1.6 equivalent to the condition that the map $P \to N$ is an effective epimorphism in $\mathrm{Shv}(\mathrm{Diff})$, so that $P \to N$ is in fact a principal $\mathcal{G}$-bundle internal to $\mathrm{Shv}(\mathrm{Diff})$. We have therefore produced a fully faithful embedding

$$\mathrm{PrnBdl}_{\mathcal{G}}(N) \hookrightarrow \mathrm{PrnBdl}_{\mathcal{G}}(\mathrm{Shv}(\mathrm{Diff}))_N.$$

It remains to prove that this inclusion is essentially surjective. Let $\mathcal{P} \twoheadrightarrow N$ be a map in $\mathrm{Shv}(\mathrm{Diff})$ equipped with the structure of a principal $\mathcal{G}$-bundle. By Theorem E.28, there is a classifying map $c\colon N \to \mathbb{B}\mathcal{G}$ and a pullback square

$$\begin{tikzcd}
\mathcal{P} \arrow[r] \arrow[d, two heads, "p"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{G}_0 \arrow[d, two heads] \\
N \arrow[r, "c"] & \mathbb{B}\mathcal{G}.
\end{tikzcd}$$

Since the map $\mathcal{G}_0 \twoheadrightarrow \mathbb{B}\mathcal{G}$ is a surjective representable submersion by Lemma 2.3.3, it follows that also the map $\mathcal{P} \twoheadrightarrow N$ is a surjective smooth submersion. In particular the stack $\mathcal{P}$ is a smooth manifold, which we will henceforth denote as P. Just as in Lemma 2.3.18 it follows that P is equipped with a left $\mathcal{G}$-action, and since the action groupoid $\mathcal{G} \ltimes P \to P$ lives in $\mathrm{Diff}_{/N}$ it satisfies condition (2) of Definition D.15. As before, condition (3) is equivalent to the condition that the map $P \to N$ is a formally principal $\mathcal{G}$-bundle and condition (1) is equivalent to the condition that $P \to N$ is an effective epimorphism in $\mathrm{Shv}(\mathrm{Diff})$. It follows that the map $P \to N$ is indeed a smooth principal $\mathcal{G}$-bundle as desired. $\square$

As a corollary, we obtain that the classifying stack $\mathbb{B}\mathcal{G}$ of a Lie groupoid $\mathcal{G}$ is a 1-stack:

Proposition 2.3.23. *Let $\mathcal{G}$ be a Lie groupoid. For every smooth manifold N, there is a natural equivalence of ∞-groupoids*

$$\mathbb{B}\mathcal{G}(N) \simeq \mathrm{PrnBdl}_{\mathcal{G}}(N).$$

In particular, $\mathbb{B}\mathcal{G}(N)$ is an ordinary groupoid for every N.

Proof. By Theorem E.28, there is an equivalence of ∞-groupoids

$$\mathbb{B}\mathcal{G}(N) = \mathrm{Hom}_{\mathrm{Shv}(\mathrm{Diff})}(N, \mathbb{B}\mathcal{G}) \simeq \mathrm{PrnBdl}_{\mathcal{G}}(\mathrm{Shv}(\mathrm{Diff}))_N.$$

By Proposition 2.3.22, the right-hand side is equivalent to the ordinary groupoid $\mathrm{PrnBdl}_{\mathcal{G}}(N)$ of smooth principal $\mathcal{G}$-bundles over N, finishing the proof. □

Combining this corollary with Corollary 2.3.8, we obtain:

Corollary 2.3.24 (Differentiable stacks are 1-stacks)**.** *Every differentiable stack $\mathcal{X}$ is a 1-stack, in the sense that it factors through the subcategory* Grpd $\subseteq$ An *of groupoids. In particular, the ∞-category* DiffStk *is a $(2,1)$-category.* □

2.4 Local properties of maps of stacks

Various properties of smooth maps between smooth manifolds have immediate generalizations to the context of maps between differentiable stacks.

Definition 2.4.1 (Local property for smooth manifolds)**.** Let P be a property of smooth maps of smooth manifolds. We say that P is a *local property* if for every pullback diagram

$$\begin{tikzcd} M' \arrow[r, "g'"] \arrow[d, "f'"'] \arrow[dr, phantom, "\lrcorner"] & M \arrow[d, "f"] \\ N' \arrow[r, two heads, "g"] & N \end{tikzcd}$$

in Diff in which g is a submersion, we have:

(1) If f has property P, then also f' has property P.

(2) If g is surjective and f' has property P, then also f has property P.

Definition 2.4.2 (Local property for stacks)**.** Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a representable morphism between stacks on Diff, and let P be a local property of smooth maps of smooth manifolds.

Then we say that *has property P* if for every representable submersion $g\colon N' \to \mathcal{Y}$ and every pullback diagram

$$\begin{tikzcd} M' \arrow[r, "g'"] \arrow[d, "f'"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{X} \arrow[d, "f"] \\ N' \arrow[r, "g", two heads] & \mathcal{Y} \end{tikzcd}$$

the smooth map $f'\colon M' \to N'$ of smooth manifolds has property P.

It is immediate from the definition that if $f\colon M \to N$ is a smooth map of smooth manifolds, then f has property P as a map of smooth manifolds if and only if it has property P when regarded as a morphism of stacks.

Before giving examples of local properties, let us address a potential point of confusion. In Definition 2.1.4, we defined the notion of a representable submersion. Now, since the property for a map between smooth manifolds to be a smooth submersion is a local property, Definition 2.4.2 gives an a priori different definition for what it means for a representable morphism of stacks to be a smooth submersion. The following lemma shows that these two notions agree in the case its source and target are differentiable stacks:

Lemma 2.4.3. *Let $\mathcal{X}$ and $\mathcal{Y}$ be differentiable stacks. Then a morphism $f\colon \mathcal{X} \to \mathcal{Y}$ is a representable submersion, in the sense of Definition 2.1.4, if and only if it is both representable and a smooth submersion in the sense of Definition 2.4.2.*

Proof. It is clear that a representable submersion is both representable and a smooth submersion, so we will prove the converse. Let $M \twoheadrightarrow \mathcal{Y}$ be a representable atlas for $\mathcal{Y}$. By part (3) of Proposition 2.1.10, the morphism f is a representable submersion if and only if its base change $M \times_{\mathcal{Y}} \mathcal{X} \to M$ is a representable submersion. Since $M \twoheadrightarrow \mathcal{Y}$ is a representable submersion and f is representable, the pullback $M \times_{\mathcal{Y}} \mathcal{X}$ is a smooth manifold. Furthermore, by assumption on f, the base change $M \times_{\mathcal{Y}} \mathcal{X} \to M$ is a smooth submersion between smooth manifolds. It follows in particular that the morphism $M \times_{\mathcal{Y}} \mathcal{X} \to M$ is a representable submersion, which finishes the proof. □

Lemma 2.4.4 (Open substacks)**.** *A morphism $\mathcal{U} \hookrightarrow \mathcal{X}$ of differentiable stacks is an open substack, in the sense of Definition 2.1.7, if and only if it is both representable and an open embedding, in the sense of Definition 2.4.2.*

Proof. The proof is completely analogous to that of Lemma 2.4.3 and will be omitted. □

In analogy with Lemma 2.4.4 we get the following more general definition of embeddings of stacks.

Definition 2.4.5 (Embeddings of stacks)**.** A smooth map $N \to M$ of smooth manifolds is called an *embedding* if it is a smooth immersion which induces a homeomorphism onto its image in M. It is called a *closed (resp. open) embedding* if the image in M is closed (resp. open).

Specializing Definition 2.4.2, we obtain the notions of *embedded*, *closed embedding* and *open embedding*.

Example 2.4.6. Other examples of properties P of smooth maps $f: M \to N$ of smooth manifolds are the following:

- f is injective, f is surjective;
- f is a submersion, f is a surjective submersion;
- f is an open map;
- f is *étale*, or a *local diffeomorphism*, meaning that for every $x \in M$ there exists an open neighborhood $x \in U \subseteq M$ such that $f|_U : U \to N$ is an open embedding;
- f is *proper*, meaning that for every compact subspace $K \subseteq N$ the preimage $f^{-1}(K) \subseteq M$ is also compact;
- f admits local sections, in the sense of Definition 2.1.2;
- f is a finite covering space, meaning that there is an open cover $\{U_i \subseteq N\}_{i \in I}$ such that every restriction $f|_{U_i} : f^{-1}(U_i) \to U_i$ is isomorphic to $U_i^{\sqcup n_i} \to U_i$ for some integer $n_i \geq 0$.

The main feature of local properties of maps of stacks is that they can be tested after pulling back along a representable surjective submersion.

Lemma 2.4.7 (Local properties are local)**.** *Let P be a local property, and consider a pullback diagram*

$$\begin{tikzcd} \mathcal{X}' \arrow[r] \arrow[d, "f'"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{X} \arrow[d, "f"] \\ \mathcal{Y}' \arrow[r, two heads, "g"] & \mathcal{Y} \end{tikzcd}$$

of stacks on Diff *such that g is an representable submersion and such that f and f' are representable.*

(1) If f has property P, then so does f'.

(2) If g is a representable surjective submersion and f' has property P, then so does f.

Proof. Part (1) is immediate from the definitions. For part (2), let $M \to \mathcal{Y}$ be a representable submersion from some smooth manifold M. By base changing the whole situation along this map, we may assume that $\mathcal{Y} = M$ is a smooth manifold. Since all four maps are representable, it follows that also $\mathcal{X}$, $\mathcal{Y}'$ and $\mathcal{Y}$ are smooth manifolds. Since P is a local property of maps between smooth manifolds, it follows that f has property P. $\square$

In practice, we use Lemma 2.4.7 mostly in the situation where $\mathcal{Y}'$ is a representable atlas for $\mathcal{Y}$:

Corollary 2.4.8. *Let $\mathcal{Y}$ be a differentiable stack with representable atlas $M \twoheadrightarrow \mathcal{Y}$ and let $f\colon \mathcal{X} \to \mathcal{Y}$ be a representable morphism of stacks. Let P be a local property of maps of smooth manifolds. Then f has property P if and only if the map $M \times_{\mathcal{Y}} \mathcal{X} \to M$ has property P.* $\square$

It follows from Corollary 2.4.8 that representable morphisms between differentiable stacks behave very similar to smooth maps between smooth manifolds. We collect some results in this direction.

Lemma 2.4.9. *Let $f\colon \mathcal{Y} \to \mathcal{X}$ be a representable submersion between differentiable stacks. Then there exists an essentially unique factorization*

$$\mathcal{Y} \twoheadrightarrow \mathcal{U} \hookrightarrow \mathcal{X}$$

of f into a surjective representable submersion $\mathcal{Y} \twoheadrightarrow \mathcal{U}$ and an open embedding $\mathcal{U} \hookrightarrow \mathcal{X}$.

We will also write $f(\mathcal{Y}) \hookrightarrow \mathcal{X}$ for the essentially unique open substack of $\mathcal{X}$ determined by the lemma, and refer to it as the *image of f*.

Proof. We may factor f as an effective epimorphism $\mathcal{Y} \twoheadrightarrow \mathcal{U}$ followed by a monomorphism $\mathcal{U} \hookrightarrow \mathcal{X}$. It remains to show that the map $\mathcal{Y} \twoheadrightarrow \mathcal{U}$ is a surjective representable submersion and the map $\mathcal{U} \hookrightarrow \mathcal{X}$ is an open embedding. By Corollary 2.4.8, this may be checked after pulling back the situation along an atlas $M \twoheadrightarrow \mathcal{X}$, and since pulling back preserves the unique factorization we may assume that $f\colon N \to M$ is a smooth submersion between smooth manifolds. In that case, the factorization is given as the factorization $N \twoheadrightarrow f(N) \hookrightarrow M$ of f through its (open) image in M. As the map $N \to f(N)$ is a surjective submersion and the inclusion $f(N) \hookrightarrow N$ is an open embedding, this proves the claim. $\square$

Lemma 2.4.10. *Consider a commutative triangle of differentiable stacks:*

$$\begin{tikzcd} & \mathcal{X} \arrow[dl, "f"'] \arrow[dr, "gf"] & \\ \mathcal{Y} \arrow[rr, "g"] & & \mathcal{Z}. \end{tikzcd}$$

(1) If both gf and g are representable, then so is f;

(2) If gf is an embedding and g is representable, then f is an embedding;

(3) If gf is proper and g is representable, then f is proper;

(4) If gf is a closed embedding and g is representable, then f is a closed embedding.

Proof. Let $O \twoheadrightarrow \mathcal{Z}$ be a representable atlas for $\mathcal{Z}$ and consider the following pullback diagram:

$$\begin{tikzcd} M \arrow[r, two heads] \arrow[d, "f'"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{X} \arrow[d, "f"] \\ N \arrow[r, two heads] \arrow[d, "g'"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{Y} \arrow[d, "g"] \\ O \arrow[r, two heads] & \mathcal{Z}. \end{tikzcd}$$

By Lemma 2.2.3, the map $N \twoheadrightarrow \mathcal{Y}$ is a representable atlas for $\mathcal{Y}$, and thus by Proposition 2.1.10 and Lemma 2.4.7, it suffices to prove that f' is representable (resp. an embedding / proper / a closed embedding). Part (1) is now immediate, as f' is a morphism between smooth manifolds and thus representable. Parts (2), (3) and (4) follow immediately from the analogous cancellation property for smooth maps between smooth manifolds, see Lemma C.7. $\square$

2.5 Vector bundles over stacks

The notion of a vector bundle over a smooth manifold admits a natural extension to the setting of stacks on Diff.

Definition 2.5.1 (Vector bundle)**.** Let $\mathcal{X}$ be a stack on Diff. For $n \geq 0$, we define an *n-dimensional vector bundle* π on $\mathcal{X}$ to be a morphism of stacks $\pi\colon \mathcal{X} \to \mathbb{B}\mathrm{GL}(n)$. The *total space* $\mathcal{E}_\pi$ of the bundle π is defined as the following pullback of stacks:

$$\begin{tikzcd} \mathcal{E}_\pi \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathbb{R}^n//\mathrm{GL}(n) \arrow[d] \\ \mathcal{X} \arrow[r, "\pi"] & \mathbb{B}\mathrm{GL}(n). \end{tikzcd}$$

We will often abuse notation and refer to a map $\mathcal{E} \to \mathcal{X}$ as a vector bundle, leaving the classifying map $\pi\colon X \to \mathbb{B}\,\mathrm{GL}(n)$ implicit in the notation. We denote the groupoid of n-dimensional vector bundles over $\mathcal{X}$ by

$$\mathrm{Vect}_n^{\simeq}(\mathcal{X}) := \mathrm{Hom}_{\mathrm{Shv}(\mathrm{Diff})}(\mathcal{X}, \mathbb{B}\,\mathrm{GL}(n)).$$

We also define

$$\mathrm{Vect}^{\simeq}(\mathcal{X}) := \mathrm{Hom}_{\mathrm{Shv}(\mathrm{Diff})}(\mathcal{X}, \bigsqcup_{n=0}^{\infty} \mathbb{B}\,\mathrm{GL}(n)).$$

This determines a limit-preserving functor $\mathrm{Vect}^{\simeq}\colon \mathrm{Shv}(\mathrm{Diff})^{\mathrm{op}} \to \mathrm{Grpd}$.

Definition 2.5.2 (Associated sphere bundle)**.** Given a vector bundle $\mathcal{E} \to \mathcal{X}$ classified by $\pi\colon \mathcal{X} \to \mathbb{B}\,\mathrm{GL}(n)$, we define its *associated sphere bundle* $S^{\mathcal{E}} \to \mathcal{X}$ as the pullback

$$\begin{tikzcd} S^{\mathcal{E}} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & S^n//\mathrm{GL}(n) \arrow[d] \\ \mathcal{X} \arrow[r, "\pi"] & \mathbb{B}\,\mathrm{GL}(n). \end{tikzcd}$$

The groupoid $\mathrm{Vect}^{\simeq}(\mathcal{X})$ is the core of a category $\mathrm{Vect}(\mathcal{X})$ of vector bundles:

Definition 2.5.3 (Category of vector bundles)**.** Consider the functor $\mathrm{Vect}\colon \mathrm{Diff}^{\mathrm{op}} \to \mathrm{Cat}$ which sends a smooth manifold M to the category of vector bundles over M. This functor satisfies the sheaf condition with respect to the open cover topology, and thus extends uniquely to a limit-preserving functor

$$\mathrm{Vect}\colon \mathrm{Shv}(\mathrm{Diff})^{\mathrm{op}} \to \mathrm{Cat}.$$

For a stack $\mathcal{X}$ on Diff, we refer to the resulting category $\mathrm{Vect}(\mathcal{X})$ as the *category of vector bundles over $\mathcal{X}$*.

Since the functor $(-)^{\simeq}\colon \mathrm{Cat} \to \mathrm{Grpd}$ which assigns to an ordinary category C its underlying groupoid $C^{\simeq}$ is a right adjoint and thus preserves limits, we see that there is a natural equivalence of groupoids

$$\mathrm{Vect}(\mathcal{X})^{\simeq} \simeq \mathrm{Vect}^{\simeq}(\mathcal{X}).$$

In case the stack $\mathcal{X}$ is a differentiable stack, we can make morphisms of vector bundles over $\mathcal{X}$ explicit as follows: given two vector bundles $\mathcal{E}_1, \mathcal{E}_1$ over $\mathcal{X}$, a morphism of vector bundles from $\mathcal{E}_1$ to $\mathcal{E}_2$ is a commutative diagram

$$\begin{tikzcd} \mathcal{E}_1 \arrow[rr] \arrow[dr] & & \mathcal{E}_2 \arrow[dl] \\ & \mathcal{X} & \end{tikzcd}$$

such that for every point $x\colon \mathrm{pt} \to \mathcal{X}$ the induced map $(\mathcal{E}_1)_x \to (\mathcal{E}_2)_x$ on fibers is a linear map between vector spaces.

2.5.1 Homotopy invariance of vector bundles

The goal of this subsection is to show that the set $\pi_0 \mathrm{Vect}(\mathcal{X})^{\simeq}$ of isomorphism classes of vector bundles on a differentiable stack $\mathcal{X}$ is a *homotopy invariant* of $\mathcal{X}$.

Remark 2.5.4. The original version of this dissertation contained a wrong result with an incorrect proof. We thank Marc Hoyois for bringing this to our attention. The corrected statement of Theorem 2.5.5 is true and can be proved using partitions of unity of separated differentiable stacks, see for instance Freed [Fre15, Theorem 15.85] for a proof sketch.

Theorem 2.5.5. *Let $\mathcal{X}$ be a separated differentiable stack and let $\pi\colon \mathcal{E} \to \mathcal{X} \times \mathbb{R}$ be a vector bundle. Then there exists an isomorphism*

$$i_0^*\mathcal{E} \cong i_1^*\mathcal{E}$$

of vector bundles over $\mathcal{X}$, where $i_r\colon \mathcal{X} \hookrightarrow \mathcal{X} \times \mathbb{R}$ denotes the inclusion of $\mathcal{X} \times \{r\}$ for $r \in \mathbb{R}$.

Definition 2.5.6. Let $f, f'\colon \mathcal{X} \to \mathcal{Y}$ be morphisms of stacks on Diff. A *homotopy* between f and f' is a map of stacks $H\colon \mathcal{X} \times \mathbb{R} \to \mathcal{Y}$ satisfying $H_0 \simeq f$ and $H_1 \simeq f'$, where $H_r\colon \mathcal{X} \to \mathcal{Y}$ denotes the composite

$$\mathcal{X} \xhookrightarrow{i_r} \mathcal{X} \times \mathbb{R} \xrightarrow{H} \mathcal{Y}.$$

We say that f is a *homotopy equivalence* if there exists a map $g\colon \mathcal{Y} \to \mathcal{X}$ such that gf and fg are homotopic to the respective identities.

Corollary 2.5.7. *Let $\mathcal{E} \to \mathcal{Y}$ be a vector bundle and let $H\colon \mathcal{X} \times \mathbb{R} \to \mathcal{Y}$ be a homotopy of morphisms $\mathcal{X} \to \mathcal{Y}$ between separated stacks. Then there is an isomorphism $H_0^*\mathcal{E} \cong H_1^*\mathcal{E}$ of vector bundles on $\mathcal{X}$.*

Proof. As $H_0^*\mathcal{E} = i_0^* H^*\mathcal{E}$ and $H_1^*\mathcal{E} = i_1^* H^*\mathcal{E}$, this is an immediate consequence of Theorem 2.5.5 applied to the vector bundle $H^*\mathcal{E}$ over $\mathcal{X} \times \mathbb{R}$. □

Corollary 2.5.8. *Assume that $f\colon \mathcal{X} \to \mathcal{Y}$ is a homotopy equivalence of separated stacks. Then for every vector bundle $\mathcal{E} \to \mathcal{X}$ there exists a vector bundle $\mathcal{E}' \to \mathcal{Y}$ with $\mathcal{E} \cong f^*(\mathcal{E}')$.*

Proof. Letting $g\colon \mathcal{Y} \to \mathcal{X}$ be a homotopy inverse and $H\colon \mathcal{X} \times \mathbb{R} \to \mathcal{X}$ a homotopy between gf and $\mathrm{id}_{\mathcal{X}}$, the previous corollary provides an isomorphism $\mathcal{E} = (\mathrm{id}_{\mathcal{X}})^*\mathcal{E} \cong (gf)^*\mathcal{E} = f^*g^*\mathcal{E}$, so we may take $\mathcal{E}' := g^*\mathcal{E}$. □

2.5.2 VB-groupoids

We will now discuss the notion of a *VB-groupoid*, which is the analogue of a vector bundle in the setting of Lie groupoids.

Definition 2.5.9 (VB-groupoid, [GM17])**.** Given a Lie groupoid $\mathcal{G}$, a *VB-groupoid over* $\mathcal{G}$ is a morphism $\pi\colon \mathcal{E} \to \mathcal{G}$ of Lie groupoids such that the maps $\pi_0\colon \mathcal{E}_0 \to \mathcal{G}_0$ and $\pi_1\colon \mathcal{E}_1 \to \mathcal{G}_1$ are vector bundles and the structure maps of $\mathcal{E}$ are morphisms of vector bundles.

In terms of differentiable stacks, VB-groupoids correspond to what are known as *2-vector bundles*, see for example [HO20] for a discussion. In order to get an actual vector bundle of differentiable stacks, one needs to assume that the VB-groupoid $\mathcal{E} \to \mathcal{G}$ is *cartesian*, in the following sense:

Definition 2.5.10 (Cartesian morphism of Lie groupoids)**.** A morphism $f\colon \mathcal{H} \to \mathcal{G}$ of Lie groupoids is called *cartesian* if the commutative squares

$$\begin{tikzcd} \mathcal{H}_1 \arrow[r, "s"] \arrow[d, "f_1"'] & \mathcal{H}_0 \arrow[d, "f_0"] \\ \mathcal{G}_1 \arrow[r, "s"] & \mathcal{G}_0 \end{tikzcd} \quad \text{and} \quad \begin{tikzcd} \mathcal{H}_1 \arrow[r, "t"] \arrow[d, "f_1"'] & \mathcal{H}_0 \arrow[d, "f_0"] \\ \mathcal{G}_1 \arrow[r, "t"] & \mathcal{G}_0 \end{tikzcd}$$

are pullback squares.[4]

Lemma 2.5.11. *Let $\pi\colon \mathcal{E} \to \mathcal{G}$ be a VB-groupoid, and assume that the morphism π is cartesian. Then the induced map $\mathbb{B}\pi\colon \mathbb{B}\mathcal{E} \to \mathbb{B}\mathcal{G}$ is naturally a vector bundle of differentiable stacks.*

Proof. By passing to nerves, the groupoid map $\pi\colon \mathcal{E} \to \mathcal{G}$ induces a morphism $N\pi\colon N\mathcal{E} \to N\mathcal{G}$ of simplicial manifolds. As π is cartesian, $N\pi$ is a cartesian natural transformation of functors $\Delta^{\mathrm{op}} \to \mathrm{Diff}$. It follows that $N\pi$ defines a section of the functor $\Delta \to \mathrm{Cat}, [n] \mapsto \mathrm{Vect}(\mathcal{G}_n)$, and thus defines an object of the limit $\lim_{[n]\in\Delta} \mathrm{Vect}(\mathcal{G}_n)$. Since the assignment $\mathcal{X} \mapsto \mathrm{Vect}(\mathcal{X})$ satisfies descent, we see that this limit is $\mathrm{Vect}(\mathbb{B}\mathcal{G})$, giving a vector bundle over $\mathbb{B}\mathcal{G}$. Since the forgetful functors $\mathrm{Vect}(\mathcal{X}) \to \mathrm{Sub}_{/\mathcal{X}}$ are compatible with pullback, it follows that the underlying submersion of this vector bundle is the map $\mathbb{B}\mathcal{E} \to \mathbb{B}\mathcal{G}$, finishing the proof. $\square$

[4]It suffices to check this for just one of the squares.

II.3 Geometry of differentiable stacks

In this chapter, we will have a closer look at the geometrical aspects of differentiable stacks. In Section 3.1, we introduce the *coarse moduli space* $|\mathcal{X}|_{\mathrm{mod}}$ of a stack $\mathcal{X}$ on Diff and show that the open subsets of $|\mathcal{X}|_{\mathrm{mod}}$ are in one-to-one correspondence with the open substacks of $\mathcal{X}$. This allows us to construct the *open complement* $\mathcal{X} \setminus \mathcal{Z}$ of a closed embedding $\mathcal{Z} \hookrightarrow \mathcal{X}$ in Section 3.2. In Section 3.3, we introduce the notion of a *separated* differentiable stack, which in terms of Lie groupoids corresponds to that of a *proper* Lie groupoid. The geometry of separated differentiable stacks is better behaved than that of arbitrary differentiable stacks, and hence separated differentiable stacks will play an important role throughout this article. In Section 3.4 we show that a morphism of separated separated differentiable stack is representable if and only if it induces injections on *isotropy groups*. In Section 3.5 we introduce the notions of relative tangent bundles and normal bundles for representable morphisms of differentiable stacks, and we will show in Section 3.6 that every embedding of separated differentiable stacks admits a tubular neighborhood. Finally, we will show in Section 3.7 that all separated differentiable stacks are locally isomorphic to a quotient stack $\mathbb{R}^n//G$ for some smooth linear action of a compact Lie group G on a Euclidean space $\mathbb{R}^n$, which leads to several important structure theorems for morphisms between separated differentiable stacks.

3.1 The coarse moduli space of a differentiable stack

We start by defining the *coarse moduli space* $|\mathcal{X}|_{\mathrm{mod}}$ of a stack $\mathcal{X}$ on Diff.

Definition 3.1.1. As the category Top of topological spaces admits colimits, the inclusion Diff $\hookrightarrow$ Top uniquely extends to a colimit-preserving functor $\mathrm{PSh}(\mathrm{Diff}) \to \mathrm{Top}$. Since an open cover $\bigsqcup_\alpha U_\alpha \twoheadrightarrow M$ of a smooth manifold M is an effective epimorphism in Top, this descends to a functor

$$|-|_{\mathrm{mod}} \colon \mathrm{Shv}(\mathrm{Diff}) \to \mathrm{Top}.$$

We call $|\mathcal{X}|_{\mathrm{mod}}$ the *coarse moduli space* of the stack $\mathcal{X}$. The functor $|-|_{\mathrm{mod}}$ admits a right adjoint $\mathrm{N}^{\mathrm{Diff}}\colon \mathrm{Top} \longrightarrow \mathrm{Shv}(\mathrm{Diff})$ given by

$$\mathrm{N}^{\mathrm{Diff}}(X)(M) := \mathrm{Hom}_{\mathrm{Top}}(M, X),$$

which equips a topological space X with its *continuous diffeology*.

Remark 3.1.2. The stack $\mathrm{N}^{\mathrm{Diff}}(X)$ is in fact *diffeological space*, that is, a concrete sheaf of sets on the site Diff. We shall not make use of this fact.

Example 3.1.3. For a smooth manifold M, we have $|M|_{\mathrm{mod}} = M$.

Example 3.1.4. For a Lie groupoid $\mathcal{G}$, the coarse moduli space of its classifying stack $\mathbb{B}\mathcal{G}$ is its given by the quotient space of $\mathcal{G}$:

$$|\mathbb{B}\mathcal{G}|_{\mathrm{mod}} \cong \mathrm{coeq}(\mathcal{G}_1 \underset{t}{\overset{s}{\rightrightarrows}} \mathcal{G}_0).$$

In particular, given the action of a Lie group G on a smooth manifold M, the coarse moduli space of the quotient stack $M//G$ is its strict quotient M/G:

$$|M//G|_{\mathrm{mod}} \cong M/G.$$

Given a stack $\mathcal{X}$ on Diff, there is an explicit construction of the topological space $|\mathcal{X}|_{\mathrm{mod}}$. We follow the treatment of Noohi [Noo05, Section 4.3], who discusses this construction in the context of topological stacks.

Construction 3.1.5. Given a stack $\mathcal{X}$ on Diff, we will define a topological space $\mathcal{X}_{\mathrm{mod}}$. The underlying set of $\mathcal{X}_{\mathrm{mod}}$ is the set of path components of the anima of global sections $\Gamma(\mathcal{X}) = \mathrm{Hom}_{\mathrm{Shv}(\mathrm{Diff})}(\mathrm{pt}, \mathcal{X})$ of $\mathcal{X}$:

$$\mathcal{X}_{\mathrm{mod}} := \pi_0(\Gamma(\mathcal{X})).$$

We define a topology on $\mathcal{X}_{\mathrm{mod}}$ as follows: a subset of $\mathcal{X}_{\mathrm{mod}}$ is open precisely if it is of the form $\pi_0(\Gamma(\mathcal{U})) \subseteq \pi_0(\Gamma(\mathcal{X}))$ for some open substack $\mathcal{U} \hookrightarrow \mathcal{X}$; this inclusion makes sense as the map $\Gamma(\mathcal{U}) \hookrightarrow \Gamma(\mathcal{X})$ is an inclusion of path components. To see that this defines a topology, observe first that arbitrary unions and finite intersections of open substacks are again open substacks: given a morphism $M \to \mathcal{X}$ from a smooth manifold M, the pullback functor $\mathrm{Shv}(\mathrm{Diff})_{/\mathcal{X}} \to \mathrm{Shv}(\mathrm{Diff})_{/M}$ preserves limits and colimits, and unions and finite intersections of open subsets of M are again open subsets. Since all morphisms of animae involved are inclusions of path components, applying π_0 commutes with finite intersections and arbitrary unions, showing that the topology on $\mathcal{X}_{\mathrm{mod}}$ is well-defined.

If $f\colon \mathcal{X} \to \mathcal{Y}$ is a map of stacks, then any open substack $\mathcal{U} \hookrightarrow \mathcal{Y}$ pulls back to an open substack $f^{-1}(\mathcal{U}) \hookrightarrow \mathcal{X}$, so the map $f_{\mathrm{mod}}\colon \mathcal{X}_{\mathrm{mod}} \to \mathcal{Y}_{\mathrm{mod}}$ induced by $\Gamma(f)\colon \Gamma(\mathcal{X}) \to \Gamma(\mathcal{Y})$ is continuous. It follows that the formation of $\mathcal{X}_{\mathrm{mod}}$ gives rise to a functor

$$(-)_{\mathrm{mod}}\colon \operatorname{Shv}(\mathrm{Diff}) \to \mathrm{Top}.$$

If $\mathcal{X} = M \in \mathrm{Diff}$ is a smooth manifold, its anima of global sections $\Gamma(M) = \operatorname{Hom}_{\mathrm{Diff}}(\mathrm{pt}, M)$ is already a set, and is in canonical bijection with the underlying set of M. Since the open substacks $\mathcal{U} \hookrightarrow M$ of M, regarded as a stack, are in one-to-one correspondence with the open subsets $U \subseteq M$, the induced topology on the set $\operatorname{Hom}_{\mathrm{Diff}}(\mathrm{pt}, M)$ is just the topology on M. We will henceforth identify M_{mod} with M.

Lemma 3.1.6. *The functor $\mathcal{X} \mapsto \mathcal{X}_{\mathrm{mod}}$ is left adjoint to the functor $\mathrm{N}^{\mathrm{Diff}}\colon \mathrm{Top} \to \operatorname{Shv}(\mathrm{Diff})$. In particular, there is a natural equivalence $\mathcal{X}_{\mathrm{mod}} \simeq |\mathcal{X}|_{\mathrm{mod}}$.*

Proof. We start by constructing the unit map $\eta\colon \mathcal{X} \to \mathrm{N}^{\mathrm{Diff}}(\mathcal{X}_{\mathrm{mod}})$ of stacks for every stack $\mathcal{X} \in \operatorname{Shv}(\mathrm{Diff})$. For $M \in \mathrm{Diff}$, the map $\mathcal{X}(M) \to \mathrm{N}^{\mathrm{Diff}}(\mathcal{X}_{\mathrm{mod}})(M) = \operatorname{Hom}_{\mathrm{Top}}(M, \mathcal{X}_{\mathrm{mod}})$ sends a map $f\colon M \to \mathcal{X}$ of stacks to the map $M = M_{\mathrm{mod}} \xrightarrow{f_{\mathrm{mod}}} \mathcal{X}_{\mathrm{mod}}$ of topological spaces. Since $(-)_{\mathrm{mod}}$ is a functor, it follows that this defines a map of stacks $\eta\colon \mathcal{X} \to \mathrm{N}^{\mathrm{Diff}}(\mathcal{X}_{\mathrm{mod}})$ which is natural in $\mathcal{X} \in \operatorname{Shv}(\mathrm{Diff})$.

We next construct the counit map $\varepsilon\colon \mathrm{N}^{\mathrm{Diff}}(T)_{\mathrm{mod}} \to T$ for every topological space T. Note that as a set we have $\mathrm{N}^{\mathrm{Diff}}(T)_{\mathrm{mod}} = \pi_0 \operatorname{Hom}_{\mathrm{Top}}(\mathrm{pt}, T) = T$. Furthermore, if $U \subseteq T$ is an open subspace, then $f^{-1}(U) \subseteq M$ is an open submanifold for every continuous map $f\colon M \to T$, and it follows that $\mathrm{N}^{\mathrm{Diff}}(U) \hookrightarrow \mathrm{N}^{\mathrm{Diff}}(T)$ is an open substack. Evaluating this map at $\mathrm{pt} \in \mathrm{Diff}$ gives the inclusion $U \subseteq T$. This shows that the topology on $\mathrm{N}^{\mathrm{Diff}}(T)_{\mathrm{mod}}$ is finer than that of T, giving a continuous map $\varepsilon\colon \mathrm{N}^{\mathrm{Diff}}(T)_{\mathrm{mod}} \to T$.

Now we show that the maps $\eta\colon \mathcal{X} \to \mathrm{N}^{\mathrm{Diff}}(\mathcal{X}_{\mathrm{mod}})$ and $\varepsilon\colon \mathrm{N}^{\mathrm{Diff}}(T)_{\mathrm{mod}} \to T$ satisfy the triangle identities:

$$\begin{tikzcd} & \mathrm{N}^{\mathrm{Diff}}(\mathrm{N}^{\mathrm{Diff}}(T)_{\mathrm{mod}}) \arrow[dr, "\varepsilon"] & \\ \mathrm{N}^{\mathrm{Diff}}(T) \arrow[ur, "\eta"] \arrow[rr, equal] & & \mathrm{N}^{\mathrm{Diff}}(T) \end{tikzcd} \quad \text{and} \quad \begin{tikzcd} & \mathrm{N}^{\mathrm{Diff}}(\mathcal{X}_{\mathrm{mod}})_{\mathrm{mod}} \arrow[dr, "\varepsilon"] & \\ \mathcal{X}_{\mathrm{mod}} \arrow[ur, "\eta"] \arrow[rr, equal] & & \mathcal{X}_{\mathrm{mod}}. \end{tikzcd}$$

For the left triangle, we may check this after mapping in from a smooth manifold M, where it is clear. The right triangle is also clear, as the second map is the identity map on underlying sets. □

Corollary 3.1.7. *For a stack $\mathcal{X}$, the underlying set of the topological space $|\mathcal{X}|_{\mathrm{mod}}$ is the set of path components of the anima of global sections of $\mathcal{X}$:*

$$|\mathcal{X}|_{\mathrm{mod}} \cong \pi_0(\Gamma(\mathcal{X})) = \pi_0 \operatorname{Hom}_{\operatorname{Shv}(\mathrm{Diff})}(\mathrm{pt}, \mathcal{X}). \qquad \square$$

Corollary 3.1.8. *Let $\mathcal{U} \hookrightarrow \mathcal{X}$ be an open substack. Then the induced map $|\mathcal{U}|_{\mathrm{mod}} \hookrightarrow |\mathcal{X}|_{\mathrm{mod}}$ is an open subspace.* □

Our next goal is to show that the above assignment $\mathcal{U} \mapsto |\mathcal{U}|_{\mathrm{mod}}$ defines a 1-to-1 correspondence between open substacks of $\mathcal{X}$ and open subspaces of its coarse moduli space $|\mathcal{X}|_{\mathrm{mod}}$.

Lemma 3.1.9 (cf. [Noo05, Lemma 4.16])**.** *Let $\mathcal{X}' \hookrightarrow \mathcal{X}$ be an embedded substack. Assume the induced map $\mathcal{X}'_{\mathrm{mod}} \to \mathcal{X}_{\mathrm{mod}}$ is a bijection of sets, or equivalently, the induced monomorphism $\Gamma(\mathcal{X}') \hookrightarrow \Gamma(\mathcal{X})$ of animae is in fact an equivalence. Then the map $\mathcal{X}' \to \mathcal{X}$ is an equivalence.*

Proof. We follow the proof of [Noo05, Lemma 4.16] in the topological context. We have to show that for every smooth manifold M, the inclusion $\mathcal{X}'(M) \hookrightarrow \mathcal{X}(M)$ of animae is in fact an equivalence, i.e., is a surjection on path components. If this is not the case, we can find a smooth manifold M and a map $f\colon M \to \mathcal{X}$ which does not lift along $\mathcal{X}' \hookrightarrow \mathcal{X}$. Define the subspace $M' \subseteq M$ by the pullback diagram

$$\begin{tikzcd} M' \arrow[r, hook] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & M \arrow[d, "f"] \\ \mathcal{X}' \arrow[r, hook] & \mathcal{X}. \end{tikzcd}$$

It follows that the embedding $M' \subseteq M$ does not have a section, and so we conclude that there is a point $w\colon \mathrm{pt} \to M$ which is not contained in M'. But this means that the point $f(w)\colon \mathrm{pt} \to \mathcal{X}$ does not lift to $\mathcal{X}'$, which is in contradiction with the assumption that the map $\pi_0(\Gamma(\mathcal{X}')) \to \pi_0(\Gamma(\mathcal{X}))$ is a bijection. □

Warning 3.1.10. Although the counit map $\mathrm{N}^{\mathrm{Diff}}(T)_{\mathrm{mod}} \to T$ is the identity on underlying sets, it is not in general a homeomorphism. For example, if $T = \mathbb{Q}$ is the set of rational numbers equipped with the subspace topology from $\mathbb{R}$, then $\mathrm{N}^{\mathrm{Diff}}(T)_{\mathrm{mod}}$ is the set $\mathbb{Q}$ equipped with the discrete topology. Topological spaces T for which the counit map $\mathrm{N}^{\mathrm{Diff}}(T)_{\mathrm{mod}} \to T$ is a homeomorphism are known as Δ-*generated topological spaces*, and it follows that the functor $\mathrm{N}^{\mathrm{Diff}}\colon \mathrm{Top} \to \mathrm{Shv}(\mathrm{Diff})$ becomes fully faithful when restricted to the Δ-generated topological spaces.

Corollary 3.1.11. *For an open substack $\mathcal{U} \hookrightarrow \mathcal{X}$, the naturality square*

$$\begin{tikzcd} \mathcal{U} \arrow[r, hook] \arrow[d, "\eta"'] & \mathcal{X} \arrow[d, "\eta"] \\ \mathrm{N}^{\mathrm{Diff}}(\mathcal{U}_{\mathrm{mod}}) \arrow[r] & \mathrm{N}^{\mathrm{Diff}}(\mathcal{X}_{\mathrm{mod}}) \end{tikzcd}$$

of the unit transformation η is a pullback square.

Proof. We have to prove that the map $\mathcal{U} \to \mathcal{X} \times_{\mathrm{N}^{\mathrm{Diff}}(\mathcal{X}_{\mathrm{mod}})} \mathrm{N}^{\mathrm{Diff}}(\mathcal{U}_{\mathrm{mod}})$ induced by the square is an equivalence. Note that this map is an open embedding, since the map $\mathcal{U} \hookrightarrow \mathcal{X}$ is an open embedding by assumption, the projection $\mathcal{X} \times_{\mathrm{N}^{\mathrm{Diff}}(\mathcal{X}_{\mathrm{mod}})} \mathrm{N}^{\mathrm{Diff}}(\mathcal{U}_{\mathrm{mod}}) \to \mathcal{X}$ is an open embedding by Corollary 3.1.8, and the class of open embeddings of stacks is closed under left cancellation. By Lemma 3.1.9 it thus suffices to show this map induces an equivalence on global sections. Note that $\Gamma(\mathrm{N}^{\mathrm{Diff}}(\mathcal{U}_{\mathrm{mod}})) = \pi_0(\Gamma(\mathcal{U}))$ and $\Gamma(\mathrm{N}^{\mathrm{Diff}}(\mathcal{X}_{\mathrm{mod}})) = \pi_0(\Gamma(\mathcal{X}))$, and since the global section functor Γ preserves pullbacks we need to show that the following square of animae is a pullback square:

$$\begin{tikzcd}
\Gamma(\mathcal{U}) \arrow[r, hook] \arrow[d] & \Gamma(\mathcal{X}) \arrow[d] \\
\pi_0(\Gamma(\mathcal{U})) \arrow[r, hook] & \pi_0(\Gamma(\mathcal{X})).
\end{tikzcd}$$

But this is clear since $\Gamma(\mathcal{U}) \hookrightarrow \Gamma(\mathcal{X})$ is an inclusion of path components. □

For a stack $\mathcal{X}$, let $\mathrm{Open}(\mathcal{X}) \subseteq \mathrm{Shv}(\mathrm{Diff})_{/\mathcal{X}}$ denote the full subcategory spanned by the open substacks $\mathcal{U} \hookrightarrow \mathcal{X}$. Similarly, for a topological space X, let $\mathrm{Open}(X) \subseteq \mathrm{Top}_{/X}$ denote the full subcategory spanned by the open subspaces $U \hookrightarrow X$. Since open embeddings are monomorphisms, it follows that both of these subcategories are in fact posets.

Corollary 3.1.12 (cf. [Noo05, Proposition 4.17])**.** *Pullback along the map $\mathcal{X} \to |\mathcal{X}|_{\mathrm{mod}}$ defines an equivalence of posets*

$$\mathrm{Open}(|\mathcal{X}|_{\mathrm{mod}}) \xrightarrow{\sim} \mathrm{Open}(\mathcal{X}),$$

with inverse given by $\mathcal{U} \mapsto |\mathcal{U}|_{\mathrm{mod}}$.

Proof. Corollary 3.1.11 shows that every open substack $\mathcal{U}$ of $\mathcal{X}$ is naturally equivalent to the inverse image of the open subspace $|\mathcal{U}|_{\mathrm{mod}} \subseteq |\mathcal{X}|_{\mathrm{mod}}$. Conversely, if $U \subseteq \mathcal{X}_{\mathrm{mod}}$ is an open subspace and $\mathcal{U} \hookrightarrow \mathcal{X}$ is its preimage in $\mathcal{X}$, it is easy to see that $U = \mathcal{U}_{\mathrm{mod}}$. □

Observation 3.1.13. Let $f \colon \mathcal{X} \twoheadrightarrow \mathcal{Y}$ be an effective epimorphism. Then the induced map $f_{\mathrm{mod}} \colon \mathcal{X}_{\mathrm{mod}} \to \mathcal{Y}_{\mathrm{mod}}$ is surjective.

Proof. We need to prove that any map $x \colon \mathrm{pt} \to \mathcal{Y}$ lifts to $\mathcal{X}$. In other words, we need to prove that the effective epimorphism $f^{-1}(\mathrm{pt}) \twoheadrightarrow \mathrm{pt}$ admits a section. But this follows immediately from the fact that effective epimorphisms have local sections, Proposition 2.1.3, and the fact that the only open cover of the point is given by the point. □

3.2 Open complements

In this section, we define the *open complement* $\mathcal{X} \setminus \mathcal{Z}$ of a closed embedding $\mathcal{Z} \hookrightarrow \mathcal{X}$.

Proposition 3.2.1. *Let $i \colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be a closed embedding of differentiable stacks. Then the induced map of topological spaces*

$$|i|_{\mathrm{mod}} \colon |\mathcal{Z}|_{\mathrm{mod}} \to |\mathcal{X}|_{\mathrm{mod}}$$

is the inclusion of a closed subspace.

Proof. Let $M \twoheadrightarrow \mathcal{X}$ be a representable atlas for $\mathcal{X}$. Form the pullback square

$$\begin{tikzcd} N \arrow[r, hook] \arrow[d, two heads] \arrow[dr, phantom, "\lrcorner"] & M \arrow[d, two heads] \\ \mathcal{Z} \arrow[r, hook, "i"] & \mathcal{X}, \end{tikzcd}$$

providing a representable atlas $N \twoheadrightarrow \mathcal{Z}$ for $\mathcal{Z}$, see Lemma 2.2.3. Let $R \subseteq M \times M$ denote the image of the map $M \times_{\mathcal{X}} M \to M \times M$. This is an equivalence relation defining $|\mathcal{X}|_{\mathrm{mod}}$: we have

$$|\mathcal{X}|_{\mathrm{mod}} = M/R.$$

Similarly we have $|\mathcal{Z}|_{\mathrm{mod}} = N/S$, where S is the image of $N \times_{\mathcal{Z}} N \to N \times N$. We claim that that S is the intersection of R and $N \times N$ inside $M \times M$. Indeed, this follows from the pullback square

$$\begin{tikzcd} N \times_{\mathcal{Z}} N \arrow[r, hook] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & M \times_{\mathcal{X}} M \arrow[d] \\ N \times N \arrow[r, hook] & M \times M. \end{tikzcd}$$

The fact that this square is a pullback square in turn follows from the pasting law of pullback squares, applied to the following squares:

$$\begin{tikzcd} N \times_{\mathcal{Z}} N \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{Z} \arrow[r] \arrow[d, "\Delta"] \arrow[dr, phantom, "\lrcorner"] & \mathcal{X} \arrow[d, "\Delta"] \arrow[dr, phantom, "\llcorner"] & M \times_{\mathcal{X}} M \arrow[l] \arrow[d] \\ N \times N \arrow[r] & \mathcal{Z} \times \mathcal{Z} \arrow[r, hook] & \mathcal{X} \times \mathcal{X} & M \times M, \arrow[l] \end{tikzcd}$$

where the middle square is a pullback square as $i \colon \mathcal{Z} \hookrightarrow \mathcal{X}$ is a monomorphism. It follows that the closed subspace $N \subseteq M$ is preserved under the equivalence relation R, and in particular its image $|\mathcal{Z}|_{\mathrm{mod}}$ in $|\mathcal{X}|_{\mathrm{mod}}$ is again a closed subspace. This finishes the proof. □

Definition 3.2.2 (Open complement)**.** Let $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be a closed embedding of differentiable stacks. We say an open embedding $j\colon \mathcal{U} \hookrightarrow \mathcal{X}$ *exhibits* $\mathcal{U}$ *as the open complement of* $\mathcal{Z}$ *in* $\mathcal{X}$ if at the level of coarse moduli spaces we have

$$|\mathcal{U}|_{\mathrm{mod}} = |\mathcal{X}|_{\mathrm{mod}} \setminus |\mathcal{Z}|_{\mathrm{mod}}.$$

By Corollary 3.1.12 and Proposition 3.2.1, such open complement exists and is unique, and we will denote it by $\mathcal{U} = \mathcal{X} \setminus \mathcal{Z}$. More explicitly, given a smooth manifold M, the subspace

$$\mathcal{U}(M) \subseteq \mathcal{X}(M) = \mathrm{Hom}_{\mathrm{Shv(Diff)}}(M, \mathcal{X})$$

is given by the collection of path components corresponding to those maps $f\colon M \to \mathcal{X}$ such that the underlying map of topological spaces $M = |M|_{\mathrm{mod}} \xrightarrow{|f|_{\mathrm{mod}}} |\mathcal{X}|_{\mathrm{mod}}$ factors through the open complement $|\mathcal{X}|_{\mathrm{mod}} \setminus |\mathcal{Z}|_{\mathrm{mod}}$.

As a consequence, we observe that every open substack of $\mathcal{Z}$ must be of the form $\mathcal{U} \cap \mathcal{Z}$ for some open substack $\mathcal{U}$ of $\mathcal{X}$:

Corollary 3.2.3. *Let* $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ *be a closed embedding of differentiable stacks. Let* $\mathcal{V} \hookrightarrow \mathcal{Z}$ *be an open embedding. Then there exists an open embedding* $\mathcal{U} \hookrightarrow \mathcal{X}$ *whose pullback along* $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ *is* $\mathcal{V} \hookrightarrow \mathcal{Z}$.

Proof. By Corollary 3.1.12, the open substack $\mathcal{V}$ of $\mathcal{Z}$ corresponds to an open subspace $|\mathcal{V}|_{\mathrm{mod}}$ of $|\mathcal{Z}|_{\mathrm{mod}}$. Since $|\mathcal{Z}|_{\mathrm{mod}} \subseteq |\mathcal{X}|_{\mathrm{mod}}$ carries the subspace topology by Proposition 3.2.1, there exists an open subspace $U \subseteq |\mathcal{X}|_{\mathrm{mod}}$ such that $U \cap |\mathcal{Z}|_{\mathrm{mod}} = |\mathcal{V}|_{\mathrm{mod}}$. The desired substack is then given by $\mathcal{U} = U \times_{|\mathcal{X}|_{\mathrm{mod}}} \mathcal{X} \subseteq \mathcal{X}$. □

3.3 Separated differentiable stacks

A well-behaved class of differentiable stacks are the *separated* differentiable stacks.

Definition 3.3.1 (Separated morphism)**.** A morphism $f\colon \mathcal{X} \to \mathcal{Y}$ of stacks on Diff is called *separated* if its diagonal $\Delta_f\colon \mathcal{X} \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$ is a proper map of stacks. We say that a stack $\mathcal{X}$ is *separated* if the morphism $\mathcal{X} \to \mathrm{pt}$ is separated. We denote by

$$\mathrm{SepStk} \subseteq \mathrm{DiffStk}$$

the full subcategory of separated differentiable stacks.

Example 3.3.2 (Proper Lie groupoids)**.** For a Lie groupoid $\mathcal{G}$, the classifying stack $\mathbb{B}\mathcal{G}$ is separated if and only if $\mathcal{G}$ is a *proper Lie groupoid* (that is, the map $(s,t)\colon \mathcal{G}_1 \to \mathcal{G}_0 \times \mathcal{G}_0$ is a proper map, Definition D.13). Indeed, this follows from Corollary 2.4.8, using the observation that the base change of the diagonal $\mathbb{B}\mathcal{G} \to \mathbb{B}\mathcal{G} \times \mathbb{B}\mathcal{G}$ along the representable atlas $\mathcal{G}_0 \times \mathcal{G}_0 \twoheadrightarrow \mathbb{B}\mathcal{G} \times \mathbb{B}\mathcal{G}$ is the map $(s,t)\colon \mathcal{G}_1 \to \mathcal{G}_0 \times \mathcal{G}_0$.

In particular, for a Lie group G the stack $\mathbb{B}G$ is separated if and only if G is compact. Similarly, the quotient stack $M//G$ of a smooth G-manifold M is separated if and only if G acts properly on M.

Lemma 3.3.3. *Let $\mathcal{X}$ and $\mathcal{Y}$ be differentiable stacks. Then any representable morphism $f\colon \mathcal{X} \to \mathcal{Y}$ is separated.*

Proof. Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a representable morphism of differentiable stacks. Let $M \twoheadrightarrow \mathcal{Y}$ be a representable atlas for $\mathcal{Y}$ and let $N := M \times_{\mathcal{Y}} \mathcal{X} \twoheadrightarrow \mathcal{X}$ denote the associated representable atlas for $\mathcal{X}$. Consider the diagram

$$\begin{tikzcd}
N \arrow[r, two heads] \arrow[d, "\Delta"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{X} \arrow[d, "\Delta"] \\
N \times_M N \arrow[r, two heads] \arrow[d, "\mathrm{pr}_1"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{X} \times_{\mathcal{Y}} \mathcal{X} \arrow[d, "\mathrm{pr}_1"] \\
N \arrow[r, two heads] & \mathcal{X}.
\end{tikzcd}$$

As the outer and bottom squares are pullbacks, so is the top square by the pasting law. The map $N \times_M N \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$ is representable and an effective epimorphism, since it is a pullback of $N \twoheadrightarrow \mathcal{X}$. As N and M are Hausdorff, the diagonal $\Delta\colon N \to N \times_M N$ is a closed embedding and hence proper. Since properness is a local property, it follows that $\mathcal{X} \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$ is also proper, showing that f is separated. $\square$

Definition 3.3.4 (Orbifold, cf. [Met03, Proposition 75, Definition 76])**.** A differentiable stack $\mathcal{X}$ is called an *étale stack* if it admits an étale atlas, i.e. there is a representable atlas $p\colon M \twoheadrightarrow \mathcal{X}$ which is a local diffeomorphism, in the sense of Definition 2.4.2. We say that $\mathcal{X}$ is an *orbifold* or a *differentiable Deligne-Mumford stack* if it is a separated étale stack.

Example 3.3.5. Let $\mathcal{G}$ be a proper *étale Lie groupoid*, meaning that the source and target maps $s,t\colon \mathcal{G}_1 \to \mathcal{G}_0$ are étale maps. Then the classifying stack $\mathbb{B}\mathcal{G}$ is a differentiable Deligne-Mumford stack. Conversely, every differentiable Deligne-Mumford stack $\mathcal{X}$ is of this form $\mathbb{B}\mathcal{G}$ for some proper étale Lie groupoid $\mathcal{G}$: take the Čech groupoid of any étale atlas $M \twoheadrightarrow \mathcal{X}$.

3.4 Isotropy groups

In this section, we introduce the *isotropy groups* of a differentiable stack $\mathcal{X}$, and show that a separated map $\mathcal{X} \to \mathcal{Y}$ of differentiable stacks is representable if and only if it induces injections at the level of isotropy groups. In particular, a separated differentiable stack is a smooth manifold if and only if its isotropy groups are trivial.

Definition 3.4.1 (Isotropy group)**.** Let $\mathcal{X}$ be a differentiable stack. A *point* of $\mathcal{X}$ is a map $x\colon \mathrm{pt} \to \mathcal{X}$ in DiffStk; we will sometimes write $x \in \mathcal{X}$ for short. The *isotropy group* G_x of x in $\mathcal{X}$ is defined as the pullback

$$\begin{tikzcd} G_x \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathrm{pt} \arrow[d, "x"] \\ \mathrm{pt} \arrow[r, "x"] & \mathcal{X}. \end{tikzcd}$$

As $G_x = \Omega_x(\mathcal{X})$ is the loop object of $\mathcal{X}$, it is automatically a group object in $\mathrm{Shv}(\mathrm{Diff})$. Note that, as a stack, G_x is equivalent to the fiber of the diagonal $\Delta\colon \mathcal{X} \to \mathcal{X} \times \mathcal{X}$ over (x,x).

Lemma 3.4.2. *For every point $x\colon \mathrm{pt} \to \mathcal{X}$ of a differentiable stack $\mathcal{X}$, the isotropy group G_x is a Lie group.*

Proof. Let $M \twoheadrightarrow \mathcal{X}$ be a representable atlas for $\mathcal{X}$. Since this atlas is surjective, the point $x\colon \mathrm{pt} \to \mathcal{X}$ may be lifted to a map $x\colon \mathrm{pt} \to M$. By forming the Čech nerve of the map $M \to \mathcal{X}$, we may present $\mathcal{X}$ as the classifying stack of a Lie groupoid $\mathcal{G}$, where $\mathcal{G}_0 = M$ and $\mathcal{G}_1 = M \times_{\mathcal{X}} M$. We claim that G_x is isomorphic to the isotropy group of $\mathcal{G}$ at $x \in M$. Indeed, this follows from the following pullback diagram:

$$\begin{tikzcd} G_x \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & t^{-1}(x) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathrm{pt} \arrow[d, "x"] \\ s^{-1}(x) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{G}_1 \arrow[r, "t"] \arrow[d, "s"] \arrow[dr, phantom, "\lrcorner"] & M \arrow[d, two heads] \\ \mathrm{pt} \arrow[r, "x"] & M \arrow[r, two heads] & \mathcal{X}. \end{tikzcd}$$

Since the isotropy groups of a Lie groupoid are Lie groups, see for example Proposition D.11, this finishes the proof. ◻

Lemma 3.4.3. *Let $\mathcal{X}$ be a separated differentiable stack. Then the isotropy group G_x is compact for every point $x\colon \mathrm{pt} \to \mathcal{X}$.*

Proof. By Corollary 2.3.8, we may assume $\mathcal{X} = \mathbb{B}\mathcal{G}$ is the classifying stack of a Lie groupoid $\mathcal{G}$. By Example 3.3.2, $\mathcal{X}$ is separated if and only if the morphism $(s,t)\colon \mathcal{G}_1 \to \mathcal{G}_0 \times \mathcal{G}_0$ is a proper map of manifolds. In particular, the fiber of this map over the point (x,x) is a compact subspace of $\mathcal{G}_1$. Since this fiber is isomorphic to the isotropy group G_x, this proves the claim. □

Proposition 3.4.4 (Crainic and Moerdijk [CM01, Theorem 1])**.** *A differentiable stack $\mathcal{X}$ is an étale stack, in the sense of Definition 3.3.4, if and only if for every point $x\colon \mathrm{pt} \to \mathcal{X}$ the isotropy group G_x is a discrete group.* □

A Lie groupoid whose isotropy groups are discrete are called *foliation groupoids*. By Proposition 2.3.14, the previous proposition is equivalent to the statement that every foliation groupoid is Morita equivalent to an étale Lie groupoid, which is the way the result is stated in [CM01].

For a separated differentiable stack $\mathcal{X}$, the isotropy groups can detect whether it is a smooth manifold:

Proposition 3.4.5 (cf. [Met03, Proposition 74])**.** *A separated differentiable stack $\mathcal{X}$ is a smooth manifold if and only if all its isotropy groups are trivial.*

Proof. If $\mathcal{X} = M$ is a smooth manifold, it is clear that all isotropy groups are trivial as the pullbacks defining them may be taken in Diff.

Conversely, let $\mathcal{X}$ be a separated differentiable stack with trivial isotropy groups, and let $M \twoheadrightarrow \mathcal{X}$ be a representable atlas for $\mathcal{X}$. Define the smooth manifold $R := M \times_{\mathcal{X}} M$, and consider the canonical map

$$R \to M \times M.$$

Since this map is a base change of the diagonal $\Delta\colon \mathcal{X} \to \mathcal{X} \times \mathcal{X}$ and $\mathcal{X}$ is separated, the map $R \to M \times M$ is proper. We claim that it is also injective. Indeed, the fiber over a pair $(x, y) \in M \times M$ is only non-empty if x and y are identified in $\mathcal{X}$, and in that case the fiber is isomorphic to the isotropy group G_x, which is assumed to be trivial. Since any proper injective map of smooth manifolds is a closed embedding, it follows that $R \hookrightarrow M \times M$ determines a closed equivalence relation on M.

Define N as the quotient space M/R, i.e. the topological space obtained from M by quotienting out the equivalence relation R. By Proposition C.6, this is again a smooth manifold, and the quotient map $M \to N$ is a surjective smooth submersion. The universal property of the stack quotient $\mathcal{X} \simeq M//R$ provides a canonical map $\mathcal{X} \to N$ making the following diagram commute:

$$\begin{tikzcd} M \arrow[r, equal] \arrow[d, two heads] & M \arrow[d] \\ \mathcal{X} \arrow[r] & N. \end{tikzcd}$$

We claim that this map $\mathcal{X} \to N$ is an equivalence, finishing the proof. Since the map $M \to N$ is an effective epimorphism by Remark 2.1.6, it will suffice to show that the induced map $\check{C}(M \twoheadrightarrow \mathcal{X}) \to \check{C}(M \twoheadrightarrow N)$ on Čech nerves is an equivalence. But this is clear: since the inclusion Diff $\hookrightarrow$ Shv(Diff) preserves finite limits, the Čech nerve of $M \to N$ is given by the Lie groupoid $(R \rightrightarrows M)$, which by definition of R is also the Čech nerve of the map $M \twoheadrightarrow \mathcal{X}$. $\square$

Corollary 3.4.6. *Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a separated morphism in* DiffStk. *The following conditions are equivalent:*

(1) The morphism f is representable;

(2) For every point $x \in \mathcal{X}$, the induced map on isotropy groups $G_x \to G_{f(x)}$ is injective.

Proof. Assume first that f is representable. Arguing as in Lemma 3.3.3, we see that the diagonal $\Delta_f\colon \mathcal{X} \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$ of f is a closed embedding, so in particular a monomorphism. But the following diagram shows that the map $G_x \to G_{f(x)}$ is a pullback of Δ_f, and thus also a monomorphism:

$$\begin{tikzcd} G_x \arrow[r] \arrow[d, hook] \arrow[dr, phantom, "\lrcorner"] & \mathcal{X} \arrow[d, hook, "\Delta_f"] & \\ G_{f(x)} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{X} \times_{\mathcal{Y}} \mathcal{X} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{Y} \arrow[d, "\Delta_{\mathcal{Y}}"] \\ \mathrm{pt} \arrow[r, "{(x,x)}"] & \mathcal{X} \times \mathcal{X} \arrow[r, "f \times f"] & \mathcal{Y} \times \mathcal{Y}. \end{tikzcd}$$

Conversely, assume that f is injective on isotropy groups. We show that f is representable. Letting $M \twoheadrightarrow \mathcal{Y}$ be a representable atlas for $\mathcal{Y}$, it will suffice to show that the pullback $\mathcal{N} := M \times_{\mathcal{Y}} \mathcal{X}$ is a smooth manifold. The isotropy groups of $\mathcal{N}$ are given by the pullback of the isotropy groups of M, $\mathcal{X}$ and $\mathcal{Y}$. But since M has trivial isotropy groups and the map $\mathcal{X} \to \mathcal{Y}$ induces injections on isotropy groups, it follows that $\mathcal{N}$ also has trivial isotropy groups. Since the map $\mathcal{N} \to M$ is separated, being a base change of f, we see that $\mathcal{N}$ is a separated differentiable stack, and thus it is a smooth manifold by Proposition 3.4.5. $\square$

Lemma 3.4.7. *Let $f\colon \mathcal{X} \hookrightarrow \mathcal{Y}$ be a monomorphism of differentiable stacks. Then for every point $x \in \mathcal{X}$, the induced map of Lie groups $G_x \to G_{f(x)}$ is an isomorphism.*

Proof. Since f is a monomorphism, its diagonal $\mathcal{X} \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$ is an equivalence, and thus we have the following pullback square of differentiable stacks:

$$\begin{tikzcd} \mathcal{X} \arrow[r, "f"] \arrow[d, "\Delta"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{Y} \arrow[d, "\Delta"] \\ \mathcal{X} \times \mathcal{X} \arrow[r, "f \times f"] & \mathcal{Y} \times \mathcal{Y}. \end{tikzcd}$$

In particular, f induces isomorphisms between the fibers of the two vertical maps. Since the induced map on fibers is the map $G_x \to G_{f(x)}$, this finishes the proof. □

3.5 Relative tangent bundles and normal bundles

In this section, we introduce the notions of *relative tangent bundle* and *normal bundle* for morphisms of differentiable stacks. We start by defining the tangent groupoid of a Lie groupoid $\mathcal{G}$.

Definition 3.5.1 (Tangent groupoid)**.** Let $\mathcal{G}$ be a Lie groupoid. We define its *tangent groupoid* $T\mathcal{G}$ as the Lie groupoid

$$T\mathcal{G} := \left(T\mathcal{G}_1 \overset{ds}{\underset{dt}{\rightrightarrows}} T\mathcal{G}_0 \right),$$

whose structure maps are the derivatives of the structure maps of $\mathcal{G}$. It comes with a natural map of Lie groupoids $\pi \colon T\mathcal{G} \to \mathcal{G}$.

The morphism $\pi \colon T\mathcal{G} \to \mathcal{G}$ is naturally a *VB-groupoid*, in the sense of Definition 2.5.9. However, it is not a *cartesian* morphism in general, in the sense of Definition 2.5.10, and so does not necessarily define a vector bundle over $\mathbb{B}\mathcal{G}$. This problem goes away when looking at *relative* versions of the construction: the relative tangent bundles, discussed in Subsection 3.5.1, and the normal bundles, discussed in Subsection 3.5.2.

3.5.1 Relative tangent bundles

Given a smooth submersion $f \colon M \to N$ between smooth manifolds, its *relative tangent bundle* is defined as the kernel of the derivative of f:

$$T_f := \ker(df \colon TM \twoheadrightarrow f^*TN) \quad \in \quad \mathrm{Vect}(M).$$

This has a direct generalization to the setting of Lie groupoids:

Definition 3.5.2 (Relative tangent groupoid)**.** Let $f\colon \mathcal{H} \to \mathcal{G}$ be a smooth submersion of Lie groupoids, meaning that $f_0\colon \mathcal{H}_0 \to \mathcal{G}_0$ and $f_1\colon \mathcal{H}_1 \to \mathcal{G}_1$ are smooth submersions. Passing to tangent groupoids gives a groupoid morphism $df\colon T\mathcal{H} \twoheadrightarrow T\mathcal{G}$ which is fiberwise linear. As it lives over f, it corresponds to a morphism $df\colon T\mathcal{H} \twoheadrightarrow f^*T\mathcal{G}$ of VB-groupoids over $\mathcal{H}$. We define the *relative tangent groupoid* Tf as the fiberwise kernel of this map:

$$Tf := \ker(df\colon T\mathcal{H} \twoheadrightarrow f^*T\mathcal{G}).$$

More explicitly, Tf_0 is the relative tangent bundle of the map $f_0\colon \mathcal{H}_0 \twoheadrightarrow \mathcal{G}_0$, Tf_1 is the relative tangent bundle of $f_1\colon \mathcal{H}_1 \twoheadrightarrow \mathcal{G}_1$, and the structure maps of Tf are obtained by restricting those from $T\mathcal{H}$. The bundle projections $Tf_0 \to \mathcal{H}_0$ and $Tf_1 \to \mathcal{H}_1$ define a morphism of Lie groupoids $Tf \to \mathcal{H}$, which is a VB-groupoid in the sense of Definition 2.5.9.

Lemma 3.5.3. *Assume the map $f\colon \mathcal{H} \to \mathcal{G}$ is a cartesian morphism of Lie groupoids, in the sense of Definition 2.5.10. Then the VB-groupoid $Tf \to \mathcal{H}$ is also cartesian.*

Proof. We have to show that the commutative squares

$$\begin{tikzcd} Tf_1 \arrow[r, "s"] \arrow[d] & Tf_0 \arrow[d] \\ \mathcal{H}_1 \arrow[r, "s"] & \mathcal{H}_0 \end{tikzcd} \qquad \text{and} \qquad \begin{tikzcd} Tf_1 \arrow[r, "t"] \arrow[d] & Tf_0 \arrow[d] \\ \mathcal{H}_1 \arrow[r, "t"] & \mathcal{H}_0 \end{tikzcd}$$

are pullback squares, which is a special case of Corollary C.13. □

Combining the previous lemma with Lemma 2.5.11 we see that the relative tangent groupoid defines a vector bundle $\mathbb{B}(Tf) \to \mathbb{B}\mathcal{H}$. We will now show that this vector bundle is Morita-invariant, in the sense that it only depends on the underlying map $\mathbb{B}f\colon \mathbb{B}\mathcal{H} \to \mathbb{B}\mathcal{G}$ of differentiable stacks.

Proposition 3.5.4. *Consider a pullback diagram of Lie groupoids*

$$\begin{tikzcd} \mathcal{H}' \arrow[r, two heads, "h"] \arrow[d, "f'"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{H} \arrow[d, "f"] \\ \mathcal{G}' \arrow[r, two heads, "g"] & \mathcal{G}, \end{tikzcd} \tag{II.3.1}$$

and assume that g and h are Morita fibrations, in the sense of Definition 2.3.10, and that f and f' are smooth submersions. Then the induced map $dh\colon Tf' \to Tf$ is again a Morita fibration.

Proof. As the map $Tf_0' \to \mathcal{H}_0'$ is the base change of $Tf_0 \to \mathcal{H}_0$ along the surjective submersion $h_0 \colon \mathcal{H}_0' \to \mathcal{H}_0$, the map $Tf_0' \to Tf_0$ is also a surjective submersion and thus dh is strongly surjective. To see that dh is also fully faithful, consider the following commutative diagram:

$$\begin{tikzcd}
Tf_1' \arrow[rr] \arrow[dr] \arrow[dd] & & Tf_0 \times Tf_0' \arrow[dr] \arrow[dd] & \\
& \mathcal{H}_1' \arrow[rr] \arrow[dd, "h_1"] & & \mathcal{H}_0' \times \mathcal{H}_0' \arrow[dd, "h_0 \times h_0"] \\
Tf_1 \arrow[rr] \arrow[dr] & & Tf_0 \times Tf_0 \arrow[dr] & \\
& \mathcal{H}_1 \arrow[rr, "(s,t)"] & & \mathcal{H}_0 \times \mathcal{H}_0
\end{tikzcd}$$

As $h \colon \mathcal{H}' \to \mathcal{H}$ is fully faithful, the front square is a pullback square in Diff. Since the given square of Lie groupoids is a pullback square, it follows from Corollary C.13 that also the left and right squares are pullback squares in Diff. By the pasting law for pullback squares, it follows that the back square is a pullback square in Diff. This finishes the proof. $\square$

Morita-invariance allows us to define the relative tangent bundle of a morphism of differentiable stacks:

Definition 3.5.5 (Relative tangent bundle)**.** Let $f \colon \mathcal{Y} \to \mathcal{X}$ be a representable submersion of differentiable stacks. We define its *relative tangent bundle* $T_f \to \mathcal{Y}$ as follows:

- Choose a representable atlas $p \colon M \twoheadrightarrow \mathcal{X}$ and let $\mathcal{G}$ denote the associated Lie groupoid given as the Čech nerve of p. We obtain an equivalence $\mathcal{X} \simeq \mathbb{B}\mathcal{G}$.

- Define an atlas $q \colon N \twoheadrightarrow \mathcal{Y}$ for $\mathcal{Y}$ as in Lemma 2.2.3 by pulling back p along f:

  $$\begin{tikzcd}
  N \arrow[r, "f_0"] \arrow[d, two heads, "q"'] \arrow[dr, phantom, "\lrcorner"] & M \arrow[d, two heads, "p"] \\
  \mathcal{Y} \arrow[r, "f"] & \mathcal{X}.
  \end{tikzcd}$$

  Let $\mathcal{H}$ denote the Čech nerve of q, so that $\mathcal{Y} \simeq \mathbb{B}\mathcal{H}$.

- There is a canonical smooth submersion of Lie groupoids $f_\bullet \colon \mathcal{H} \to \mathcal{G}$, which is cartesian by Lemma E.16. We let T_f be the classifying stack of the relative tangent groupoid $Tf_\bullet$:

  $$T_f := \mathbb{B}T(f_\bullet).$$

- The structure map $Tf_\bullet \to \mathcal{H}$ induces a morphism of stacks $T_f \to \mathcal{Y}$, which by Lemma 3.5.3 and Lemma 2.5.11 admits a canonical structure of a vector bundle.

Lemma 3.5.6. *Up to equivalence, the definition of the vector bundle $T_f \to \mathcal{Y}$ is independent of the choice of atlas $M \twoheadrightarrow \mathcal{X}$.*

Proof. Let $M \twoheadrightarrow \mathcal{X}$ and $M' \twoheadrightarrow \mathcal{X}$ be two representable atlases of $\mathcal{X}$. Going through the above construction gives two vector bundles $\mathbb{B}(Tf_\bullet) \to \mathcal{Y}$ and $\mathbb{B}(Tf'_\bullet) \to \mathcal{Y}$, which we need to show are isomorphic.

Define $M'' := M \times_{\mathcal{X}} M'$, so that the map $M'' \twoheadrightarrow \mathcal{X}$ is another representable atlas. Letting $\mathcal{G}$, $\mathcal{G}'$ and $\mathcal{G}'$ denote the Čech nerves of the maps $M \twoheadrightarrow \mathcal{X}$, $M' \twoheadrightarrow \mathcal{X}$ and $M'' \twoheadrightarrow \mathcal{X}$, we obtain morphisms of Lie groupoids

$$\mathcal{G} \leftarrow \mathcal{G}'' \to \mathcal{G}'$$

which by Proposition 2.3.14 are Morita fibrations. Applying the above construction we obtain three relative tangent Lie groupoids $Tf_\bullet$, $Tf'_\bullet$ and $Tf''_\bullet$. They are connected via morphisms of Lie groupoids

$$Tf_\bullet \leftarrow Tf''_\bullet \to Tf'_\bullet,$$

which by Proposition 3.5.4 are Morita fibrations and thus induce equivalences on differentiable stacks. It follows that $\mathbb{B}(Tf_\bullet) \simeq \mathbb{B}(Tf'_\bullet)$ finishing the proof. □

Remark 3.5.7. The construction of the relative tangent bundle T_f we have provided is somewhat unsatisfactory, as it is not defined intrinsically in terms of the differentiable stacks but requires a (non-coherent) choice of atlas. A more satisfactory definition would start by giving a fully functorial description of the relative tangent bundle construction at the level of smooth manifolds, and then use descent to extend this construction to all representable morphisms of stacks on Diff. We will not spell out the details of such a construction.

3.5.2 Normal bundles

We will now move to the construction of the normal bundle of an embedding of stacks, which is to a large extent analogous to that of the relative tangent bundle. Recall that the normal bundle N_i of an embedding $i\colon M \hookrightarrow N$ is defined as the cokernel of the derivative of i:

$$N_i := \operatorname{coker}(if\colon TM \hookrightarrow f^*TN) \quad \in \quad \operatorname{Vect}(M).$$

This admits a direct generalization to the setting of Lie groupoids:

Definition 3.5.8 (Normal groupoid)**.** Let $i\colon \mathcal{H} \to \mathcal{G}$ be an immersion of Lie groupoids, meaning that $i_0\colon \mathcal{H}_0 \to \mathcal{G}_0$ and $i_1\colon \mathcal{H}_1 \to \mathcal{G}_1$ are immersions of smooth manifolds. Consider its derivative $di\colon T\mathcal{H} \hookrightarrow T\mathcal{G}$ on tangent groupoids, which as before we may regard as a morphism of VB-groupoids $T\mathcal{H} \to i^*T\mathcal{G}$ over $\mathcal{H}$. We define the *normal groupoid Ni of i* as the fiberwise cokernel of this map:

$$Ni := \operatorname{coker}(di\colon T\mathcal{H} \hookrightarrow i^*T\mathcal{G}).$$

This means that Ni_0 is the normal bundle of the map $i_0\colon \mathcal{H}_0 \to \mathcal{G}_0$, Ni_1 is the normal bundle of the map $i_1\colon \mathcal{H}_1 \to \mathcal{G}_1$, and the structure maps of Ni are obtained as a quotient of the structure maps of $i^*T\mathcal{G}$. The structure maps of the normal bundles determine a VB-groupoid $Ni \to \mathcal{H}$.

Lemma 3.5.9. *If $i\colon \mathcal{H} \to \mathcal{G}$ is a cartesian morphism of Lie groupoids, then also the VB-groupoid $Ni \to \mathcal{H}$ is cartesian.*

Proof. Just like in Lemma 3.5.3, this follows from Corollary C.13. □

Remark 3.5.10 (Linearization of a Lie groupoid)**.** Let $\mathcal{G}$ be a Lie groupoid and let $S \subseteq \mathcal{G}_0$ be a saturated submanifold, in the sense that S is closed under the $\mathcal{G}$-action, Definition D.12. Define the Lie groupoid $\mathcal{G}_S$ over S as the restriction of $\mathcal{G}$ to S:

$$(\mathcal{G}_S)_1 = \{g \in \mathcal{G}_1 \mid s(g), t(g) \in S\} = s^{-1}(S) = t^{-1}(S),$$

where the last two equalities use the fact that S is saturated. There is an inclusion of Lie groupoids $i\colon \mathcal{G}_S \hookrightarrow \mathcal{G}$, and the fact that S is saturated guarantees that i is a cartesian morphism. In this case, the normal groupoid Ni of i is known in the literature as the *linearization of $\mathcal{G}$ at S* and is denoted $N_S\mathcal{G}$. Its space of objects is $N_S = \operatorname{coker}(TS \to T\mathcal{G}_0|_S)$, the normal bundle of S inside $\mathcal{G}_0$.

Example 3.5.11. Assume that $x \in \mathcal{G}_0$ is a fixed point of $\mathcal{G}$, and let $S = \{x\}$. Then $\mathcal{N}_S\mathcal{G}$ is the action groupoid $G_x \ltimes T_x\mathcal{G}_0$ of the isotropy group action on the tangent space of $\mathcal{G}_0$ at x.

Example 3.5.12. Assume that $\mathcal{G} = G \ltimes M$ is the action groupoid for some Lie group G and some smooth G-manifold M. Let $S = \mathcal{O}_x$ denote the orbit of a point $x \in M$ and let $N_x := T_xM/T_xS$. Then $\mathcal{N}_S\mathcal{G}$ is isomorphic to the action groupoid $G \ltimes (G \times_{G_x} N_x)$.

Morita equivalent Lie groupoids have Morita equivalent linearizations:

Proposition 3.5.13 ([HF19, Proposition 6.4.1])**.** *Consider a pullback of Lie groupoids as in* (II.3.1) *and assume that g and h are Morita fibrations and that f and f' are immersions. Then the induced morphism of Lie groupoids $Nf' \to Nf$ is a Morita fibration.*

Proof. The proof is entirely analogous to that of Proposition 3.5.4 and we will leave it to the reader. □

Definition 3.5.14 (Normal bundle)**.** Let $i\colon \mathcal{S} \hookrightarrow \mathcal{X}$ be an embedding of differentiable stacks. We define a *normal bundle* $\mathcal{N}_i \to \mathcal{S}$ as follows:

- Choose a representable atlas $p\colon M \twoheadrightarrow \mathcal{X}$ and let $\mathcal{G}$ denote the associated Lie groupoid given as the Čech nerve of p. We obtain an equivalence $\mathcal{X} \simeq \mathbb{B}\mathcal{G}$.
- Define the submanifold $S \subseteq M$ via the pullback square

$$\begin{tikzcd} S \arrow[r, hook] \arrow[d, two heads, "p_S"'] \arrow[dr, phantom, "\lrcorner", very near start] & M \arrow[d, two heads, "p"] \\ \mathcal{S} \arrow[r, hook, "i"] & \mathcal{X}. \end{tikzcd}$$

  This determines a Lie groupoid $\mathcal{H} = \mathcal{G}_S \overset{i_\bullet}{\hookrightarrow} \mathcal{G}$ presenting $\mathcal{S}$.
- We define the differentiable stack $\mathcal{N}_i$ as the classifying stack of the normal groupoid $Ni_\bullet = \mathcal{N}_S\mathcal{G}$. The structure map $\mathcal{N}_S\mathcal{G} \to \mathcal{G}_S$ is cartesian by Lemma 3.5.9 and thus induces a vector bundle of differentiable stacks $\mathcal{N}i \to \mathcal{S}$ by Lemma 2.5.11.

Lemma 3.5.15. *Up to equivalence, the definition of $\mathcal{N}_i \to \mathcal{S}$ is independent of the choice of atlas.*

Proof. This follows from Proposition 3.5.13. The proof is entirely analogous to that of Lemma 3.5.6 and will be omitted. □

3.5.3 Compatibilities

We record various compatibilities between normal bundles and relative tangent bundles.

Lemma 3.5.16 (Normal bundle of composite embedding)**.** *Given embeddings $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ and $j\colon \mathcal{X} \hookrightarrow \mathcal{Y}$, there is a canonical short exact sequence of vector bundles over $\mathcal{Z}$:*

$$\mathcal{N}_i \to \mathcal{N}_{ji} \to i^*\mathcal{N}_j.$$

Proof. When $\mathcal{Z}$, $\mathcal{X}$ and $\mathcal{Y}$ are smooth manifolds, this is immediate from the definitions using the second isomorphism theorem for groups. This directly implies the analogous statement for cartesian morphisms of Lie groupoids, which then gives the desired result by picking an atlas for $\mathcal{Y}$ and choosing the atlases for $\mathcal{Z}$ and $\mathcal{X}$ via pullback. □

Lemma 3.5.17 (Normal bundle of pullback)**.** *Consider a pullback diagram of differentiable stacks*

$$\begin{array}{ccc} \mathcal{Z}' & \xrightarrow{g} & \mathcal{Z} \\ {\scriptstyle i'}\downarrow & \lrcorner & \downarrow{\scriptstyle i} \\ \mathcal{X}' & \xrightarrow{f} & \mathcal{X}, \end{array}$$

where i and i' are embeddings and f and g are representable submersions. Then there is a preferred isomorphism $\mathcal{N}_{i'} \cong g^*\mathcal{N}_i$ of vector bundles over $\mathcal{Z}'$.

Proof. If all four of the stacks are smooth manifolds, this is Corollary C.13. By choosing a representable atlas for $\mathcal{X}$ and choosing the atlases for the other three stacks via pullback, this directly implies the general claim. □

Lemma 3.5.18 (Normal bundle of isotopic maps)**.** *Let $i\colon \mathcal{Z}\times\mathbb{R} \to \mathcal{X}$ be a map of stacks such that the map $(i,\mathrm{id})\colon \mathcal{Z}\times\mathbb{R} \to \mathcal{X}\times\mathbb{R}$ is an embedding. Then there is an isomorphism of vector bundles*

$$\mathcal{N}_{i_0} \cong \mathcal{N}_{i_1} \quad \in \quad \mathrm{Vect}(\mathcal{Z}).$$

Proof. Let $\mathcal{N}_{(i,\mathrm{id})} \to \mathcal{Z}\times\mathbb{R}$ denote the normal bundle of (i,id). As the embeddings maps $i_0, i_1\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ are base changes of the map (i,id) along the inclusions $\mathcal{X}\times\{0\} \hookrightarrow \mathcal{X}\times\mathbb{R}$ and $\mathcal{X}\times\{1\} \hookrightarrow \mathcal{X}\times\mathbb{R}$, Lemma 3.5.17 provides isomorphisms of vector bundles

$$\mathcal{N}_{i_0} \cong \mathcal{N}_{(i,\mathrm{id})}|_0 \qquad \text{and} \qquad \mathcal{N}_{i_1} \cong \mathcal{N}_{(i,\mathrm{id})}|_1.$$

But by Corollary 2.5.8, the vector bundle $\mathcal{N}_{(i,\mathrm{id})}$ over $\mathcal{Z}\times\mathbb{R}$ is isomorphic to $\mathcal{E}\times\mathbb{R}$ for some vector bundle $\mathcal{E}$ over $\mathcal{Z}$, and thus $\mathcal{N}_{i_0} \cong \mathcal{E} \cong \mathcal{N}_{i_1}$. □

Proposition 3.5.19. *Let $f\colon \mathcal{Y} \to \mathcal{X}$ be a representable submersion of differentiable stacks. Then there is an isomorphism $T_f \cong N_{\Delta_f}$ of vector bundles over $\mathcal{Y}$, where $\Delta_f\colon \mathcal{Y} \to \mathcal{Y}\times_{\mathcal{X}}\mathcal{Y}$ is the diagonal of f.*

Proof. We prove the lemma at progressing levels of generality.

Step 1: We first prove the claim when $\mathcal{X}$ is a point, so that $\mathcal{Y} = M$ is a smooth manifold and $T_f = TM$ is the tangent bundle of M. By Lemma C.12, the derivatives of the two projections $\mathrm{pr}_1, \mathrm{pr}_2\colon M\times M \to M$ induce an isomorphism of vector bundles $T(M\times M) \cong TM\times TM$ over $M\times M$. Under this isomorphism, the differential $d\Delta_M\colon TM \to T(M\times M)$ corresponds to the diagonal $\Delta_{TM}\colon TM \to TM\times TM$. Regarding this as a map of vector bundles over M given fiberwise by the diagonal $\Delta_{T_xM}\colon T_xM \to T_xM\times T_xM$, its cokernel in $\mathrm{Vect}(M)$ admits an explicit isomorphism to TM given by the composite $TM \xrightarrow{(1,-1)} TM\times TM \twoheadrightarrow \mathrm{coker}(d\Delta_M)$. Since this cokernel is N_{Δ_M}, this finishes the proof.

Step 2: Next we show the claim when $\mathcal{X} = N$ is a smooth manifold, so that $f\colon M \to N$ is a smooth submersion between smooth manifolds. Observe that the isomorphisms from Step 1 give rise to a commutative square of vector bundles over M:

$$\begin{tikzcd} TM \arrow[r] \arrow[d, "\cong"'] & f^*T_N \arrow[d, "\cong"] \\ N_{\Delta_M} \arrow[r] & f^*N_{\Delta_N}. \end{tikzcd}$$

Since T_f is defined as the fiber of the top map, it suffices to show that N_{Δ_f} is the fiber of the bottom map. To this end, consider the following diagram:

$$\begin{tikzcd} M \arrow[r, "\Delta_f"] \arrow[dr, "\Delta_M"'] & M \times_N M \arrow[r, "g"] \arrow[d, hook, "i"'] \arrow[dr, phantom, "\lrcorner"] & N \arrow[d, hook, "\Delta_N"] \\ & M \times M \arrow[r, "f \times f"] & N \times N, \end{tikzcd}$$

where i is the inclusion. By Lemma 3.5.16, there is the following short exact sequence of vector bundles over M:

$$N_{\Delta_f} \to N_{\Delta_M} \to \Delta_f^* N_i.$$

But by Lemma 3.5.17, the vector bundel N_i is isomorphic to $g^*N_{\Delta_N}$, and since $f = g \circ \Delta_f$ it follows that $\Delta_f^* N_i \cong f^*N_{\Delta_N}$. This finishes the proof of Step 2.

Step 3: We now show the general statement. By choosing an atlas for $\mathcal{X}$, we may assume that $\mathcal{X} = \mathbb{B}\mathcal{G}$ is the classifying stack of a Lie groupoid. By pulling back this atlas to $\mathcal{Y}$ we may similarly assume that $\mathcal{Y} = \mathbb{B}\mathcal{H}$ and that f is induced by a cartesian morphism $f\colon \mathcal{H} \to \mathcal{G}$ of Lie groupoids. It will thus suffice to show that the relative tangent groupoid of f is isomorphic to the normal groupoid of its diagonal. This follows from Step 2. □

Lemma 3.5.20. *Consider a commutative diagram of differentiable stacks*

$$\begin{tikzcd} \mathcal{S} \arrow[r, hook, "i"] \arrow[dr, "g"'] & \mathcal{Y} \arrow[d, "f"] \\ & \mathcal{X}, \end{tikzcd}$$

where f and g are representable submersions and i is an embedding. Then there is a short exact sequence of vector bundles

$$T_g \to i^*T_f \to \mathcal{N}_i.$$

Proof. As in the previous proofs, we may reduce to the case where the three stacks are smooth manifolds, say S, Y and X. In this case the statement follows from the fact that forming vertical kernels commutes with forming horizontal cokernels in the following diagram of vector bundles over S:

$$\begin{tikzcd}
T_g \arrow[r, hook] \arrow[d, hook] & i^*T_f \arrow[r, two heads] \arrow[d, hook] & N_i \arrow[d, equal] \\
TS \arrow[r, hook] \arrow[d, two heads] & i^*TY \arrow[r, two heads] \arrow[d, two heads] & N_i \arrow[d] \\
g^*TX \arrow[r, equal] & i^*f^*TX \arrow[r] & 0.
\end{tikzcd}$$

□

3.6 Tubular neighborhoods

In this section, we introduce the notion of a *tubular neighborhood* of an embedding of differentiable stacks. We further show that under separatedness assumptions such tubular neighborhoods always exist.

Definition 3.6.1 (Tubular neighborhood)**.** Let $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be an embedding of differentiable stacks. A *tubular neighborhood of $\mathcal{Z}$ in $\mathcal{X}$* consists of the data of a commutative diagram

$$\begin{tikzcd}
 & \mathcal{Z} \arrow[dl, hook, bend right, "i"'] \arrow[d, hook] \arrow[dr, hook, bend left, "s_0"] & \\
\mathcal{X} & \mathcal{U} \arrow[l, hook', "j"'] \arrow[r, hook, "j'"] & \mathcal{N}_i
\end{tikzcd}$$

where the maps j and j' are open embeddings of differentiable stacks, and where s_0 is the zero section of the bundle projection $\mathcal{N}_i \to \mathcal{Z}$ of the normal bundle of i.

We might sometimes refer to the data of a tubular neighborhood by just mentioning the open substack $\mathcal{U}$ of $\mathcal{X}$, leaving the rest of the data implicit.

The main input for the existence of tubular neighborhoods is the *linearization theorem for proper Lie groupoids*, due to Zung [Zun06] and Weinstein [Wei02] and later refined and clarified by [CS13; PPT14; HF18].

Definition 3.6.2. Let $\mathcal{G}$ be a Lie groupoid and let $S \subseteq M$ a saturated submanifold. We say that $\mathcal{G}$ is *linearizable around* S if there exist open sets $S \subseteq U \subseteq M$ and $S \subseteq V \subseteq \mathcal{N}_S$ and an isomorphism of Lie groupoids

$$\mathcal{G}|_U \cong \mathcal{N}_S(\mathcal{G})|_V$$

which is the identity on $\mathcal{G}_S$. We say that $\mathcal{G}$ is *linearizable* if it is linearizable around any saturated submanifold.

The notation $\mathcal{G}|_U$ in this definition stands for the *restriction* of the Lie groupoid to the open subspace U, see Example D.7.

Theorem 3.6.3 (Linearization theorem for proper Lie groupoids, [Wei02; Zun06; CS13; PPT14], [HF18, Corollary 5.3.3])**.** *Proper Lie groupoids are linearizable.*

We will use the linearization theorem to show the existence of tubular neighborhoods. We need to following auxiliary lemma:

Lemma 3.6.4. *Let $\mathcal{G} = (\mathcal{G}_1 \rightrightarrows M)$ be a Lie groupoid, let $U \subseteq M$ be an open subspace and let $\mathcal{G}|_U$ be the restriction of $\mathcal{G}$ to U, as in Example D.7. Then the inclusion $\mathcal{G}|_U \hookrightarrow \mathcal{G}$ of Lie groupoids induces an open embedding*

$$\mathbb{B}(\mathcal{G}|_U) \hookrightarrow \mathbb{B}\mathcal{G}$$

at the level of classifying stacks.

Proof. Let $U' := \mathcal{G} \cdot U \subseteq M$ be the *$\mathcal{G}$-saturation of U*, that is, the union of all orbits of $\mathcal{G}$ intersecting U non-trivially. Then U' defines an open subspace of the coarse moduli space $|\mathbb{B}\mathcal{G}|_{\mathrm{mod}} = \mathrm{coeq}(\mathcal{G}_1 \rightrightarrows M)$ of $\mathbb{B}\mathcal{G}$, and hence defines an open substack $\mathcal{U} \subseteq \mathbb{B}\mathcal{G}$, fitting in a pullback square as follows:

$$\begin{tikzcd} U' \arrow[r, hook] \arrow[d, two heads] \arrow[dr, phantom, "\lrcorner", very near start] & M \arrow[d, two heads] \\ \mathcal{U} \arrow[r, hook] & \mathbb{B}\mathcal{G}. \end{tikzcd}$$

We claim that the composite $U \hookrightarrow U' \twoheadrightarrow \mathcal{U}$ is a representable atlas for $\mathcal{U}$. By Lemma 2.2.6 it suffices to check this after pulling back along the atlas $U' \twoheadrightarrow \mathcal{U}$, where this follows from the fact that every $x \in U'$ is of the form gu for some $g \in \mathcal{G}_1$ and $u \in U$. We conclude that $\mathcal{U}$ is equivalent to the classifying stack of the Čech nerve of the map $U \twoheadrightarrow \mathcal{U}$, which is precisely the restricted Lie groupoid $\mathcal{G}|_U$. As the resulting open embedding $\mathbb{B}(\mathcal{G}|_U) \hookrightarrow \mathbb{B}\mathcal{G}$ agrees with the map induced by the inclusion $\mathcal{G}|_U \hookrightarrow \mathcal{G}$ of Lie groupoids, this finishes the proof. □

Corollary 3.6.5 (Existence of tubular neighborhoods)**.** *Let $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be an embedding of separated stacks. Then i admits a tubular neighborhood.*

Proof. By choosing a representable atlas for $\mathcal{X}$, we may assume that $\mathcal{X} = \mathbb{B}\mathcal{G}$ is the classifying stack of a Lie groupoid $\mathcal{G}$. As $\mathcal{X}$ is separated, $\mathcal{G}$ is proper, see Example 3.3.2. The embedding i determines a saturated embedded submanifold $S \subseteq \mathcal{G}_0$. By the linearization theorem 3.6.3 there exist open neighborhoods $S \subseteq U \subseteq \mathcal{G}_0$ and $S \subseteq V \subseteq \mathcal{N}_S$ together with an an isomorphism of Lie groupoids $\mathcal{G}|_U \cong \mathcal{N}_S(\mathcal{G})|_V$. We define

$$\mathcal{U} := \mathbb{B}(\mathcal{G}|_U) \hookrightarrow \mathbb{B}\mathcal{G} = \mathcal{X} \quad \text{and} \quad \mathcal{V} := \mathbb{B}(\mathcal{N}_S(\mathcal{G})|_V) \hookrightarrow \mathbb{B}\mathcal{N}_S(\mathcal{G}) = \mathcal{N}_i.$$

By Lemma 3.6.4 these are open neighborhoods of $\mathcal{Z} = \mathbb{B}\mathcal{G}_S$. There is an equivalence $\mathcal{U} \simeq \mathcal{V}$ which is the identity on $\mathcal{Z}$. This finishes the proof. □

We record two useful corollaries of the existence of tubular neighborhoods.

Corollary 3.6.6. *Every embedding $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ of separated differentiable stacks factors as a composite*

$$\mathcal{Z} \overset{i'}{\hookrightarrow} \mathcal{U} \overset{j}{\to} \mathcal{X}$$

of a closed embedding i' followed by an open embedding j, where the map i' admits a retraction $f\colon \mathcal{U} \to \mathcal{Z}$. □

Corollary 3.6.7 (Extension of representable submersions)**.** *Let $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be an embedding of separated differentiable stacks. For every representable submersion $\mathcal{Y} \to \mathcal{Z}$, there exists a pullback square*

$$\begin{tikzcd} \mathcal{Y} \arrow[r] \arrow[d, "f"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{Y}' \arrow[d, "f'"] \\ \mathcal{Z} \arrow[r, hook, "i"] & \mathcal{X} \end{tikzcd}$$

where f' is also a representable submersion.

Proof. Choosing a factorization of i as in Corollary 3.6.6, we may define $\mathcal{Y}' := \mathcal{U} \times_{\mathcal{Z}} \mathcal{Y}$, and define f' as the composite $\mathcal{U} \times_{\mathcal{Z}} \mathcal{Y} \xrightarrow{\mathrm{pr}_{\mathcal{U}}} \mathcal{U} \overset{j}{\hookrightarrow} \mathcal{X}$. □

Remark 3.6.8 (Linearization of representable submersions)**.** It is possible to prove the following variant of Corollary 3.6.5: for every pullback diagram

$$\begin{tikzcd} \mathcal{Z}' \arrow[r, hook, "i'"] \arrow[d, "f|_{\mathcal{Z}'}"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{X}' \arrow[d, "f"] \\ \mathcal{Z} \arrow[r, hook, "i"] & \mathcal{X} \end{tikzcd}$$

where i and i' are embeddings and f is a representable submersion, one may choose tubular neighbhorhoods $\mathcal{Z} \subseteq \mathcal{U} \subseteq \mathcal{X}$, $\mathcal{Z} \subseteq \mathcal{V} \subseteq \mathcal{N}_i$ and $\mathcal{Z}' \subseteq \mathcal{U}' \subseteq \mathcal{X}'$, $\mathcal{Z}' \subseteq \mathcal{V}' \subseteq \mathcal{N}_{i'}$ fitting in a commutative diagram as follows:

$$\begin{tikzcd} \mathcal{N}_{i'} \arrow[d, "df"'] & \mathcal{V}' \arrow[l, hook'] \arrow[r, "\simeq"] & \mathcal{U}' \arrow[r, hook] & \mathcal{X}' \arrow[d, "f"] \\ \mathcal{N}_i & \mathcal{V} \arrow[l, hook'] \arrow[r, "\simeq"] & \mathcal{U} \arrow[r, hook] & \mathcal{X}. \end{tikzcd}$$

Just as in the proof of Corollary 3.6.5, this can be deduced from the analogous linearization result for proper Lie groupoids, which was proved by Hoyo and Fernandes [HF19, Theorem 4.2.3].

3.6.1 Relative tubular neighborhoods

The notion of tubular neighborhood admits a *relative* version, in which the embedding $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ lives over some base stack $\mathcal{S}$.

Definition 3.6.9 (Relative tubular neighborhood)**.** Consider a commutative diagram of differentiable stacks

$$\begin{tikzcd}
\mathcal{Z} \arrow[rr, "i"] \arrow[dr, "g"'] & & \mathcal{X} \arrow[dl, "f"] \\
& \mathcal{S}, &
\end{tikzcd}$$

where the morphisms f and g are representable submersions and the morphism i is an embedding. A *tubular neighborhood of $\mathcal{Z}$ in $\mathcal{X}$ relative to $\mathcal{S}$* is a commutative diagram

$$\begin{tikzcd}
& \mathcal{Z} \arrow[dl, hook', "i"'] \arrow[d, hook] \arrow[dr, hook, "s_0"] & \\
\mathcal{X} \arrow[dr, "f"'] & \mathcal{U} \arrow[l, hook', "j"'] \arrow[r, hook, "j'"] \arrow[d] & \mathcal{N}_i \arrow[dl, "g'"] \\
& \mathcal{S}, &
\end{tikzcd}$$

where the maps j and j' are open embeddings of differentiable stacks, and g' is the composite of the bundle projection $\mathcal{N}_i \to \mathcal{Z}$ followed by the map $g\colon \mathcal{Z} \to \mathcal{S}$.

Relative tubular neighborhoods always exist if the stack $\mathcal{X}$ is separated: as in Corollary 3.6.5, one may reduce to the analogous statement in the context of Lie groupoids, where the proof strategy of [HF18, Corollary 5.3.3] applies. In the case that $\mathcal{S}$, $\mathcal{Z}$ and $\mathcal{X}$ are smooth manifold, this has been proved by [Mat12]. Unfortunately, we were unable to find a reference for the general relative statement for differentiable stacks. We will not give a complete proof either, but we will give a detailed proof sketch indicating the main ingredients of a proof. We thank Shachar Carmeli, Florian Naef and Maarten Mol for useful discussions.

Proposition 3.6.10 (Existence of relative tubular neighborhoods)**.** *Let $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be an embedding of differentiable stacks over some base stack $\mathcal{S}$, as in Definition 3.6.9, and assume that $\mathcal{X}$ is separated. Then i admits a tubular neighborhood relative to $\mathcal{S}$.*

Detailed proof sketch. Broadly speaking, the proof proceeds just like the ordinary proof of the existence of tubular neighborhoods for differentiable manifolds: one chooses a Riemannian metric and shows that the resulting exponential map restricts to a diffeomorphism on a small open neighborhood inside the normal bundle of the embedding. The subtlety lies in making precise the notions of 'Riemannian metric' and 'exponential map' in the context of differentiable stacks $\mathcal{X}$ relative to some base stack $\mathcal{S}$. We will indicate one possible approach, which follows the usual two-step procedure of reducing to the setting of smooth manifolds.

Step 1: One first considers the case where the base stack $\mathcal{S}$ is a smooth manifold S, so that also $\mathcal{Z} = Z$ and $\mathcal{X} = X$ are smooth manifolds. By Lemma 3.5.20, the normal bundle of i fits in a short exact sequence of vector bundles

$$T_g \hookrightarrow i^* T_f \to N_i$$

over Z. Observe that the Lie bracket on TX restricts to T_f by naturality of the Lie bracket, so that T_f becomes a Lie algebroid over X with the inclusion $T_f \hookrightarrow TX$ serving as the anchor map. If we choose a Riemannian metric on X, we may restrict it to the relative tangent bundle T_f, thus turning T_f into a *Riemannian Lie algebroid*. Generalizing the usual notion of connections on smooth manifolds, there is a notion of a *connection* on a Lie algebroid, and one can show that every Riemannian Lie algebroid admits a unique connection which is both *metric* and *torsion-free*; this is known as the *Levi-Civita-connection*, see [Bou11, Section 3.1]. One obtains a corresponding notion of *geodesics* with respect to this connection. In the case of the relative tangent bundle T_f, it follows from [Bou11, Proposition 3.2] that the geodesics with respect to the Levi-Civita connection on T_f are precisely those curves $\gamma\colon [t_0, t_1] \to X$ such that the composite $f \circ \gamma\colon [t_0, t_1] \to S$ is constant, say with value $s \in S$, and such that the resulting curve $\gamma\colon [t_0, t_1] \to X_s = f^{-1}(s)$ into the fiber of f is a geodesic when X_s is equipped with the Riemannian metric inherited from X. The usual construction of the exponential map carries over to this relative setting and produces a *relative exponential map*

$$\exp_f\colon \mathcal{E}_f \to X,$$

where $\mathcal{E}_f \subseteq T_f$ is the open neighborhood of the zero section on which the relevant geodesics are defined. From the fact that the relative geodesics live in a single fiber, it follows the map $\exp_f$ lives over S, in the sense that $f(\exp(x, v)) = f(x)$ for every relative tangent vector v at $x \in X$.

Since $i^* T_f$ admits a Riemannian metric, we can embed N_i into $i^* T_f$ as the orthogonal complement of T_g. We may restrict the relative exponential map to the intersection $N_i \cap$

$i^*\mathcal{E}_f \subseteq i^*T_f$, and a calculation of the derivative of this restriction shows that it is a local diffeomorphism. An adaptation of the standard argument for Riemannian manifolds shows that this map becomes an actual diffeomorphism when restricted to a sufficiently small open neighborhood of the zero section in N_i, see for example [Lan95, Theorem IV.5.1]. This finishes Step 1.

Step 2: In the general case, we may choose an atlas for $\mathcal{S}$ and equip $\mathcal{Z}$ and $\mathcal{X}$ with the atlases obtained by pullback, as in Lemma 2.2.3. This means that we may represent $\mathcal{S}$, $\mathcal{Z}$ and $\mathcal{X}$ by Lie groupoids $\mathcal{K}$, $\mathcal{H}$ and $\mathcal{G}$, and the maps i, f and g correspond to cartesian morphisms of Lie groupoids $i\colon \mathcal{H} \hookrightarrow \mathcal{G}$, $g\colon \mathcal{Z} \to \mathcal{K}$ and $f\colon \mathcal{G} \to \mathcal{K}$.

Our goal is to reduce the claim to the statement from Step 1. We will use the proof strategy from [HF18, Theorem 5.3.1]. Since $\mathcal{X}$ is separated, the corresponding Lie groupoid $\mathcal{G}$ is proper by Example 3.3.2. It was shown by Hoyo and Fernandes [HF18, Theorem 4.3.4] that every proper Lie groupoid $\mathcal{G}$ admits a *simplicial Riemannian metric*[1]: a collection of Riemannian metrics on each of the smooth manifolds $\mathcal{G}_n$ for $[n] \in \Delta$ such that for every injective map $\varphi\colon [n] \to [m]$ in Δ the induced map $\mathcal{G}_\varphi\colon \mathcal{G}_m \to \mathcal{G}_n$ is a Riemannian submersion. We may restrict these Riemannian metrics to the relative tangent bundles $Tf_n \to \mathcal{G}_n$. By Lemma 3.5.3, the relative tangent groupoid $Tf_\bullet \to \mathcal{G}$ is cartesian, so that for every map $[n] \to [m]$ in Δ we obtain a pullback square

$$\begin{tikzcd} Tf_m \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{G}_m \arrow[d] \\ Tf_n \arrow[r] & \mathcal{G}_n. \end{tikzcd}$$

Due to the uniqueness of the geodesics, it follows that the relative exponential maps $\exp_{f_n}\colon \mathcal{E}_{f_n} \to \mathcal{G}_n$ produced in Step 1 are all compatible: for every injection $\varphi\colon [n] \to [m]$ in Δ, the square

$$\begin{tikzcd} \mathcal{E}_{f_m} \arrow[r, "\exp_{f_m}"] \arrow[d, "d\mathcal{G}_\varphi"'] & \mathcal{G}_m \arrow[d, "\mathcal{G}_\varphi"] \\ \mathcal{E}_{f_n} \arrow[r, "\exp_{f_n}"] & \mathcal{G}_n \end{tikzcd}$$

commutes. As in [HF18, Theorem 5.3.1], one can show that there exist compatible open neighborhood $U_n \subseteq N_{i_n} \cap i_n^*\mathcal{E}_{f_n}$ of the zero section on which the relative exponential map $\exp_{f_n}$ restricts to a diffeomorphism, where the word 'compatible' means that the U_n form a simplicial diagram in Diff constituting a Lie groupoid. Passing to classifying stacks then produces the desired relative tubular neighborhood. □

[1] In fact, del Hoyo and Fernandes restrict attention to *2-metrics*. We will work with simplicial metrics for ease of exposition.

3.7 Local structure of separated differentiable stacks

An important consequence of the linearization theorem for proper Lie groupoids mentioned above is the simple local structure of separated differentiable stacks: locally they look like a quotient stack $V//G$ for a compact Lie group G and an orthogonal G-representation V. We shall discuss this result and deduce various consequences. The starting point is the following theorem on the local structure of proper Lie groupoids:

Theorem 3.7.1 ([CS13, Corollary 3.9], [PPT14, Corollary 3.11]). *Let $\mathcal{G}$ be a proper Lie groupoid over M, let $x \in M$ and let $\mathcal{O} = \mathcal{O}_x \subseteq M$ be the orbit of x. Then there exists an open neighborhood $\mathcal{O} \subseteq U \subseteq M$ such that the restriction $\mathcal{G}|_U$ is Morita equivalent to the action groupoid $G_x \ltimes \mathcal{N}_x$, where $\mathcal{N}_x = T_xM/T_x(\mathcal{O})$ is the normal tangent space at x of the orbit $\mathcal{O}$ inside M. Under this Morita equivalence, the orbit $\mathcal{O} \subseteq M$ corresponds to the point $0 \in \mathcal{N}_x$.* □

We deduce from this the local structure theorem for separated differentiable stacks:

Theorem 3.7.2 (Local structure of separated differentiable stacks). *Let $\mathcal{X}$ be a separated differentiable stack and let $x \in \mathcal{X}$ be a point of $\mathcal{X}$. Then there exists an open substack $\mathcal{U} \subseteq \mathcal{X}$ containing x, a G_x-representation V, and an equivalence*

$$\mathcal{U} \simeq V//G_x$$

sending the point $x \in \mathcal{U}$ to the point $0 \in V//G_x$.

Proof. Choose an atlas $f\colon M \twoheadrightarrow \mathcal{X}$, and let $\mathcal{G} := \check{C}(f)$ denote the associated Čech groupoid. The point x of $\mathcal{X}$ can be lifted to a point x of M. Let $\mathcal{O} = \mathcal{O}_x$ denote the orbit of x in M, which is an embedded submanifold by Proposition D.11, and let $\mathcal{N}_x := T_xM/T_x\mathcal{O}$ denote the normal space x of $\mathcal{O}$ inside M. By Theorem 3.7.1, there exist an open neighbhorhood U of $\mathcal{O}$ in M such that the restricted Lie groupoid $\mathcal{G}|_U$ is Morita equivalent to the action groupoid $G_x \ltimes \mathcal{N}_x$ and such that the orbit $\mathcal{O}$ corresponds to the zero-vector $0 \in \mathcal{N}_x$ under this Morita equivalence. By Proposition 2.3.14, this Morita equivalence induces an equivalence of stacks

$$\mathbb{B}(\mathcal{G}|_U) \simeq \mathbb{B}(G_x \ltimes \mathcal{N}_x) = \mathcal{N}_x//G_x$$

which sends $x \in \mathbb{B}(\mathcal{G}|_U)$ to $0 \in \mathcal{N}_x//G_x$. By Lemma 3.6.4, the inclusion $\mathcal{G}|_U \hookrightarrow \mathcal{G}$ then induces an open embedding

$$\mathcal{N}_x//G_x \simeq \mathbb{B}(\mathcal{G}|_U) \hookrightarrow \mathbb{B}\mathcal{G} \simeq \mathcal{X},$$

finishing the proof. □

Using the structure theorem, one can generalize various results about the differential geometry of smooth G-manifolds for a compact Lie group G to the setting of separated differentiable stacks.

3.7.1 Factorization of proper maps

We will show that every proper map of separated differentiable stacks can locally be factored as a closed embedding followed by a proper representable submersion.

Proposition 3.7.3. *Let $p\colon \mathcal{X} \to \mathcal{Y}$ be a proper map between separated differentiable stacks. Around every point $y \in \mathcal{Y}$ there exists an open neighborhood $\mathcal{U} \subseteq \mathcal{Y}$ such that the restriction $p_{\mathcal{U}}\colon \mathcal{X}_{\mathcal{U}} \to \mathcal{U}$ factors as*

$$\mathcal{X}_{\mathcal{U}} \xhookrightarrow{i} S^{\mathcal{E}} \xrightarrow{\pi} \mathcal{U},$$

where $\pi\colon \mathcal{E} \to \mathcal{U}$ is a vector bundle and where i is a closed embedding factoring through $\mathcal{E} \hookrightarrow S^{\mathcal{E}}$.

Proof. Let $G = G_y$ denote the isotropy group of $\mathcal{Y}$ at y. By Theorem 3.7.2, there is an open neighborhood $\mathcal{U}'$ of y in $\mathcal{Y}$ of the form $V//G$ for some orthogonal G-representation V. The restriction $\mathcal{X}_{\mathcal{U}'}$ of $\mathcal{X}$ is then of the form $M//G$ for some G-manifold M, and the map p is induced by a G-equivariant proper map $q\colon M \to V$. Let $D \subseteq V$ denote the open unit disk inside V. Since the closure of D inside V is compact and the map q is proper, it follows that the closure of the preimage $p'^{-1}(D) \subseteq M$ is also compact. Hence by [Bre72, Theorem VI.4.1], there exists a G-representation W and a G-equivariant closed embedding $q^{-1}(D) \hookrightarrow W$. Setting $\mathcal{U} = D//G \hookrightarrow \mathcal{Y}$, we thus obtain a closed embedding $\mathcal{X}_{\mathcal{U}} = q^{-1}(D)//G \hookrightarrow W//G$.

Define a vector bundle $\mathcal{E} \to \mathcal{U}$ via the following pullback square:

$$\begin{tikzcd}
\mathcal{E} \arrow[r] \arrow[d, "\pi"'] \arrow[dr, phantom, "\lrcorner"] & W//G \arrow[d] \\
\mathcal{U} = D//G \arrow[r] & \mathbb{B}G.
\end{tikzcd}$$

Then we get $S^{\mathcal{E}} \simeq S^W//G \times_{\mathbb{B}G} \mathcal{U}$. The maps $\mathcal{X}_{\mathcal{U}} \to \mathcal{U}$ and $\mathcal{X}_{\mathcal{U}} \hookrightarrow W//G \hookrightarrow S^W//G$ then determine a commutative diagram

$$\begin{tikzcd}
\mathcal{X}_{\mathcal{U}} \arrow[r, hook, "i"] \arrow[dr, "p_{\mathcal{U}}"'] & S^{\mathcal{E}} \arrow[d, "\pi"] \\
& \mathcal{U}.
\end{tikzcd}$$

The map i is an embedding, and it is proper by Lemma 2.4.10, since the map $p_{\mathcal{U}}$ is proper. Since any proper embedding is a closed embedding, this finishes the proof. $\square$

3.7.2 Locally extending sections

As our final result in this chapter, we show that a partial section of a representable submersion defined on a closed substack can locally be extended to a section defined on an open neighborhood of the closed substack. Furthermore, this section can be chosen to be the zero section of a vector bundle. We start with an auxiliary lemma.

Lemma 3.7.4. *Let G be a compact Lie group, let $f: Y \to X$ be a G-equivariant submersion between smooth G-manifolds and let $s: X \to Y$ be a section of f. For every $x \in X$, there exist G-invariant open neighborhoods $x \in N \subseteq X$ and $s(x) \in E \subseteq Y$ such that:*

- *We have $f(E) \subseteq N$ and $s(N) \subseteq E$;*
- *The restriction $f|_E: E \to N$ is isomorphic to a G-equivariant vector bundle with zero-section $s|_N: N \to E$.*

Proof. Let $H := G_x \subseteq G$ denote the isotropy group of x in X, which is also the isotropy group of $s(x)$ in Y. Consider the closed submanifold $S := s(X) \subseteq Y$. The restriction $p|_S: S \to X$ is a diffeomorphism and in particular a smooth submersion. By [PW19, Corollary 2.8], there exist orthogonal H-representations V, W and W' and G-equivariant open embeddings $\Phi: G \times_H (V \times W \times W') \hookrightarrow Y$ and $\Psi: G \times_H V \hookrightarrow X$ such that:

- We have $\Theta([e, 0, 0, 0]) = s(x)$ and $\Psi([e, 0]) = x$;
- The map
$$\Theta: G \times_H (V \times W) \to M, \quad [(g, v, w)] \mapsto \Phi[(g, v, w, 0)]$$
has image in $S = s(X)$ and comprises a G-equivariant diffeomorphism onto an open neighborhood of the orbit Gx in S.
- The following diagram commutes:

$$\begin{tikzcd}
G \times_H (V \times W \times W') \arrow[r, hook, "\Phi"] \arrow[d, "\mathrm{id}_G \times \mathrm{pr}_V"'] & M \arrow[d, "f"] \\
G \times_H V \arrow[r, hook, "\Psi"] & N.
\end{tikzcd}$$

Since $p|_S\colon S \to X$ is a diffeomorphism, it follows that $W = 0$ is the trivial H-representation. Observe that the left vertical map is a G-equivariant vector bundle. Defining $s_0\colon G \times_H V \to G \times_H (V \times W \times W')$ by $t([g,v]) = [g,v,0,0]$, we see that the diagram

$$\begin{tikzcd} G \times_H (V \times W \times W') \arrow[r, hook, "\Phi"] & M \\ G \times_H V \arrow[u, "s_0"] \arrow[r, hook, "\Psi"] & N \arrow[u, "s"'] \end{tikzcd}$$

commutes. Since s_0 is the zero section of the vector bundle, this proves the claim. □

Proposition 3.7.5. *Consider a commutative diagram of separated differentiable stacks*

$$\begin{tikzcd} & \mathcal{Y} \arrow[d, "p"] \\ \mathcal{Z} \arrow[ur, hook, "t"] \arrow[r, hook, "i"] & \mathcal{X}, \end{tikzcd}$$

where p is a representable submersion and i is a closed embedding. Then for every point $z \in \mathcal{Z}$, there is a commutative diagram

$$\begin{tikzcd} & \mathcal{E} \arrow[r, hook, "j_{\mathcal{E}}"] \arrow[d, shift left, "\pi"] & \mathcal{Y} \arrow[d, "p"] \\ z \in \mathcal{Z} \cap \mathcal{N} \arrow[ur, hook, "t"] \arrow[r, hook, "i"] & \mathcal{N} \arrow[u, shift left, "s_0"] \arrow[r, hook, "j_{\mathcal{N}}"] & \mathcal{X}, \end{tikzcd}$$

where $j_{\mathcal{E}}\colon \mathcal{E} \hookrightarrow \mathcal{Y}$ is an open neighborhood of $t(z)$ inside $\mathcal{Y}$, $j_{\mathcal{N}}\colon \mathcal{N} \hookrightarrow \mathcal{X}$ is an open neighborhood of z inside $\mathcal{X}$, and the map $\pi\colon \mathcal{E} \to \mathcal{N}$ is a vector bundle with zero section $s_0\colon \mathcal{N} \to \mathcal{E}$.

Proof. We will prove the statement in three steps:

Step 1 We first reduce to the case where $\mathcal{X} \simeq X//G$ is the quotient stack of a smooth action of a compact Lie group G on a smooth manifold X, and the point $z \in \mathcal{X}$ lifts to a G-fixed point $z' \in X$;

Step 2 We then further reduce to the case where the partial section $s\colon \mathcal{Z} \to \mathcal{Y}$ of p is the restriction along $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ of a section $s\colon \mathcal{X} \to \mathcal{Y}$ of p;

Step 3 In this case, we prove that, locally in $\mathcal{Y}$, s can be chosen to be the zero-section of a vector bundle.

Step 1. Let $G := G_z$ denote the isotropy group of z in $\mathcal{X}$. By Theorem 3.7.2, we can find an open neighborhood $\mathcal{U}$ of z in $\mathcal{X}$ of the form $X//G$ for some smooth G-manifold X such that the point $z \in \mathcal{X}$ lifts to a G-fixed point $z' \in X$. The restrictions $\mathcal{Z}_{\mathcal{U}}$ and $\mathcal{Y}_{\mathcal{U}}$ over $\mathcal{U}$ are of the form $Z//G$ and $Y//G$ for smooth G-manifolds Z and Y-respectively. As z' is G-fixed in X, it follows that z' is also G-fixed in Z and thus $t(z)$ is G-fixed in Y.

Step 2. By Step 1, it will suffice to prove the analogous statement in the setting of smooth G-manifolds. So assume given a commutative triangle

$$\begin{tikzcd} & Y \arrow[d, "p"] \\ Z \arrow[ur, "s"] \arrow[r, hook, "i"] & X \end{tikzcd}$$

of smooth G-manifolds, and a G-fixed point $z \in Z$. We will show that the partial section t of p extends to a section $s\colon U \to Y$ on some open neighborhood U of Z inside X.

By [PW19, Proposition 2.7], there exist orthogonal G-representations V and W, and G-equivariant embeddings $\Phi\colon V \times W \hookrightarrow Y$ and $\Psi\colon V \hookrightarrow X$ such that $\Phi(0,0) = t(z)$, $\Psi(0) = i(z)$, and the following diagram commutes:

$$\begin{tikzcd} V \times W \arrow[r, hook, "\Phi"] \arrow[d, "\mathrm{pr}_V"'] & Y \arrow[d, "p"] \\ V \arrow[r, hook, "\Psi"] & X. \end{tikzcd}$$

By scaling down the embedding Ψ if needed, we may assume that the section $t\colon Z \cap V \to Y$ factors through $V \times W \hookrightarrow Y$. The second component $Z \to W$ can be extended to a small open neighborhood U of $Z \cap V$ inside X, which leads to a section $s\colon U \to U \times W$ extending t.

Step 3: From Step 2, we have obtained a commutative diagram

$$\begin{tikzcd} & U' \arrow[r, hook] \arrow[d, shift left, "p|_{U'}"] & Y \arrow[d, "p"] \\ z \in Z \cap U \arrow[ur, "t"] \arrow[r, hook, "i"] & U \arrow[r, hook] \arrow[u, shift left, "s"] & X \end{tikzcd}$$

of smooth G-manifolds. By Lemma 3.7.4, there exist G-invariant open neighborhoods $z \in N \subseteq U$ and $t(z) \in E \subseteq U'$ such that $p(E) \subseteq N$, $s(N) \subseteq E$, and the restriction $p|_E\colon E \to N$ is isomorphic to a G-equivariant vector bundle with zero-section $s|_N\colon N \to E$. In other words, we obtain a commutative diagram

$$\begin{tikzcd} & E \arrow[r, hook] \arrow[d, shift left, "\pi"] & U' \arrow[r, hook] \arrow[d, shift left, "p|_{U'}"] & Y \arrow[d, "p"] \\ z \in Z \cap N \arrow[ur, "t"] \arrow[r, hook, "i"] & N \arrow[r, hook] \arrow[u, shift left, "s_0"] & U \arrow[r, hook] \arrow[u, shift left, "s"] & X, \end{tikzcd}$$

where π is a G-equivariant vector bundle with zero-section s_0. Passing to quotient stacks then gives the claim. $\square$

II.4 Genuine sheaves

In this chapter, we will introduce for every separated differentiable stack $\mathcal{X}$ the ∞-category $\mathrm{H}(\mathcal{X})$ of *genuine sheaves of animae on* $\mathcal{X}$ and the ∞-category $\mathrm{SH}(\mathcal{X})$ of *genuine sheaves of spectra on* $\mathcal{X}$.

The ∞-categories $\mathrm{H}(\mathcal{X})$ and $\mathrm{SH}(\mathcal{X})$ may be thought of as 'genuine' refinements of the ∞-categories $\mathrm{Shv}(\mathcal{X})$ and $\mathrm{Shv}(\mathcal{X};\mathrm{Sp})$ of ordinary sheaves of animae/spectra on $\mathcal{X}$. While the latter two ∞-categories are easy to define using descent, they do not fully capture all of the "stacky" behavior of $\mathcal{X}$. To illustrate this, consider the case of a classifying stack $\mathcal{X} = \mathbb{B}G$ of a compact Lie group G. Using descent for the effective epimorphism $\mathrm{pt} \twoheadrightarrow \mathbb{B}G$, one can show that sheaves on $\mathbb{B}G$ correspond to animae/spectra with a G-action: there are equivalences of ∞-categories

$$\mathrm{Shv}(\mathbb{B}G) \simeq \mathrm{An}^{BG} \qquad \text{and} \qquad \mathrm{Shv}(\mathbb{B}G;\mathrm{Sp}) \simeq \mathrm{Sp}^{BG},$$

see Lemma 4.1.21 below. In equivariant homotopy theory, one frequently wants to work instead with the more refined ∞-categories An_G and Sp_G of *genuine G-animae* (a.k.a. 'G-spaces') and *genuine G-spectra*, which retain additional geometric fixed point data for closed subgroups of G. Forcing descent with respect to the map $\mathrm{pt} \twoheadrightarrow \mathbb{B}G$ destroys this geometric information, since it inverts all G-equivariant maps which are an equivalence after forgetting the G-action.

The goal of this chapter is to extend the above story to the case of an arbitrary separated differentiable stack $\mathcal{X}$ by introducing a notion of a 'genuine sheaf' on $\mathcal{X}$, which compared to an ordinary sheaf on $\mathcal{X}$ retains additional isotropy information. The resulting ∞-categories $\mathrm{H}(\mathcal{X})$ and $\mathrm{SH}(\mathcal{X})$ come equipped with forgetful functors to $\mathrm{Shv}(\mathcal{X})$ and $\mathrm{Shv}(\mathcal{X};\mathrm{Sp})$, respectively, which are simultaneously localizing and colocalizing, inverting precisely those maps which become equivalences after forgetting all this isotropy information. When $\mathcal{X}$ is the classifying stack $\mathbb{B}G$, these two functors specialize to the forgetful functors $\mathrm{An}_G \to \mathrm{An}^{BG}$ and $\mathrm{Sp}_G \to \mathrm{Sp}^{BG}$ discussed above. In the case of a smooth manifold M, which has trivial isotropy groups, there is no difference between genuine sheaves and ordinary sheaves.

The approach we will take for defining genuine sheaves closely resembles the definition of the motivic categories $\mathrm{H}(S)$ and $\mathrm{SH}(S)$ for a scheme S, introduced by Morel and Voevodsky [MV99] in their development of *motivic homotopy theory* (also referred to as $\mathbb{A}^1$-*homotopy theory*). The starting point is the observation that there is an equivalence between the ∞-category An of animae and the ∞-category $\mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff})$ of homotopy invariant sheaves of animae on the site of smooth manifolds. An analogue for a separated differentiable stack $\mathcal{X}$ is obtained by replacing the site Diff of smooth manifolds by the site $\mathrm{Sub}_{/\mathcal{X}}$ of representable submersions over $\mathcal{X}$, where we think of a representable submersion $\mathcal{Y} \to \mathcal{X}$ as an '$\mathcal{X}$-indexed family of smooth manifolds'. The ∞-category $\mathrm{H}(\mathcal{X})$ of genuine sheaves of animae on $\mathcal{X}$ is then defined as the ∞-category of homotopy invariant sheaves on $\mathrm{Sub}_{/\mathcal{X}}$. The ∞-category $\mathrm{SH}(\mathcal{X})$ of genuine sheaves of *spectra* on $\mathcal{X}$ is defined, at least locally in $\mathcal{X}$, by monoidally inverting the sphere bundles $S^{\mathcal{E}}$ in $\mathrm{H}(\mathcal{X})_*$, where $\mathcal{E}$ runs over all vector bundles over $\mathcal{X}$.

The chapter is organized as follows. In Section 4.1, we introduce the site $\mathrm{Sub}_{/\mathcal{X}}$ for a differentiable stack $\mathcal{X}$ and study the ∞-category of sheaves on $\mathrm{Sub}_{/\mathcal{X}}$. In Sections 4.2 and 4.3, we introduce the ∞-categories $\mathrm{H}(\mathcal{X})$ and $\mathrm{SH}(\mathcal{X})$ of genuine sheaves of animae/spectra, respectively. In Section 4.4, we establish the connection with equivariant homotopy theory by producing equivalences of ∞-categories $\mathrm{H}(\mathbb{B}G) \simeq \mathrm{An}_G$ and $\mathrm{SH}(\mathbb{B}G) \simeq \mathrm{Sp}_G$. Finally, we show in Section 4.5 that the assignments $\mathcal{X} \mapsto \mathrm{H}(\mathcal{X})$ and $\mathcal{X} \mapsto \mathrm{SH}(\mathcal{X})$ admit universal properties, phrased in terms of the notion of a *pullback formalism* $\mathcal{C}\colon \mathrm{DiffStk}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$ due to Drew and Gallauer [DG22].

4.1 Sheaves on $\mathrm{Sub}_{/\mathcal{X}}$

In this section, we study the ∞-category $\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ of sheaves on the category $\mathrm{Sub}_{/\mathcal{X}}$ of representable submersions over $\mathcal{X}$.

Definition 4.1.1. Let $\mathcal{X} \in \mathrm{DiffStk}$ be a differentiable stack. We let $\mathrm{Sub}_{/\mathcal{X}} \subseteq \mathrm{DiffStk}_{/\mathcal{X}}$ denote the full subcategory spanned by the representable submersions $\mathcal{Y} \to \mathcal{X}$.

Recall that for a Lie groupoid $\mathcal{G}$, the ∞-category $\mathrm{Sub}_{/\mathbb{B}\mathcal{G}}$ of representable submersions into the classifying stack $\mathbb{B}\mathcal{G}$ is naturally equivalent to the category $\mathrm{Diff}_{\mathcal{G}}$ of smooth $\mathcal{G}$-manifolds, see Corollary 2.3.19. In particular, $\mathrm{Sub}_{/\mathcal{X}}$ is an ordinary category for every differentiable stack $\mathcal{X}$.

The category $\mathrm{Sub}_{/\mathcal{X}}$ admits a Grothendieck topology given by the open covers:

Definition 4.1.2 (Open covers)**.** Given a differentiable stack $\mathcal{Y}$, a collection of morphisms $j_\alpha\colon \mathcal{U}_\alpha \to \mathcal{Y}$ in DiffStk is called an *open cover of* $\mathcal{Y}$ if the following two conditions are satisfied:

(1) Each morphism $j_\alpha\colon \mathcal{U}_\alpha \hookrightarrow \mathcal{Y}$ is an open embedding of differentiable stacks;

(2) The morphism $\bigsqcup_\alpha \mathcal{U}_\alpha \to \mathcal{Y}$ is an effective epimorphism in $\mathrm{Shv}(\mathrm{Diff})$.

A sieve $\{f_\beta\colon \mathcal{Y}_\beta \to \mathcal{Y}\}$ in DiffStk is said to be a *covering sieve* of $\mathcal{Y}$ if it contains an open cover of $\mathcal{Y}$.

Proposition 4.1.3. *The covering sieves of Definition 4.1.2 equip* DiffStk *with the structure of a Grothendieck topology.*

Proof. The identity on $\mathcal{Y}$ is an open cover, hence generates a covering sieve of $\mathcal{Y}$. If $\{f_j\colon \mathcal{Y}_j \to \mathcal{Y}\}$ is a covering sieve of $\mathcal{Y}$ containing an open cover $\{\iota_i\colon \mathcal{U}_i \hookrightarrow \mathcal{Y}\}$ and $\mathcal{Y}' \to \mathcal{Y}$ is a morphism of differentiable stacks, then the pullback sieve $\{\mathcal{Y}_j \times_{\mathcal{Y}} \mathcal{Y}' \to \mathcal{Y}'\}$ on $\mathcal{Y}'$ contains the open cover $\{\mathcal{V}_i := \mathcal{U}_i \times_{\mathcal{Y}} \mathcal{Y}' \hookrightarrow \mathcal{Y}'\}$, where we use that $\bigsqcup_i \mathcal{V}_i \to \mathcal{Y}'$ is an effective epimorphism since it is a base change of the map $\bigsqcup_i \mathcal{U}_i \to \mathcal{Y}$. Finally, let $\{\iota_i\colon \mathcal{U}_i \to \mathcal{Y}\}$ be an open cover of $\mathcal{Y}$ and let $\{f_j\colon \mathcal{Y}_j \to \mathcal{Y}\}$ be an arbitrary sieve. Assume that for every i, the pullback sieve $\{\mathcal{U}_i \times_{\mathcal{Y}} \mathcal{Y}_j \to \mathcal{U}_i\}$ is a covering sieve, so that it contains an open cover $\{\mathcal{V}_{i,j} \hookrightarrow \mathcal{U}_i\}$. We claim that the collection of composites $\{\mathcal{V}_{i,j} \hookrightarrow \mathcal{U}_i \hookrightarrow \mathcal{Y}\}$ forms an open cover of $\mathcal{Y}$, contained in the original sieve. Indeed, these maps are open embeddings and the composite map

$$\bigsqcup_i \bigsqcup_j \mathcal{V}_{i,j} \twoheadrightarrow \bigsqcup_i \mathcal{U}_i \twoheadrightarrow \mathcal{Y}$$

is an effective epimorphism. □

The open covers in DiffStk directly induce a Grothendieck topology on $\mathrm{Sub}_{/\mathcal{X}}$:

Definition 4.1.4. We define a Grothendieck topology on $\mathrm{Sub}_{/\mathcal{X}}$ in which a collection of morphisms $\{f_i\colon \mathcal{Y}_i \to \mathcal{Y}\}$ in $\mathrm{Sub}_{/\mathcal{X}}$ is a cover of $\mathcal{Y}$ if and only if it is cover of $\mathcal{Y}$ in the ∞-category DiffStk, in the sense of Definition 4.1.2. Given a presentable ∞-category $\mathcal{C}$, we let $\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}; \mathcal{C})$ denote the ∞-category of $\mathcal{C}$-sheaves on the resulting site $(\mathrm{Sub}_{/\mathcal{X}}, \mathrm{open})$.

The sheaf condition can be made very explicit: a presheaf on $\mathrm{Sub}_{/\mathcal{X}}$ is a sheaf if and only if it satisfies a form of excision and turns countable unions of open substacks into limits. More precisely:

Proposition 4.1.5 (cf. [BBP19, Theorem 5.1], [ADH21, Theorem 3.6.1]). *Let $\mathcal{X}$ be a differentiable stack and let $\mathcal{C}$ be a presentable ∞-category. Then a $\mathcal{C}$-valued presheaf $\mathcal{F}\colon \mathrm{Sub}_{/\mathcal{X}}^{\mathrm{op}} \to \mathcal{C}$ is a sheaf if and only if $\mathcal{F}$ satisfies the following conditions:*

(1) The object $\mathcal{F}(\emptyset)$ is terminal in $\mathcal{C}$;

(2) For every representable submersion $\mathcal{Y} \to \mathcal{X}$ and every pair of open substacks $\mathcal{U}, \mathcal{V} \subseteq \mathcal{Y}$ satisfying $\mathcal{Y} = \mathcal{U} \cup \mathcal{V}$, the induced square

$$\begin{tikzcd}
\mathcal{F}(\mathcal{Y}) \arrow[r] \arrow[d] & \mathcal{F}(\mathcal{V}) \arrow[d] \\
\mathcal{F}(\mathcal{U}) \arrow[r] & \mathcal{F}(\mathcal{U} \cap \mathcal{V})
\end{tikzcd}$$

is a pullback square in $\mathcal{C}$;

(3) For every representable submersion $\mathcal{Y} \to \mathcal{X}$ and every $\mathbb{N}$-indexed sequence of open substacks

$$\mathcal{U}_0 \subseteq \mathcal{U}_1 \subseteq \cdots \subseteq \mathcal{Y}$$

such that $\mathcal{Y} = \bigcup_{n\geq 0} \mathcal{U}_n = \mathcal{Y}$, the induced morphism

$$\mathcal{F}(\mathcal{Y}) \to \lim_{n\geq 0} \mathcal{F}(\mathcal{U}_n)$$

is an equivalence in $\mathcal{C}$.

Proof. Given a representable submersion $\mathcal{Y} \to \mathcal{X}$, any open substack $\mathcal{U}$ of $\mathcal{Y}$ is naturally an object of $\mathrm{Sub}_{/\mathcal{X}}$ via the composite $\mathcal{U} \hookrightarrow \mathcal{Y} \to \mathcal{X}$, and this provides a functor $\mathrm{Open}(\mathcal{Y}) \to \mathrm{Sub}_{/\mathcal{X}}$. Observe that $\mathcal{F}$ is a sheaf if and only if for every $\mathcal{Y}$ the composite functor

$$\mathrm{Open}(\mathcal{Y})^{\mathrm{op}} \to \mathrm{Sub}_{/\mathcal{X}}^{\mathrm{op}} \xrightarrow{\mathcal{F}} \mathcal{C}$$

is a sheaf with respect to the open cover topology on $\mathrm{Open}(\mathcal{Y})$. Recall from Corollary 3.1.12 that the poset $\mathrm{Open}(\mathcal{Y})$ of open substacks of $\mathcal{Y}$ is equivalent to the poset $\mathrm{Open}(|\mathcal{Y}|_{\mathrm{mod}})$ of open subspaces of the coarse moduli space $|\mathcal{Y}|_{\mathrm{mod}}$. Since $\mathcal{Y}$ admits a representable atlas by a smooth manifold, the topological space $|\mathcal{Y}|_{\mathrm{mod}}$ is the quotient of a smooth manifold, and thus is in particular *hereditarily Lindelöf*: every open subspace $U \subseteq |\mathcal{Y}|_{\mathrm{mod}}$ is Lindelöf, meaning that every open cover of U admits a countable subcover. It is a general fact that a presheaf $\mathcal{F}\colon \mathrm{Open}(X)^{\mathrm{op}} \to \mathcal{C}$ on a hereditarily Linfelöf topological space X is a sheaf if and only if it satisfies the conditions analogous to (1), (2) and (3), see for example [ADH21, Proposition 3.6.6]. Applying this result to $X = |\mathcal{Y}|_{\mathrm{mod}}$ finishes the proof. $\square$

Corollary 4.1.6. *The subcategory* $\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \subseteq \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}})$ *is closed under* ω_1*-filtered colimits.*

Proof. This is immediate from Proposition 4.1.5 and the fact that ω_1-filtered colimits commute with countable (that is, ω_1-small) limits in the ∞-category of animae by [Lur09, Proposition 5.3.3.3]. □

4.1.1 Pullback and pushforward functors

Let $f\colon \mathcal{X}' \to \mathcal{X}$ be a morphism of differentiable stacks. Our goal in this subsection is to construct the *pullback* and *pushforward* functors

$$f^*\colon \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \rightleftarrows \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}'}) : f_*,$$

and prove various basic properties about them.

We start with the construction of f_*. Since base changes of representable submersions exist in DiffStk and are again representable submersions, there is a functor $\mathrm{Sub}_{/\mathcal{X}} \to \mathrm{Sub}_{/\mathcal{X}'}$ given by $\mathcal{Y} \mapsto \mathcal{Y} \times_{\mathcal{X}} \mathcal{X}'$. Precomposition with this functor then defines a pushforward functor at the level of presheaf categories:

$$\begin{aligned} f_*\colon \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}'}) &\to \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}}), \\ f_*\mathcal{F}(\mathcal{Y}) &:= \mathcal{F}(\mathcal{Y} \times_{\mathcal{X}} \mathcal{X}'). \end{aligned}$$

Lemma 4.1.7. *The functor* f_* *restricts to sheaves:*

$$f_*\colon \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}'}) \to \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}).$$

Proof. Observe that the functor $- \times_{\mathcal{X}} \mathcal{X}'\colon \mathrm{Sub}_{/\mathcal{X}} \to \mathrm{Sub}_{/\mathcal{X}'}$ preserves covering sieves. Since coverings consists of open embeddings, $\mathrm{Sub}_{/\mathcal{X}}$ admits pullbacks along open embeddings and the functor $- \times_{\mathcal{X}} \mathcal{X}'$ preserves such pullbacks, it follows that this is a continuous functor in the sense of Definition E.39, and thus the restriction functor f_* preserves sheaves. □

The functor $f_*\colon \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}'}) \to \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ admits a left adjoint

$$f^*\colon \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}'})$$

given as the composite $\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \hookrightarrow \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{PSh}(\mathrm{Sub}_{\mathcal{X}/}) \xrightarrow{L_{\mathrm{open}}} \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}'})$, where the middle functor is left Kan extension along the functor $- \times_{\mathcal{X}} \mathcal{X}'\colon \mathrm{Sub}_{/\mathcal{X}} \to \mathrm{Sub}_{/\mathcal{X}'}$. In case f is a representable submersion, f^* admits a further left adjoint $f_\sharp$:

Lemma 4.1.8. *Let $f\colon \mathcal{X}' \to \mathcal{X}$ be a representable submersion.*

(1) The functor $-\times_{\mathcal{X}} \mathcal{X}'\colon \mathrm{Sub}_{/\mathcal{X}} \to \mathrm{Sub}_{/\mathcal{X}'}$ admits a left adjoint $f_\sharp\colon \mathrm{Sub}_{/\mathcal{X}'} \to \mathrm{Sub}_{/\mathcal{X}}$ given by postcomposition with f.

(2) The functor $f_\sharp\colon \mathrm{Sub}_{/\mathcal{X}'} \to \mathrm{Sub}_{/\mathcal{X}}$ is a continuous functor, so that restriction along it preserves sheaves.

(3) The functor $f^*\colon \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}'})$ admits a left adjoint

$$f_\sharp\colon \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}'}) \to \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$$

which on representables restricts to $f_\sharp\colon \mathrm{Sub}_{/\mathcal{X}'} \to \mathrm{Sub}_{/\mathcal{X}}$.

Proof. Assertion (1) is clear. For assertion (2), note that $f_\sharp$ preserves covering sieves and preserves all pullbacks that exist in $\mathrm{Sub}_{/\mathcal{X}'}$. In particular, the functor $\mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}'})$ given by restriction along $f_\sharp$ preserves sheaves. As $f_\sharp$ is left adjoint to $-\times_{\mathcal{X}} \mathcal{X}'\colon \mathrm{Sub}_{/\mathcal{X}} \to \mathrm{Sub}_{/\mathcal{X}'}$, restriction along $f_\sharp$ is left adjoint to the functor $f_*\colon \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}'}) \to \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$, and therefore is equivalent to f^*. But this means that f^* admits a left adjoint given by the composite

$$\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}'}) \hookrightarrow \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}'}) \to \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}}) \xrightarrow{L_{\mathrm{open}}} \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}),$$

where the middle functor is left Kan extension along $f_\sharp$. Since the left Kan extension reduces to the functor $f_\sharp$ on representable objects, the last claim follows. This finishes the proof. $\square$

To have more control over the functoriality of the construction $\mathcal{X} \mapsto \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$, we now provide an alternative description of the pullback functors f^* in terms of the ∞-topos $\mathrm{Shv}(\mathrm{DiffStk})$ of sheaves on the site DiffStk from Proposition 4.1.3.

Lemma 4.1.9. *For a differentiable stack $\mathcal{X}$, the inclusion $\mathrm{Sub}_{/\mathcal{X}} \hookrightarrow \mathrm{DiffStk}_{/\mathcal{X}}$ induces a fully faithful functor*

$$\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \hookrightarrow \mathrm{Shv}(\mathrm{DiffStk}_{/\mathcal{X}}) \simeq \mathrm{Shv}(\mathrm{DiffStk})_{/\mathcal{X}},$$

given by left Kan extension followed by sheafification. The essential image is the subcategory of $\mathrm{Shv}(\mathrm{DiffStk})_{/\mathcal{X}}$ generated under colimits by the representable submersions $f\colon \mathcal{Y} \to \mathcal{X}$. For a morphism $f\colon \mathcal{X}' \to \mathcal{X}$, the following diagram commutes:

$$\begin{tikzcd}
\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \arrow[r, "f^*"] \arrow[d, hook] & \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}'}) \arrow[d, hook] \\
\mathrm{Shv}(\mathrm{DiffStk})_{/\mathcal{X}} \arrow[r, "f^*"] & \mathrm{Shv}(\mathrm{DiffStk})_{/\mathcal{X}'}
\end{tikzcd}$$

where the bottom map is given by pullback along f in Shv(DiffStk).

Proof. Given a representable submersion $\mathcal{Y} \to \mathcal{X}$ and an open cover $\{\mathcal{U}_i \hookrightarrow \mathcal{Y}\}$ of $\mathcal{Y}$, each of the composite morphisms $\mathcal{U}_i \to \mathcal{Y} \to \mathcal{X}$ is again a representable submersion. It follows that the inclusion $\mathrm{Sub}_{/\mathcal{X}} \hookrightarrow \mathrm{DiffStk}_{/\mathcal{X}}$ is a cocontinuous functor. It follows from Corollary E.47 that the right Kan extension functor along $\mathrm{Sub}_{/\mathcal{X}} \hookrightarrow \mathrm{DiffStk}_{/\mathcal{X}}$ preserves sheaves and is fully faithful.

Since this inclusion also preserves covering sieves, it is also a continuous functor, so that restriction along it preserves sheaves. It follows that this restriction functor admits a further left adjoint given by the functor in the statement. As this is a double left adjoint of a fully faithful functor, it is itself fully faithful, proving the first claim. The description of the essential image is immediate.

To see that the claimed commutative diagram exists, observe that all four functors preserve colimits, and thus it suffices to prove the claim at on representables. But here the claim is clear: by definition, the functor f^* restricts on representables to the functor $- \times_{\mathcal{X}} \mathcal{X}' \colon \mathrm{Sub}_{/\mathcal{X}} \to \mathrm{Sub}_{/\mathcal{X}'}$, which in turn is just defined as a restriction of the functor $- \times_{\mathcal{X}} \mathcal{X}' \colon \mathrm{DiffStk}_{/\mathcal{X}} \to \mathrm{DiffStk}_{/\mathcal{X}'}$. ◻

By the previous lemma, we can turn the assigment $\mathcal{X} \mapsto \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ into a functor $\mathrm{DiffStk}^{\mathrm{op}} \to \mathrm{Cat}_\infty$ by regarding it as a subfunctor of the slice functor $\mathcal{X} \mapsto \mathrm{Shv}(\mathrm{SepStk})_{/\mathcal{X}}$.

Lemma 4.1.10. *The functor* $\mathrm{DiffStk}^{\mathrm{op}} \to \mathrm{Cat}_\infty$ *given by* $\mathcal{X} \mapsto \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ *is a sheaf of* ∞*-categories on* DiffStk.

Proof. It follows directly from Remark 4.5.2 below and Theorem E.3 that the assignment $\mathcal{X} \mapsto \mathrm{Shv}(\mathrm{SepStk})_{/\mathcal{X}}$ is a sheaf of ∞-categories on SepStk. Since the assignment $\mathcal{X} \mapsto \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ is a subfunctor of this functor, it thus remains to show that the condition for an object to be contained in this subcategory can be checked locally in $\mathcal{X}$. To this end, let $f \colon A \to \mathcal{X}$ be an object of $\mathrm{Shv}(\mathrm{SepStk})_{/\mathcal{X}}$ satisfying the assumption that each of the base changes $A \times_{\mathcal{X}} \mathcal{U}_i$ can be written as iterated colimits of representable submersion $\mathcal{Y}_{i,j} \to \mathcal{U}_i$. By descent, it follows that the map $A \to \mathcal{X}$ can be written as an iterated colimit of the composites $\mathcal{Y}_{i,j} \to \mathcal{U}_i \hookrightarrow \mathcal{X}$. Since these are again representable submersions, this finishes the proof of the claim. ◻

Lemma 4.1.11. *The inclusion* $\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \hookrightarrow \mathrm{Shv}(\mathrm{DiffStk}_{/\mathcal{X}})$ *preserves finite products.*

Proof. In both source and target finite products commute with colimits in each variable. In particular, the statement may be checked at the level of representables. There it is clear as the category $\mathrm{Sub}_{/\mathcal{X}}$ admits products and these are preserved by the inclusion $\mathrm{Sub}_{/\mathcal{X}} \hookrightarrow \mathrm{DiffStk}_{/\mathcal{X}}$. □

Corollary 4.1.12 (Symmetric monoidality)**.** *For every morphism $f: \mathcal{X}' \to \mathcal{X}$ of differentiable stacks, the pullback functor $f^*: \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}'})$ preserves finite products. In particular, the assignment $\mathcal{X} \mapsto \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ defines a limit-preserving functor $\mathrm{DiffStk}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$.*

Proof. The first statement follows from Lemma 4.1.11 and Lemma 4.1.9. The second statement is then a direct consequence from Lemma 4.1.10 as the forgetful functor $\mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}) \to \mathrm{Cat}_\infty$ preserves limits. □

4.1.2 Sheaves on the coarse moduli space

In Section 3.1 we assigned to every differentiable stack $\mathcal{X}$ a topological space $|\mathcal{X}|_{\mathrm{mod}}$, the coarse moduli space of $\mathcal{X}$. The goal of this section is to relate its sheaf ∞-topos $\mathrm{Shv}(|\mathcal{X}|_{\mathrm{mod}})$ to the ∞-topos $\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$.

Recall from Corollary 3.1.12 that the poset $\mathrm{Open}(|\mathcal{X}|_{\mathrm{mod}})$ of open subsets of the coarse moduli space is equivalent to the poset $\mathrm{Open}(\mathcal{X})$ of open substacks of $\mathcal{X}$, giving an equivalence

$$\mathrm{Shv}(|\mathcal{X}|_{\mathrm{mod}}) = \mathrm{Shv}(\mathrm{Open}(|\mathcal{X}|_{\mathrm{mod}})) \simeq \mathrm{Shv}(\mathrm{Open}(\mathcal{X})).$$

Henceforth, we will work with the sheaf category $\mathrm{Shv}(\mathrm{Open}(\mathcal{X}))$ rather than $\mathrm{Shv}(|\mathcal{X}|_{\mathrm{mod}})$. Since every open embedding $\mathcal{U} \hookrightarrow \mathcal{X}$ is in particular a representable submersion, there is an inclusion of categories $\iota: \mathrm{Open}(\mathcal{X}) \hookrightarrow \mathrm{Sub}_{/\mathcal{X}}$.

Proposition 4.1.13. *Let $\mathcal{X}$ be a differentiable stack.*

(1) The inclusion $\iota: \mathrm{Open}(\mathcal{X}) \hookrightarrow \mathrm{Sub}_{/\mathcal{X}}$ is a cocontinuous functor, in the sense of Definition E.43. In particular, right Kan extension along ι preserves sheaves and defines a fully faithful geometric morphism of ∞-topoi

$$\iota_*: \mathrm{Shv}(\mathrm{Open}(\mathcal{X})) \hookrightarrow \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}).$$

(2) The inclusion $\iota: \mathrm{Open}(\mathcal{X}) \hookrightarrow \mathrm{Sub}_{/\mathcal{X}}$ admits a left adjoint $\mathrm{im}: \mathrm{Sub}_{/\mathcal{X}} \to \mathrm{Open}(\mathcal{X})$ which is also cocontinuous. In particular, the restriction functor $-\circ\iota: \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{PSh}(\mathrm{Open}(\mathcal{X}))$ preserves sheaves, and thus restricts to a functor

$$\iota^*: \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{Shv}(\mathrm{Open}(\mathcal{X}))$$

which is left adjoint to ι_*.

(3) *The functor ι^* admits a fully faithful left-exact left adjoint*

$$\iota_\sharp\colon \operatorname{Shv}(\operatorname{Open}(\mathcal{X})) \hookrightarrow \operatorname{Shv}(\operatorname{Sub}_{/\mathcal{X}}).$$

(4) *The functor ι^* preserves limits and colimits.*

(5) *For every morphism $f\colon \mathcal{Y} \to \mathcal{X}$ of differentiable stacks, there are preferred commutative diagrams as follows:*

$$\begin{tikzcd} \operatorname{Shv}(\operatorname{Sub}_{/\mathcal{Y}}) \arrow[r, "\iota^*"] \arrow[d, "f_*"'] & \operatorname{Shv}(\operatorname{Open}(\mathcal{Y})) \arrow[d, "f_*"] \\ \operatorname{Shv}(\operatorname{Sub}_{/\mathcal{X}}) \arrow[r, "\iota^*"] & \operatorname{Shv}(\operatorname{Open}(\mathcal{X})), \end{tikzcd} \qquad \begin{tikzcd} \operatorname{Shv}(\operatorname{Open}(\mathcal{X})) \arrow[r, "\iota_\sharp"] \arrow[d, "f^*"'] & \operatorname{Shv}(\operatorname{Sub}_{/\mathcal{X}}) \arrow[d, "f^*"] \\ \operatorname{Shv}(\operatorname{Open}(\mathcal{Y})) \arrow[r, "\iota_\sharp"] & \operatorname{Shv}(\operatorname{Sub}_{/\mathcal{Y}}). \end{tikzcd}$$

(6) *For every sheaf $\mathcal{F} \in \operatorname{Shv}(\operatorname{Open}(\mathcal{X}))$, the sheaf $\iota_\sharp\mathcal{F} \in \operatorname{Shv}(\operatorname{Sub}_{/\mathcal{X}})$ is given at a representable submersion $f\colon \mathcal{Y} \to \mathcal{X}$ by*

$$(\iota_\sharp\mathcal{F})(\mathcal{Y}) \simeq \Gamma_{\mathcal{Y}}(f^*\mathcal{F}),$$

the anima of global sections of the pullback sheaf $f^*\mathcal{F} \in \operatorname{Shv}(\operatorname{Open}(\mathcal{Y}))$.

Proof. We start with part (1). The fact that ι is cocontinuous is clear as any open cover in $\operatorname{Sub}_{\mathcal{X}}$ of an open substack $\mathcal{U} \hookrightarrow \mathcal{X}$ consists of smaller open substacks $\mathcal{V} \hookrightarrow \mathcal{U} \hookrightarrow \mathcal{X}$. The 'in particular' follows from Corollary E.47.

Now we prove part (2). By Lemma 2.4.9, every representable submersion $f\colon \mathcal{Y} \to \mathcal{X}$ factors uniquely as a representable surjective submersion $\mathcal{Y} \twoheadrightarrow f(\mathcal{Y})$ followed by an open embedding $f(\mathcal{Y}) \hookrightarrow \mathcal{X}$. Since effective epimorphisms and monomorphisms are part of a factorization system on $\operatorname{Shv}(\operatorname{Diff})$, it follows immediately that sending $(f\colon \mathcal{Y} \to \mathcal{X}) \in \operatorname{Sub}_{/\mathcal{X}}$ to $\operatorname{im}(f) := (f(\mathcal{Y}) \hookrightarrow \mathcal{X}) \in \operatorname{Open}(\mathcal{X})$ defines a left adjoint im of ι.

To see that im is cocontinuous, consider a representable submersion $f\colon \mathcal{Y} \to \mathcal{X}$ and consider a collection of open substacks $\mathcal{U}_i \hookrightarrow f(\mathcal{Y})$ covering its image in $\mathcal{X}$. Then the preimages $\mathcal{V}_i := f^{-1}(\mathcal{U}_i) \subseteq \mathcal{Y}$ form an open cover of $\mathcal{Y}$ satisfying $f(\mathcal{V}_i) = \mathcal{U}_i \subseteq f(\mathcal{Y})$. This proves that $\operatorname{im}\colon \operatorname{Sub}_{/\mathcal{X}} \to \operatorname{Open}(\mathcal{X})$ is cocontinuous.

As in part (1), it follows from Corollary E.47 that right Kan extension along im preserves sheaves and defines a geometric morphism of ∞-topoi. Since right Kan extension along im is equivalent to restriction along ι, this proves (2).

For (3), we know that $\iota^* \simeq \operatorname{im}_*$ admits a left-exact left adjoint $\iota_\sharp \simeq \operatorname{im}^*$ by part (2). Furthermore, since ι_* is fully faithful, it follows that also $\iota_\sharp$ is fully faithful.

Part (4) is immediate, since ι^* admits both a left and a right adjoint.

For part (5), the right square is obtained from the left square by passing to left adjoints. The left square is obtained by restriction along the following commutative diagram:

$$\begin{tikzcd} \mathrm{Open}(\mathcal{Y}) \arrow[r, "\iota"] \arrow[d, "\mathcal{X}\times_{\mathcal{Y}}-"'] & \mathrm{Sub}_{/\mathcal{Y}} \arrow[d, "\mathcal{X}\times_{\mathcal{Y}}-"] \\ \mathrm{Open}(\mathcal{X}) \arrow[r, "\iota"] & \mathrm{Sub}_{/\mathcal{X}}\,. \end{tikzcd}$$

For part (6), observe that the value of $\iota_\sharp\mathcal{F}$ at $\mathcal{Y}$ is the same as the anima of global sections of the sheaf $f^*\iota_\sharp\mathcal{F} \in \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{Y}})$, as the functor $f^*\colon \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{Y}})$ is given by precomposition with the functor $f\circ -\colon \mathrm{Sub}_{/\mathcal{Y}} \to \mathrm{Sub}_{/\mathcal{X}}$. By part (5), this is the same as the global sections of the sheaf $\iota_\sharp f^*\mathcal{F} \in \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{Y}})$, which in turn is the same as the global sections of $\iota^*\iota_\sharp f^*\mathcal{F} \in \mathrm{Shv}(\mathrm{Open}(\mathcal{Y}))$. As $\iota_\sharp$ is fully faithful, the latter sheaf is equivalent to $f^*\mathcal{F}$, proving the claim. $\square$

Corollary 4.1.14. *Let $\mathcal{X}$ be a differentiable stack. For every representable submersion $f\colon \mathcal{Y} \to \mathcal{X}$, consider the composite*

$$\iota^*\circ f^*\colon\ \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \xrightarrow{f^*} \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{Y}}) \xrightarrow{\iota^*} \mathrm{Shv}(\mathrm{Open}(\mathcal{Y})).$$

Then the functors $\iota^*\circ f^*$ preserve all limits and colimits and are jointly conservative.

Proof. The functor f^* preserves limits and colimits by Lemma 4.1.8 and the functor ι^* by Proposition 4.1.13. It thus remains to show these functors are jointly conservative. But this is clear: for a sheaf $\mathcal{F} \in \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$, the value of $\mathcal{F}$ at some object $(f\colon \mathcal{Y}\to\mathcal{X}) \in \mathrm{Shv}(\mathrm{Sub}_{\mathcal{Y}})$ is equivalent to the anima of global sections of $(\iota^*\circ f^*)\mathcal{F} = \iota^*f^*\mathcal{F}$:

$$\mathcal{F}(\mathcal{Y}) \simeq \Gamma_{\mathcal{Y}}(\iota^*f^*\mathcal{F}).$$ $\square$

The previous result can be used to prove properties of the ∞-category $\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ by reducing them to analogous properties of the ∞-categories $\mathrm{Shv}(\mathrm{Open}(\mathcal{Y}))$. The following result on hypercompleteness of $\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ is an important illustration of this method. We thank Marc Hoyois for useful discussions concerning the proof of this result.

Proposition 4.1.15. *For a separated differentiable stack $\mathcal{X}$, the ∞-topos* $\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ *is hypercomplete.*

Proof. Let $\varphi\colon \mathcal{F}\to\mathcal{F}'$ be an ∞-connected morphism in $\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$. We have to show that φ is an equivalence. By Corollary 4.1.14, we may show that the image of φ under the composite

$$\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \xrightarrow{f^*} \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{Y}}) \xrightarrow{\iota^*} \mathrm{Shv}(\mathrm{Open}(\mathcal{Y}))$$

is an equivalence for every $\mathcal{Y} \in \mathrm{Sub}_{/\mathcal{X}}$. As $\iota^* \circ f^*$ preserves limits and colimits, it preserves ∞-connected morphisms, and it follows that the morphism $\iota^* f^* \varphi$ is again ∞-connected. Hence it will suffice to show that the ∞-topos $\mathrm{Shv}(\mathrm{Open}(\mathcal{Y}))$ is hypercomplete.

As $\mathcal{X}$ is separated by assumption and the map $f\colon \mathcal{Y} \to \mathcal{X}$ is representable and thus separated by Lemma 3.3.3, it follows that also $\mathcal{Y}$ is separated. By Theorem 3.7.2, $\mathcal{Y}$ is locally isomorphic to a quotient stack $M//G$ for a compact Lie group G and a smooth G-manifold M. As $\mathrm{Shv}(\mathrm{Open}(\mathcal{Y}))$ satisfies descent for open covers, a morphism in $\mathrm{Shv}(\mathrm{Open}(\mathcal{Y}))$ is an equivalence if and only if it is so locally in $\mathcal{Y}$, and we may thus assume that $\mathcal{Y}$ is of the form $M//G$.

Observe that pullback along the quotient map $M \twoheadrightarrow M//G$ defines an inclusion of posets $\mathrm{Open}(M//G) \hookrightarrow \mathrm{Open}(M)$ whose image consists of the G-invariant open subspaces of M. This inclusion admits a left adjoint given by sending an open subspace U to the G-saturation $G \cdot U = \{g \cdot u \mid g \in G, u \in U\}$, and this functor preserves open coverings. In particular, the inclusion $\mathrm{Open}(M//G) \hookrightarrow \mathrm{Open}(M)$ is a morphism of sites by Example E.42, and the resulting pullback functor $\mathrm{Shv}(\mathrm{Open}(M//G)) \hookrightarrow \mathrm{Shv}(\mathrm{Open}(M)) = \mathrm{Shv}(M)$ preserves ∞-connected morphisms by Remark E.51. It will thus suffice to show that $\mathrm{Shv}(M)$ is hypercomplete. This is a special case of [Lur09, Theorem 7.2.3.6, Corollary 7.2.1.12], see also Example E.54. □

4.1.3 Ordinary sheaves on differentiable stacks

In this subsection, we recall the definition of the ∞-category $\mathrm{Shv}(\mathcal{X})$ of (ordinary) sheaves on a differentiable stack $\mathcal{X}$, and compare it with the ∞-topos $\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$. The material of this subsection will not play a significant role in the remainder of the article and may be skipped on first reading.

Lemma 4.1.16. *Let C be a presentable ∞-category. Then the functor* $\mathrm{Shv}(-;C)\colon \mathrm{Diff}^{\mathrm{op}} \to \mathrm{Cat}_\infty$ *if a sheaf of ∞-categories with respect to the open cover topology on* Diff.

Proof. Any open cover of a smooth manifold M consists of objects of $\mathrm{Open}(M)$ and hence it suffices that for every M the composite

$$\mathrm{Open}(M)^{\mathrm{op}} \hookrightarrow (\mathrm{Diff}_{/M})^{\mathrm{op}} \xrightarrow{\mathrm{fgt}} \mathrm{Diff}^{\mathrm{op}} \xrightarrow{\mathrm{Shv}(-;C)} \mathrm{Cat}_\infty$$

is a sheaf of ∞-categories on M. For every open $U \subseteq M$, the inclusion $\mathrm{Shv}(U) \hookrightarrow \mathrm{Shv}(M)$ induces an equivalence $\mathrm{Shv}(U) \xrightarrow{\sim} \mathrm{Shv}(M)_{/U}$, and thus we obtain an equivalence

$$\mathrm{Shv}(U;C) \simeq \mathrm{Shv}(M)_{/U} \otimes C,$$

where the tensor product is Lurie's tensor product of presentable ∞-categories. It is clear that for a smaller open subset $V \subseteq U \subseteq M$ the restriction functor $(-)|_V\colon \mathrm{Shv}(U) \to \mathrm{Shv}(V)$ corresponds under this equivalence to the functor $\mathrm{Shv}(M)_{/U} \to \mathrm{Shv}(M)_{/V}$ given by pullback along the inclusion $V \hookrightarrow U$, hence we see that the above composite is equivalent to the composite

$$\mathrm{Open}(M)^{\mathrm{op}} \xrightarrow{\mathrm{Shv}(M)_{/-}} \mathrm{Pr}^{\mathrm{R}} \xrightarrow{-\otimes C} \mathrm{Pr}^{\mathrm{R}} \xrightarrow{\mathrm{fgt}} \mathrm{Cat}_\infty.$$

Since the ∞-topos $\mathrm{Shv}(M)$ satisfies descent, the first functor sends covering sieves to limits. As the functors $-\otimes C\colon \mathrm{Pr}^{\mathrm{R}} \to \mathrm{Pr}^{\mathrm{R}}$ and $\mathrm{fgt}\colon \mathrm{Pr}^{\mathrm{R}} \to \mathrm{Cat}_\infty$ preserves limits, this finishes the proof. $\square$

Definition 4.1.17 (Sheaves on a stack)**.** Let C be a presentable ∞-category. By the previous lemma, the functor $\mathrm{Shv}(-;C)\colon \mathrm{Diff}^{\mathrm{op}} \to \mathrm{Cat}_\infty$ uniquely extends to a limit-preserving functor

$$\mathrm{Shv}(-;C)\colon \mathrm{Shv}(\mathrm{Diff})^{\mathrm{op}} \to \mathrm{Cat}_\infty.$$

For a stack $\mathcal{X}$ on Diff, we refer to $\mathrm{Shv}(\mathcal{X};C)$ as the *∞-category of C-valued sheaves on $\mathcal{X}$*.

Remark 4.1.18. We will frequently refer to a sheaf on $\mathcal{X}$ as an *ordinary sheaf*, to emphasize the contrast with the notion of a *genuine sheaf* which will be introduced in Section 4.2.

There are comparison functors between $\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ and $\mathrm{Shv}(\mathcal{X})$ analogous to the ones for $\mathrm{Shv}(|\mathcal{X}|_{\mathrm{mod}})$ from Proposition 4.1.13:

Construction 4.1.19. Given $\mathcal{X}$ be a differentiable stack, we construct a functor

$$\gamma^*\colon \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{Shv}(\mathcal{X})$$

which is natural in $\mathcal{X}$. Since the assignment $\mathcal{X} \mapsto \mathrm{Shv}(\mathcal{X})$ is by definition right Kan extended from the subcategory $\mathrm{Diff} \subseteq \mathrm{DiffStk}$, it suffices to define this when $\mathcal{X} = M$ is a smooth manifold, naturally in M. In this case we let γ^* be the functor $\iota^*\colon \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{Shv}(M)$ from Proposition 4.1.13.

Since the ∞-category $\mathrm{Shv}(\mathcal{X})$ is a limit of $\mathrm{Shv}(M)$ where M ranges over all smooth manifolds M equipped with a map of stacks $f\colon M \to \mathcal{X}$, the functor γ^* is essentially determined by the existence of commutative squares

$$\begin{tikzcd} \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \arrow[r, "\gamma^*"] \arrow[d, "f^*"'] & \mathrm{Shv}(\mathcal{X}) \arrow[d, "f^*"] \\ \mathrm{Shv}(\mathrm{Sub}_{/M}) \arrow[r, "\iota^*"] & \mathrm{Shv}(M), \end{tikzcd}$$

naturally in M.

Proposition 4.1.20. *Let $\mathcal{X}$ be a differentiable stack.*

(1) The functor $\gamma^*\colon \operatorname{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \to \operatorname{Shv}(\mathcal{X})$ admits a fully faithful right adjoint

$$\gamma_*\colon \operatorname{Shv}(\mathcal{X}) \hookrightarrow \operatorname{Shv}(\mathrm{Sub}_{/\mathcal{X}})).$$

(2) The functor γ^* admits a fully faithful left adjoint

$$\gamma_\sharp\colon \operatorname{Shv}(\mathcal{X}) \hookrightarrow \operatorname{Shv}(\mathrm{Sub}_{/\mathcal{X}})).$$

(3) For every morphism $g\colon \mathcal{Y} \to \mathcal{X}$ of differentiable stacks, there are preferred commutative diagrams as follows:

$$\begin{tikzcd} \operatorname{Shv}(\mathrm{Sub}_{/\mathcal{Y}}) \arrow[r, "\gamma^*"] \arrow[d, "g_*"'] & \operatorname{Shv}(\mathcal{Y}) \arrow[d, "g_*"] \\ \operatorname{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \arrow[r, "\gamma^*"] & \operatorname{Shv}(\mathcal{X}), \end{tikzcd} \qquad \begin{tikzcd} \operatorname{Shv}(\mathcal{X}) \arrow[r, hook, "\gamma_\sharp"] \arrow[d, "g^*"'] & \operatorname{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \arrow[d, "g^*"] \\ \operatorname{Shv}(\mathcal{Y}) \arrow[r, hook, "\gamma_\sharp"] & \operatorname{Shv}(\mathrm{Sub}_{/\mathcal{Y}}). \end{tikzcd}$$

(4) For every sheaf $\mathcal{F} \in \operatorname{Shv}(\mathcal{X})$, the sheaf $\gamma_\sharp\mathcal{F} \in \operatorname{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ is given at a representable submersion $f\colon \mathcal{Y} \to \mathcal{X}$ by

$$(\gamma_\sharp\mathcal{F})(\mathcal{Y}) \simeq \Gamma_{\mathcal{Y}}(f^*\mathcal{F}),$$

the anima of global sections of the pullback sheaf $f^*\mathcal{F} \in \operatorname{Shv}(\mathcal{Y})$.

Proof. For parts (1) and (2), we first show that the functor γ^* preserves limits and colimits. Let $N \twoheadrightarrow \mathcal{X}$ be a representable atlas for $\mathcal{X}$. By descent there is an equivalence $\operatorname{Shv}(\mathcal{X}) \simeq \lim_{[n]\in\Delta} \operatorname{Shv}(N^{\times^n_{\mathcal{X}}})$. Since each map $N^{\times^n_{\mathcal{X}}} \to \mathcal{X}$ is a representable submersion, it will suffice to show that for every representable submersion $f\colon M \to \mathcal{X}$, the composite

$$\operatorname{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \xrightarrow{\gamma^*} \operatorname{Shv}(\mathcal{X}) \xrightarrow{f^*} \operatorname{Shv}(M)$$

preserves limits and colimits. But by definition of γ^*, this functor agrees with the composite

$$\operatorname{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \xrightarrow{f^*} \operatorname{Shv}(\mathrm{Sub}_{/M}) \xrightarrow{\iota^*} \operatorname{Shv}(M),$$

which preserve limits and colimits by Corollary 4.1.14. It follows in particular that γ^* admits both a right adjoint γ_* and a left adjoint $\gamma_\sharp$.

For part (3) follows directly from part (5) of Proposition 4.1.13, since the both functors g_* commute with the pullback functors f^* for representable submersions $M \to \mathcal{X}$.

We may now prove that the functors γ_* and $\gamma_\sharp$ are fully faithful, finishing the proof of parts (1) and (2). It suffices to prove that $\gamma_\sharp$ is fully faithful, as then so is its double right adjoint γ_*. To show that the unit $\mathrm{id} \to \gamma^*\gamma_\sharp$ is an equivalence, it suffices to do so after applying the pullback functor $f^*\colon \mathrm{Shv}(\mathcal{X}) \to \mathrm{Shv}(M)$ for every representable submersion $f\colon M \to \mathcal{X}$. To this end, consider the following diagram:

$$\begin{tikzcd}
\mathrm{Shv}(\mathcal{X}) \arrow[r, "\gamma_\sharp"] \arrow[d, "f^*"'] & \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \arrow[r, "\gamma^*"] \arrow[d, "f^*"] & \mathrm{Shv}(\mathcal{X}) \arrow[d, "f^*"] \\
\mathrm{Shv}(M) \arrow[r, "\iota_\sharp"] & \mathrm{Shv}(\mathrm{Sub}_{/M}) \arrow[r, "\iota^*"] & \mathrm{Shv}(M).
\end{tikzcd}$$

The right square commutes by definition of γ^* while the left square commutes by part (3) we have just proved. We thus obtain an equivalence $f^*\gamma^*\gamma_\sharp \simeq \iota^*\iota_\sharp f^*$ and the claim thus follows from fully faithfulness of $\iota_\sharp$.

Finally, part (4) follows from (3) just as in the proof of part (6) of Proposition 4.1.13. □

The following result, which identifies $\mathcal{C}$-valued sheaves on the classifying stack $\mathbb{B}G$ with objects in $\mathcal{C}$ with a G-action, was pointed out to the author by Dustin Clausen.

Lemma 4.1.21 (Sheaves on classifying stacks)**.** *Let G be a Lie group and let $\mathcal{C}$ be a presentable ∞-category. Then there is an equivalence*

$$\mathrm{Shv}(\mathbb{B}G;\mathcal{C}) \simeq \mathcal{C}^{BG}$$

between the ∞-category of $\mathcal{C}$-valued sheaves on the classifying stack $\mathbb{B}G$ and the ∞-category of $\mathcal{C}$-valued local system on the classifying space $BG \in \mathrm{An}$.

Proof. It will suffice to show the claim when $\mathcal{C}$ is the ∞-category of animae, as the general case may be obtained by tensoring both sides with $\mathcal{C}$ in Pr^{L}. By definition, the stack $\mathbb{B}G$ is the colimit of the simplicial diagram $\Delta^{\mathrm{op}} \to \mathrm{Shv}(\mathrm{Diff})$, $[n] \mapsto G^n$, hence by descent there is an equivalence

$$\mathrm{Shv}(\mathbb{B}G) \simeq \lim_{[n]\in\Delta} \mathrm{Shv}(G^n).$$

For a topological space X, let $\mathrm{Loc}(X) \subseteq \mathrm{Shv}(X)$ denote the subcategory of *locally constant sheaves*, as in [Lur17, Definition A.1.12]. By [Lur17, Theorem A.1.15], there is a functorial equivalence $\mathrm{Loc}(X) \simeq \mathrm{An}^{\mathrm{shp}(X)}$, where $\mathrm{shp}(X) \in \mathrm{An}$ denotes the *shape* of X. Since G is a smooth manifold, its shape agrees with its homotopy type, and thus we obtain an equivalence

$$\lim_{[n]\in\Delta} \mathrm{Loc}(G^n) \simeq \lim_{[n]\in\Delta} \mathrm{An}^{G^n} \simeq \mathrm{An}^{\mathrm{colim}_{[n]\in\Delta^{\mathrm{op}}} G^n} \simeq \mathrm{An}^{BG}.$$

To finish the proof, it thus remains to observe that the inclusion

$$\lim_{[n]\in\Delta} \mathrm{Loc}(G^n) \hookrightarrow \lim_{[n]\in\Delta} \mathrm{Shv}(G^n)$$

is essentially surjective: given an object $(\mathcal{F}_n)_{[n]\in\Delta}$ of the target, the sheaf $\mathcal{F}_n \in \mathrm{Shv}(G^n)$ is the pullback along $G^n \to \mathrm{pt}$ of the sheaf $\mathcal{F}_0 \in \mathrm{Shv}(\mathrm{pt}) = \mathrm{Loc}(\mathrm{pt})$ and thus is a locally constant sheaf. □

4.2 Genuine sheaves of animae

In this section, we will introduce and study the ∞-category $\mathrm{H}(\mathcal{X})$ of genuine sheaves of animae on a differentiable stack $\mathcal{X}$: homotopy invariant sheaves of animae on the site $\mathrm{Sub}_{/\mathcal{X}}$.

4.2.1 Homotopy invariant presheaves

We start by defining the notion of a *homotopy invariant presheaf* and study the homotopy localization functor $L_{\mathbb{R}}$. Throughout this subsection, we fix a differentiable stack $\mathcal{X}$ and a presentable ∞-category $\mathcal{C}$.

Definition 4.2.1 (Homotopy invariance)**.** A $\mathcal{C}$-valued presheaf $\mathcal{F} \in \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}};\mathcal{C})$ is said to be *homotopy invariant*[1] if for every object $\mathcal{Y} \in \mathrm{Sub}_{/\mathcal{X}}$, the projection map $\mathrm{pr}\colon \mathcal{Y}\times\mathbb{R} \to \mathcal{Y}$ induces an equivalence $\mathrm{pr}^*\colon \mathcal{F}(\mathcal{Y}) \xrightarrow{\sim} \mathcal{F}(\mathcal{Y}\times\mathbb{R})$ in $\mathcal{C}$. We denote by

$$\mathrm{PSh}^{\mathrm{htp}}(\mathrm{Sub}_{/\mathcal{X}};\mathcal{C}) \subseteq \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}};\mathcal{C})$$

the full subcategory spanned by the homotopy invariant presheaves.

Since a presheaf $\mathcal{F}$ is homotopy invariant if and only if it is local with respect to the projections $\mathrm{pr}\colon \mathcal{Y}\times\mathbb{R} \to \mathcal{Y}$, the ∞-category $\mathrm{PSh}^{\mathrm{htp}}(\mathrm{Sub}_{/\mathcal{X}};\mathcal{C})$ is a localization of $\mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}};\mathcal{C})$ at these maps and in particular $\mathrm{PSh}^{\mathrm{htp}}(\mathrm{Sub}_{/\mathcal{X}};\mathcal{C})$ is a presentable ∞-category. We recall an explicit description of the localization functor, due to Morel and Voevodsky [MV99] and already known to Suslin. Our discussion takes inspiration from [ADH21, Chapter 5].

Construction 4.2.2 (The Morel–Suslin–Voevodsky construction)**.** We define the *algebraic n-simplex* Δ^n_{alg} as the hyperplane in $\mathbb{R}^{n+1}$ defined by

$$\Delta^n_{\mathrm{alg}} := \{(x_0,x_1,\dots,x_n) \in \mathbb{R}^{(n+1)} \mid \sum_{i=0}^{n} x_i = 1\}.$$

[1]This is sometimes also referred to as $\mathbb{R}$-*invariance* or *concordance invariance*.

The algebraic simplices form a cosimplicial smooth manifold $\Delta^\bullet_{\mathrm{alg}}\colon \Delta \to \mathrm{Diff}$. Given a presheaf $\mathcal{F} \in \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}};\mathcal{C})$, we define another presheaf $L_{\mathbb{R}}\mathcal{F}$ by

$$L_{\mathbb{R}}\mathcal{F}(\mathcal{Y}) := \mathrm{colim}_{[n]\in\Delta^{\mathrm{op}}}\mathcal{F}(\mathcal{Y}\times\Delta^n_{\mathrm{alg}}),$$

the geometric realization in $\mathcal{C}$ of the simplicial object $[n] \mapsto \mathcal{F}(\mathcal{Y}\times\Delta^n_{\mathrm{alg}})$. This defines a functor

$$L_{\mathbb{R}}\colon \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}};\mathcal{C}) \to \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}};\mathcal{C}).$$

As $\Delta^0_{\mathrm{alg}} \cong \mathrm{pt}$, there is a canonical map of presheaves $\eta_{\mathcal{F}}\colon \mathcal{F} \to L_{\mathbb{R}}\mathcal{F}$.

Proposition 4.2.3. *For any presheaf $\mathcal{F} \in \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}};\mathcal{C})$, the presheaf $L_{\mathbb{R}}\mathcal{F}$ is homotopy invariant. Furthermore, the resulting functor*

$$L_{\mathbb{R}}\colon \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}};\mathcal{C}) \to \mathrm{PSh}^{\mathrm{htp}}(\mathrm{Sub}_{/\mathcal{X}};\mathcal{C})$$

is left adjoint to the inclusion $\mathrm{PSh}^{\mathrm{htp}}(\mathrm{Sub}_{/\mathcal{X}};\mathcal{C}) \hookrightarrow \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}};\mathcal{C})$, with unit given by the map $\eta_{\mathcal{F}}\colon \mathcal{F} \to L_{\mathbb{R}}\mathcal{F}$.

Proof. The proof is in essence identical to that of [ADH21, Proposition 5.1.2], where the case $\mathcal{X} = \mathrm{pt}$ is treated. We spell out the core ingredients of the proof and leave some details to the reader.

First we show that the presheaf $L_{\mathbb{R}}\mathcal{F}$ is homotopy invariant. Given a smooth manifold $\mathcal{Y}$, we have to show that the functor $\mathrm{pr}^*\colon L_{\mathbb{R}}\mathcal{F}(\mathcal{Y}) \to L_{\mathbb{R}}\mathcal{F}(\mathcal{Y}\times\mathbb{R})$ induced by the projection $\mathrm{pr}\colon \mathcal{Y}\times\mathbb{R} \to \mathcal{Y}$ is an equivalence. We claim that an inverse is given by the map $i_0^*\colon L_{\mathbb{R}}\mathcal{F}(\mathcal{Y}\times\mathbb{R}) \to L_{\mathbb{R}}\mathcal{F}(\mathcal{Y})$, where $i_0\colon \mathcal{Y} \hookrightarrow \mathcal{Y}\times\mathbb{R}$ denotes the inclusion of $\mathcal{Y}\times\{0\}$. Since $\mathrm{pr}\circ i_0 = \mathrm{id}$, it remains to show that $\mathrm{pr}^* i_0^* \simeq \mathrm{id}$. This is a standard argument which goes back at least to [MV99]: the composite map

$$\mathcal{Y}\times\mathbb{R}\times\Delta^\bullet_{\mathrm{alg}} \xrightarrow{\mathrm{pr}\times\mathrm{id}} \mathcal{Y}\times\Delta^\bullet_{\mathrm{alg}} \xrightarrow{i_0\times\mathrm{id}} \mathcal{Y}\times\mathbb{R}\times\Delta^\bullet_{\mathrm{alg}}$$

is simplicially homotopic to the identity of the simplicial object $\mathcal{Y}\times\mathbb{R}\times\Delta^\bullet_{\mathrm{alg}}$ and since simplicially homotopic maps between simplicial objects induce homotopic maps on realizations this provides the desired homotopy $\mathrm{pr}^* i_0^* \simeq \mathrm{id}^* = \mathrm{id}$.

In order to show that the map $\eta_{\mathcal{F}}\colon \mathcal{F} \to L_{\mathbb{R}}\mathcal{F}$ is the unit of an adjunction, we apply [Lur09, Proposition 5.2.7.4]. Note that if $\mathcal{F}$ is already homotopy invariant, each of the maps $\mathcal{F}(\mathcal{Y}) \to \mathcal{F}(\mathcal{Y}\times\Delta^n_{\mathrm{alg}})$ is an equivalence and hence the map $\eta_{\mathcal{F}}$ is an equivalence. For arbitrary $\mathcal{F}$, we showed above that $L_{\mathbb{R}}\mathcal{F}$ is homotopy invariant, and thus the map $\eta_{L_{\mathbb{R}}\mathcal{F}}\colon L_{\mathbb{R}}\mathcal{F} \to L_{\mathbb{R}}L_{\mathbb{R}}\mathcal{F}$ is an equivalence for arbitrary $\mathcal{F}$. It follows that also the map

$L_{\mathbb{R}}\eta_{\mathcal{F}}\colon L_{\mathbb{R}}\mathcal{F} \to L_{\mathbb{R}}L_{\mathbb{R}}\mathcal{F}$ is an equivalence, as this agrees with $\eta_{L_{\mathbb{R}}\mathcal{F}}$ up to swapping the two indices $[n]$ and $[m]$ in the double colimit

$$L_{\mathbb{R}}L_{\mathbb{R}}\mathcal{F} = \operatorname{colim}_{[n]\in\Delta^{\mathrm{op}}} \operatorname{colim}_{[m]\in\Delta^{\mathrm{op}}} \mathcal{F}(\mathcal{Y}\times\Delta^n_{\mathrm{alg}}\times\Delta^m_{\mathrm{alg}}).$$

This shows that the conditions of part (3) of [Lur09, Proposition 5.2.7.4] are satisfied, finishing the proof. □

Corollary 4.2.4 (cf. Hoyois [Hoy17, Proposition 3.4])**.** *Assume that geometric realizations commute with finite products in C (e.g. C is an ∞-topos or C is stable). Then the localization functor $L_{\mathbb{R}}$ preserves finite products.*

Proof. Since the functor $\mathcal{F} \mapsto \mathcal{F}(-\times\Delta^n_{\mathrm{alg}})$ preserves limits and geometric realizations commute with finite products in $\mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}};C)$, this is clear from the definition of $L_{\mathbb{R}}$. □

Corollary 4.2.5 (cf. Hoyois [Hoy17, Proposition 3.4])**.** *When $C = \mathrm{An}$ is the ∞-category of animae, the localization functor $L_{\mathbb{R}}\colon \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{PSh}^{\mathrm{htp}}(\mathrm{Sub}_{/\mathcal{X}})$ is locally cartesian: for a pair of maps $A \to B$ and $C \to B$ in $\mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}})$ such that A and B are homotopy invariant, the exchange map*

$$L_{\mathbb{R}}(A\times_B C) \to A\times_B L_{\mathbb{R}}(C)$$

is an equivalence in $\mathrm{PSh}^{\mathrm{htp}}(\mathrm{Sub}_{/\mathcal{X}})$.

Proof. We have to show that for all $\mathcal{Y}\in\mathrm{Sub}_{/\mathcal{Y}}$ the square

$$\begin{tikzcd}
\operatorname{colim}_{[n]\in\Delta^{\mathrm{op}}}\Big(A(\mathcal{Y}\times\Delta^n_{\mathrm{alg}})\times_{B(\mathcal{Y}\times\Delta^n_{\mathrm{alg}})} C(\mathcal{Y}\times\Delta^n_{\mathrm{alg}})\Big) \arrow[r] \arrow[d] & A(\mathcal{Y}) \arrow[d] \\
\operatorname{colim}_{[n]\in\Delta^{\mathrm{op}}} C(\mathcal{Y}\times\Delta^n_{\mathrm{alg}}) \arrow[r] & B(\mathcal{Y})
\end{tikzcd}$$

is a pullback square. As A and B are homotopy invariant, the top left corner is equivalent to the realization of the simplicial object $[n]\mapsto A(\mathcal{Y})\times_{B(\mathcal{Y})} C(\mathcal{Y}\times\Delta^n_{\mathrm{alg}})$, and thus this is an instance of universality of colimits in $\mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}})$. □

4.2.2 Genuine sheaves of animae

The following is the main definition of this section.

Definition 4.2.6 (Genuine sheaves)**.** Let $\mathcal{X}$ be a differentiable stack. A *genuine sheaf* on $\mathcal{X}$ is a sheaf $\mathcal{F} \in \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ which is homotopy invariant as a presheaf, in the sense of Definition 4.2.1. We denote by

$$\mathrm{H}(\mathcal{X}) := \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Sub}_{/\mathcal{X}}) \subseteq \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$$

the full subcategory spanned by the genuine sheaves on $\mathcal{X}$. More generally, if C is a presentable ∞-category, we let

$$\mathrm{H}(\mathcal{X};C) := \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Sub}_{/\mathcal{X}};C) \subseteq \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}};C)$$

be the subcategory of homotopy invariant C-valued sheaves on $\mathrm{Sub}_{/\mathcal{X}}$. Note that there is an equivalence $\mathrm{H}(\mathcal{X},C) \simeq \mathrm{H}(\mathcal{X}) \otimes C$.

As in the case of presheaves, $\mathrm{H}(\mathcal{X})$ is a localization of $\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ at the projections $\mathcal{Y} \times \mathbb{R} \to \mathcal{Y}$, and the inclusion admits a left adjoint

$$L_{\mathrm{htp}} \colon \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{H}(\mathcal{X}),$$

which we will refer to as the *homotopy localization functor*. We will study this functor in more detail in Subsection 4.2.3 below.

Unsurprisingly, a consequence of forcing homotopy invariance is that fiberwise homotopic maps over $\mathcal{X}$ become equivalent in $\mathrm{H}(\mathcal{X})$.

Definition 4.2.7. Let $\mathcal{X}$ be a differentiable stack, let $\mathcal{Y}, \mathcal{Z} \in \mathrm{Sub}_{/\mathcal{X}}$ be submersions over $\mathcal{X}$, and let $f_0, f_1 \colon \mathcal{Y} \to \mathcal{Z}$ be two maps in $\mathrm{Sub}_{/\mathcal{X}}$. A *homotopy over* $\mathcal{X}$ is a homotopy $f \colon \mathcal{Y} \times \mathbb{R} \to \mathcal{Z}$ in $\mathrm{Sub}_{/\mathcal{X}}$ whose restriction to $i_r \colon \mathcal{Y} \times \{r\} \hookrightarrow \mathcal{Y} \times \mathbb{R}$ is the map f_r, for $r \in \{0,1\}$.

A map $f \colon \mathcal{Y} \to \mathcal{Z}$ in $\mathrm{Sub}_{/\mathcal{X}}$ is a *strict homotopy equivalence* if there is a map $g \colon \mathcal{Y} \to \mathcal{Z}$ over $\mathcal{X}$ such that the maps gf and fg are homotopic over $\mathcal{X}$ to the respective identities, in the sense of Definition 2.5.6.

Lemma 4.2.8. *Let $\mathcal{X}$ be a differentiable stack and let $f \colon \mathcal{Y} \times \mathbb{R} \to \mathcal{Z}$ be a homotopy over $\mathcal{X}$ of maps $\mathcal{Y} \to \mathcal{Z}$ in* $\mathrm{Sub}_{/\mathcal{X}}$. *Then the maps $L_{\mathrm{htp}}(f_0), L_{\mathrm{htp}}(f_1) \colon L_{\mathrm{htp}}(\mathcal{Y}) \to L_{\mathrm{htp}}(\mathcal{Z})$ are equivalent as maps in* $\mathrm{H}(\mathcal{X})$.

Proof. It will suffice to prove this in the case that $\mathcal{Z} = \mathcal{Y} \times \mathbb{R}$ and f is the identity, so that f_0 and f_1 are the inclusions $i_r \colon \mathcal{Y} \times \{r\} \hookrightarrow \mathcal{Y} \times \mathbb{R}$. In that case, $L_{\mathrm{htp}}(i_0)$ and $L_{\mathrm{htp}}(i_1)$ are both inverse to the equivalence $L_{\mathrm{htp}}(\mathrm{pr}) \colon L_{\mathrm{htp}}(\mathcal{Y} \times \mathbb{R}) \xrightarrow{\sim} L_{\mathrm{htp}}(\mathcal{Y})$, and are hence equivalent. $\square$

Corollary 4.2.9. *Let $f\colon \mathcal{Y} \to \mathcal{Z}$ be a strict homotopy equivalence over $\mathcal{X}$. Then the morphism $L_{\mathrm{htp}}(f)\colon L_{\mathrm{htp}}(\mathcal{Y}) \to L_{\mathrm{htp}}(\mathcal{Z})$ is an equivalence in* $\mathrm{H}(\mathcal{X})$. □

Example 4.2.10. Every vector bundle $\mathcal{E} \to \mathcal{Y}$ over $\mathcal{X}$ is a strict homotopy equivalence in $\mathrm{Sub}_{/\mathcal{X}}$.

4.2.3 The homotopy localization functor

The homotopy localization functor $L_{\mathrm{htp}}\colon \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{H}(\mathcal{X})$ admits an explicit formula as a transfinite iteration of the localization functor $L_{\mathbb{R}}$ and the sheafification functor L_{open}. The goal of this subsection is to make this precise and to deduce some consequences.

Construction 4.2.11. For an ordinal α, let $[\alpha]$ denote the linearly ordered set of ordinals $\beta \leq \alpha$. We will construct for every ordinal α a functor

$$L\colon [\alpha] \to \mathrm{Fun}(\mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}}), \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}})),$$

which we informally denote by $\beta \mapsto L^{\beta}$ for $\beta \leq \alpha$. We proceed via transfinite induction:

- If $\alpha = 0$, we set $L^0 = \mathrm{id}$.
- If $\alpha = \beta + 1$ is a sucessor ordinal, we define

  $$L^{\beta+1} := L_{\mathrm{open}} \circ L_{\mathbb{R}} \circ L^{\beta}\colon \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}}).$$

  The units $\mathrm{id} \to L_{\mathrm{open}}$ and $\mathrm{id} \to L_{\mathbb{R}}$ define a natural transformation $L^{\beta} \to L^{\beta+1}$ which thus defines the extension $L\colon [\beta+1] \to \mathrm{Fun}(\mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}}), \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}}))$.
- If α is a limit ordinal, we define

  $$L^{\alpha} := \mathrm{colim}_{\beta<\alpha} L^{\beta}$$

  and let $L\colon [\alpha] = \alpha^{\triangleright} \to \mathrm{Fun}(\mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}}), \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}}))$ denote the cocone for this colimit.

We may think of the functor L^{α} as the α-fold power of the functor $L^1 = L_{\mathrm{open}} \circ L_{\mathbb{R}}$.

Lemma 4.2.12. *For every ordinal α, the functor $L^{\alpha}\colon \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}})$ preserves finite products and is locally cartesian.*

Proof. We proceed via transfinite induction. For $\alpha = 0$ we have $L^\alpha = \mathrm{id}$ and the claim is obvious. When $\alpha = \beta + 1$ is a successor ordinal, we have $L^\alpha = L_{\mathrm{open}} \circ L_{\mathbb{R}} \circ L^\beta$ and the claim follows from a combination of the induction hypothesis, Corollary 4.2.4 and left-exactness of the sheafification functor L_{open}. When α is a limit ordinal, the claim is immediate as finite limits commute with filtered colimits in $\mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}})$. □

Lemma 4.2.13. *For any uncountable limit ordinal α, the functor $L^\alpha\colon \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{PSh}(\mathrm{Shv}_{/\mathcal{X}})$ lands in the subcategory $\mathrm{H}(\mathcal{X})$ of homotopy invariant sheaves. Moreover, the transformation $\mathrm{id} = L^0 \to L^\alpha$ exhibits the functor L^α as a left adjoint to the inclusion $\mathrm{H}(\mathcal{X}) \hookrightarrow \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}})$.*

Proof. We start by showing that L^α lands in homotopy invariant sheaves. It follows immediately from Corollary 4.1.6 that $L^\alpha\mathcal{F}$ is a sheaf for every preheaf $\mathcal{F} \in \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}})$, as for every $\beta < \alpha$ the functor $L^{\beta+1} = L_{\mathrm{open}} \circ L_{\mathbb{R}} \circ L^\beta$ lands in sheaves. Since $L_{\mathbb{R}}^2 \simeq L_{\mathbb{R}}$, we obtain an equivalent definition of L^α if for successor ordinals we had defined $L^{\beta+1} = L_{\mathbb{R}} \circ L_{\mathrm{open}} \circ L_{\mathbb{R}} \circ L^\beta$, and since homotopy invariant presheaves are closed under colimits in $\mathrm{PSh}(\mathrm{Shv}_{/\mathcal{X}})$ it follows that L^α lands in homotopy invariant presheaves. This proves the first claim.

To prove that L^α is left adjoint to the inclusion, let $\mathcal{F}, \mathcal{G} \in \mathrm{PSh}(\mathrm{Shv}_{/\mathcal{X}})$ be presheaves and assume that $\mathcal{G}$ is a homotopy invariant sheaf. We will prove by transfinite induction that for every ordinal $\beta \leq \alpha$, the map

$$\mathrm{Hom}_{\mathrm{PSh}(\mathrm{Shv}_{/\mathcal{X}})}(L^\beta\mathcal{F}, \mathcal{G}) \to \mathrm{Hom}_{\mathrm{PSh}(\mathrm{Shv}_{/\mathcal{X}})}(\mathcal{F}, \mathcal{G})$$

given by precomposition with $\mathcal{F} \to L^\beta\mathcal{F}$ is an equivalence. For $\beta = 0$ we have $L^\beta\mathcal{F} = \mathcal{F}$ and there is nothing to show. When $\beta = \gamma + 1$ is a successor ordinal, the claim follows from the induction hypothesis for γ, the fact that $L^\beta\mathcal{F} = L_{\mathrm{open}}(L_{\mathbb{R}}(L^\gamma\mathcal{F}))$ and the fact that $\mathcal{G}$ is a homotopy invariant sheaf. When β is a limit ordinal, this is immediate from the induction hypothesis.

Taking $\beta = \alpha$ then finishes the proof. □

Corollary 4.2.14. *The functor $L_{\mathrm{htp}}\colon \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{H}(\mathcal{X})$ preserves finite products and is locally cartesian: for every pair of maps $A \to B$ and $C \to B$ in $\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ such that A and B are homotopy invariant, the exchange map*

$$L_{\mathrm{htp}}(A \times_B C) \to A \times_B L_{\mathrm{htp}}(C)$$

is an equivalence in $\mathrm{H}(\mathcal{X})$.

Proof. Let α be an uncountable ordinal. By Lemma 4.2.13 there is an equivalence $L_{\mathrm{htp}} \simeq L^\alpha$, and thus the claim follows from Lemma 4.2.12. □

4.2.4 Pullback and pushforward functors

The ∞-categories $\mathrm{H}(\mathcal{X})$ come equipped with pullback and pushforward functors analogous to those for the ∞-categories $\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$.

Lemma 4.2.15. *For a morphism $f\colon \mathcal{X}' \to \mathcal{X}$ of differentiable stacks, the pushforward functor $f_*\colon \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}'}) \to \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ preserves homotopy invariant sheaves:*

$$f_*\colon \mathrm{H}(\mathcal{X}') \to \mathrm{H}(\mathcal{X}).$$

Proof. The functor f_* is given by precomposition with the base change functor $-\times_{\mathcal{X}} \mathcal{X}'\colon \mathrm{Sub}_{/\mathcal{X}} \to \mathrm{Sub}_{/\mathcal{X}'}$, which sends the projection map $\mathcal{Y}\times\mathbb{R} \to \mathcal{Y}$ to the projection $(\mathcal{Y}\times_{\mathcal{X}} \mathcal{X}')\times\mathbb{R} \to \mathcal{Y}\times_{\mathcal{X}}\mathcal{X}'$. The claim follows immediately. □

The pushforward functor $f_*\colon \mathrm{H}(\mathcal{X}') \to \mathrm{H}(\mathcal{X})$ admits a left adjoint

$$f^*\colon \mathrm{H}(\mathcal{X}) \to \mathrm{H}(\mathcal{X}')$$

given as the composite $\mathrm{H}(\mathcal{X}) \hookrightarrow \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \xrightarrow{f^*} \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}'}) \xrightarrow{L_{\mathrm{htp}}} \mathrm{H}(\mathcal{X}')$. Here we abuse notation by writing both f^* for the pullback functor both before and after applying homotopy localization; when potential confusion arises we make the source and target of the functor f^* explicit. In case f is a representable submersion, the two functors in fact agree:

Lemma 4.2.16. *For every representable submersion $f\colon \mathcal{Y} \to \mathcal{X}$, the pullback functor $f^*\colon \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{Y}})$ preserves homotopy invariant sheaves.*

Proof. The functor f^* is given by precomposition with the forgetful functor $\mathrm{Sub}_{/\mathcal{Y}} \to \mathrm{Sub}_{/\mathcal{X}}$. Since this preserves the projection maps $\mathcal{Z}\times\mathbb{R} \to \mathcal{Z}$, the claim follows. □

Corollary 4.2.17. *For every representable submersion $f\colon \mathcal{Y} \to \mathcal{X}$, the pullback functor $f^*\colon \mathrm{H}(\mathcal{X}) \to \mathrm{H}(\mathcal{Y})$ admits a left adjoint $f_\sharp\colon \mathrm{H}(\mathcal{Y}) \to \mathrm{H}(\mathcal{X})$ given as the composite*

$$\mathrm{H}(\mathcal{Y}) \hookrightarrow \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{Y}}) \xrightarrow{f_\sharp} \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \xrightarrow{L_{\mathrm{htp}}} \mathrm{H}(\mathcal{X}).$$ □

Again, we abuse notation by writing $f_\sharp$ for the left adjoint to f^* both before and after applying homotopy localization.

Proposition 4.2.18 (Symmetric monoidality)**.** *For a morphism $f\colon \mathcal{X}' \to \mathcal{X}$ of differentiable stacks, the pullback functor $f^*\colon \mathrm{H}(\mathcal{X}) \to \mathrm{H}(\mathcal{X}')$ preserves finite products.*

Proof. By Corollary 4.2.14, this follows directly from Corollary 4.1.12. □

Proposition 4.2.19 (Smooth base change)**.** *Consider a pullback square of differentiable stacks*

$$\begin{tikzcd} \mathcal{Y}' \arrow[r, "h"] \arrow[d, "f'"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{Y} \arrow[d, "f"] \\ \mathcal{X}' \arrow[r, "g"] & \mathcal{X}, \end{tikzcd}$$

where f (and thus f') is a representable submersion. Then the Beck-Chevalley map

$$\mathrm{BC}_\sharp \colon f'_\sharp h^* \Rightarrow g^* f_\sharp$$

of functors $\mathrm{H}(\mathcal{Y}) \to \mathrm{H}(\mathcal{X}')$ *is an equivalence.*

Proof. The map is obtained by applying the homotopy localization functor

$$L_{\mathrm{htp}} \colon \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}'}) \to \mathrm{H}(\mathcal{X}')$$

to the analogous exchange map $f'_\sharp h^* \Rightarrow g^* f_\sharp$ as functors from $\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{Y}})$ to $\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}'})$, and thus it will suffice to show the latter is an equivalence. As all functors in sight preserve colimits, it suffices to check this on representables $\mathcal{Z} \in \mathrm{Sub}_{/\mathcal{Y}}$. But in this case, this is simply the equivalence $\mathcal{Z} \times_{\mathcal{Y}} \mathcal{Y}' \xrightarrow{\sim} \mathcal{Z} \times_{\mathcal{X}} \mathcal{X}'$ over $\mathcal{X}'$, obtained from the pullback square. $\square$

Corollary 4.2.20. *Let $j \colon \mathcal{U} \hookrightarrow \mathcal{X}$ be an open embedding of differentiable stacks. Then the functors*

$$j_\sharp \colon \mathrm{H}(\mathcal{U}) \to \mathrm{H}(\mathcal{X}) \qquad \textit{and} \qquad j_* \colon \mathrm{H}(\mathcal{U}) \to \mathrm{H}(\mathcal{X})$$

are fully faithful.

Proof. Applying Proposition 4.2.19 to the pullback square

$$\begin{tikzcd} \mathcal{U} \arrow[r, equal] \arrow[d, equal] \arrow[dr, phantom, "\lrcorner"] & \mathcal{U} \arrow[d, "j"] \\ \mathcal{U} \arrow[r, "j"] & \mathcal{X} \end{tikzcd}$$

shows that the unit $\mathrm{id} \to j^* j_\sharp$ is an equivalence, showing that $j_\sharp$ is fully faithful. The claim for j_* follows immediately. $\square$

Proposition 4.2.21 (Smooth projection formula)**.** *Let $f \colon \mathcal{Y} \to \mathcal{X}$ be a representable submersion of differentiable stacks. Then for all objects $A \in \mathrm{H}(\mathcal{X})$ and $B \in \mathrm{H}(\mathcal{Y})$, the exchange map*

$$\mathrm{PF}_\sharp \colon f_\sharp(f^* A \times B) \to A \times f_\sharp B$$

is an equivalence in $\mathrm{H}(\mathcal{X})$

Proof. By Corollary 4.2.14, it suffices to prove the analogous statement for the functor $f_\sharp\colon \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{Y}}) \to \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$. As both sides preserve colimits in A and B, we may assume that $A \in \mathrm{Sub}_{/\mathcal{X}}$ and $B \in \mathrm{Sub}_{/\mathcal{Y}}$ are representables. In that case, this map is simply the canonical equivalence

$$(A \times_{\mathcal{X}} \mathcal{Y}) \times_{\mathcal{Y}} B \xrightarrow{\sim} A \times_{\mathcal{X}} B. \qquad \square$$

Proposition 4.2.22 (Homotopy invariance)**.** *Let $\pi\colon \mathcal{E} \to \mathcal{X}$ be a vector bundle over a differentiable stack $\mathcal{X}$. Then the functor $\pi^*\colon \mathrm{H}(\mathcal{X}) \to \mathrm{H}(\mathcal{E})$ is fully faithful.*

Proof. We show that for every object $A \in \mathrm{H}(\mathcal{X})$, the counit map $\pi_\sharp\pi^*A \to A$ is an equivalence. Since both sides preserve colimits in the variable A, it suffices to prove this when A is a representable $(f\colon \mathcal{Y} \to \mathcal{X}) \in \mathrm{Sub}_{/\mathcal{X}}$. In that case, there is an equivalence $\pi_\sharp\pi^*A \simeq L_{\mathrm{htp}}(\mathcal{E} \times_{\mathcal{X}} \mathcal{Y})$, and the counit is given by the map $L_{\mathrm{htp}}(\mathrm{pr}_{\mathcal{Y}})\colon L_{\mathrm{htp}}(\mathcal{E} \times_{\mathcal{X}} \mathcal{Y}) \to L_{\mathrm{htp}}(\mathcal{Y})$ induced by the projection onto $\mathcal{Y}$. Since this projection is a strict homotopy equivalence, the claim thus follows from Corollary 4.2.9. $\square$

We end the section by making precise the functoriality of the assignment $\mathcal{X} \mapsto \mathrm{H}(\mathcal{X})$ and showing that it forms a sheaf of ∞-categories on DiffStk.

Construction 4.2.23. We will turn the assignment $\mathcal{X} \mapsto \mathrm{H}(\mathcal{X})$ into a functor $\mathrm{DiffStk}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$. Consider again the functor $\mathrm{Shv}(\mathrm{Sub}_{/-})\colon \mathrm{DiffStk}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$ from Corollary 4.1.12. As the value at the terminal object $\mathrm{pt} \in \mathrm{DiffStk}$ is $\mathrm{Shv}(\mathrm{Diff})$, this functor admits a canonical enhancement to a functor $\mathrm{Shv}(\mathrm{Sub}_{/-})\colon \mathrm{DiffStk}^{\mathrm{op}} \to \mathrm{CAlg}_{\mathrm{Shv}(\mathrm{Diff})}(\mathrm{Pr}^{\mathrm{L}})$. To obtain $\mathrm{H}(\mathcal{X})$ from $\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$, we have to invert the morphisms $\mathcal{Y} \times \mathbb{R} \to \mathcal{Y}$, but note that these morphisms are obtained by tensoring $\mathcal{Y} \in \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ with the map $\mathbb{R} \to \mathrm{pt}$ in $\mathrm{Shv}(\mathrm{Diff})$. As localizing the ∞-category $\mathrm{Shv}(\mathrm{Diff})$ at the map $\mathbb{R} \to \mathrm{pt}$ gives the ∞-category An by Proposition 4.2.27, it follows that for every differentiable stack $\mathcal{X}$ there is an equivalence

$$\mathrm{H}(\mathcal{X}) \simeq \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \otimes_{\mathrm{Shv}(\mathrm{Diff})} \mathrm{An} \quad \in \quad \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}).$$

For every morphism $f\colon \mathcal{X}' \to \mathcal{X}$, the pullback functor $f^*\colon \mathrm{H}(\mathcal{X}) \to \mathrm{H}(\mathcal{X}')$ is obtained by localizing the pullback functor $f^*\colon \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}'})$, and it follows that the functor

$$\mathrm{H}(-) := \mathrm{Shv}(\mathrm{Sub}_{/-}) \otimes_{\mathrm{Shv}(\mathrm{Diff})} \mathrm{An}\colon \mathrm{DiffStk}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$$

is a functorial extension of the assignment $\mathcal{X} \mapsto \mathrm{H}(\mathcal{X})$.

Lemma 4.2.24. *The functor $\mathcal{X} \mapsto \mathrm{H}(\mathcal{X})$ is a sheaf of ∞-categories on* DiffStk.

Proof. It suffices to show that for every differentiable stack $\mathcal{X}$, the composition of H with the forgetful functor $\mathrm{Open}(\mathcal{X})^{\mathrm{op}} \to \mathrm{DiffStk}^{\mathrm{op}}$ is a sheaf of ∞-categories on $\mathrm{Open}(\mathcal{X})$. As each inclusion $j\colon \mathcal{U} \hookrightarrow \mathcal{V}$ in $\mathrm{Open}(\mathcal{X})$ is a representable submersion, it follows from Lemma 4.2.16 that the pullback functor $j^*\colon \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{V}}) \to \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{U}})$ sends the subcategory $\mathcal{H}(\mathcal{V})$ into the subcategory $\mathcal{H}(\mathcal{U})$. This determines a unique subdiagram $\mathrm{H}(-) \subseteq \mathrm{Shv}(\mathrm{Sub}_{/-})$ on $\mathrm{Open}(\mathcal{X})^{\mathrm{op}}$. Since each composite

$$\mathrm{H}(\mathcal{U}) \hookrightarrow \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \xrightarrow{L_{\mathrm{htp}}} \mathrm{H}(\mathcal{U})$$

is an equivalence, the functoriality of this subdiagram agrees with the restricted functoriality from the functor $\mathrm{H}\colon \mathrm{DiffStk}^{\mathrm{op}} \to \mathrm{Cat}_\infty$. Since the assignment $\mathcal{U} \mapsto \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{U}})$ is a sheaf of ∞-categories by Lemma 4.1.10, it will thus suffice to prove the following statement: given an open cover $\{\mathcal{U}_i \hookrightarrow \mathcal{X}\}$ of a differentiable stack $\mathcal{X}$, a sheaf $\mathcal{F} \in \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ is homotopy invariant if and only if its restriction to every $\mathcal{U}_i$ is homotopy invariant. But this is immediate from the definition. $\square$

4.2.5 From ordinary sheaves to genuine sheaves

In this subsection, we will show that the forgetful functor $\gamma^*\colon \mathrm{H}(\mathcal{X}) \to \mathrm{Shv}(\mathcal{X})$ from genuine sheaves to ordinary sheaves, defined in Construction 4.1.19, exhibits $\mathrm{Shv}(\mathcal{X})$ as both a localization as well as a colocalization of $\mathrm{H}(\mathcal{X})$. In particular, every ordinary sheaf on a differentiable stack $\mathcal{X}$ naturally gives rise to two distinct genuine sheaves on $\mathcal{X}$.

Notation 4.2.25. For a differentiable stack $\mathcal{X}$, we abuse notation and write $\gamma^*\colon \mathrm{H}(\mathcal{X}) \to \mathrm{Shv}(\mathcal{X})$ for the restriction of the functor $\gamma^*\colon \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{Shv}(\mathcal{X})$ from Construction 4.1.19.

Proposition 4.2.26. *Let $\mathcal{X}$ be a differentiable stack. Then the fully faithful functor $\gamma_\sharp\colon \mathrm{Shv}(\mathcal{X}) \hookrightarrow \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ from Proposition 4.1.20 lands in the subcategory of genuine sheaves:*

$$\gamma_\sharp\colon \mathrm{Shv}(\mathcal{X}) \hookrightarrow \mathrm{H}(\mathcal{X}).$$

In particular, the functor $\gamma^*\colon \mathrm{H}(\mathcal{X}) \to \mathrm{Shv}(\mathcal{X})$ admits a fully faithful left adjoint.

Proof. Given an ordinary sheaf $\mathcal{F} \in \mathrm{Shv}(\mathcal{X})$, we show that the associated sheaf $\gamma_\sharp\mathcal{F} \in \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ is homotopy invariant. Recall from Proposition 4.1.20 that, for a representable submersion $f\colon \mathcal{Y} \to \mathcal{X}$, the ∞-groupoid $(\iota_\sharp\mathcal{F})(\mathcal{Y})$ is obtained by first pulling back $\mathcal{F}$ to a sheaf $f^*\mathcal{F} \in \mathrm{Shv}(\mathcal{Y})$ over $\mathcal{Y}$, and then passing to global sections:

$$(\iota_\sharp\mathcal{F})(\mathcal{Y}) \simeq \Gamma_{\mathcal{Y}}(f^*\mathcal{F}).$$

In a similar way, one sees that the restriction map $\mathrm{pr}^*\colon (\iota_\sharp\mathcal{F})(\mathcal{Y}) \to (\iota_\sharp\mathcal{F})(\mathcal{Y}\times\mathbb{R})$ induced by the projection map $\mathrm{pr}\colon \mathcal{Y}\times\mathbb{R} \to \mathcal{Y}$ is equivalent to the map $\Gamma_{\mathcal{Y}}(f^*\mathcal{F}) \to \Gamma_{\mathcal{Y}\times\mathbb{R}}((f\circ \mathrm{pr})^*\mathcal{F}) = \Gamma_{\mathcal{Y}}(\mathrm{pr}_*\mathrm{pr}^* f^*\mathcal{F})$ induced by the unit map $\mathrm{id} \to \mathrm{pr}_*\mathrm{pr}^*$. It will thus suffice to show that this unit map is an equivalence, i.e., that the pullback functor

$$\mathrm{pr}^*\colon \mathrm{Shv}(\mathcal{Y}) \to \mathrm{Shv}(\mathcal{Y}\times\mathbb{R})$$

is fully faithful. Since both sides satisfy descent in the variable $\mathcal{Y}$, it will suffice to prove this in the case where $\mathcal{Y} = M$ is a smooth manifold, which is a special case of [Lur17, Lemma A.2.9]. This finishes the proof. □

As we will see in Section 4.4, the functor $\gamma^*\colon \mathrm{H}(\mathcal{X}) \to \mathrm{Shv}(\mathcal{X})$ is not an equivalence in general: the ∞-category $\mathrm{H}(\mathcal{X})$ of genuine sheaves captures more geometric fixed point data than the ∞-category $\mathrm{Shv}(\mathcal{X})$ of ordinary sheaves. The following result shows that this difference between genuine and ordinary sheaves vanishes when $\mathcal{X}$ has trivial isotropy:

Proposition 4.2.27. *Let M be a smooth manifold. Then the functor*

$$\gamma^*\colon \mathrm{H}(M) \to \mathrm{Shv}(M)$$

from Notation 4.2.25 is an equivalence of ∞-categories.

Proof. Since γ^* admits a fully faithful left adjoint $\gamma_\sharp$, it will suffice to show that the functor γ^* is conservative. To this end, let $\varphi\colon \mathcal{F} \to \mathcal{G}$ be a morphism of genuine sheaves on $\mathcal{X}$, i.e., a morphism of homotopy invariant sheaves on $\mathrm{Sub}_{/\mathcal{X}}$, and assume that the induced map $\varphi(U)\colon \mathcal{F}(U) \to \mathcal{G}(U)$ is an equivalence of animae for every open subspace $U \subseteq M$. By homotopy invariance of $\mathcal{F}$ and $\mathcal{G}$, it follows that also the map $\varphi(U\times\mathbb{R}^n)\colon \mathcal{F}(U\times\mathbb{R}^n) \to \mathcal{G}(U\times\mathbb{R}^n)$ is an equivalence for every $U \in \mathrm{Open}(M)$ and $n \in \mathbb{N}$. Consider now an arbitrary smooth submersion $f\colon Y \to M$ and let $\mathcal{U}$ be the poset of open subspaces V of Y for which the restriction $f|_V\colon V \to f(V) =: U$ is isomorphic in $\mathrm{Sub}_{/M}$ to the projection $\mathrm{pr}\colon U\times\mathbb{R}^n \to U$ for some natural number n. By Theorem C.8, $\mathcal{U}$ forms a *complete open cover* of Y, in the sense of Definition E.58, and thus it follows from Proposition E.60 that Y admits a hypercover consisting of elements of $\mathcal{U}$. Since $\mathrm{Shv}(\mathrm{Sub}_{/M})$ is hypercomplete by Proposition 4.1.15 and since the map $\varphi(V)\colon \mathcal{F}(V) \to \mathcal{G}(V)$ is an equivalence for every $V \in \mathcal{U}$, we deduce that also the map $\varphi(Y)\colon \mathcal{F}(V) \to \mathcal{G}(V)$ is an equivalence. Since this holds for every $Y \in \mathrm{Sub}_{/M}$, it follows that the map φ is an equivalence in $\mathrm{H}(M)$, finishing the proof. □

Remark 4.2.28. By taking $M = \mathrm{pt}$ in the previous proposition, we recover the well-known fact that the constant sheaf functor $\Gamma^*\colon \mathrm{An} \to \mathrm{Shv}(\mathrm{Diff})$ induces an equivalence $\mathrm{An} \simeq \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff})$. We refer to [ADH21, Proposition 4.3.1] for a more direct proof of this fact.

We saw in Proposition 4.2.26 that the functor $\gamma^*\colon \mathrm{H}(\mathcal{X}) \to \mathrm{Shv}(\mathcal{X})$ admits a fully faithful left adjoint $\gamma_\sharp$ and thus exhibits $\mathrm{Shv}(\mathcal{X})$ as a colocalization of $\mathrm{H}(\mathcal{X})$. The next lemma shows that it also admits a full faithful *right* adjoint:

Lemma 4.2.29. *Let $\mathcal{X}$ be a differentiable stack. Then the functor $\gamma^*\colon \mathrm{H}(\mathcal{X}) \to \mathrm{Shv}(\mathcal{X})$ admits a fully faithful right adjoint*

$$\gamma_*\colon \mathrm{Shv}(\mathcal{X}) \hookrightarrow \mathrm{H}(\mathcal{X}).$$

Proof. To show that γ^* admits a right adjoint, it suffices to show that it preserves small colimits. Choosing a representable atlas $f\colon M \twoheadrightarrow \mathcal{X}$ for $\mathcal{X}$, the functor $f^*\colon \mathrm{Shv}(\mathcal{X}) \to \mathrm{Shv}(M)$ is conservative and preserves colimits, so it suffices to show that the composite $f^* \circ \gamma^*$ preserves colimits. By naturality of γ^*, this composite is equivalent to the composite

$$\mathrm{H}(\mathcal{X}) \xrightarrow{f^*} \mathrm{H}(M) \xrightarrow{\gamma^*} \mathrm{Shv}(M).$$

The first functor preserves colimits as it admits a right adjoint f_*, and the second functor preserves colimits as it is an equivalence by Proposition 4.2.27. This shows that γ^* preserves colimits and thus admits a right adjoint γ_*. As γ_* is a double right adjoint of the fully faithful functor $\gamma_\sharp\colon \mathrm{Shv}(\mathcal{X}) \hookrightarrow \mathrm{H}(\mathcal{X})$, it follows that also γ_* is fully faithful, finishing the proof. □

We may now make precise the heuristic mentioned in the introduction of this chapter that a morphism is inverted by the localization functor $\gamma^*\colon \mathrm{H}(\mathcal{X}) \to \mathrm{Shv}(\mathcal{X})$ if and only if it becomes an equivalence after forgetting the isotropy information:

Corollary 4.2.30. *Let $\mathcal{X}$ be a differentiable stack and let $f\colon M \twoheadrightarrow \mathcal{X}$ be a representable atlas for $\mathcal{X}$. Then a morphism $\varphi\colon \mathcal{F} \to \mathcal{G}$ of genuine sheaves on $\mathcal{X}$ is sent to an equivalence under the functor $\gamma^*\colon \mathrm{H}(\mathcal{X}) \to \mathrm{Shv}(\mathcal{X})$ if and only if the morphism $f^*\gamma\colon f^*\mathcal{F} \to f^*\mathcal{G}$ in $\mathrm{H}(M)$ is an equivalence.*

Proof. By Proposition 4.2.27, the bottom horizontal map in the commutative diagram

$$\begin{tikzcd} \mathrm{H}(\mathcal{X}) \arrow[r, "\gamma^*"] \arrow[d, "f^*"'] & \mathrm{Shv}(\mathcal{X}) \arrow[d, "f^*"] \\ H(M) \arrow[r, "\gamma^*", "\sim"'] & \mathrm{Shv}(M) \end{tikzcd}$$

is an equivalence. As the functor $f^*\colon \mathrm{Shv}(\mathcal{X}) \to \mathrm{Shv}(M)$ is conservative, the claim follows. □

4.3 Genuine sheaves of spectra

In this section, we define for every separated differentiable stack $\mathcal{X}$ an ∞-category $\mathrm{SH}(\mathcal{X})$ of *genuine sheaves of spectra on $\mathcal{X}$*.

4.3.1 Pointed genuine sheaves

Fix a differentiable stack $\mathcal{X}$.

Definition 4.3.1 (Pointed genuine sheaves)**.** We define the ∞-category $\mathrm{H}_\bullet(\mathcal{X})$ of *pointed genuine sheaves on $\mathcal{X}$* as the ∞-category of pointed objects in $\mathrm{H}(\mathcal{X})$:

$$\mathrm{H}_\bullet(\mathcal{X}) := \mathrm{H}(\mathcal{X})_*.$$

Observe that this is equivalent to the ∞-category $\mathrm{H}(\mathcal{X}, \mathrm{An}_*)$ of An_*-valued genuine sheaves on $\mathcal{X}$.

A priori, the ∞-category $\mathrm{H}_\bullet(\mathcal{X})$ contains two kinds of spheres S^n: the geometric spheres, obtained by regarding S^n as a smooth manifold, and the homotopical spheres, obtained by regarding S^n as an ∞-groupoid. We will show that they are in fact equivalent.

Lemma 4.3.2. *For any vector bundle $p\colon \mathcal{E} \to \mathcal{X}$, there is a cofiber sequence in* $\mathrm{H}(\mathcal{X})$

$$L_{\mathrm{htp}}(\mathcal{E} \setminus s(\mathcal{X})) \hookrightarrow L_{\mathrm{htp}}(\mathcal{E}) \to L_{\mathrm{htp}}(S^{\mathcal{E}}).$$

Proof. There are pullback squares

$$\begin{tikzcd}
\mathcal{E} \setminus s(\mathcal{E}) \arrow[r, hook] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{E} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & S^{\mathcal{E}} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{X} \arrow[d, "p"] \\
(\mathbb{R}^n \setminus \{0\})//\mathrm{GL}(n) \arrow[r, hook] & \mathbb{R}^n//\mathrm{GL}(n) \arrow[r] & S^n//\mathrm{GL}(n) \arrow[r] & \mathbb{B}\mathrm{GL}(n).
\end{tikzcd}$$

By descent, it will suffice to prove the statement for the universal case $\mathbb{R}^n//\mathrm{GL}(n) \to \mathbb{B}\mathrm{GL}(n)$. Identify the subspace $\mathbb{R}^n \subseteq S^n$ with the open complement of the point $\infty \in S^n$. Then $\mathbb{R}^n$ and $S^n \setminus \{0\}$ form an open cover of S^n whose intersection is $\mathbb{R}^n \setminus \{0\}$. By passing to quotient stacks, we obtain an analogous open cover of $S^n//\mathrm{GL}(n)$. By descent, we thus obtain a pushout square

$$\begin{tikzcd}
L_{\mathrm{htp}}(\mathbb{R}^n \setminus \{0\}//\mathrm{GL}(n)) \arrow[r] \arrow[d] \arrow[dr, phantom, "\ulcorner"] & L_{\mathrm{htp}}(\mathbb{R}^n//\mathrm{GL}(n)) \arrow[d] \\
L_{\mathrm{htp}}(S^n \setminus \{0\}//\mathrm{GL}(n)) \arrow[r] & L_{\mathrm{htp}}(S^n//\mathrm{GL}(n))
\end{tikzcd}$$

in $\mathrm{H}(\mathbb{B}\,\mathrm{GL}(n))$. As $S^n \setminus \{0\} \cong \mathbb{R}^n$ is strictly homotopy equivalent to the point, the lower left corner is terminal in $\mathrm{H}(\mathbb{B}\,\mathrm{GL}(n))$, and hence this pushout square constitutes the desired cofiber sequence. $\square$

Corollary 4.3.3. *For every natural number $n \geq 0$, the following two objects in $\mathrm{H}_\bullet(\mathcal{X})$ are equivalent:*

(1) The geometric n-sphere, defined as the image of $S^n \in \mathrm{Diff}$ under the functor

$$\mathrm{Diff} \xrightarrow{\mathcal{X}\times -} \mathrm{Sub}_{/\mathcal{X}} \hookrightarrow \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \xrightarrow{L_{\mathrm{htp}}} \mathrm{H}(\mathcal{X}),$$

with base point given by the map $\{\infty\} \to S^n$.

(2) The homotopical n-sphere, defined as the image of $S^n \in \mathrm{An}_*$ under the unique symmetric monoidal colimit-preserving functor $\mathrm{An}_* \to \mathrm{H}_\bullet(\mathcal{X})$.

Proof. This is clear when $n = 0$. For $n > 1$, this follows by induction: applying Lemma 4.3.2 to the vector bundle $\mathcal{X} \times \mathbb{R}^n$ and using the homotopy equivalences $\mathbb{R}^n \setminus \{0\} \simeq S^{n-1}$ and $\mathbb{R}^n \simeq \mathrm{pt}$, we obtain a cofiber sequence

$$L_{\mathrm{htp}}(\mathcal{X} \times S^{n-1}) \to * \to L_{\mathrm{htp}}(\mathcal{X} \times S^n),$$

showing that $L_{\mathrm{htp}}(\mathcal{X} \times S^n) \simeq \Sigma(L_{\mathrm{htp}}(\mathcal{X} \times S^{n-1}))$. $\square$

4.3.2 Genuine sheaves of spectra on global quotient stacks

We will now define the ∞-category $\mathrm{SH}(\mathcal{X})$ in the case where $\mathcal{X}$ is a *global quotient stack*, i.e. of the form $M//G$ for some compact Lie group G and some smooth G-manifold M. Equivalently, by Corollary 2.3.21, this means that the stack $\mathcal{X}$ comes equipped with a representable map $\mathcal{X} \to \mathbb{B}G$ for some compact Lie group G.

Definition 4.3.4. Let $\mathcal{X}$ be a global quotient stack, and let $\mathrm{Sph}(\mathcal{X}) \subseteq \mathrm{H}_\bullet(\mathcal{X})$ denote the subcategory spanned by the sphere bundles $L_{\mathrm{htp}}(S^{\mathcal{E}})$ for all vector bundles $\mathcal{E} \to \mathcal{X}$. We define the ∞-category $\mathrm{SH}(\mathcal{X})$ of *genuine sheaves of spectra over $\mathcal{X}$* by formally inverting the objects of $\mathrm{Sph}(\mathcal{X})$:

$$\mathrm{SH}(\mathcal{X}) := \mathrm{H}_\bullet(\mathcal{X})[\mathrm{Sph}(\mathcal{X})^{-1}] \quad \in \quad \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}).$$

More explicitly, this means that $\mathrm{SH}(\mathcal{X})$ comes equipped with a functor $\Sigma^\infty\colon \mathrm{H}_\bullet(\mathcal{X}) \to \mathrm{SH}(\mathcal{X})$ such that for any presentable ∞-category $\mathcal{E}$ precomposition with Σ^∞ induces an inclusion of path components

$$\mathrm{Hom}_{\mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})}(\mathrm{SH}(\mathcal{X}), \mathcal{E}) \to \mathrm{Hom}_{\mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})}(\mathrm{H}_\bullet(\mathcal{X}), \mathcal{E})$$

hitting those symmetric monoidal left adjoints $F\colon \mathrm{H}_\bullet(\mathcal{X}) \to \mathcal{E}$ which sends every sphere bundle $L_{\mathrm{htp}}(S^{\mathcal{E}}) \in \mathrm{H}_\bullet(\mathcal{X})$ to an invertible object in $\mathcal{E}$.

We let $\Sigma^\infty_+\colon \mathrm{H}(\mathcal{X}) \to \mathrm{SH}(\mathcal{X})$ denote the composite

$$\mathrm{H}(\mathcal{X}) \xrightarrow{(-)_+} \mathrm{H}_\bullet(\mathcal{X}) \xrightarrow{\Sigma^\infty} \mathrm{SH}(\mathcal{X}).$$

Remark 4.3.5. We shall show in Proposition 4.4.17 below that for a compact Lie group G the ∞-category $\mathrm{SH}(\mathbb{B}G)$ of genuine sheaves of spectra over the classifying stack $\mathbb{B}G$ is equivalent to the ∞-category Sp_G of genuine G-spectra, explaining the terminology.

Lemma 4.3.6. *For every global quotient stack $\mathcal{X}$, the ∞-category $\mathrm{SH}(\mathcal{X})$ is stable.*

Proof. As $\mathrm{SH}(\mathcal{X})$ is stable if and only if the homotopical 1-sphere S^1 is invertible, this is a direct consequence from Corollary 4.3.3, since the object $L_{\mathrm{htp}}(\mathcal{X}\times S^1) \in H_\bullet(\mathcal{X})$ is the sphere bundle of the vector bundle $\mathcal{X}\times\mathbb{R} \to \mathcal{X}$ and thus invertible in $\mathrm{SH}(\mathcal{X})$. □

Warning 4.3.7. The ∞-category $\mathrm{SH}(\mathcal{X})$ should not be confused with the ∞-category $\mathrm{H}(\mathcal{X};\mathrm{Sp})$ of Sp-valued genuine sheaves on $\mathcal{X}$, which is obtained from $\mathrm{H}_\bullet(\mathcal{X})$ by only inverting the object $L_{\mathrm{htp}}(\mathcal{X}\times S^1)$.

Lemma 4.3.8. *For every vector bundle $\mathcal{E} \to \mathcal{X}$, the sphere bundle $S = L_{\mathrm{htp}}(S^{\mathcal{E}}) \in \mathrm{H}_\bullet(\mathcal{X})$ is a symmetric object, in the sense that the cyclic permutation $\sigma_{123}\colon S\wedge S\wedge S \to S\wedge S\wedge S$ is homotopic to the identity.*

Proof. It suffices to prove this in the universal case $\mathcal{E} = \mathbb{R}^n//\mathrm{GL}(n) \to \mathbb{B}\,\mathrm{GL}(n)$. Note that the twist map $\tau = \left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)\colon \mathbb{R}^n\times\mathbb{R}^n \to \mathbb{R}^n\times\mathbb{R}^n$ is homotopic to the linear map $\left(\begin{smallmatrix}1&0\\0&-1\end{smallmatrix}\right)$ via the rotation homotopy $t \mapsto \left(\begin{smallmatrix}-\sin(\frac{\pi}{2}t)&\cos(\frac{\pi}{2}t)\\\cos(\frac{\pi}{2}t)&\sin(\frac{\pi}{2}t)\end{smallmatrix}\right)$. It follows that the cyclic permutation $\sigma_{123}\colon \mathbb{R}^n\times\mathbb{R}^n\times\mathbb{R}^n \to \mathbb{R}^n\times\mathbb{R}^n\times\mathbb{R}^n$ is homotopic to the identity. As all of the linear maps in the homotopy are invertible, they extend to maps $S^{n+n+n} \to S^{n+n+n}$, giving the claim. □

Proposition 4.3.9. *Let $f\colon \mathcal{Y} \to \mathcal{X}$ be a representable morphism of global quotient stacks. Then the functor*

$$\mathrm{H}(\mathcal{Y}) \otimes_{\mathrm{H}(\mathcal{X})} \mathrm{SH}(\mathcal{X}) \to \mathrm{SH}(\mathcal{Y}),$$

obtained by the tensoring up the functor $\Sigma^\infty_+\colon \mathrm{H}(\mathcal{Y}) \to \mathrm{SH}(\mathcal{Y})$ from $\mathrm{H}(\mathcal{X})$ to $\mathrm{SH}(\mathcal{Y})$, is an equivalence.

Proof. By assumption we may write $\mathcal{X}$ in the form $N//G$ for a compact Lie group G and a smooth G-manifold N. Since f is representable, it follows that we may write $\mathcal{Y}$ as $M//G$ for some smooth G-manifold M and that f is induced by a G-equivariant map $M \to N$.

Observe that if we know the statement for both maps $M//G \to \mathbb{B}G$ and $N//G \to \mathbb{B}G$, it also follows for f. This means we may assume that $\mathcal{X} = \mathbb{B}G$ and $\mathcal{Y} = M//G$.

Let $E \to M$ be a G-equivariant vector bundle over M. By [Seg68, Proposition 2.4] there exists a finite-dimensional G-representation V and a G-equivariant vector bundle $E' \to M$ such that there is an isomorphism $E \oplus E' \cong V \times M$ as vector bundles over M. This implies that $S^{V\times M} \simeq S^E \otimes S^{E'}$, and thus S^E is a factor of the pullback of $S^V \in \mathrm{H}_\bullet(\mathbb{B}G)$ along the map $M//G \to \mathbb{B}G$. Since this holds for all $E \in \mathrm{Vect}^G(M)$, the claim now follows from Lemma I.2.45 and Lemma I.2.44 in Part I. □

Corollary 4.3.10 (Sheaves of spectra on smooth manifolds)**.** *When $\mathcal{X} = M$ is a smooth manifold, there is an equivalence*

$$\mathrm{SH}(M) \simeq \mathrm{Shv}(M; \mathrm{Sp})$$

between the ∞-categories of genuine and ordinary sheaves of spectra on M.

Proof. Recall the equivalence $\mathrm{H}(M) \simeq \mathrm{Shv}(M)$ from Proposition 4.2.27. By Proposition 4.3.9 we thus have $\mathrm{SH}(M) \simeq \mathrm{Shv}(M) \otimes_{\mathrm{An}} \mathrm{Sp} = \mathrm{Shv}(M; \mathrm{Sp})$. □

We will now make precise the functoriality of the assignment $\mathcal{X} \mapsto \mathrm{SH}(\mathcal{X})$ and show that it is a sheaf of ∞-categories.

Construction 4.3.11. Let $\mathrm{QtStk} \subseteq \mathrm{DiffStk}$ denote the full subcategory of global quotient stacks. We will turn the assignment $\mathcal{X} \mapsto \mathrm{SH}(\mathcal{X})$ into a functor $\mathrm{SH}\colon \mathrm{QtStk}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$. Recall from Definition 2.41 in Chapter I.2 the ∞-category $\mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})_{\mathrm{aug}}$ of augmented presentably symmetric monoidal ∞-categories: presentably symmetric monoidal ∞-categories C equipped with a fully faithful subcategory S. The assignment $\mathcal{X} \mapsto \mathrm{H}_\bullet(\mathcal{X})$ admits a lift to a functor $\mathrm{H}_\bullet\colon \mathrm{QtStk}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})_{\mathrm{aug}}$ by equipping $\mathrm{H}_\bullet(\mathcal{X})$ with the subcategory $\mathrm{Sph}(\mathcal{X})$ of sphere bundles from Definition 4.3.4; this is well-defined as the pullback functor f^* sends sphere bundles to sphere bundles for any morphism of stacks $f\colon \mathcal{Y} \to \mathcal{X}$. The functor $\mathrm{SH}\colon \mathrm{QtStk}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$ is now given as the composite

$$\mathrm{SH}\colon \mathrm{QtStk}^{\mathrm{op}} \xrightarrow{\mathrm{H}_\bullet} \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})_{\mathrm{aug}} \xrightarrow{\mathcal{L}} \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}),$$

where $\mathcal{L}$ is the formal inversion functor $(C, S) \mapsto C[S^{-1}]$ from Lemma 2.42 in Chapter I.2.

More explicitly, given a morphism $f\colon \mathcal{Y} \to \mathcal{X}$ of global quotient stacks, the functor $f^*\colon \mathrm{SH}(\mathcal{X}) \to \mathrm{SH}(\mathcal{Y})$ is the unique symmetric monoidal colimit-presering left adjoint which makes the following diagram commute:

$$\begin{tikzcd}
\mathrm{H}(\mathcal{X}) \arrow[r, "f^*"] \arrow[d, "\Sigma^\infty_+"'] & \mathrm{H}(\mathcal{Y}) \arrow[d, "\Sigma^\infty_+"] \\
\mathrm{SH}(\mathcal{X}) \arrow[r, dashed, "f^*"] & \mathrm{SH}(\mathcal{Y}).
\end{tikzcd}$$

Lemma 4.3.12. *The functor* $\mathrm{SH}\colon \mathrm{QtStk}^{\mathrm{op}} \to \mathrm{Cat}_\infty$ *is a sheaf of* ∞*-categories on* QtStk.

Proof. It suffices to show that for every compact Lie group G, the functor $\mathrm{Diff}_G^{\mathrm{op}} \to \mathrm{Cat}_\infty$ sending M to $\mathrm{SH}(M//G)$ is a sheaf of ∞-categories on Diff_G. By Proposition 4.3.9, this functor is equivalent to the functor $M \mapsto \mathrm{H}_\bullet(M//G) \otimes_{\mathrm{H}_\bullet(\mathbb{B}G)} \mathrm{SH}(\mathbb{B}G)$. As the objects $L_{\mathrm{htp}}(S^V) \in H(\mathbb{B}G)$ are symmetric objects by Lemma 4.3.8, the ∞-category $\mathrm{SH}(\mathbb{B}G)$ may be expressed as a colimit in Pr^{L} of the sequence

$$\mathrm{H}_\bullet(\mathbb{B}G) \xrightarrow{-\otimes S^{V_1}} \mathrm{H}_\bullet(\mathbb{B}G) \xrightarrow{-\otimes S^{V_2}} \mathrm{H}_\bullet(\mathbb{B}G) \xrightarrow{-\otimes S^{V_3}} \dots,$$

where $V_1, V_2, V_3, \dots$ is a sequence of irreducible G-representations containing every irreducible G-representation infinitely many times, see [Rob15, Corollary 2.22] and [Hoy17, Section 6.1]. It follows that for every smooth G-manifold M the ∞-category $\mathrm{SH}(M//G)$ is expressed as the colimit in Pr^{L} of the sequence

$$\mathrm{H}_\bullet(M//G) \xrightarrow{-\otimes S^{V_1}} \mathrm{H}_\bullet(M//G) \xrightarrow{-\otimes S^{V_2}} \mathrm{H}_\bullet(M//G) \xrightarrow{-\otimes S^{V_3}} \dots,$$

which means that the underlying ∞-category of $\mathrm{SH}(M//G)$ is a limit in Cat_∞ of the diagram of right adjoints $\underline{\mathrm{Hom}}(S^{V_i}, -)\colon \mathrm{H}_\bullet(M//G) \to \mathrm{H}_\bullet(M//G)$. Since the functor $\mathrm{H}_\bullet(-//G)\colon \mathrm{Diff}_G^{\mathrm{op}} \to \mathrm{Cat}_\infty$ is a sheaf of ∞-categories by Lemma 4.2.24 and limits commute with limits, it follows that also the functor $\mathrm{SH}(-//G)\colon \mathrm{Diff}_G^{\mathrm{op}} \to \mathrm{Cat}_\infty$ is a sheaf of ∞-categories. This finishes the proof. $\square$

4.3.3 Genuine sheaves of spectra on separated stacks

We will now define the ∞-category $\mathrm{SH}(\mathcal{X})$ for an arbitrary separated differentiable stack $\mathcal{X}$. Observe that every global quotient stack $\mathcal{X} = M//G$ is in particular separated, since we assume that the Lie group G is compact. This provides an inclusion of QtStk into the $(2,1)$-category SepStk of separated differentiable stacks.

Lemma 4.3.13. *Every sheaf of* ∞*-categories on* QtStk *extends uniquely to a sheaf of* ∞*-categories on* SepStk *via right Kan extension.*

Proof. Since every open substack of a global quotient stack is again a global quotient stack, right Kan extension provides a fully faithful functor from sheaves on QtStk to sheaves on SepStk, see Corollary E.47. Since every separated differentiable stack is locally a global quotient stack by Theorem 3.7.2, this functor is an equivalence. □

Definition 4.3.14 (Genuine sheaves of spectra)**.** We define the functor

$$\mathrm{SH}\colon \mathrm{SepStk}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$$

as the unique extension of the functor $\mathrm{SH}\colon \mathrm{QtStk}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$ from Construction 4.3.11. For a separated differentiable stack $\mathcal{X}$ we refer to the ∞-category $\mathrm{SH}(\mathcal{X})$ as the *∞-category of genuine sheaves of spectra on $\mathcal{X}$*. We similarly define a functor $\Sigma^\infty_+\colon \mathrm{H}(\mathcal{X}) \to \mathrm{SH}(\mathcal{X})$ by extension from QtStk.

The ∞-category $\mathrm{SH}(\mathcal{X})$ admits the following explicit description:

$$\mathrm{SH}(\mathcal{X}) \simeq \lim_{\mathcal{U}\in\mathrm{Open}^{\mathrm{QtStk}}(\mathcal{X})} \mathrm{SH}(\mathcal{U}) \quad \in \quad \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}).$$

Here the limit runs over the poset $\mathrm{Open}^{\mathrm{QtStk}}(\mathcal{X})$ of open substacks $\mathcal{U} \subseteq \mathcal{X}$ of $\mathcal{X}$ which are global quotient stacks.

Proposition 4.3.15. *Let $f\colon \mathcal{Y} \to \mathcal{X}$ be a representable morphism of separated differentiable stacks. Then the functor*

$$\mathrm{H}(\mathcal{Y}) \otimes_{\mathrm{H}(\mathcal{X})} \mathrm{SH}(\mathcal{X}) \to \mathrm{SH}(\mathcal{Y}),$$

obtained by the tensoring up the functor $\Sigma^\infty_+\colon \mathrm{H}(\mathcal{Y}) \to \mathrm{SH}(\mathcal{Y})$ from $\mathrm{H}(\mathcal{X})$ to $\mathrm{SH}(\mathcal{Y})$, is an equivalence.

Proof. By Proposition 4.3.9 this holds after pulling back along any inclusion $\mathcal{U} \hookrightarrow \mathcal{X}$ of a global quotient stack $\mathcal{U}$. The general statement then follows by descent. □

Proposition 4.3.16. *Let $f\colon \mathcal{Y} \to \mathcal{X}$ be a representable submersion of separated differentiable stacks.*

(1) (Left adjoint) The pullback functor $f^*\colon \mathrm{SH}(\mathcal{Y}) \to \mathrm{SH}(\mathcal{X})$ admits a left adjoint, denoted by $f_\sharp\colon \mathrm{SH}(\mathcal{X}) \to \mathrm{SH}(\mathcal{Y})$;

(2) (Smooth base change) These left adjoints satisfy smooth base change, as in Proposition 4.2.19;

(3) (Smooth projection formula) These left adjoints satisfy the smooth projection formula, as in Proposition 4.2.21.

Proof. By Proposition 4.3.15, the functor $f^*\colon \mathrm{SH}(\mathcal{Y}) \to \mathrm{SH}(\mathcal{X})$ is given by tensoring the functor $f^*\colon \mathrm{H}(\mathcal{Y}) \to \mathrm{H}(\mathcal{X})$ with $\mathrm{SH}(\mathcal{X})$ over $\mathrm{H}(\mathcal{X})$. By Corollary 4.2.17 the latter functor admits a left adjoint $f_\sharp\colon \mathrm{H}(\mathcal{X}) \to \mathrm{H}(\mathcal{Y})$ which is $\mathrm{H}(\mathcal{X})$-linear by Proposition 4.2.21, and thus we may tensor it with $\mathrm{SH}(\mathcal{X})$ over $\mathrm{H}(\mathcal{X})$ to obtain a $\mathrm{SH}(\mathcal{X})$-linear left adjoint $f_\sharp\colon \mathrm{H}(\mathcal{X}) \to \mathrm{H}(\mathcal{Y})$, proving parts (1) and (3). For part (2), consider a pullback square of separated differentiable stacks

$$\begin{tikzcd}
\mathcal{Y}' \arrow[r, "h"] \arrow[d, "f'"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{Y} \arrow[d, "f"] \\
\mathcal{X}' \arrow[r, "g"] & \mathcal{X}.
\end{tikzcd}$$

By Proposition 4.2.19, the induced square

$$\begin{tikzcd}
\mathrm{H}(\mathcal{X}) \arrow[r, "f^*"] \arrow[d, "g^*"'] & \mathrm{H}(\mathcal{Y}) \arrow[d, "h^*"] \\
\mathrm{H}(\mathcal{X}') \arrow[r, "f'^*"] & \mathrm{H}(\mathcal{X})
\end{tikzcd}$$

is horizontally left adjointable. By tensoring with $\mathrm{SH}(\mathcal{X})$ over $\mathrm{H}(\mathcal{X})$ and using the equivalence $\mathrm{H}(\mathcal{Y}) \otimes_{\mathrm{H}(\mathcal{X})} \mathrm{SH}(\mathcal{X}) \to \mathrm{SH}(\mathcal{Y})$ from Proposition 4.3.15, we see that the top square in the following diagram is horizontally left adjointable:

$$\begin{tikzcd}
\mathrm{SH}(\mathcal{X}) \arrow[r, "f^*"] \arrow[d, "g^*"'] & \mathrm{SH}(\mathcal{Y}) \arrow[d, "h^*"] \\
\mathrm{H}(\mathcal{X}') \otimes_{\mathrm{H}(\mathcal{X})} \mathrm{SH}(\mathcal{X}) \arrow[r, "f'^*"] \arrow[d] & \mathrm{SH}(\mathcal{X}) \otimes_{\mathrm{H}(\mathcal{X})} \mathrm{SH}(\mathcal{X}) \arrow[d] \\
\mathrm{H}(\mathcal{X}') \otimes_{\mathrm{H}(\mathcal{X}')} \mathrm{SH}(\mathcal{X}') \arrow[r, "f'^*"] \arrow[d, equal] & \mathrm{SH}(\mathcal{X}) \otimes_{\mathrm{H}(\mathcal{X}')} \mathrm{SH}(\mathcal{X}') \arrow[d, "\simeq"] \\
\mathrm{SH}(\mathcal{X}') \arrow[r, "f'^*"] & \mathrm{SH}(\mathcal{Y}').
\end{tikzcd}$$

The middle square is also clearly horizontally left adjointable, and the equivalence on the bottom right is another instance of Proposition 4.3.15. It follows that the outer square is horizontally left adjointable, which is what we needed to prove. □

4.4 Comparison with genuine equivariant spectra

Recall from the introduction of this chapter that our main motivation for introducing the ∞-categories $\mathrm{H}(\mathcal{X})$ and $\mathrm{SH}(\mathcal{X})$ of genuine sheaves of animae/spectra on a differentiable stack $\mathcal{X}$ is its relation to equivariant homotopy theory. For a compact Lie group G, there are ∞-categories An_G and Sp_G of *genuine G-animae* (a.k.a. 'G-spaces') and *genuine G-spectra*, respectively. The goal of this section is to show that there are equivalences of ∞-categories

$$\mathrm{H}(\mathbb{B}G) \simeq \mathrm{An}_G \qquad \text{and} \qquad \mathrm{SH}(\mathbb{B}G) \simeq \mathrm{Sp}_G,$$

explaining the terminology 'genuine sheaves'. We thank Marc Hoyois and Sil Linskens for useful discussions and valuable input concerning the results of this section.

We start by analyzing the ∞-category $\mathrm{H}(\mathbb{B}G)$. Recall from Corollary 2.3.20 that the category $\mathrm{Sub}_{/\mathbb{B}G}$ of representable submersions over the classifying stack $\mathbb{B}G$ is equivalent to the category Diff_G of smooth G-manifolds. As a consequence, we get an equivalence

$$H(\mathbb{B}G) = \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Sub}_{/\mathbb{B}G}) \xrightarrow{\sim} \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G)$$

between the ∞-category of genuine sheaves on $\mathbb{B}G$ and the ∞-category of homotopy invariant sheaves on the site Diff_G, where the Grothendieck topology is given by G-equivariant open covers. We shall show that the latter is equivalent to An_G in two steps:

(a) We first show that the ∞-category $\mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G)$ is equivalent to a presheaf category, indexed by the full subcategory $\mathrm{Orb}'_G \subseteq \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G)$ spanned by the objects $L_{\mathrm{htp}}(G/H)$;

(b) We then produce a comparison functor $R\colon \mathrm{An}_G \to \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G)$ and show that it preserves colimits and restricts to an equivalence $\mathrm{Orb}_G \xrightarrow{\sim} \mathrm{Orb}'_G$.

By combining (a) and (b), and using Elmendorf's theorem, it follows that R is an equivalence, leading to the desired equivalence $H(\mathbb{B}G) \simeq \mathrm{An}_G$. The equivalence $\mathrm{SH}(\mathbb{B}G) \simeq \mathrm{Sp}_G$ is then immediate an immediate consequence. We shall now spell out these steps in more detail.

4.4.1 Genuine sheaves form a presheaf category

We start by showing that the ∞-category $\mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G)$ is equivalent to a presheaf category.

Proposition 4.4.1. *The evaluation functors*

$$\mathrm{ev}_{G/H}\colon \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G) \to \mathrm{An}$$

for closed subgroups $H \leqslant G$ are jointly conservative.

Proof. Let $\varphi\colon \mathcal{F} \to \mathcal{F}'$ be a morphism of homotopy invariant sheaves on Diff_G and assume that $\varphi(G/H)\colon \mathcal{F}(G/H) \to \mathcal{F}'(G/H)$ is an equivalence for every closed subgroup $H \leqslant G$.

We have to show that $\varphi(M)\colon \mathcal{F}(M) \to \mathcal{F}'(M)$ is an equivalence for every smooth G-manifold. To this end, recall that M is locally of the form $G \times_H \mathbb{R}^n$ for some closed subgroup $H \subseteq G$ and some H-action on $\mathbb{R}^n$, and thus it follows from Proposition E.60 that M admits a hypercover by G-manifolds of this form. Since the ∞-topos $\mathrm{Shv}(\mathrm{Diff}_G)$ is hypercomplete by Proposition 4.1.15, it follows that M is a colimit of objects of the form $G \times_H \mathbb{R}^n$. Consequently, it will suffice to show that the map

$$\varphi(G \times_H \mathbb{R}^n)\colon \mathcal{F}(G \times_H \mathbb{R}^n) \to \mathcal{F}'(G \times_H \mathbb{R}^n)$$

is an equivalence for all $H \leqslant G$ and all $n \in \mathbb{N}$. But since $\mathcal{F}$ and $\mathcal{F}'$ are homotopy invariant and $G \times_H \mathbb{R}^n$ is G-equivariantly homotopy equivalent to G/H, this follows directly from the assumption on φ. $\square$

Corollary 4.4.2. *The ∞-category $\mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G)$ is generated under colimits by the objects $L_{\mathrm{htp}}(G/H)$ for all closed subgroups $H \leqslant G$.*

Proof. By adjunction and the Yoneda lemma, there is a natural equivalence

$$\mathrm{ev}_{G/H} \simeq \mathrm{Hom}_{\mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G)}(L_{\mathrm{htp}}(G/H), -)$$

of functors $\mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G) \to \mathrm{An}$. By Proposition 4.4.1, these functors are conservative. It follows that the objects $L_{\mathrm{htp}}(G/H)$ generate $\mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G)$ under colimits, see for example [Yan22, Corollary 2.5]. $\square$

Definition 4.4.3 (Fixed point functor)**.** For a closed subgroup $H \leqslant G$, consider the functor $G/H \times -\colon \mathrm{Diff} \to \mathrm{Diff}_G$ from smooth manifolds to smooth G-manifolds. We suggestively denote by

$$(-)^H\colon \mathrm{PSh}(\mathrm{Diff}_G) \to \mathrm{PSh}(\mathrm{Diff})$$

the functor given by precomposition with this functor, and refer to it as the *H-fixed point functor*.

Remark 4.4.4. For a smooth G-manifold M, the sheaf $M^H\colon \mathrm{Diff}^{\mathrm{op}} \to \mathrm{An}$ is a subsheaf of the representable sheaf on M, given at a smooth manifold N by those smooth maps $N \to M$ which factor through the subset of H-fixed points $M^H \subseteq M$. This justifies the notation and terminology for the functor $(-)^H$.

The functor $(-)^H$ is fully compatible with sheafification and homotopy localization:

Proposition 4.4.5. *Let $H \leqslant G$ be a closed subgroup.*

(1) The functor $(-)^H\colon \mathrm{PSh}(\mathrm{Diff}_G) \to \mathrm{PSh}(\mathrm{Diff})$ *preserves sheaves.*

(2) The functor $(-)^H$ *commutes with sheafification:*

$$\begin{tikzcd} \mathrm{PSh}(\mathrm{Diff}_G) \arrow[r, "(-)^H"] \arrow[d, "L_{\mathrm{open}}"'] & \mathrm{PSh}(\mathrm{Diff}) \arrow[d, "L_{\mathrm{open}}"] \\ \mathrm{Shv}(\mathrm{Diff}_G) \arrow[r, "(-)^H"] & \mathrm{Shv}(\mathrm{Diff}). \end{tikzcd}$$

(3) The functor $(-)^H\colon \mathrm{PSh}(\mathrm{Diff}_G) \to \mathrm{PSh}(\mathrm{Diff})$ *preserves homotopy invariant presheaves.*

(4) The functor $(-)^H$ *commutes with homotopy localization:*

$$\begin{tikzcd} \mathrm{PSh}(\mathrm{Diff}_G) \arrow[r, "(-)^H"] \arrow[d, "L_{\mathbb{R}}"'] & \mathrm{PSh}(\mathrm{Diff}) \arrow[d, "L_{\mathbb{R}}"] \\ \mathrm{PSh}^{\mathrm{htp}}(\mathrm{Diff}_G) \arrow[r, "(-)^H"] & \mathrm{PSh}^{\mathrm{htp}}(\mathrm{Diff}). \end{tikzcd}$$

(5) The functor $(-)^H\colon \mathrm{PSh}(\mathrm{Diff}_G) \to \mathrm{PSh}(\mathrm{Diff})$ *restricts to a functor*

$$(-)^H\colon \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G) \to \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff})$$

which preserves limits and colimits.

Proof. Part (1) is clear from the fact that the functor $G/H \times -\colon \mathrm{Diff} \to \mathrm{Diff}_G$ preserves pullbacks along open embeddings and that it sends open covers to G-equivariant open covers.

For part (2), it suffices by Corollary E.45 to show that the functor $G/H \times -\colon \mathrm{Diff} \to \mathrm{Diff}_G$ is a cocontinuous functor of sites, in the sense of Definition E.43. This is immediate from the fact that for a smooth manifold M, the G-equivariant subsets of $G/H \times M$ are precisely those of the form $G/H \times U$ for open subsets $U \subseteq M$.

Part (3) is obvious from the definition of homotopy invariance. Part (4) is immediate from the formula for the homotopy localization functor provided in Construction 4.2.2.

For part (5), the H-fixed point functor restricts to homotopy invariant sheaves by parts (1) and (3). At the level of presheaves, the H-fixed point functor $(-)^H\colon \mathrm{PSh}(\mathrm{Diff}_G) \to \mathrm{PSh}(\mathrm{Diff})$ preserves limits and colimits as these are computed pointwise. It follows at once that also the functor $(-)^H\colon \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G) \to \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff})$ preserves limits as these are computed in presheaves. Since colimits in homotopy invariant sheaves are computed by first computing the colimit in presheaves and then reflecting back into homotopy invariant sheaves, it then follows from parts (2) and (4) that the functor $(-)^H\colon \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G) \to \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff})$ also preserves colimits, finishing the proof. □

Corollary 4.4.6. *For every closed subgroup $H \leqslant G$, the evaluation functor*

$$\mathrm{ev}_{G/H}\colon \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G) \to \mathrm{An}$$

preserves colimits.

Proof. This evaluation functor is given by the composite

$$\mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G) \xrightarrow{(-)^H} \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}) \xrightarrow{\Gamma_*} \mathrm{An},$$

where Γ_* denotes the global section functor. The first functor preserves colimits by part (5) of Proposition 4.4.5. The second functor preserves colimits because it is an equivalence: it is the right adjoint of the equivalence $\gamma_\sharp\colon \mathrm{An} \xrightarrow{\sim} \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff})$ of Proposition 4.2.27, cf. Remark 4.2.28. $\square$

Corollary 4.4.7. *Let* $\mathrm{Orb}'_G \subseteq \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G)$ *denote the full subcategory spanned by the objects* $L_{\mathrm{htp}}(G/H)$*. Then the unique colimit-preserving extension*

$$\mathrm{PSh}(\mathrm{Orb}'_G) \to \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G)$$

of the inclusion $\mathrm{Orb}'_G \hookrightarrow \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G)$ *is an equivalence.*

Proof. The objects $L_{\mathrm{htp}}(G/H)$ are completely compact (or *tiny*) by Corollary 4.4.6: by adjunction and representability there is an equivalence of functors

$$\mathrm{Hom}_{\mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G)}(L_{\mathrm{htp}}(G/H), -) \simeq \mathrm{ev}_{G/H}\colon \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G) \to \mathrm{An}.$$

Furthermore, the objects $L_{\mathrm{htp}}(G/H)$ generate $\mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G)$ under colimits by Corollary 4.4.2. The statement is now an instance of [Lur09, Corollary 5.1.6.11]. $\square$

4.4.2 Comparison with genuine equivariant animae

Having established that $\mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G)$ is a presheaf category, we will now move to the construction of a comparison functor $R\colon \mathrm{An}_G \to \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G)$ and prove that it is an equivalence.

Definition 4.4.8 (Equivariant homotopy type)**.** Consider the composite functor

$$\Pi_G\colon \mathrm{Diff}_G \xrightarrow{\mathrm{fgt}} \mathrm{Top}_G \to \mathrm{An}_G,$$

where the first functor is the forgetful functor from smooth G-manifolds to topological G-spaces and where the second is the Dwyer-Kan localization at the weak G-homotopy equivalences defining the ∞-category An_G of genuine G-animae (a.k.a 'G-spaces'). Observe that Π_G sends open covers of smooth G-manifolds to colimit-diagrams, and hence it uniquely extends to a colimit-preserving functor

$$\Pi_G\colon \mathrm{Shv}(\mathrm{Diff}_G) \to \mathrm{An}_G.$$

We refer to the functor Π_G as the *G-equivariant homotopy type functor*.

Definition 4.4.9. As Π_G is colimit-preserving, it admits a right adjoint, which we will denote by $R\colon \mathrm{An}_G \to \mathrm{Shv}(\mathrm{Diff}_G)$. It follows immediately from the adjunction that for a genuine G-anima X, the sheaf $R(X)\colon \mathrm{Diff}_G^{\mathrm{op}} \to \mathrm{An}$ is given by

$$R(X)(M) = \mathrm{Hom}_{\mathrm{An}_G}(\Pi_G(M), X).$$

Observation 4.4.10. The functor $\Pi_G\colon \mathrm{Diff}_G \to \mathrm{An}_G$ is homotopy invariant: for a smooth G-manifold M we have $\Pi_G(M\times\mathbb{R}) \xrightarrow{\sim} \Pi_G(M)$. It follows that the functor $R\colon \mathrm{An}_G \to \mathrm{Shv}(\mathrm{Diff}_G)$ lands in homotopy invariant sheaves, so that R defines a functor

$$R\colon \mathrm{An}_G \to \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G).$$

Lemma 4.4.11. *The functor $R\colon \mathrm{An}_G \to \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G)$ preserves colimits.*

Proof. Since the evaluation functors $\mathrm{ev}_{G/H}\colon \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G) \to \mathrm{An}$ are jointly conservative by Proposition 4.4.1 and preserve colimits by Corollary 4.4.6, it will suffice to show that each of the composites $\mathrm{ev}_{G/H}\circ R$ preserves colimits. Observe that this composite is given by sending X to

$$R(X)(G/H) = \mathrm{Hom}_{\mathrm{An}_G}(\Pi_G(G/H), X) = \mathrm{Hom}_{\mathrm{An}_G}(G/H, X) = X^H,$$

the anima of H-fixed points of X. This functor is well-known to preserve colimits. $\square$

To show that R is an equivalence, it will suffice to show that R restricts to an equivalence between $\mathrm{Orb}_G \subseteq \mathrm{An}_G$ and $\mathrm{Orb}'_G \subseteq \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G)$. The main ingredient for this will be the comparison between the simplicial enrichments of Diff_G and Top_G.

Construction 4.4.12. Recall from Construction 4.2.2 the cosimplicial object $\Delta_{\mathrm{alg}}\colon \Delta \to \mathrm{Diff}_G$, sending $[n]$ to the algebraic n-simplex Δ^n_{alg} equipped with the trivial G-action. It gives rise to a simplicial enrichment of Diff_G given by

$$\mathrm{Hom}^{\Delta}_{\mathrm{Diff}_G}(M,N)_n := \mathrm{Hom}_{\mathrm{Diff}_G}(M\times\Delta^n_{\mathrm{alg}}, N).$$

In a similar way, we obtain a simplicial enrichment of Top_G:

$$\mathrm{Hom}^{\Delta}_{\mathrm{Top}_G}(X,Y)_n := \mathrm{Hom}_{\mathrm{Top}_G}(X \times \Delta^n_{\mathrm{alg}}, Y).$$

The forgetful functor $\mathrm{Diff}_G \to \mathrm{Top}_G$ induces a morphism of simplicial sets

$$\varphi\colon \mathrm{Hom}^{\Delta}_{\mathrm{Diff}_G}(M,N) \to \mathrm{Hom}^{\Delta}_{\mathrm{Top}_G}(M,N),$$

natural in M and N.

Our next goal is to show that the map φ is a simplicial homotopy equivalence of Kan simplicial sets.

Lemma 4.4.13. *For every $n,k \geq 0$, every simplicial subset $A \subseteq \Delta^n \times \Delta^k$ and every solid commutative diagram of simplicial sets*

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, hook] & \mathrm{Hom}^{\Delta}_{\mathrm{Diff}_G}(M,N) \arrow[d, "\varphi"] \\ \Delta^n \times \Delta^k \arrow[r, "H"'] \arrow[ur, dashed, "\tilde{H}"] & \mathrm{Hom}^{\Delta}_{\mathrm{Top}_G}(M,N), \end{tikzcd}$$

there exists a diagonal filler $\tilde{H}$ which makes the top triangle commute strictly and which makes the bottom triangle commute up to a homotopy which is constant on A.

Proof. The map H corresponds to a G-equivariant continuous map $H\colon M \times \Delta^n_{\mathrm{alg}} \times \Delta^k_{\mathrm{alg}} \to N$, and the map f expresses that H is a smooth map when restricted to the (closed) subset of $M \times \Delta^n_{\mathrm{alg}} \times \Delta^k_{\mathrm{alg}}$ corresponding to $A \subseteq \Delta^n \times \Delta^k$. By [Bre72, Theorem VI.4.2], H is G-equivariantly homotopic to a smooth map $\tilde{H}\colon M \times \Delta^n_{\mathrm{alg}} \times \Delta^k_{\mathrm{alg}} \to N$ which agrees with H on the subset of $M \times \Delta^n_{\mathrm{alg}} \times \Delta^k_{\mathrm{alg}}$ corresponding to $A \subseteq \Delta^n \times \Delta^k$, and the homotopy can be chosen to be constant on this subset. $\square$

Proposition 4.4.14. *For every two smooth G-manifolds M and N, the map*

$$\varphi\colon \mathrm{Hom}^{\Delta}_{\mathrm{Diff}_G}(M,N) \to \mathrm{Hom}^{\Delta}_{\mathrm{Top}_G}(M,N)$$

is a simplicial homotopy equivalence of Kan simplicial sets.

Proof. We first show that the source and target are Kan simplicial sets. This is clear for $\mathrm{Hom}^{\Delta}_{\mathrm{Top}_G}(M,N)$. For $\mathrm{Hom}^{\Delta}_{\mathrm{Diff}_G}(M,N)$, this follows immediately from Lemma 4.4.13 by taking $k=0$ and letting $A = \Lambda^n_l \subseteq \Delta^n$ be a horn.

To show that the map φ is a simplicial homotopy equivalence, it will thus suffice to show that φ induces a bijection on path components and isomorphisms on all simplicial homotopy groups. Surjectivity follows from Lemma 4.4.13 using $A = \partial\Delta^n \subseteq \Delta^n$. Injectivity follows from Lemma 4.4.13 using $A = \Delta^n \times \partial\Delta^1 \subseteq \Delta^n \times \Delta^1$. $\square$

The above comparsion map leads to an equivalence between $L_{\mathrm{htp}}(G/H)$ and $R(G/H)$.

Construction 4.4.15. We produce an equivalence $\psi\colon L_{\mathrm{htp}}(G/H) \to R(G/H)$ of homotopy invariant sheaves on Diff_G. Let M be a smooth G-manifold. By Proposition 4.2.3, there is an equivalence

$$L_{\mathrm{htp}}(G/H)(M) \simeq |\mathrm{Hom}^{\Delta}_{\mathrm{Diff}_G}(M, G/H)| \quad \in \quad \mathrm{An}.$$

Furthermore, since the ∞-category An_G is the homotopy coherent nerve of the simplicially enriched category CW_G of G-CW-complexes, which is a full sipmlicial subcategory of Top_G, we have by [HK20] a natural equivalence

$$R(G/H)(M) \simeq \mathrm{Hom}_{\mathrm{An}_G}(\Pi_G(M), G/H) \simeq |\mathrm{Hom}^{\Delta}_{\mathrm{Top}_G}(M, G/H)| \quad \in \quad \mathrm{An}.$$

The map $\varphi\colon \mathrm{Hom}^{\Delta}_{\mathrm{Diff}_G}(M, G/H) \to \mathrm{Hom}^{\Delta}_{\mathrm{Top}_G}(M, G/H)$ from Construction 4.4.12 thus produces the desired map $\psi\colon L_{\mathrm{htp}}(G/H) \to R(G/H)$, which is an equivalence by Proposition 4.4.14.

Theorem 4.4.16. *Let G be a Lie group. Then the functor $R\colon \mathrm{An}_G \to \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G) = \mathrm{H}(\mathbb{B}G)$ is an equivalence of ∞-categories.*

Proof. Consider the full subcategory $\mathrm{Orb}_G \subseteq \mathrm{An}_G$ spanned by the orbits G/H and the full subcategory $\mathrm{Orb}'_G \subseteq \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G)$ spanned by the objects $L_{\mathrm{htp}}(G/H)$. The equivalence $R(G/H) \simeq L_{\mathrm{htp}}(G/H)$ of Construction 4.4.15 shows that the functor R restricts to a functor $R|\colon \mathrm{Orb}_G \to \mathrm{Orb}'_G$. Consider now the following diagram:

$$\begin{tikzcd}
\mathrm{PSh}(\mathrm{Orb}_G) \arrow[r] \arrow[d, "\simeq"'] & \mathrm{PSh}(\mathrm{Orb}'_G) \arrow[d, "\simeq"] \\
\mathrm{An}_G \arrow[r, "R"] & \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}_G).
\end{tikzcd}$$

The top horizontal functor is the colimit-extension of $R|$. The vertical functors are the colimit-extensions of the respective inclusions; the left vertical functor is an equivalence by Elmendorf's theorem [Elm83] while the right vertical functor is an equivalence by Corollary 4.4.7. It thus remains to show that the restriction $R|\colon \mathrm{Orb}_G \to \mathrm{Orb}'_G$ is an equivalence of ∞-categories. The equivalence $R(G/H) \simeq L_{\mathrm{htp}}(G/H)$ shows it is essentially surjective. For fully faithfulness, let $H \leqslant G$ be a closed subgroup and X a genuine G-anima, and consider the following diagram:

$$\begin{tikzcd}
\mathrm{Hom}_{\mathrm{An}_G}(G/H, X) \arrow[r, "R"] \arrow[dr] & \mathrm{Hom}_{\mathrm{Shv}(\mathrm{Diff})}(R(G/H), R(X)) \arrow[d, "-\circ\psi"] \\
& \mathrm{Hom}_{\mathrm{Shv}(\mathrm{Diff}_G)}(L_{\mathrm{htp}}(G/H), R(X)).
\end{tikzcd}$$

We have to show that the horizontal map is an equivalence. Since ψ is an equivalence, we may equivalently show that the diagonal composite is an equivalence. We may further compose this diagonal with the following chain of equivalences:

$$\mathrm{Hom}_{\mathrm{Shv}(\mathrm{Diff}_G)}(L_{\mathrm{htp}}(G/H), R(X)) \simeq R(X)(G/H) \simeq \mathrm{Hom}_{\mathrm{An}_G}(G/H, X).$$

The resulting map $\mathrm{Hom}(G/H, X) \to \mathrm{Hom}(G/H, X)$ is natural in X and thus by the Yoneda-lemma it is given by precomposition with some endomorphism $G/H \to G/H$. Since every such endomorphism is an equivalence, this finishes the proof. □

4.4.3 Comparison with genuine equivariant spectra

Having established the equivalence between $\mathrm{H}(\mathbb{B}G)$ and An_G in Theorem 4.4.16, we will now move to its stable analogue. The equivalence $R\colon \mathrm{An}_G \xrightarrow{\sim} \mathrm{H}(\mathbb{B}G)$ induces an equivalence $R_*\colon \mathrm{An}_{G,*} \xrightarrow{\sim} \mathrm{H}(\mathbb{B}G)_* = \mathrm{H}_\bullet(\mathbb{B}G)$. This functor sends the representation sphere S^V of a G-representation V to the sphere bundle $S^{V//G}$ of the vector bundle $V//G \to \mathbb{B}G$ associated to V, and thus induces a functor

$$\mathrm{Sp}_G = \mathrm{An}_{G,*}[\{(S^V)^{-1}\}] \to \mathrm{H}_\bullet(\mathbb{B}G)[\{(S^{\mathcal{E}})^{-1}\}] = \mathrm{SH}(\mathbb{B}G).$$

Proposition 4.4.17. *Let G be a compact Lie group. Then the functor* $\mathrm{Sp}_G \to \mathrm{SH}(\mathbb{B}G)$ *is an equivalence of ∞-categories.*

Proof. This is immediate as every vector bundle $\mathcal{E} \to \mathbb{B}G$ is of the form $V//G \to \mathbb{B}G$ for some G-representation V. □

4.5 Pullback formalisms and universal properties

In this section, we study the assignments $\mathcal{X} \mapsto \mathrm{H}(\mathcal{X})$ and $\mathcal{X} \mapsto \mathrm{SH}(\mathcal{X})$ and show that they can be characterized by universal properties. The universal properties will be phrased within the setting of *pullback formalisms* on SepStk, a concept we will introduce momentarily. We will show:

(1) The assignment $\mathcal{X} \mapsto \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ is the initial pullback formalism on SepStk, see Proposition 4.5.12;

(2) The assignment $\mathcal{X} \mapsto \mathrm{H}(\mathcal{X})$ is the initial pullback formalism on SepStk satisfying homotopy invariance, see Proposition 4.5.21;

(3) The assignment $\mathcal{X} \mapsto \mathrm{SH}(\mathcal{X})$ is the initial pullback formalism on SepStk satisfying both homotopy invariance and *genuinely stability*, see Proposition 4.5.27.

These results will not be used later in this paper and thus may be skipped on first reading.

4.5.1 Sheaves of ∞-categories on SepStk

We start by making explicit the notion of a *sheaf of ∞-categories on* SepStk.

Definition 4.5.1. A *sheaf of ∞-categories on* SepStk is a functor $\mathcal{C}\colon \mathrm{SepStk}^{\mathrm{op}} \to \mathrm{Cat}_\infty$ which satisfies the sheaf condition with respect to the open cover topology: for every open cover $\{\mathcal{U}_i\}_{i\in I}$ of a separated differentiable stack $\mathcal{X}$, the map

$$\mathcal{C}(\mathcal{X}) \to \lim\Big(\textstyle\prod_{i\in I} \mathcal{C}(\mathcal{U}_i) \rightrightarrows \prod_{i,j\in I} \mathcal{C}(\mathcal{U}_i \times_{\mathcal{X}} \mathcal{U}_j) \overrightarrow{\rightrightarrows} \ \cdots \ \Big)$$

is an equivalence. Similarly, a *sheaf of presentably symmetric monoidal ∞-categories* is a functor $\mathcal{C}\colon \mathrm{SepStk}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$ satisfying the sheaf condition. Since the forgetful functor $\mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}) \to \mathrm{Cat}_\infty$ preserves limits, this may be tested on underlying ∞-categories.

Remark 4.5.2. A sheaf of ∞-categories on SepStk may equivalently be encoded by a limit-preserving functor $\mathrm{Shv}(\mathrm{SepStk})^{\mathrm{op}} \to \mathrm{Cat}_\infty$. In other words, letting $\mathcal{B}$ denote the ∞-topos $\mathrm{Shv}(\mathrm{SepStk})$, this is the same data as a *$\mathcal{B}$-category* in the sense of [Mar21; MW21], see also Chapter I.2.

Example 4.5.3. The three assignments $\mathcal{X} \mapsto \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$, $\mathcal{X} \mapsto \mathrm{H}(\mathcal{X})$ and $\mathcal{X} \mapsto \mathrm{SH}(\mathcal{X})$ form sheaves of presentably symmetric monoidal ∞-categories on SepStk. For the first two, see Corollary 4.1.12 and Lemma 4.2.24. For the third, we showed in Lemma 4.3.12 that the assignment $\mathcal{X} \mapsto \mathrm{SH}(\mathcal{X})$ is a sheaf of ∞-categories on QtStk, and this then uniquely extends to a sheaf of ∞-categories on all of SepStk.

Lemma 4.5.4 (Disjoint union)**.** *Let $\mathcal{C}$ be a sheaf of ∞-categories on* SepStk.

(1) The ∞-category $\mathcal{C}(\emptyset)$ is equivalent to the terminal ∞-category;

(2) For every collection of differentiable stacks $\{\mathcal{X}_i\}_{i\in I}$, the pullback functors $\mathcal{C}(\bigsqcup_{i\in I}\mathcal{X}_i) \to \mathcal{C}(\mathcal{X}_i)$ induced by the inclusions $\mathcal{X}_i \hookrightarrow \bigsqcup_{i\in I}\mathcal{X}_i$ constitute an equivalence of ∞-categories

$$\mathcal{C}(\bigsqcup_{i\in I}\mathcal{X}_i) \to \prod_{i\in I}\mathcal{C}(\mathcal{X}_i).$$

Proof. Part (1) follows from applying the sheaf condition to the empty cover of the empty stack. For part (2), note that the disjoint union $\bigsqcup_{i\in I}\mathcal{X}_i$ is covered by the collection of open substacks $\mathcal{X}_i \hookrightarrow \bigsqcup_{i\in I}\mathcal{X}_i$, which have empty intersection. By part (1), the sheaf condition for this cover thus provides the claimed equivalence. □

4.5.2 Pullback formalisms on differentiable stacks

We now introduce the notion of a *pullback formalism* on the site SepStk. Our definition is essentially a specialization of that of Drew and Gallauer [DG22], except for some differences in conventions explained in Remark 4.5.6 below.

Definition 4.5.5 (Pullback formalism, cf. [DG22, Definition 2.11])**.** Let $C\colon \mathrm{SepStk}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$ be a sheaf of presentably symmetric monoidal ∞-categories on SepStk. We will say that C is a *pullback formalism* on SepStk if the following conditions are satisfied:

(1) For every representable submersion $f\colon \mathcal{Y} \to \mathcal{X}$ of differentiable stacks, the pullback functor $f^*\colon C(\mathcal{X}) \to C(\mathcal{Y})$ admits a left adjoint $f_\sharp\colon C(\mathcal{Y}) \to C(\mathcal{X})$;

(2) (Smooth base change) For every pullback square

$$\begin{tikzcd} \mathcal{Y}' \arrow[r, "h"] \arrow[d, "f'"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{Y} \arrow[d, "f"] \\ \mathcal{X}' \arrow[r, "g"] & \mathcal{X} \end{tikzcd}$$

of differentiable stacks, where f (and thus f') is a representable submersion, the Beck-Chevalley transformation

$$\mathrm{BC}_\sharp\colon f_\sharp h^* \to g^* i'_\sharp\colon C(\mathcal{Y}) \to C(\mathcal{X}')$$

is an equivalence;

(3) (Smooth projection formula) For every representable submersion $f\colon \mathcal{Y} \to \mathcal{X}$ of differentiable stacks, every $A \in C(\mathcal{X})$ and every $B \in C(\mathcal{Y})$, the projection formula map

$$\mathrm{PF}_\sharp\colon f_\sharp(f^*A \otimes B) \to A \otimes f_\sharp B \quad \in \quad C(\mathcal{X})$$

is an equivalence.

Let $\mathcal{D}$ be another pullback formalism on SepStk. A natural transformation $F\colon C \to \mathcal{D}$ of functors $\mathrm{SepStk}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$ is called a *morphism of pullback formalisms* if for every representable submersion $f\colon \mathcal{Y} \to \mathcal{X}$ of differentiable stacks, the naturality square

$$\begin{tikzcd} C(\mathcal{Y}) \arrow[r, "F_{\mathcal{Y}}"] \arrow[d, "f^*"'] & \mathcal{D}(\mathcal{Y}) \arrow[d, "f^*"] \\ C(\mathcal{X}) \arrow[r, "F_{\mathcal{X}}"] & \mathcal{D}(\mathcal{X}) \end{tikzcd}$$

is vertically left adjointable, in the sense that the Beck-Chevalley transformation $f_\sharp \circ F_{\mathcal{X}} \Rightarrow F_{\mathcal{Y}} \circ f_\sharp \colon C(\mathcal{X}) \to \mathcal{D}(\mathcal{Y})$ is an equivalence. We let

$$\mathrm{PB}(\mathrm{SepStk}) \subseteq \mathrm{Fun}(\mathrm{SepStk}^{\mathrm{op}}, \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}))$$

denote the (non-full) subcategory spanned by the pullback formalisms on SepStk and the morphisms of pullback formalisms.

Remark 4.5.6. In contrast to [DG22], we include both the presentability condition as well as the descent condition with respect to the Grothendieck topology on SepStk into the definition of a pullback formalism. Hence the ∞-category $\mathrm{PB}(\mathrm{SepStk})$ would be written as $\mathrm{PB}^{\mathrm{pr}}_{\mathrm{open}}(\mathrm{SepStk})$ in [DG22]. Closely related is also the notion of a $(*,\sharp,\otimes)$-*formalism* by Khan and Ravi [KR21, Definition 5.5].

Remark 4.5.7. By Drew and Gallauer [DG22, Proposition 2.12, Proposition 4.4, Proposition 5.6], the (very large) ∞-category $\mathrm{PB}(\mathrm{SepStk})$ is presentable and the inclusion $\mathrm{PB}(\mathrm{SepStk}) \hookrightarrow \mathrm{Fun}(\mathrm{SepStk}^{\mathrm{op}}, \mathrm{CAlg}(\mathrm{Cat}_\infty))$ admits a left adjoint.

Example 4.5.8. Each of the three functors $\mathcal{X} \mapsto \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$, $\mathcal{X} \mapsto \mathrm{H}(\mathcal{X})$ and $\mathcal{X} \mapsto \mathrm{SH}(\mathcal{X})$ is a pullback formalism, see Corollary 4.2.17, Proposition 4.2.19, Proposition 4.2.21 and Proposition 4.3.16.

The condition for a sheaf of ∞-categories on SepStk to be a pullback formalism may be neatly formulated in the language of parametrized category theory.

Construction 4.5.9. Let $\mathcal{B}$ denote the ∞-topos $\mathrm{Shv}(\mathrm{SepStk})$. By Remark 4.5.2, a sheaf C of presentably symmetric monoidal ∞-categories on SepStk may equivalently be encoded by a fiberwise presentably symmetric monoidal $\mathcal{B}$-category. In particular, the sheaf $\mathcal{X} \mapsto \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ from Example 4.5.3 corresponds to a $\mathcal{B}$-category $\mathbf{U}$. By Lemma 4.1.9 this defines a parametrized subcategory of $\mathcal{X} \mapsto \mathrm{Shv}(\mathrm{SepStk})_{/\mathcal{X}}$, and thus it defines a *class of $\mathcal{B}$-groupoids* in the sense of [MW21, Remark 2.7.5]. We say that a morphism $f\colon \mathfrak{Y} \to \mathfrak{X}$ in $\mathrm{Shv}(\mathrm{SepStk})$ is a *morphism in* $\mathbf{U}$ if it is an object of the subcategory $\mathbf{U}(\mathfrak{X}) \subseteq \mathrm{Shv}(\mathrm{SepStk})_{/\mathfrak{X}}$.

Lemma 4.5.10. *With the above notations, let* $C\colon \mathrm{Shv}(\mathrm{SepStk})^{\mathrm{op}} = \mathcal{B}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$ *be a fiberwise presentably symmetric monoidal* $\mathcal{B}$*-category. Then its restriction to* $\mathrm{SepStk}^{\mathrm{op}}$ *is a pullback formalism if and only if the* $\mathcal{B}$*-category* C *admits* $\mathbf{U}$*-colimits and the tensor product* $\mathcal{B}$*-functor* $-\otimes-\colon C\times C \to C$ *preserves* $\mathbf{U}$*-colimits in both variables.*

Proof. We refer to [MW21, Definition 4.1.3] for the definition of parametrized colimits, and to [MW21, Example 4.1.9] for the characterization of parametrized colimits along $\mathcal{B}$-groupoids. It follows from this characterization that C admits $\mathbf{U}$-colimits if and only if the following two conditions hold:

(1') For every morphism $f\colon \mathfrak{Y} \to \mathfrak{X}$ in $\mathbf{U}$, the pullback functor $f^*\colon C(\mathfrak{X}) \to C(\mathfrak{Y})$ admits a left adjoint $f_\sharp\colon C(\mathfrak{Y}) \to C(\mathfrak{X})$;

(2') For every pullback square

$$\begin{tikzcd} \mathfrak{Y}' \arrow[r, "h"] \arrow[d, "f'"'] \arrow[dr, phantom, "\lrcorner"] & \mathfrak{Y} \arrow[d, "f"] \\ \mathfrak{X}' \arrow[r, "g"] & \mathfrak{X} \end{tikzcd}$$

in $\mathrm{Shv}(\mathrm{SepStk})$ where f (and thus f') is in $\mathbf{U}$, the Beck-Chevalley transformation $f_\sharp h^* \to g^* i'_\sharp$ is an equivalence.

Similarly, the condition that the $\mathcal{B}$-functor $-\otimes-\colon C\times C \to C$ to preserve $\mathbf{U}$-colimits in both variables is characterized by the following condition:

(3') For every $f\colon \mathfrak{Y} \to \mathfrak{X}$ in $\mathbf{U}$, every $A \in C(\mathfrak{X})$ and every $B \in C(\mathfrak{Y})$, the projection formula map $f_\sharp(f^*A \otimes B) \to A \otimes f_\sharp B$ is an equivalence.

Since $\mathbf{U}$ is generated under fiberwise colimits by the representable submersions $f\colon \mathcal{Y} \to \mathcal{X}$, Lemma 4.1.9, and since $\mathcal{B} = \mathrm{Shv}(\mathrm{SepStk})$ is the localization of a presheaf category, it follows from [MW22, Proposition A.2.1] and [MW21, Corollary 3.2.10] that it suffices to check the conditions (1')-(3') in the case where $f\colon \mathcal{Y} \to \mathcal{X}$ is a representable submersion between differentiable stacks. But these are precisely conditions (1)-(3) in the definition of a pullback formalism, finishing the proof. ◻

Given an ∞-topos $\mathcal{B}$ and a class of $\mathcal{B}$-groupoids $\mathbf{U}$, Martini and Wolf [MW22, Section 8] defined a *tensor product of* $\mathbf{U}$*-cocomplete* $\mathcal{B}$*-categories*, giving rise to a symmetric monoidal ∞-category $\mathrm{Cat}^{\mathbf{U}\text{-cc}}(\mathcal{B})$ of $\mathbf{U}$-cocomplete $\mathcal{B}$-categories. The above characterization of pullback formalisms can then be reformulated as follows:

Corollary 4.5.11. *Let $\mathcal{B} = \mathrm{Shv}(\mathrm{SepStk})$ and let $\mathbf{U}'$ be the class of $\mathcal{B}$-categories containing both $\mathbf{U}$ and the locally constant $\mathcal{B}$-categories. There is a fully faithful inclusion*

$$\mathrm{PB}(\mathrm{SepStk}) \hookrightarrow \mathrm{CAlg}(\mathrm{Cat}^{\mathbf{U}'-\mathrm{cc}}(\mathcal{B}))$$

whose essential image consists of those C such that $C(\mathcal{X})$ is presentable for every $\mathcal{X} \in$ SepStk.

Proof. Recall that a $\mathcal{B}$-category $K\colon \mathcal{B}^{\mathrm{op}} \to \mathrm{Cat}_\infty$ is called *locally constant* if it is equivalent to the sheafification of a constant functor with value some ∞-category K. A $\mathcal{B}$-category C admits K-indexed parametrized colimits if and only if each ∞-category $C(\mathcal{X})$ admits K-indexed colimits and each pullback functo f^* preserves K-indexed colimits, see [MW21, Example 4.2.6]. In particular, a commutative algebra in $\mathrm{Cat}^{\mathbf{U}'-\mathrm{cc}}(\mathcal{B})$ consists of symmetric monoidal $\mathcal{B}$-category $C\colon \mathcal{B}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Cat}_\infty)$ which, in addition to satisfying the conditions (1)-(3) from Definition 4.5.5, also factors through the subcategory $\mathrm{CAlg}(\mathrm{Cat}_\infty^{\mathrm{cc}})$ of symmetric monoidal ∞-categories admitting small colimits and whose tensor product preserves small colimits in each variable. Since the condition on a transformation $F\colon C \to \mathcal{D}$ of functors $\mathrm{SepStk}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$ to satisfy the Beck-Chevalley condition in Definition 4.5.5 corresponds to the condition for the associated $\mathcal{B}$-functor to preserve $\mathbf{U}$-indexed colimits, it follows that ∞-category PB(SepStk) is equivalent to the subcategory of $\mathrm{CAlg}(\mathrm{Cat}^{\mathbf{U}'-\mathrm{cc}}(\mathcal{B}))$ spanned by those functors $\mathcal{B}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Cat}_\infty)$ which in fact factor through the full subcategory $\mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}) \subseteq \mathrm{CAlg}(\mathrm{Cat}_\infty^{\mathrm{cc}})$. This finishes the proof. □

4.5.3 The initial pullback formalism

Proposition 4.5.12. *The assignment $\mathcal{X} \mapsto \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ is the initial pullback formalism on* SepStk.

Proof. Let $\mathcal{B} = \mathrm{Shv}(\mathrm{SepStk})$ and recall from the previous subsection that we defined the $\mathcal{B}$-category $\mathbf{U}$ as the $\mathcal{B}$-category corresponding to the pullback formalism $\mathcal{X} \mapsto \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$. By Corollary 4.5.11, it will now suffice to prove that $\mathbf{U}$ is an initial object of the ∞-category $\mathrm{CAlg}(\mathrm{Cat}^{\mathbf{U}'-\mathrm{cc}}(\mathcal{B}))$, or equivalently that $\mathbf{U}$ is the monoidal unit of the symmetric monoidal ∞-category $\mathrm{Cat}^{\mathbf{U}'-\mathrm{cc}}(\mathcal{B})$. But since $\mathbf{U}$ is already closed under fiberwise colimits, we see that $\mathbf{U}$ is the smallest $\mathcal{B}$-subcategory of $\Omega_{\mathcal{B}}$ containing the point and closed under $\mathbf{U}'$-colimits, and thus the theorem is a consequence of [MW21, Theorem 7.1.11]. □

Notation 4.5.13. Let C be a pullback formalism on SepStk. By the previous proposition, there exists a unique morphism $h^C\colon\ \mathrm{Shv}(\mathrm{Sub}_{/-}) \Rightarrow C$ of pullback formalisms. In particular, for every differentiable stack $\mathcal{X}$, there is a symmetric monoidal left adjoint

$$h^C_{\mathcal{X}} \colon \operatorname{Shv}(\operatorname{Sub}_{/\mathcal{X}}) \to C(\mathcal{X}).$$

Because these functors are required to commute with the functors $f_\sharp$ for all representable submersions f, it follows that $h^C_{\mathcal{X}}$ is given at a representable object $(f\colon \mathcal{Y} \to \mathcal{X}) \in \operatorname{Sub}_{/\mathcal{X}}$ by

$$h^C_{\mathcal{X}}(\mathcal{Y}) = f_\sharp \mathbb{1}_{\mathcal{Y}} \quad \in \quad C(\mathcal{X}).$$

We will often just write $h_{\mathcal{X}}(\mathcal{Y})$ for this object when C is clear from context.

Definition 4.5.14. A pullback formalism C on SepStk is called *pointed* if the ∞-category $C(\mathcal{X})$ is pointed for every differentiable stack $\mathcal{X}$. We say that C is *stable* if $C(\mathcal{X})$ is stable for every differentiable stack $\mathcal{X}$.

Remark 4.5.15. For a morphism $f\colon \mathcal{X} \to \mathcal{Y}$ of differentiable stacks, the pullback functor $f^*\colon C(\mathcal{X}) \to C(\mathcal{Y})$ preserves all colimits. Hence if C is pointed, f^* is a pointed functor. Similarly, if C is stable the pullback functors are exact.

Corollary 4.5.16. *The assignment* $\mathcal{X} \mapsto \operatorname{Shv}(\operatorname{Sub}_{/\mathcal{X}}; \mathrm{An}_*)$ *is the initial pointed pullback formalism on* SepStk. *The assignment* $\mathcal{X} \mapsto \operatorname{Shv}(\operatorname{Sub}_{/\mathcal{X}}; \mathrm{Sp})$ *is the initial stable pullback formalism on* SepStk.

Proof. These pullback functors are obtained from $\mathcal{X} \mapsto \operatorname{Shv}(\operatorname{Sub}_{/\mathcal{X}})$ by pointwise tensoring with An_* and Sp, respectively, and hence the result follows from Proposition 4.5.12 and the fact that the functors $-\otimes \mathrm{An}_*\colon \operatorname{CAlg}(\mathrm{Pr}^{\mathrm{L}}) \to \operatorname{CAlg}(\mathrm{Pr}^{\mathrm{L}}_*)$ and $-\otimes \mathrm{Sp}\colon \operatorname{CAlg}(\mathrm{Pr}^{\mathrm{L}}) \to \operatorname{CAlg}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$ are left adjoint to the respective inclusions. □

4.5.4 Homotopy invariance

We introduce the notion of homotopy invariance for a pullback formalism, and show that the assignment $\mathcal{X} \mapsto \mathrm{H}(\mathcal{X})$ is the initial pullback formalism on SepStk satisfying homotopy invariance.

Definition 4.5.17 (Homotopy invariance)**.** We say that a pullback formalism C on SepStk satisfies *homotopy invariance* if for every differentiable stack $\mathcal{X}$ the pullback functor $\mathrm{pr}^*\colon C(\mathcal{X}) \to C(\mathcal{X} \times \mathbb{R})$ associated to the projection $\mathrm{pr}\colon \mathcal{X} \times \mathbb{R} \to \mathcal{X}$ is fully faithful.

Warning 4.5.18. We emphasize that homotopy invariance does not imply that the ∞-category $C(\mathcal{X})$ is a homotopy invariant of the stack $\mathcal{X}$.

The condition of homotopy invariance is in essence the condition that the real line becomes contractible in C:

Lemma 4.5.19. *A pullback formalism C on* SepStk *satisfies homotopy invariance if and only if the object $h_{\mathrm{pt}}(\mathbb{R}) \in C(\mathrm{pt})$ is terminal.*

Proof. Note that C satisfies homotopy invariance if and only if for every object $A \in C(\mathcal{X})$ the counit map $\mathrm{pr}_\sharp \mathrm{pr}^* A \to A$ is an equivalence. By the smooth projection formula and smooth base change, one sees that there is an equivalence $\mathrm{pr}_\sharp \mathrm{pr}^* A \simeq h_{\mathcal{X}}(\mathcal{X} \times \mathbb{R}) \otimes A$, and the counit map is obtained by tensoring A with the projection $h_{\mathcal{X}}(\mathcal{X} \times \mathbb{R}) \to h_{\mathcal{X}}(\mathcal{X})$. Since this map is the pullback along $\mathcal{X} \to \mathrm{pt}$ of the map of $h_{\mathrm{pt}}(\mathbb{R}) \to h_{\mathrm{pt}}(\mathrm{pt}) = *$ in $C(\mathrm{pt})$, we get the claim. □

For a separated differentiable stack $\mathcal{X}$, consider the symmetric monoidal functor

$$\mathrm{Shv}(\mathrm{Diff}) \xrightarrow{p^*} \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}) \xrightarrow{h_{\mathcal{X}}} C(\mathcal{X}),$$

where $p \colon \mathcal{X} \to \mathrm{pt}$ is the map to the point. Homotopy invariance of C demands that this functor inverts the map $\mathbb{R} \to \mathrm{pt}$, or in other words, that it factors through the homotopy localization functor $L_{\mathrm{htp}} \colon \mathrm{Shv}(\mathrm{Diff}) \to \mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff}) \simeq \mathrm{An}$. Conversely, this means that we can enforce homotopy invariance by tensoring with An over Shv(Diff):

Lemma 4.5.20. *The inclusion* $\mathrm{PB}^{\mathrm{htp}}(\mathrm{SepStk}) \hookrightarrow \mathrm{PB}(\mathrm{SepStk})$ *of homotopy invariant pullback formalisms into all pullback formalisms admits a left adjoint $C \mapsto C^{\mathrm{htp}}$ given by*

$$C^{\mathrm{htp}}(\mathcal{X}) := C(\mathcal{X}) \otimes_{\mathrm{Shv}(\mathrm{Diff})} \mathrm{An}.$$

Proof. As argued above, C satisfies homotopy invariance if and only if the Shv(Diff)-algebra $C(\mathcal{X})$ in Pr^{L} is the restriction of an $\mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff})$-algebra in Pr^{L}, and a left adjoint to the inclusion $\mathrm{Pr}^{\mathrm{L}} \simeq \mathrm{CAlg}_{\mathrm{Shv}^{\mathrm{htp}}(\mathrm{Diff})} \hookrightarrow \mathrm{CAlg}_{\mathrm{Shv}(\mathrm{Diff})}(\mathrm{Pr}^{\mathrm{L}})$ is given by $- \otimes_{\mathrm{Shv}(\mathrm{Diff})} \mathrm{An}$. It thus remains to show that C^{htp} is in fact a pullback formalism.

The fact that C^{htp} is a sheaf of ∞-categories can be proved just like in Lemma 4.2.24. The left adjoints $f_\sharp$ for C^{htp} are inherited from C, as are smooth base change and the smooth projection formula, where for the latter we invoke Corollary 4.2.14. It follows that C^{htp} is a pullback formalism, which satisfies homotopy invariance by Lemma 4.5.19. □

Proposition 4.5.21. *The assignment $\mathcal{X} \mapsto \mathrm{H}(\mathcal{X})$ is the initial homotopy invariant pullback formalism on* SepStk.

Proof. As the functor $\mathrm{H}(-)$ was defined in Construction 4.2.23 as C^{htp} for $C(\mathcal{X}) = \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$, this is immediate from Lemma 4.5.20 and Proposition 4.5.12. □

It follows that for every homotopy invariant pullback formalism C on SepStk there is a unique morphism $\mathrm{H} \to C$ of pullback formalisms. We will abuse notation and denote this morphism again by h^C, just like in Notation 4.5.13. So for every $\mathcal{X} \in \mathrm{SepStk}$, there is a symmetric monoidal left adjoint

$$h^C_{\mathcal{X}} \colon \mathrm{H}(\mathcal{X}) \to C(\mathcal{X});$$

it sends the object $L_{\mathrm{htp}}(\mathcal{Y})$ to $h^C_{\mathcal{X}} = g_\sharp \mathbb{1}_{\mathcal{Y}}$ for every $(g \colon \mathcal{Y} \to \mathcal{X}) \in \mathrm{Sub}_{/\mathcal{X}}$. We will often write $h_{\mathcal{X}}(\mathcal{Y})$ for $h^C_{\mathcal{X}}(\mathcal{Y})$ when C is clear from context.

Corollary 4.5.22. *Let $\mathcal{X}$ be a differentiable stack and let C be a pullback formalism on* SepStk *satisfying homotopy invariance.*

(1) If $f \colon \mathcal{Y} \times \mathbb{R} \to \mathcal{Z}$ is a homotopy over $\mathcal{X}$ of maps $\mathcal{Y} \to \mathcal{Z}$ in $\mathrm{Sub}_{/\mathcal{X}}$*, then the maps $h_{\mathcal{X}}(f_0), h_{\mathcal{X}}(f_1) \colon h_{\mathcal{X}}(\mathcal{Y}) \to h_{\mathcal{X}}(\mathcal{Z})$ are equivalent as maps in $C(\mathcal{X})$;*

(2) Let $f \colon \mathcal{Y} \to \mathcal{Z}$ be a strict homotopy equivalence over $\mathcal{X}$. Then the morphism

$$h_{\mathcal{X}}(f) \colon h_{\mathcal{X}}(\mathcal{Y}) \to h_{\mathcal{X}}(\mathcal{Z})$$

is an equivalence in $C(\mathcal{X})$.

Proof. This is immediate from Lemma 4.2.8 Corollary 4.2.9. □

Corollary 4.5.23. *Assume that C is a homotopy invariant pullback formalism on* SepStk. *Then for every vector bundle $\pi \colon \mathcal{E} \to \mathcal{X}$, the functor $\pi^* \colon C(\mathcal{X}) \to C(\mathcal{E})$ is fully faithful.*

Proof. We need to show that the counit map $\pi_\sharp \pi^* \to \mathrm{id}$ is a homotopy equivalence. By the smooth projection formula, this map is given by tensoring with $h_{\mathcal{X}}(\mathcal{E}) \to h_{\mathcal{X}}(\mathcal{X})$. As the map $\mathcal{E} \to \mathcal{X}$ is a strict homotopy equivalence, this map is an equivalence by Corollary 4.5.22. □

Corollary 4.5.24. *The assignment $\mathcal{X} \mapsto \mathrm{H}_\bullet(\mathcal{X})$ is the initial pointed homotopy invariant pullback formalism on* SepStk.

Proof. This is immediate from Proposition 4.5.21. □

4.5.5 Genuine stability

We introduce the notion of genuine stability for a pullback formalism C, and show that the assignment $\mathcal{X} \mapsto \mathrm{SH}(\mathcal{X})$ of genuine sheaves of spectra form the universal example of a pullback formalism which satisfies both homotopy invariance and genuinely stability.

Notation 4.5.25 (Sphere bundles in a pullback formalism)**.** Let C be a pointed pullback formalism on SepStk satisfying homotopy invariance. Let $\mathcal{E} \to \mathcal{X}$ be a vector bundle of separated differentiable stacks. Then we define the *sphere bundle of* $\mathcal{E}$ *in* C as the cofiber

$$S^{\mathcal{E}} := \mathrm{cofib}(h_{\mathcal{X}}(\mathcal{X}) \to h_{\mathcal{X}}(S^{\mathcal{E}})) \quad \in \quad C(\mathcal{X}),$$

where the map is induced from the canonical section $\mathcal{X} \to S^{\mathcal{E}}$ at ∞.

Definition 4.5.26 (Genuine stability)**.** A pullback formalism C on SepStk is called *genuinely stable* if it is pointed and for every vector bundle $\pi\colon \mathcal{E} \to \mathcal{X}$ its sphere bundle $S^{\mathcal{E}} \in C(\mathcal{X})$ is monoidally invertible. In this case, we let $S^{-\mathcal{E}}$ denote a monoidal inverse of $S^{\mathcal{E}}$.

Note that every morphism $F\colon C \to \mathcal{D}$ of pullback formalisms preserves the sphere bundles $S^{\mathcal{E}}$. In particular, if C is genuinely stable, then so is $\mathcal{D}$.

Proposition 4.5.27. *The assignment* $\mathcal{X} \mapsto \mathrm{SH}(\mathcal{X})$ *is the initial homotopy invariant genuinely stable pullback formalism on* SepStk.

Proof. Observe that a pullback formalism C on SepStk is homotopy invariant or genuinely stable if and only if its restriction to the subcategory QtStk of global quotient stacks is homotopy invariant or genuinely stable, hence we may as well restrict to pullback formalisms on QtStk. In that case, SH is obtained from $\mathrm{H}_\bullet$ by inverting the sphere bundles, hence the statement is a consequence of Corollary 4.5.24. □

II.5 Localization sequences

Our goal in this chapter is to prove a version of the localization theorem for genuine sheaves on differentiable stacks. This roughly states that, for a closed embedding $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ with open complement $j\colon \mathcal{U} \hookrightarrow \mathcal{X}$, the inclusions

$$\mathrm{SH}(\mathcal{Z}) \xhookrightarrow{i_*} \mathrm{SH}(\mathcal{X}) \xhookleftarrow{j_*} \mathrm{SH}(\mathcal{U})$$

form a *recollement* in the sense of [Lur17, Definition A.8.1]. Our proof strategy will closely follow that of [Hoy17, Section 4.3] and [Kha19], which in turn are based on the original reference [MV99, Theorem 3.2.21]. Other sources are [CD19, Section 2.3] and [Vol21, Section 4].

5.1 The localization axiom

In the spirit of [CD19, Section 2.3], we start by introducing the localization axiom for an arbitrary pullback formalism $C\colon \mathrm{SepStk}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$ on SepStk (see Definition 4.5.5).

Lemma 5.1.1. *Let C be a pointed pullback formalism on* SepStk *and let $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be a closed embedding of differentiable stacks with complementary open embedding $j\colon \mathcal{U} \to \mathcal{X}$. Then the following properties are satisfied:*

(a1) The unit $\mathrm{id} \implies j^* j_\sharp$ *is an equivalence;*

(a2) The counit $j^* j_* \implies \mathrm{id}$ *is an equivalence;*

(a3) The functor $j_\sharp\colon C(\mathcal{U}) \to C(\mathcal{X})$ *is fully faithful;*

(a4) The functor $j_*\colon C(\mathcal{U}) \to C(\mathcal{X})$ *is fully faithful;*

(b1) The composite functor $i^* j_\sharp\colon C(\mathcal{U}) \to C(\mathcal{Z})$ *is zero;*

(b2) The composite functor $j^* i_*\colon C(\mathcal{Z}) \to C(\mathcal{U})$ *is zero;*

(c) The composite map $j_\sharp j^* \xRightarrow{\text{counit}} \mathrm{id} \xRightarrow{\text{unit}} i_* i^*$ *admits a preferred null-homotopy.*

Proof. Note that (a1)-(a4) are all equivalent, and also (b1) and (b2) are equivalent. Properties (a1) and (b1) follow from smooth base change, applied to the following two pullback squares:

$$\begin{tikzcd} \mathcal{U} \arrow[r, equal] \arrow[d, equal] \arrow[dr, phantom, "\lrcorner"] & \mathcal{U} \arrow[d, hook, "j"] \\ \mathcal{U} \arrow[r, hook, "j"] & \mathcal{X} \end{tikzcd} \qquad \begin{tikzcd} \emptyset \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{U} \arrow[d, hook, "j"] \\ \mathcal{Z} \arrow[r, hook, "i"] & \mathcal{X}. \end{tikzcd}$$

For (c), note that by naturality the counit $j_\sharp j^* \to \mathrm{id}$ and the unit $\mathrm{id} \to i_* i^*$ give rise to a commutative square

$$\begin{tikzcd} j_\sharp j^* \arrow[r] \arrow[d] & \mathrm{id} \arrow[d] \\ j_\sharp j^* i_* i^* \arrow[r] & i_* i^*. \end{tikzcd}$$

By (b2) the lower left corner is the zero functor, hence this square produces the required null-homotopy. □

Definition 5.1.2 (Localization axiom, cf. [CD19, Section 2.3])**.** Let C be a pointed pullback formalism on SepStk. Let $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be a closed embedding of differentiable stacks with complementary open embedding $j\colon \mathcal{U} \to \mathcal{X}$. We say that C *satisfies the localization*[1] *axiom for i,* (Loc$_i$)*,* if the following two conditions are satisfied:

(1) The functor $i_*\colon C(\mathcal{Z}) \to C(\mathcal{X})$ is conservative;

(2) The sequence

$$j_\sharp j^* \xRightarrow{\text{counit}} \mathrm{id} \xRightarrow{\text{unit}} i_* i^*,$$

equipped with the null-homotopy from (c) above, is a cofiber sequence of natural transformations $C(\mathcal{X}) \to C(\mathcal{X})$.

We say that C *satisfies the localization axiom* if C satisfies (Loc$_i$) for every closed embedding $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ of separated differentiable stacks.

The localization property implies that the functor i_* is well-behaved: it is fully faithful, satisfies base change, satisfies the projection formula and commutes with the pushforward functors $f_\sharp$ for representable submersions f:

[1]This is sometimes referred to as the *gluing axiom*.

Proposition 5.1.3 (Properties closed pushforwards)**.** *Let C be a pointed pullback formalism on* SepStk *which satisfies property* (Loc_i) *for a closed embedding $i\colon \mathcal{Z} \to \mathcal{X}$. Then the following conditions are satisfied:*

(1) (Fully faithfulness) The functor $i_*\colon C(\mathcal{Z}) \to C(\mathcal{X})$ is fully faithful.

(2) (Closed projection formula) For objects $A \in C(\mathcal{Z})$ and $B \in C(\mathcal{X})$, the exchange map

$$\mathrm{PF}_*\colon i_*(A) \otimes B \to i_*(A \otimes i^* B)$$

is an equivalence in $C(\mathcal{X})$.

(3) (Closed base change) For every pullback square of differentiable stacks

$$\begin{tikzcd} \mathcal{Z}' \arrow[r, hook, "i'"] \arrow[d, "f'"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{X}' \arrow[d, "f"] \\ \mathcal{Z} \arrow[r, hook, "i"] & \mathcal{X}, \end{tikzcd}$$

the exchange map

$$\mathrm{BC}_*\colon f^* i_* \Rightarrow i'_* f'^*\colon C(\mathcal{Z}) \to C(\mathcal{X}')$$

is an equivalence.

(4) (Smooth-closed base change) Consider a pullback square of differentiable stacks

$$\begin{tikzcd} \mathcal{Z}' \arrow[r, hook, "i'"] \arrow[d, "f'"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{Y} \arrow[d, "f"] \\ \mathcal{Z} \arrow[r, hook, "i"] & \mathcal{X}, \end{tikzcd}$$

where f is a representable submersion. Assume that C satisfies property $(\mathrm{Loc}_{i'})$*. Then the exchange map*

$$\mathrm{BC}_{\sharp,*}\colon f_\sharp i'_* \Rightarrow i_* f'_\sharp\colon C(\mathcal{Z}') \to C(\mathcal{X})$$

is an equivalence.

Proof. To see that i_* is fully faithful, we need to show that the counit map $i^* i_* Z \to Z$ is an equivalence for all $Z \in C(\mathcal{Z})$. Since i_* is conservative, it suffices to show that $i_* i^* i_* Z \to i_* Z$ is an equivalence, and by the triangle identities it will suffice to show that the unit map $i_* Z \to i_* i^* i_* Z$ is an equivalence. By (Loc_i), there is a cofiber sequence

$$j_\sharp j^* i_* Z \to i_* Z \to i_* i^* i_* Z$$

in $\mathcal{C}(\mathcal{X})$. But by (b2), the object $j_\sharp j^* i_* Z$ is zero, and thus the map $i_* Z \to i_* i^* i_* Z$ is an equivalence. This proves part (1).

We now show part (2). As i_* is fully faithful, the unit map $A \to i^* i_* A$ is an equivalence and thus it suffices to prove the claim when $A = i^* A'$ for some object $A' \in \mathcal{C}(\mathcal{X})$. In this case, consider the following diagram:

$$\begin{tikzcd}
j_\sharp j^*(A') \otimes B \arrow[r] & A' \otimes B \arrow[r] \arrow[dd, equal] & i_* i^*(A') \otimes B \arrow[d, "\mathrm{PF}_*"] \\
j_\sharp(j^*(A') \otimes j^* B) \arrow[u, "\simeq"', "\mathrm{PF}_\sharp"] \arrow[d, "\simeq"'] & & i_*(i^* A \otimes i^* B) \arrow[d, "\simeq"] \\
j_\sharp j^*(A' \otimes B) \arrow[r] & A' \otimes B \arrow[r] & i_* i^*(A' \otimes B).
\end{tikzcd}$$

The diagram is commutative by Lemma F.18. By assumption, the top and bottom sequences are cofiber sequences. The left vertical composite is an equivalence by the smooth projection formula, and thus also the right vertical composite is an equivalence. It follows that the map $i_* i^*(A') \otimes B \to i_*(i^* A' \otimes i^* B)$ is an equivalence, proving part (2).

The proofs of parts (3) and (4) are similar to that of part (2), and hence we will be more brief. For (3), it suffices by fully faithfulness of i_* to show that the composite $f^* i_* i^* \Rightarrow i'_* f'^* i^* \simeq i'_* i'^* f^*$ is an equivalence. This follows from the following diagram of cofiber sequences:

$$\begin{tikzcd}
f^* j_\sharp j^* \arrow[r] \arrow[d, "\simeq"'] & f^* \arrow[r] \arrow[d, equal] & f^* i_* i^* \arrow[d] \\
j'_\sharp j'^* f^* \arrow[r] & f^* \arrow[r] & i'_* i'^* f^*,
\end{tikzcd}$$

where the left equivalence is obtained from smooth base change. The diagram commutes by Lemma F.5. For (4), it suffices to show that the composite $f_\sharp i'_* i'^* \Rightarrow i_* f'_\sharp i'^* \simeq i_* i^* f_\sharp$ is an equivalence. This follows from the following diagram of cofiber sequences:

$$\begin{tikzcd}
f_\sharp j'_\sharp j'^* \arrow[r] \arrow[d, "\simeq"'] & f_\sharp \arrow[r] \arrow[d, equal] & f_\sharp i'_* i'^* \arrow[d] \\
j_\sharp j^* f_\sharp \arrow[r] & f_\sharp \arrow[r] & i_* i^* f_\sharp,
\end{tikzcd}$$

where the left equivalence is obtained from smooth base change. Again the diagram commutes by Lemma F.5. ◻

The proof strategy from the previous lemma also applies to the following very general statement:

Lemma 5.1.4. *Let $\mathcal{C}$ and $\mathcal{D}$ be pointed pullback formalisms on* SepStk *which satisfy property* (Loc$_i$) *for a closed embedding $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$. Let $F\colon \mathcal{C} \to \mathcal{D}$ be a morphism of pullback formalisms. Then the Beck-Chevalley transformation*

$$F_{\mathcal{X}} \circ i_* \Rightarrow i_* \circ F_{\mathcal{Z}} \quad \in \quad \mathrm{Fun}(\mathcal{C}(\mathcal{Z}), \mathcal{D}(\mathcal{X}))$$

is an equivalence.

Proof. As $i_*\colon \mathcal{C}(\mathcal{Z}) \hookrightarrow \mathcal{C}(\mathcal{X})$ is fully faithful, it suffices to show that the map $F_{\mathcal{X}} i_* i^* \to i_* F_{\mathcal{Z}} i^*$ is an equivalence. As F is a morphism of pullback formalisms, the latter functor is equivalent to $i_* i^* F_{\mathcal{X}}$. We may now contemplate the following two cofiber sequences:

$$\begin{tikzcd}
F_{\mathcal{X}} j_\sharp j^* \arrow[r] \arrow[d, "\simeq"'] & F_{\mathcal{X}} \arrow[r] \arrow[d, equal] & F_{\mathcal{X}} i_* i^* \arrow[d] \\
j_\sharp j^* F_{\mathcal{X}} \arrow[r] & F_{\mathcal{X}} \arrow[r] & i_* i^* F_{\mathcal{X}}.
\end{tikzcd}$$

The vertical equivalence on the left is given by the equivalences $F_{\mathcal{X}} j_\sharp \simeq j_\sharp F_{\mathcal{Z}}$ and $F_{\mathcal{Z}} j^* \simeq j^* F_{\mathcal{Z}}$. As the diagram commutes by Lemma F.5, it follows that also the right vertical map on cofibers is an equivalence, finishing the proof. □

Corollary 5.1.5 (Closed exceptional pullback)**.** *For a closed embedding $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$, the functor $i_*\colon \mathcal{C}(\mathcal{Z}) \hookrightarrow \mathcal{C}(\mathcal{X})$ preserves colimits, and thus admits a right adjoint*

$$i^!\colon \mathcal{C}(\mathcal{X}) \to \mathcal{C}(\mathcal{Z}),$$

called the exceptional pullback functor.

Proof. Given an indexing ∞-category, we have to show that i_* preserves I-indexed colimits. This is a special case of Lemma 5.1.4 applied to the morphism of pullback formalisms $\mathrm{colim}_I\colon \mathcal{C}^I \to \mathcal{C}$. □

In the stable setting, we obtain the following alternative characterization of the localization axiom:

Lemma 5.1.6. *Let $\mathcal{C}$ be a stable pullback formalism on* SepStk *and let $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be a closed embedding with open complement $j\colon \mathcal{U} \to \mathcal{X}$. Then $\mathcal{C}$ satisfies* (Loc$_i$) *if and only if the following two conditions hold:*

(d) The functor $i_*\colon \mathcal{C}(\mathcal{Z}) \to \mathcal{C}(\mathcal{X})$ is fully faithful;

(e) The functors $j^*\colon \mathcal{C}(\mathcal{X}) \to \mathcal{C}(\mathcal{U})$ and $i^*\colon \mathcal{C}(\mathcal{X}) \to \mathcal{C}(\mathcal{Z})$ are jointly conservative.

Proof. First assume that C satisfies the localization axiom (Loc_i) for i. We have seen in Proposition 5.1.3 that i_* is fully faithful, proving (d). To see that the functors j^* and i^* are jointly conservative, let $f\colon X \to Y$ be a morphism in $C(\mathcal{X})$ such that i^*f and j^*f are equivalences. It follows that $j_\sharp j^* f$ and i_*i^*f are equivalences, and hence so is f by the bifiber sequence $j_\sharp j^* f \to f \to i_*i^*f$. This proves (e).

Conversely, assume that conditions (d) and (e) are satisfied. We will show that C satisfies (Loc_i). As every fully faithful functor is conservative, it remains to show that for every $X \in C(\mathcal{X})$, the sequence

$$j_\sharp j^* X \to X \to i_* i^* X$$

is a cofiber sequence in $C(\mathcal{X})$. By (e), we may test this after pulling back along i and j. Using (a1), (b1), (b2) and (d), the pulled back sequences are equivalent to

$$j^*X \to j^*X \to 0 \qquad \text{and} \qquad 0 \to i^*X \to i^*X,$$

which are clearly cofiber sequences. □

Remark 5.1.7 (Stable recollement)**.** If C is a stable pullback formalism on SepStk which satisfies the localization axiom for $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$, then Lemma 5.1.6 shows that the full subcategories $i_*\colon C(\mathcal{Z}) \hookrightarrow C(\mathcal{X})$ and $j_*\colon C(\mathcal{U}) \hookrightarrow C(\mathcal{Z})$ of $C(\mathcal{X})$ are part of a *stable recollement*, in the sense of [Cal+20, Definition A.2.9]:

$$\begin{tikzcd} C(\mathcal{Z}) \arrow[r, "i_*"] & C(\mathcal{X}) \arrow[l, bend right, "i^*"'] \arrow[l, bend left, "i^!"] \arrow[r, "j^*"] & C(\mathcal{U}). \arrow[l, bend right, "j_\sharp"'] \arrow[l, bend left, "j_*"] \end{tikzcd}$$

5.2 The localization theorem for genuine sheaves

In the previous section, we introduced the localization axiom for a pointed pullback formalism C on SepStk. The goal of this section is to verify this axiom in the case $C = \mathrm{H}_\bullet$ of pointed genuine sheaves.

5.2.1 Exactness of the closed pushforward functor

Let $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be a closed embedding of differentiable stacks. In this subsection, we will show that the pushforward functor $i_*\colon \mathrm{H}(\mathcal{Z}) \to \mathrm{H}(\mathcal{X})$ preserves all *weakly contractible* colimits, meaning those colimits indexed by weakly contractible[2] ∞-categories. This exactness property of i_* follows from a general criterion on a morphism of sites. We follow Khan [Kha19, Section 3.1].

Definition 5.2.1 ([Kha19, Definition 3.1.5])**.** Let $\mathbf{C}$ and $\mathbf{D}$ be essentially small ∞-categories equipped with Grothendieck topologies $\tau_{\mathbf{C}}$ and $\tau_{\mathbf{D}}$. Assume that $\mathbf{D}$ admits an initial object $\emptyset_{\mathbf{D}}$. A functor $u\colon \mathbf{C} \to \mathbf{D}$ is called *topologically quasi-cocontinuous* if the following condition holds:

> For every $c \in \mathbf{C}$ and every $\tau_{\mathbf{D}}$-covering sieve $R' \hookrightarrow y(u(c))$ in $\mathbf{D}$, the sieve $R \hookrightarrow y(c)$, generated by morphisms $c' \to c$ such that either $u(c')$ is initial or $y(u(c')) \to y(u(c))$ factors through $R' \hookrightarrow y(u(c))$, is a covering in $\mathbf{C}$.

Lemma 5.2.2 ([Kha19, Lemma 3.1.6])**.** *Consider a topologically quasi-cocontinuous functor $u\colon \mathbf{C} \to \mathbf{D}$, as in Definition 5.2.1. Assume further that the following two conditions on $\mathbf{D}$ are satisfied:*

(1) The initial object $\emptyset_{\mathbf{D}}$ is strict: given an object $d \in \mathbf{D}$, any morphism $d \to \emptyset_{\mathbf{D}}$ is invertible.

(2) For any object $d \in \mathbf{D}$, the sieve $\emptyset_{\mathrm{PSh}(\mathbf{D})} \hookrightarrow y(d)$ is a covering in $\mathbf{D}$ if and only if d is initial (where $\emptyset_{\mathrm{PSh}(\mathbf{D})}$ denotes the initial object of $\mathrm{PSh}(\mathbf{D})$).

Then the functor $\mathrm{Shv}_{\tau_{\mathbf{D}}}(\mathbf{D}) \to \mathrm{Shv}_{\tau_{\mathbf{C}}}(\mathbf{C})$ given by the assignment $\mathcal{F} \mapsto L_{\tau_{\mathbf{C}}}(u^*(\mathcal{F}))$ commutes with contractible colimits. $\square$

We apply this to the situation of differentiable stacks.

Proposition 5.2.3. *Let $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be a closed embedding of differentiable stacks. Then the functor $i_*\colon \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{Z}}) \to \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$ commutes with contractible colimits.*

Proof. We apply Lemma 5.2.2 to the morphism of sites $i^*\colon \mathrm{Sub}_{/\mathcal{X}} \to \mathrm{Sub}_{/\mathcal{Z}}$. It is clear that the initial object $\emptyset \to \mathcal{Z}$ of $\mathrm{Sub}_{/\mathcal{Z}}$ is strict and that the only object of $\mathrm{Sub}_{/\mathcal{Z}}$ covered by the empty sieve is the empty stack. It thus remains to show that i^* is topologically quasi-cocontinuous.

Let $f\colon \mathcal{Y} \to \mathcal{X}$ be a representable submersion of differentiable stacks and consider an open cover $\{\mathcal{V}_i \hookrightarrow i^*\mathcal{Y}\}_{i\in I}$ of the pullback $i^*\mathcal{Y} \to \mathcal{Z}$. We have to show that the sieve

$$T = \{\mathcal{W} \hookrightarrow \mathcal{Y} \mid i^*\mathcal{W} = \emptyset \text{ or } i^*\mathcal{W} \hookrightarrow \mathcal{V}_i \text{ for some } i \in I\}$$

[2] Recall that an ∞-category I is called *weakly contractible* if the ∞-groupoid $|I|$ formed by inverting all morphisms in I is contractible

covers $\mathcal{Y}$. For every $i \in I$, let $\mathcal{W}_i \hookrightarrow \mathcal{Y}$ be an open substack such that $\mathcal{V}_i = i^*\mathcal{W}_i \subseteq i^*\mathcal{Y}$; this exists by Corollary 3.2.3. Further, let $j\colon \mathcal{U} \hookrightarrow \mathcal{X}$ denote the open complement of $\mathcal{Z}$. It is clear that the inclusion $\mathcal{Y}_{\mathcal{U}} \hookrightarrow \mathcal{Y}$ and the inclusions $\mathcal{W}_i \hookrightarrow \mathcal{Y}$ are all in T and that they cover $\mathcal{Y}$, proving the claim. $\square$

Corollary 5.2.4. *Let* $\mathrm{PSh}_\emptyset(\mathrm{Sub}_{/\mathcal{Z}}) \subseteq \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{Z}})$ *and* $\mathrm{PSh}_\emptyset(\mathrm{Sub}_{/\mathcal{X}}) \subseteq \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}})$ *denote the full subcategories spanned by those presheaves sending $\emptyset$ to $*$. Then the following diagram commutes:*

$$\begin{tikzcd}
\mathrm{PSh}_\emptyset(\mathrm{Sub}_{/\mathcal{Z}}) \arrow[r, "i_*"] \arrow[d, "L_{\mathrm{open}}"'] & \mathrm{PSh}_\emptyset(\mathrm{Sub}_{/\mathcal{X}}) \arrow[d, "L_{\mathrm{open}}"] \\
\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{Z}}) \arrow[r, "i_*"] & \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}}).
\end{tikzcd}$$

Proof. By Proposition 5.2.3 both composites preserve contractible colimits. Since they agree on $\mathrm{Sub}_{/\mathcal{Z}}$, the claim is an immediate consequence of the fact that $\mathrm{PSh}_\emptyset(\mathrm{Sub}_{/\mathcal{Z}})$ is freely generated under weakly contractible colimits by $\mathrm{Sub}_{/\mathcal{Z}}$: a presheaf on $\mathrm{Sub}_{/\mathcal{Z}}$ is in $\mathrm{PSh}_\emptyset(\mathrm{Sub}_{/\mathcal{Z}})$ if and only if its category of elements is weakly contractible. $\square$

Lemma 5.2.5. *Let $f\colon \mathcal{Z} \to \mathcal{X}$ be a morphism of differentiable stacks. Then the functor* $f_*\colon \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{Z}}) \to \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}})$ *preserves $L_{\mathbb{R}}$-local morphisms.*

Proof. As f_* preserves colimits, it suffices to show that it sends morphisms of the form $\mathrm{pr}\colon \mathcal{Y} \times \mathbb{R} \to \mathcal{Y}$ to $L_{\mathbb{R}}$-local morphisms, where $\mathcal{Y} \in \mathrm{Sub}_{/\mathcal{Z}}$. The projection map admits a section given by the map $i_0\colon \mathcal{Y} \to \mathcal{Y} \times \mathbb{R}$, and the composite $i_0 \circ \mathrm{pr}\colon \mathcal{Y} \times \mathbb{R} \to \mathcal{Y} \times \mathbb{R}$ is homotopic to the identity map. It will thus suffice to show that f_* sends homotopic maps to homotopic maps. But this is clear: if $H\colon \mathcal{Y}' \times \mathbb{R} \to \mathcal{X}'$ is a homotopy between two maps H_0 and H_1, then a homotopy between $f_*(H_0)$ and $f_*(H_1)$ is given by the composite $f_*(\mathcal{Y}') \times \mathbb{R} \to f_*(\mathcal{Y}' \times \mathbb{R}) \xrightarrow{f_*(H)} f_*(\mathcal{X}')$, where the first map is the projection formula map. $\square$

Corollary 5.2.6. *Let $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be a closed embedding of differentiable stacks. Then the functor* $i_*\colon \mathrm{H}(\mathcal{Z}) \to \mathrm{H}(\mathcal{X})$ *preserves weakly contractible colimits.*

Proof. Colimits in $\mathrm{H}(\mathcal{X})$ may be computed by first forming them in $\mathrm{PSh}_\emptyset(\mathrm{Sub}_{/\mathcal{X}})$ and then applying the homotopy localization functor $L_{\mathrm{htp}}\colon \mathrm{PSh}_\emptyset(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{H}(\mathcal{X})$. Since the functor $i_*\colon \mathrm{PSh}_\emptyset(\mathrm{Sub}_{/\mathcal{Z}}) \to \mathrm{PSh}_\emptyset(\mathrm{Sub}_{/\mathcal{X}})$ preserves weakly contractible colimits, it will thus suffice to show that the followimg diagram commutes:

$$\begin{tikzcd}
\mathrm{PSh}_\emptyset(\mathrm{Sub}_{/\mathcal{Z}}) \arrow[r, "i_*"] \arrow[d, "L_{\mathrm{open}}"'] & \mathrm{PSh}_\emptyset(\mathrm{Sub}_{/\mathcal{X}}) \arrow[d, "L_{\mathrm{open}}"] \\
\mathrm{H}(\mathcal{Z}) \arrow[r, "i_*"] & \mathrm{H}(\mathcal{X}).
\end{tikzcd}$$

This follows from Corollary 5.2.4 and Lemma 5.2.5, using the fact from Lemma 4.2.13 that L_{htp} is an ω_1-indexed colimit of iterations of L_{open} and $L_{\mathbb{R}}$. □

Corollary 5.2.7 (Exceptional pushforward)**.** *Let $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be a closed embedding of differentiable stacks. Then the functor $i_*\colon \mathrm{H}_\bullet(\mathcal{Z}) \to \mathrm{H}_\bullet(\mathcal{X})$ preserves all small colimits, and thus admits a right adjoint $i^!\colon \mathrm{H}_\bullet(\mathcal{X}) \to \mathrm{H}_\bullet(\mathcal{Z})$.*

Proof. Since $\mathrm{H}_\bullet(\mathcal{Z})$ and $\mathrm{H}_\bullet(\mathcal{X})$ are pointed, it is clear that i_* preserves the initial object. By Corollary 5.2.6 it preserves all contractible colimits. The claim follows: given any small ∞-category I and a diagram $F\colon I \to \mathrm{H}(\mathcal{Z})_\bullet$, we may lift F to a functor $\widetilde{F}\colon I^{\triangleleft} \to \mathrm{H}(\mathcal{Z})_\bullet$ which sends the cone point to the zero object. Note that $\widetilde{F}$ has the same colimit as F, but has a weakly contractible indexing diagram. As i_* preserves the initial object, we have $i_* \circ F^{\triangleleft} = (i_* \circ F)^{\triangleleft}$, and it follows that i_* preserves the colimit of F. □

5.2.2 Presheaves of $\mathcal{Z}$-trivialized morphisms

To streamline the proof of the localization theorem for $C = \mathrm{H}_\bullet$, we introduce some auxiliary notation regarding certain presheaves of $\mathcal{Z}$-trivialized morphisms. Our definitions, statements and proofs are direct analogues of those of Khan [Kha19, Section 4.1] and go back to Morel and Voevodsky [MV99].

In the following, we denote for every differentiable stack $\mathcal{X}$ the Yoneda embedding of $\mathrm{Sub}_{/\mathcal{X}}$ by $h_{\mathcal{X}}\colon \mathrm{Sub}_{/\mathcal{X}} \hookrightarrow \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}})$.

Definition 5.2.8. Let $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be a closed embedding with open complement $j\colon \mathcal{U} \hookrightarrow \mathcal{X}$. For any $\mathcal{Y} \in \mathrm{Sub}_{/\mathcal{X}}$, we define the presheaf $h_{\mathcal{X}}^{\mathcal{Z}}(\mathcal{Y}) \in \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}})$ as the following pushout:

$$\begin{tikzcd}
h_{\mathcal{X}}(\mathcal{Y}_{\mathcal{U}}) \arrow[r] \arrow[d] & h_{\mathcal{X}}(\mathcal{Y}) \arrow[d] \\
h_{\mathcal{X}}(\mathcal{U}) \arrow[r] \arrow[ur, phantom, "\ulcorner"] & h_{\mathcal{X}}^{\mathcal{Z}}(\mathcal{Y}).
\end{tikzcd}$$

Note that for $\mathcal{W} \in \mathrm{Sub}_{/\mathcal{X}}$, the anima $\Gamma(\mathcal{W}, h_{\mathcal{X}}(\mathcal{U}))$ is either empty or contractible depending on whether $\mathcal{W}_{\mathcal{Z}}$ is empty or not. It follows that the anima of sections $\Gamma(\mathcal{W}, h_{\mathcal{X}}^{\mathcal{Z}}(\mathcal{Y}))$ is contractible when $\mathcal{W}_{\mathcal{Z}}$ is empty, and otherwise is given by the set $\mathrm{Hom}_{/\mathcal{X}}(\mathcal{W}, \mathcal{Y})$.

Since i^* commutes with colimits, we have $i^*(h^{\mathcal{Z}}_{\mathcal{X}}(\mathcal{Y})) \simeq h_{\mathcal{Z}}(\mathcal{Y}_{\mathcal{Z}})$.

Construction 5.2.9 (Presheaf of $\mathcal{Z}$-trivialized morphisms)**.** Let $\mathcal{Y} \in \mathrm{Sub}_{/\mathcal{X}}$ and let $t\colon \mathcal{Z} \hookrightarrow \mathcal{Y}$ be a morphism over $\mathcal{X}$, i.e., a partially defined section of $\mathcal{Y} \to \mathcal{X}$. We define the presheaf $h^{\mathcal{Z}}_{\mathcal{X}}(\mathcal{Y},t) \in \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}})$, referred to as the *presheaf of $\mathcal{Z}$-trivialized morphisms*, as the following pullback:

$$\begin{tikzcd} h^{\mathcal{Z}}_{\mathcal{X}}(\mathcal{Y},t) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & h^{\mathcal{Z}}_{\mathcal{X}}(\mathcal{Y}) \arrow[d, "u_i^*"] \\ \mathrm{pt}_{\mathcal{X}} \arrow[r, "\tau"] & i_*i^*h^{\mathcal{Z}}_{\mathcal{X}}(\mathcal{Y}). \end{tikzcd}$$

Here the right vertical map is the unit map and the bottom horizontal map is given by

$$\mathrm{pt}_{\mathcal{X}} \simeq i_*h_{\mathcal{Z}}(\mathcal{Z}) \xrightarrow{t} i_*h_{\mathcal{Z}}(\mathcal{Y}_{\mathcal{Z}}) \simeq i_*i^*h^{\mathcal{Z}}_{\mathcal{X}}(\mathcal{Y}).$$

In other words, the anima $\Gamma(\mathcal{W}, h^{\mathcal{Z}}_{\mathcal{X}}(\mathcal{Y},t))$ is contractible when $\mathcal{W}_{\mathcal{Z}}$ is empty, and otherwise is given by the fiber of the restriction map

$$\mathrm{Hom}_{/\mathcal{X}}(\mathcal{W},\mathcal{Y}) \to \mathrm{Hom}_{/\mathcal{Z}}(\mathcal{W}_{\mathcal{Z}},\mathcal{Y}_{\mathcal{Z}})$$

at the point defined by the composite $\mathcal{W}_{\mathcal{Z}} \to \mathcal{Z} \xrightarrow{t} \mathcal{X}_{\mathcal{Z}}$.

Remark 5.2.10. For a representable submersion $p\colon \mathcal{X}' \to \mathcal{X}$, there is a natural equivalence

$$p^*h^{\mathcal{Z}}_{\mathcal{X}}(\mathcal{Y},t) \simeq h^{p^*\mathcal{Z}}_{\mathcal{X}'}(p^*\mathcal{Y}, p^*t),$$

where $p^*t\colon p^*\mathcal{Z} \to p^*\mathcal{Y}$ is the base change of t along p. This is immediate from the fact that p^* preserves limits and colimits.

Our goal in this subsection will be to show that the presheaf $h^{\mathcal{Z}}_{\mathcal{X}}(\mathcal{Y},t)$ is weakly contractible, see Proposition 5.2.13 below.

Lemma 5.2.11. *Let $\pi\colon \mathcal{E} \to \mathcal{X}$ be a vector bundle with zero section $s_0\colon \mathcal{X} \hookrightarrow \mathcal{E}$. Let $t\colon \mathcal{Z} \to \mathcal{E}$ denote the composite $t = s_0 \circ i$. Then the presheaf $h^{\mathcal{Z}}_{\mathcal{X}}(\mathcal{E},t)$ is weakly contractible.*

Proof. The map π induces a map $\pi\colon h^{\mathcal{Z}}_{\mathcal{X}}(\mathcal{E},t) \to h^{\mathcal{Z}}_{\mathcal{X}}(\mathcal{X},i)$, and conversely s_0 induces a map $s_0\colon h^{\mathcal{Z}}_{\mathcal{X}}(\mathcal{X},i) \to h^{\mathcal{Z}}_{\mathcal{X}}(\mathcal{E},t)$. We have $\pi \circ s_0 = \mathrm{id}$, and $s_0 \circ \pi \simeq \mathrm{id}$ via the homotopy

$$\mathbb{R} \times h^{\mathcal{Z}}_{\mathcal{X}}(\mathcal{E},t) \to h^{\mathcal{Z}}_{\mathcal{X}}(\mathcal{E},t), \quad (a,f) \mapsto af.$$

We thus have a weak equivalence $h^{\mathcal{Z}}_{\mathcal{X}}(\mathcal{E},t) \simeq h^{\mathcal{Z}}_{\mathcal{X}}(\mathcal{X},i)$. The claim now follows since the presheaf $h^{\mathcal{Z}}_{\mathcal{X}}(\mathcal{X},i)$ is the terminal presheaf. $\square$

Lemma 5.2.12. *Let $j' \colon \mathcal{V} \hookrightarrow \mathcal{Y}$ be an open embedding in $\mathrm{Sub}_{/\mathcal{X}}$. Let $t_{\mathcal{V}} \colon \mathcal{Z} \to \mathcal{V}$ be a partial section of the map $\mathcal{V} \to \mathcal{X}$ and let $t_{\mathcal{Y}} = j' \circ t_{\mathcal{V}}$. Then the induced map*

$$j' \colon h_{\mathcal{X}}^{\mathcal{Z}}(\mathcal{V}, t_{\mathcal{V}}) \to h_{\mathcal{X}}^{\mathcal{Z}}(\mathcal{Y}, t_{\mathcal{Y}})$$

in $\mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}})$ is inverted by the sheafification functor $L_{\mathrm{open}} \colon \mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}}) \to \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$.

Proof. We will show the map $L_{\mathrm{open}}(j')$ is both a monomorphism as well as an effective epimorphism in $\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$.

Step 1: We first show that $L_{\mathrm{open}}(j')$ is a monomorphism. Since the sheafification preserves pullbacks, we may show that the map $j' \colon h_{\mathcal{X}}^{\mathcal{Z}}(\mathcal{V}, t_{\mathcal{V}}) \to h_{\mathcal{X}}^{\mathcal{Z}}(\mathcal{Y}, t_{\mathcal{Y}})$ is a monomorphism. Given $\mathcal{W} \in \mathrm{Sub}_{/\mathcal{X}}$, we will show that the map

$$\Gamma(\mathcal{W}, h_{\mathcal{X}}^{\mathcal{Z}}(\mathcal{V}, t_{\mathcal{V}})) \to \Gamma(\mathcal{W}, h_{\mathcal{X}}^{\mathcal{Z}}(\mathcal{Y}, t_{\mathcal{Y}}))$$

of animae is a monomorphism. This is clear if $\mathcal{W}_{\mathcal{Z}}$ is empty, so assume it is not. Then this map is the map induced on fibers in the following diagram:

$$\begin{tikzcd}
\Gamma(\mathcal{W}, h_{\mathcal{X}}^{\mathcal{Z}}(\mathcal{V}, t_{\mathcal{V}})) \arrow[r] \arrow[d, dashed] & \mathrm{Map}_{/\mathcal{X}}(\mathcal{W}, \mathcal{V}) \arrow[r] \arrow[d] & \mathrm{Map}_{/\mathcal{Z}}(\mathcal{W}_{\mathcal{Z}}, \mathcal{V}_{\mathcal{Z}}) \arrow[d] \\
\Gamma(\mathcal{W}, h_{\mathcal{X}}^{\mathcal{Z}}(\mathcal{Y}, t_{\mathcal{Y}})) \arrow[r] & \mathrm{Map}_{/\mathcal{X}}(\mathcal{W}, \mathcal{Y}) \arrow[r] & \mathrm{Map}_{/\mathcal{Z}}(\mathcal{W}_{\mathcal{Z}}, \mathcal{Y}_{\mathcal{Z}}).
\end{tikzcd}$$

Since the middle and right vertical morphisms are monomorphisms of animae, so is the induced map on fibers.

Step 2: We show that the map $L_{\mathrm{open}}(j')$ is an effective epimorphism in $\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$. It suffices to show that for every $\mathcal{W} \in \mathrm{Sub}_{/\mathcal{X}}$, any $\mathcal{W}$-section of $h_{\mathcal{X}}^{\mathcal{Z}}(\mathcal{Y}, t)$ can locally be lifted along j'. We may again assume that $\mathcal{W}_{\mathcal{Z}}$ is non-empty, since the claim is clear otherwise. In that case, let f be a $\mathcal{W}$-section of $h_{\mathcal{X}}^{\mathcal{Z}}(\mathcal{Y}, t)$, i.e., a $\mathcal{Z}$-trivialized morphism $f \colon \mathcal{W} \to \mathcal{Y}$. We claim that the substacks

$$\mathcal{W}_{\mathcal{U}} \hookrightarrow \mathcal{W}, \qquad \mathcal{W}_{\mathcal{V}} \hookrightarrow \mathcal{W}$$

cover $\mathcal{W}$. Indeed, $\mathcal{W}_{\mathcal{U}}$ is the open complement of the closed substack $\mathcal{W}_{\mathcal{Z}}$, and the map $f_{\mathcal{Z}} \colon \mathcal{W}_{\mathcal{Z}} \to \mathcal{Y}_{\mathcal{Z}}$ factors through $\mathcal{Z}$ by assumption, and thus in particular through $\mathcal{V}_{\mathcal{Z}}$.

It will thus suffice to show that both $f_{\mathcal{U}}$ as well as $f_{\mathcal{V}}$ admit lifts along j'. For $f_{\mathcal{V}}$ this is true by construction, as $\mathcal{Y}_{\mathcal{V}} = \mathcal{V}$. For $f_{\mathcal{U}}$ this is automatic as $\mathcal{U}_{\mathcal{Z}} = \emptyset$. □

Proposition 5.2.13. *Let $i \colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be a closed embedding of separated differentiable stacks with open complement $j \colon \mathcal{U} \hookrightarrow \mathcal{X}$. Let $p \colon \mathcal{Y} \to \mathcal{X}$ be a representable submersion, and let $t \colon \mathcal{Z} \hookrightarrow \mathcal{Y}$ be a section of p over $\mathcal{Z}$. Then the presheaf $h_{\mathcal{X}}^{\mathcal{Z}}(\mathcal{Y}, t)$ is weakly contractible, in the sense that its image in $\mathrm{H}(\mathcal{X})$ is the terminal object.*

Proof. The question is local in $\mathcal{X}$, in light of Remark 5.2.10. Around points in $\mathcal{U} = \mathcal{X} \setminus \mathcal{Z}$ the statement is trivially true, so it suffices to prove the claim locally around points $z \in \mathcal{Z}$. By Proposition 3.7.5, we can therefore reduce to the case where there is a commutative diagram

$$\begin{tikzcd} & \mathcal{E} \arrow[r, hook, "j"] \arrow[d, shift left, "\pi"] & \mathcal{Y} \arrow[d, "p"] \\ \mathcal{Z} \arrow[ur, hook, "t"] \arrow[r, hook, "i"] & \mathcal{X} \arrow[u, shift left, "s_0"] \arrow[r, equal] & \mathcal{X}, \end{tikzcd}$$

where j is an open embedding and the map $\pi\colon \mathcal{E} \to \mathcal{X}$ is a vector bundle with zero section $s_0\colon \mathcal{X} \to \mathcal{E}$. By Lemma 5.2.12, the map j induces a weak equivalence

$$j\colon h_{\mathcal{X}}^{\mathcal{Z}}(\mathcal{E}, t) \xrightarrow{\sim} h_{\mathcal{X}}^{\mathcal{Z}}(\mathcal{Y}, t).$$

But by Lemma 5.2.11, the presheaf $h_{\mathcal{X}}^{\mathcal{Z}}(\mathcal{E}, t)$ is weakly contractible, finishing the proof. $\square$

5.2.3 Proof of the localization theorem

We are now ready for the proof of the localization theorem for genuine sheaves. Let $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be a closed embedding with open complement $j\colon \mathcal{U} \hookrightarrow \mathcal{X}$. For any sheaf $\mathcal{F} \in \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$, consider the functorial commutative square

$$\begin{tikzcd} j_\sharp j^*(\mathcal{F}) \arrow[r, "c_j^\sharp"] \arrow[d, "u_i^*"] & \mathcal{F} \arrow[d, "u_i^*"] \\ j_\sharp j^* i_* i^*(\mathcal{F}) \arrow[r, "c_j^\sharp"] & i_* i^*(\mathcal{F}), \end{tikzcd}$$

where $c_j^\sharp\colon j_\sharp j^* \to \mathrm{id}$ is the adjunction counit and $u_i^*\colon \mathrm{id} \to i_* i^*$ is the adjunction unit. By smooth base change, $j^* i_*$ is the constant functor with value the terminal object $\mathcal{U} \in \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{U}})$. Hence the lower left corner $j_\sharp j^* i_* i^*(\mathcal{F})$ of the diagram is simply $j_\sharp \mathcal{U} \in \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})$.

Theorem 5.2.14 (Localization theorem for genuine sheaves)**.** *Let $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be a closed embedding of separated differentiable stacks with open complement $j\colon \mathcal{U} \hookrightarrow \mathcal{X}$. Then for every $\mathcal{F} \in \mathrm{H}(\mathcal{X})$ the square*

$$\begin{tikzcd} j_\sharp j^*(\mathcal{F}) \arrow[r, "c_j^\sharp"] \arrow[d] & \mathcal{F} \arrow[d, "u_i^*"] \\ j_\sharp(\mathcal{U}) \arrow[r] & i_* i^*(\mathcal{F}) \end{tikzcd}$$

is cocartesian in $\mathrm{H}(\mathcal{X})$.

Proof. Given an object $\mathcal{F} \in \mathrm{H}(\mathcal{X})$, we have to show that the square is weakly cocartesian in $\mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}})$, in the sense of **??**. Observe that $\mathcal{F}$ can be written as a weakly contractible colimit of representables, indexed by the ∞-category

$$(\mathrm{Sub}_{/\mathcal{X}})_{/\mathcal{F}} = \mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})_{/\mathcal{F}} \times_{\mathrm{Shv}(\mathrm{Sub}_{/\mathcal{X}})} \mathrm{Sub}_{/\mathcal{X}},$$

which is weakly contractible as it has an initial object $\emptyset \to \mathcal{F}$. By Corollary 5.2.6, all four functors in the square commute with contractible colimits, and thus it suffices to prove the claim when $\mathcal{F} = L_{\mathrm{htp}}(\mathcal{Y})$ for some representable submersion $f\colon \mathcal{Y} \to \mathcal{X}$. In other words, we have to show that canonical map

$$h_{\mathcal{X}}^{\mathcal{Z}}(\mathcal{Y}) = h_{\mathcal{X}}(\mathcal{Y}) \sqcup_{h_{\mathcal{X}}(\mathcal{Y}_{\mathcal{U}})} h_{\mathcal{X}}(\mathcal{U}) \to i_* h_{\mathcal{Z}}(\mathcal{Y}_{\mathcal{Z}})$$

is a weak equivalence of presheaves on $\mathrm{Sub}_{/\mathcal{X}}$. By universality of colimits in $\mathrm{PSh}(\mathrm{Sub}_{/\mathcal{X}})$, it will suffice to prove that for any other $(p\colon \mathcal{W} \to \mathcal{X}) \in \mathrm{Sub}_{/\mathcal{X}}$ and any morphism $h_{\mathcal{X}}(\mathcal{W}) \to i_* h_{\mathcal{Z}}(\mathcal{Y}_{\mathcal{Z}})$, the base change

$$h_{\mathcal{X}}^{\mathcal{Z}}(\mathcal{Y}) \times_{i_* h_{\mathcal{Z}}(\mathcal{Y}_{\mathcal{Z}})} h_{\mathcal{X}}(\mathcal{W}) \to h_{\mathcal{X}}(\mathcal{W})$$

is a weak equivalence. Note that we have $h_{\mathcal{X}}(\mathcal{W}) \simeq p_\sharp h_{\mathcal{W}}(\mathcal{W})$, and thus by the smooth projection formula we may identify this morphism with

$$p_\sharp(p^* h_{\mathcal{X}}^{\mathcal{Z}}(\mathcal{Y}) \times_{p^* i_* h_{\mathcal{Z}}(\mathcal{Y}_{\mathcal{Z}})} h_{\mathcal{W}}(\mathcal{W})) \to p_\sharp h_{\mathcal{X}}(\mathcal{W}).$$

Consider the pullback square

$$\begin{tikzcd} \mathcal{W}_{\mathcal{Z}} \arrow[r, hook, "k"] \arrow[d, "q"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{W} \arrow[d, "p"] \\ \mathcal{Z} \arrow[r, hook, "i"] & \mathcal{X}. \end{tikzcd}$$

By smooth base change we have $p^* i_* \simeq k_* q^*$, and under this equivalence, the above morphism is the image of $p_\sharp$ of the morphism

$$h_{\mathcal{W}}^{\mathcal{W}_{\mathcal{Z}}}(\mathcal{Y} \times_{\mathcal{X}} \mathcal{W}) \times_{k_* h_{\mathcal{W}_{\mathcal{Z}}}((\mathcal{Y} \times_{\mathcal{X}} \mathcal{W})_{\mathcal{Z}})} h_{\mathcal{W}}(\mathcal{W}) \to h_{\mathcal{W}}(\mathcal{W}).$$

But the source of this morphism is precisely the presheaf $h_{\mathcal{W}}^{\mathcal{W}_{\mathcal{Z}}}(\mathcal{Y} \times_{\mathcal{X}} \mathcal{W}, t_{\mathcal{W}})$, where $t_{\mathcal{W}}\colon \mathcal{Z} \times_{\mathcal{X}} \mathcal{W} \to \mathcal{Y} \times_{\mathcal{X}} \mathcal{W}$ is the base change of $t\colon \mathcal{Z} \to \mathcal{Y}$ along $p\colon \mathcal{W} \to \mathcal{X}$. We conclude by Proposition 5.2.13. $\square$

Remark. During the revision process of this dissertation, it was pointed out to the author by Marc Hoyois that the above proof contains a a subtle mistake that goes back to the proof of Morel and Voevodsky: the smooth projection formula for pullbacks (as opposed to products) of presheaves on $\mathrm{Sub}_{/\mathcal{X}}$ does not a priori hold, because $\mathrm{Sub}_{/\mathcal{X}}$ does not have all pullbacks. We will provide a fix of this proof in future work.

Corollary 5.2.15. *For every closed embedding $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ of separated differentiable stacks with open complement $j\colon \mathcal{U} \hookrightarrow \mathcal{X}$ and every pointed genuine sheaf $\mathcal{F} \in \mathrm{H}_\bullet(\mathcal{X})$, the sequence*

$$j_\sharp j^*\mathcal{F} \to \mathcal{F} \to i_*i^*\mathcal{F}$$

is a cofiber sequence in $\mathrm{H}_\bullet(\mathcal{X})$.

Proof. For the purpose of this proof, let us make a distinction in notation between the functors $j_\sharp^{\mathrm{H}}\colon \mathrm{H}(\mathcal{U}) \hookrightarrow \mathrm{H}(\mathcal{X})$ and $j_\sharp^{\mathrm{H}_\bullet}\colon \mathrm{H}_\bullet(\mathcal{U}) \hookrightarrow \mathrm{H}_\bullet(\mathcal{X})$. We may express $j_\sharp^{\mathrm{H}_\bullet}$ in terms of $j_\sharp^{\mathrm{H}}$ as follows: for a pointed object $\mathcal{G} \in \mathrm{H}_\bullet(\mathcal{U}) = \mathrm{H}(\mathcal{U})_*$, regarded as a morphism $\mathcal{U} \to \mathcal{G}$ in $\mathrm{H}(\mathcal{U})$, we have a pushout square

$$\begin{tikzcd}
j_\sharp^{\mathrm{H}}(\mathcal{U}) \arrow[r] \arrow[d] \arrow[dr, phantom, "\ulcorner"] & j_\sharp^{\mathrm{H}}(\mathcal{G}) \arrow[d] \\
y(\mathcal{X}) \arrow[r] & j_\sharp^{\mathrm{H}_\bullet}(\mathcal{G}).
\end{tikzcd}$$

In particular, the object $j_\sharp^{\mathrm{H}_\bullet} j^*\mathcal{F}$ is given by the top left pushout square in the following commutative diagram:

$$\begin{tikzcd}
j_\sharp^{\mathrm{H}}(\mathcal{U}) \arrow[r] \arrow[d] \arrow[dr, phantom, "\ulcorner"] & y(\mathcal{X}) \arrow[d] & \\
j_\sharp j^*\mathcal{F} \arrow[r] \arrow[d] & j_\sharp^{\mathrm{H}_\bullet} j^*\mathcal{F} \arrow[r] \arrow[d] & \mathcal{F} \arrow[d] \\
j_\sharp(\mathcal{U}) \arrow[r] & y(\mathcal{X}) \arrow[r] & i_*i^*\mathcal{F}.
\end{tikzcd}$$

It follows by the pasting rule for pushout squares that the bottom left square is also a pushout. As the outer bottom rectangle is a pushout by Theorem 5.2.14, it follows from another application of the pasting rule that also the bottom right square is a pushout square. This square is precisely the underlying null-homotopy of the sequence $j_\sharp^{\mathrm{H}_\bullet}\mathcal{F} \to \mathcal{F} \to i_*i^*\mathcal{F}$ in the statement, so since the forgetful functor $\mathrm{H}_\bullet(\mathcal{X}) \to \mathrm{H}(\mathcal{X})$ preserves pushouts this finishes the proof. □

Theorem 5.2.16 (Localization theorem for pointed genuine sheaves)**.** *The pullback formalism $C = \mathrm{H}_\bullet$ satisfies the localization axiom.*

Proof. In light of Corollary 5.2.15, it remains to show that for every closed embedding $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$, the pushforward functor $i_*\colon \mathrm{H}_\bullet(\mathcal{Z}) \hookrightarrow \mathrm{H}_\bullet(\mathcal{X})$ is conservative. This may be tested after forgetting the base point. So let $\varphi\colon \mathcal{F}_1 \to \mathcal{F}_2$ be a morphism in $\mathrm{H}(\mathcal{Z})$ such that the induced map $i_*(\varphi)$ is invertible in $\mathrm{H}(\mathcal{X})$. We need to show that φ itself is already an equivalence, i.e., that for every $\mathcal{Y} \in \mathrm{Sub}_{/\mathcal{Z}}$ the map $\varphi(\mathcal{Y})\colon \mathcal{F}_1(\mathcal{Y}) \to \mathcal{F}_2(\mathcal{Y})$ is an equivalence. By Corollary 3.6.7, there exists an object $\mathcal{Y}' \in \mathrm{Sub}_{/\mathcal{X}}$ such that $\mathcal{Y} \simeq \mathcal{Z} \times_{\mathcal{X}} \mathcal{Y}'$. It thus then follows that $\mathcal{F}_k(\mathcal{Y}) \simeq \mathcal{F}_k(\mathcal{Z} \times_{\mathcal{X}} \mathcal{Y}') = i^*\mathcal{F}_k(\mathcal{Y}')$ and similarly $\varphi(\mathcal{Y}) \simeq i_*\varphi(\mathcal{Y}')$. This proves the claim. □

Corollary 5.2.17. *The functors* $i_*\colon \mathrm{H}_\bullet(\mathcal{Z}) \to \mathrm{H}_\bullet(\mathcal{X})$ *for closed embeddings between separated differentiable stacks are fully faithful and satisfy the closed projection formula, closed base change and smooth-closed base change.*

Proof. This is a special case of Proposition 5.1.3. □

Corollary 5.2.18 (Localization theorem for genuine sheaves of spectra)**.** *The pullback formalism* $\mathcal{C} = \mathrm{SH}$ *satisfies the localization axiom.*

Proof. In light of Lemma 5.1.4, this is an immediate consequence of Theorem 5.2.16. □

II.6 Relative Poincaré duality for differentiable stacks

The goal of this chapter is to establish a relative version of Poincaré duality for separated differentiable stacks: for every proper representable submersion $f\colon \mathcal{Y} \to \mathcal{X}$ of separated differentiable stacks, we will establish an equivalence

$$f_\sharp(-) \simeq f_*(- \otimes \omega_f)$$

of functors $\mathrm{SH}(\mathcal{Y}) \to \mathrm{SH}(\mathcal{X})$, where $\omega_f \in \mathrm{SH}(\mathcal{Y})$ is the *relative dualizing sheaf* of f. Our treatment is similar to that of Hoyois [Hoy17], who proves the analogous result in the context of equivariant motivic homotopy theory.

6.1 Statement of Main Theorem

The proof of relative Poincaré duality only relies on three crucial properties satisfied by the ∞-categories $\mathrm{SH}(\mathcal{X})$: homotopy invariance, the localization axiom, and the stability axiom. We have chosen to write all constructions and proofs of this chapter in this level of generality, as we feel that removing the specifics of the construction of $\mathrm{SH}(\mathcal{X})$ from the discussion clarifies the nature of the argument. Following [KR21, Definition 5.5], we will refer to these three properties as the *Voevodsky conditions*, as they were first singled out by Voevodsky in the case of schemes.

Definition 6.1.1 (Voevodsky conditions)**.** Consider a pullback formalism $\mathcal{C}\colon \mathrm{SepStk}^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$ on SepStk, in the sense of Definition 4.5.5. We say that $\mathcal{C}$ *satisfies the Voevodsky conditions* if it satisfies homotopy invariance, genuine stability and the localization axiom:

(1) (Homotopy invariance, Definition 4.5.17) For every separated differentiable stack $\mathcal{X}$, the pullback functor $\mathrm{pr}^*\colon \mathcal{C}(\mathcal{X}) \to \mathcal{C}(\mathcal{X}\times\mathbb{R})$ associated to the projection map $\mathrm{pr}\colon \mathcal{X}\times\mathbb{R} \to \mathcal{X}$ is fully faithful;

(2) (Genuine stability, Definition 4.5.26) The pullback formalism C is pointed and for every vector bundle $\pi\colon \mathcal{E} \to \mathcal{X}$ over a separated differentiable stack $\mathcal{X}$, the associated sphere bundle $S^{\mathcal{E}} \in C(\mathcal{X})$ is monoidally invertible.

(3) (Localization axiom, Definition 5.1.2) For every closed embedding $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ with open complement $j\colon \mathcal{U} \hookrightarrow \mathcal{Y}$, the functors $i_*\colon C(\mathcal{Z}) \hookrightarrow C(\mathcal{X})$ and $j_*\colon C(\mathcal{U}) \hookrightarrow C(\mathcal{X})$ are fully faithful, and there is a cofiber sequence $j_\sharp j^* \to \mathrm{id} \to i_* i^*$ in $\mathrm{Fun}(C(\mathcal{X}), C(\mathcal{X}))$. Equivalently, $C(\mathcal{X})$ is a recollement of the full subcategories $C(\mathcal{U})$ and $C(\mathcal{Z})$.

Example 6.1.2. The pullback formalism $C = \mathrm{SH}$ satisfies the Voevodsky conditions: homotopy invariance and genuine stability hold by definition, while the localization axiom holds by Corollary 5.2.18.

For the remainder of the section, we fix a pullback formalism C satisfying the Voevodsky conditions. We will now give the statement of relative Poincaré duality in this level of generality.

Definition 6.1.3 (Dualizing object)**.** Let $f\colon \mathcal{Y} \to \mathcal{X}$ be a representable submersion. We define its *dualizing object* $\omega_f \in C(\mathcal{Y})$ as

$$\omega_f := \mathrm{pr}_{1\sharp}\Delta_* \mathbb{1}_{\mathcal{Y}} \quad \in \quad C(\mathcal{Y}),$$

where $\mathrm{pr}_1\colon \mathcal{Y} \times_{\mathcal{X}} \mathcal{Y} \to \mathcal{Y}$ is the projection to the first factor and $\Delta\colon \mathcal{Y} \to \mathcal{Y} \times_{\mathcal{X}} \mathcal{Y}$ is the diagonal of f.

We will see below in Corollary 6.2.11 that the object ω_f is the incarnation in $C(\mathcal{Y})$ of the tangent sphere bundle S^{T_f} of f over $\mathcal{Y}$, i.e. the fiberwise one-point compactification of the relative tangent bundle $T_f \to \mathcal{Y}$ of f.

Remark 6.1.4 (Twist functor)**.** We refer to the functor $-\otimes\omega_f\colon C(\mathcal{Y}) \to C(\mathcal{Y})$ as a 'twist functor'. By the smooth and closed projection formulas, it is equivalent to the composite $\mathrm{pr}_{1\sharp}\Delta_*\colon C(\mathcal{Y}) \to C(\mathcal{Y})$:

$$X \otimes \omega_f = X \otimes \mathrm{pr}_{1\sharp}\Delta_* \mathbb{1}_{\mathcal{Y}} \simeq \mathrm{pr}_{1\sharp}(\mathrm{pr}_1^* X \otimes \Delta_* \mathbb{1}_{\mathcal{Y}}) \simeq \mathrm{pr}_{1\sharp}\Delta_*(\Delta^* \mathrm{pr}_1^* X \otimes \mathbb{1}_{\mathcal{Y}}) \simeq \mathrm{pr}_{1\sharp}\Delta_* X.$$

Construction 6.1.5 (Poincaré duality map, cf. [CD19, Section 2.4.b])**.** Given a representable submersion $f\colon \mathcal{Y} \to \mathcal{X}$, we construct a natural transformation $\mathfrak{p}_f\colon f_\sharp(-) \to f_*(-\otimes\omega_f)$, called the *Poincaré duality map*. Consider the following pullback square:

$$\begin{tikzcd}
\mathcal{Y} \times_{\mathcal{X}} \mathcal{Y} \arrow[r, "\mathrm{pr}_2"] \arrow[d, "\mathrm{pr}_1"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{Y} \arrow[d, "f"] \\
\mathcal{Y} \arrow[r, "f"] & \mathcal{X}.
\end{tikzcd}$$

There is a *double Beck-Chevalley transformation* $\mathrm{BC}_{\sharp,*}\colon f_\sharp \mathrm{pr}_{2*} \to f_* \mathrm{pr}_{1\sharp}$ associated to this diagram, see Appendix F.2 for details. We then define $\mathfrak{p}_f$ as the following composite:

$$\mathfrak{p}_f\colon f_\sharp \simeq f_\sharp \mathrm{pr}_{2*}\Delta_* \xrightarrow{\mathrm{BC}_{\sharp,*}} f_* \mathrm{pr}_{1\sharp}\Delta_* \simeq f_*(-\otimes \omega_f),$$

where the last equivalence holds by Remark 6.1.4.

Remark 6.1.6. There seems to be no standard terminology for the transformation $\mathfrak{p}_f$. Cisinski and Déglise use the term 'purity equivalence'. Bachmann and Hoyois [BH21, Section 5.4] use the term 'ambidexterity equivalence'. Although we certainly think this map as a geometric form of ambidexterity, we prefer to reserve the term 'ambidexterity' for the homotopical notion introduced in Part I. As explained in the introduction, we prefer the terminology 'relative Poincaré duality': we want to think of the representable submersion $f\colon \mathcal{Y} \to \mathcal{X}$ as an $\mathcal{X}$-indexed family of smooth manifolds over $\mathcal{X}$, and we think of the functors $f_\sharp$ and f_* as the $\mathcal{X}$-indexed homology and cohomology, respectively, of this family.

The following is our main theorem, which is a refined version of Theorem A stated in the introduction:

Theorem 6.1.7 (Relative Poincaré duality)**.** *Let C be a pullback formalism on* SepStk *satisfying the Voevodsky conditions. Let $f\colon \mathcal{Y} \to \mathcal{X}$ be a proper representable submersion between separated differentiable stacks. Then the transformation*

$$\mathfrak{p}_f\colon f_\sharp(-) \to f_*(-\otimes \omega_f)$$

is an equivalence of functors $C(\mathcal{Y}) \to C(\mathcal{X})$.

The proof strategy for Theorem 6.1.7 is a generalization of the usual proof of Atiyah duality for a compact smooth manifold M. A crucial ingredient in the proof is the construction of a *Pontryagin-Thom collapse map* associated to a closed embedding of differentiable stacks, discussed in Section 6.2. To simplify the required bookkeeping in the proof of Theorem 6.1.7, we introduce in Section 6.3 the notion of a *kernel operator*, which is closely related to the notion of Fourier-Mukai transforms in algebraic geometry. Using these ingredients, we give a proof of Theorem 6.1.7 in Section 6.4. Finally, we discuss various important consequences of relative Poincaré duality in Section 6.5, like relative Atiyah duality, proper base change and smooth-proper base change.

6.2 Relative Thom spaces and the Pontryagin-Thom construction

Fixing a pullback formalism C satisfying the Voevodsky conditions, we start by introducing the analogues of Thom spaces and the Pontryagin-Thom construction in C. Throughout, we fix a base stack $\mathcal{S}$, and all other stacks are assumed to live in the category $\mathrm{Sub}_{/\mathcal{S}}$: every differentiable stack $\mathcal{X}$ comes equipped with a representable submersion $g_{\mathcal{X}}\colon \mathcal{X} \to \mathcal{S}$ and every morphism $f\colon \mathcal{Y} \to \mathcal{X}$ is assumed to commute with the structure maps to $\mathcal{S}$.

6.2.1 Relative Thom spaces

We start by defining relative Thom spaces.

Definition 6.2.1 (Relative Thom space)**.** Consider a commutative diagram

$$\begin{tikzcd} \mathcal{Z} \arrow[rr, "i"] \arrow[dr, "g_{\mathcal{Z}}"'] & & \mathcal{X} \arrow[dl, "g_{\mathcal{X}}"] \\ & \mathcal{S}, & \end{tikzcd}$$

where $g_{\mathcal{Z}}$ and $g_{\mathcal{X}}$ are representable submersions and i is a closed embedding. We define the *(relative) Thom space of $\mathcal{Z}$ in $\mathcal{X}$ over $\mathcal{S}$* as

$$\mathrm{Th}_{\mathcal{S}}(\mathcal{X},\mathcal{Z}) := (g_{\mathcal{X}})_{\sharp} i_* \mathbb{1}_{\mathcal{Z}} \quad \in \quad C(\mathcal{S}).$$

We might sometimes write $\mathrm{Th}^{C}_{\mathcal{S}}(\mathcal{X},\mathcal{Z})$ to emphasize the dependence on C.

Remark 6.2.2 (Change of base stack)**.** If $h\colon \mathcal{S} \to \mathcal{S}'$ is a representable submersion of differentiable stacks, then every differentiable stack over $\mathcal{S}$ may be regarded as a differentiable stack over $\mathcal{S}'$ by composing with h. It is clear from the definitions that we have

$$\mathrm{Th}_{\mathcal{S}'}(\mathcal{X},\mathcal{Z}) \simeq h_{\sharp}\,\mathrm{Th}_{\mathcal{S}}(\mathcal{X},\mathcal{Z}) \quad \in \quad C(\mathcal{S}').$$

Example 6.2.3. If $\mathcal{Z} = \mathcal{X}$ and $i = \mathrm{id}_{\mathcal{X}}$, we get $\mathrm{Th}_{\mathcal{S}}(\mathcal{X},\mathcal{X}) = h_{\mathcal{S}}(\mathcal{X})$.

Example 6.2.4. Given a representable submersion $f\colon \mathcal{Y} \to \mathcal{X}$, the dualizing object $\omega_f \in C(\mathcal{Y})$ is an example of a relative Thom space: we have

$$\omega_f = \mathrm{pr}_{1\sharp}\Delta_* \mathbb{1}_{\mathcal{Y}} = \mathrm{Th}_{\mathcal{Y}}(\mathcal{Y} \times_{\mathcal{X}} \mathcal{Y}, \mathcal{Y}) \quad \in \quad C(\mathcal{Y}),$$

where the diagonal $\Delta\colon \mathcal{Y} \to \mathcal{Y} \times_{\mathcal{X}} \mathcal{Y}$ of f is a closed embedding which we regard as a morphism in $\mathrm{Sub}_{/\mathcal{X}}$ via the first projection map $\mathrm{pr}_1\colon \mathcal{Y} \times_{\mathcal{X}} \mathcal{Y} \to \mathcal{Y}$.

The relative Thom space $\mathrm{Th}_{\mathcal{S}}(\mathcal{X},\mathcal{Z})$ may be thought of as a homotopy quotient of the stack $\mathcal{X}$ by the open substack $\mathcal{X}\setminus\mathcal{Z}$, taken fiberwise over $\mathcal{S}$. This is made precise by the following lemma:

Lemma 6.2.5. *For a closed embedding $i\colon \mathcal{Z}\hookrightarrow\mathcal{X}$ in $\mathrm{Sub}_{/\mathcal{X}}$, there is a preferred cofiber sequence*

$$h_{\mathcal{S}}(\mathcal{X}\setminus\mathcal{Z})\to h_{\mathcal{S}}(\mathcal{X})\to \mathrm{Th}_{\mathcal{S}}(\mathcal{X},\mathcal{Z})\quad\in\quad C(\mathcal{S}).$$

Proof. Let $j\colon \mathcal{X}\setminus\mathcal{Z}\hookrightarrow\mathcal{X}$ denote the open complement of i. By the localization axiom, there is a preferred cofiber sequence

$$j_\sharp\mathbb{1}_{\mathcal{X}\setminus\mathcal{Z}}\to\mathbb{1}_{\mathcal{X}}\to i_*\mathbb{1}_{\mathcal{Z}}\quad\in\quad C(\mathcal{X}).$$

Applying the colimit-preserving functor $(g_{\mathcal{X}})_\sharp\colon C(\mathcal{X})\to C(\mathcal{Z})$, we get a cofiber sequence

$$(g_{\mathcal{X}\setminus\mathcal{Z}})_\sharp\mathbb{1}_{\mathcal{X}\setminus\mathcal{Z}}\to(g_{\mathcal{X}})_\sharp\mathbb{1}_{\mathcal{X}}\to(g_{\mathcal{X}})_\sharp i_*\mathbb{1}_{\mathcal{Z}}=\mathrm{Th}_{\mathcal{S}}(\mathcal{X},\mathcal{Z})\quad\in\quad C(\mathcal{Z}).$$

Since $h_{\mathcal{S}}(\mathcal{X}\setminus\mathcal{Z})=(g_{\mathcal{X}\setminus\mathcal{Z}})_\sharp\mathbb{1}_{\mathcal{X}\setminus\mathcal{Z}}$ and $h_{\mathcal{S}}(\mathcal{X})=(g_{\mathcal{X}})_\sharp\mathbb{1}_{\mathcal{X}}$, this finishes the proof. □

We will frequently be interested in the case where $\mathcal{S}=\mathcal{Z}$, so that $i\colon\mathcal{S}\hookrightarrow\mathcal{X}$ is the section of a representable submersion $g\colon\mathcal{X}\to\mathcal{S}$. In the case g is a vector bundle, the relative Thom space is given by the associated sphere bundle:

Corollary 6.2.6. *Let $\pi\colon\mathcal{E}\to\mathcal{S}$ be a vector bundle and let $s\colon\mathcal{S}\to\mathcal{E}$ be its zero section. Then the Thom space of s over $\mathcal{S}$ is the sphere bundle associated to $\mathcal{E}$:*

$$\mathrm{Th}_{\mathcal{S}}(\mathcal{E},\mathcal{S})\simeq S^{\mathcal{E}}\quad\in\quad C(\mathcal{S}).$$

Proof. Consider the unique morphism $h\colon \mathrm{H}\to C$ of pullback formalisms. The induced functor $h_{\mathcal{S}}\colon\mathrm{H}(\mathcal{S})\to C(\mathcal{S})$ preserves colimits and sends the object $L_{\mathrm{htp}}(\mathcal{Y})$ to $h_{\mathcal{S}}(\mathcal{Y})$ for every $\mathcal{Y}\in\mathrm{Sub}_{/\mathcal{S}}$. By Lemma 4.3.2 we thus obtain for every vector bundle $\pi\colon\mathcal{E}\to\mathcal{S}$ a pushout square

$$\begin{tikzcd}
h_{\mathcal{S}}(\mathcal{E}\setminus\mathcal{S}) \arrow[r] \arrow[d] \arrow[dr, phantom, "\ulcorner"] & h_{\mathcal{S}}(\mathcal{E}) \arrow[d] \\
h_{\mathcal{S}}(\mathcal{S}) \arrow[r] & h_{\mathcal{S}}(\mathcal{S}^{\mathcal{E}})
\end{tikzcd}$$

in $C(\mathcal{S})$, which gives rise to an equivalence between the cofibers in $C(\mathcal{S})$ of the two horizontal maps. As the object $S^{\mathcal{E}}$ was defined in Notation 4.5.25 to be the cofiber of the bottom map and the relative Thom space $\mathrm{Th}_{\mathcal{S}}(\mathcal{E},\mathcal{S})$ is by Lemma 6.2.5 the cofiber of the top map, this finishes the proof. □

We prove various basic properties of the relative Thom space construction. We start by showing that the relative Thom space $\mathrm{Th}_{\mathcal{S}}(\mathcal{X},\mathcal{Z})$ only depends on an arbitrarily small open neighborhood of $\mathcal{Z}$ in $\mathcal{X}$.

Lemma 6.2.7 (Invariance under open neighborhoods)**.** *Let $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be a closed embedding over $\mathcal{S}$, and let $j\colon \mathcal{U} \hookrightarrow \mathcal{X}$ be an open embedding over $\mathcal{S}$ such that i factors through a map $i'\colon \mathcal{Z} \hookrightarrow \mathcal{U}$. Then there is a preferred equivalence*

$$\mathrm{Th}_{\mathcal{S}}(\mathcal{U},\mathcal{Z}) \xrightarrow{\sim} \mathrm{Th}_{\mathcal{S}}(\mathcal{X},\mathcal{Z}) \quad \in \quad \mathcal{C}(\mathcal{S}).$$

Proof. Applying smooth-closed base change to the pullback square

$$\begin{tikzcd} \mathcal{Z} \arrow[r, "i'"] \arrow[d, equal] \arrow[dr, phantom, "\lrcorner"] & \mathcal{U} \arrow[d, "j"] \\ \mathcal{Z} \arrow[r, "i"] & \mathcal{X}, \end{tikzcd}$$

we obtain an equivalence $j_\sharp i'_* \xrightarrow{\sim} i_*$. We thus obtain the desired equivalence as the following composite:

$$\mathrm{Th}_{\mathcal{S}}(\mathcal{U},\mathcal{Z}) = (g_{\mathcal{U}})_\sharp i'_* \mathbb{1}_{\mathcal{Z}} = (g_{\mathcal{X}})_\sharp j_\sharp i'_* \mathbb{1}_{\mathcal{Z}} \xrightarrow{\sim} (g_{\mathcal{X}})_\sharp i_* \mathbb{1}_{\mathcal{Z}} = \mathrm{Th}_{\mathcal{S}}(\mathcal{X},\mathcal{Z}). \qquad \square$$

Due to the existence of relative tubular neighborhoods from Proposition 3.6.10, the previous lemma implies that the relative Thom space of a closed embedding $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ in fact only depends on its normal bundle:

Corollary 6.2.8. *Let $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be a closed embedding over $\mathcal{S}$ and let $\mathcal{N}_i \to \mathcal{Z}$ be its normal bundle. There is an equivalence*

$$\mathrm{Th}_{\mathcal{S}}(\mathcal{X},\mathcal{Z}) \simeq \mathrm{Th}_{\mathcal{S}}(\mathcal{N}_i,\mathcal{Z}) \quad \in \quad \mathcal{C}(\mathcal{S}).$$

Proof. By Proposition 3.6.10, the substack $\mathcal{Z}$ admits a tubular neighborhood $\mathcal{U}$ inside $\mathcal{X}$ relative to $\mathcal{S}$, meaning that there exists a commutative diagram

$$\begin{tikzcd} & \mathcal{Z} \arrow[dl, hook', "i"'] \arrow[d, hook] \arrow[dr, hook, "s_0"] & \\ \mathcal{X} & \mathcal{U} \arrow[l, hook', "j"'] \arrow[r, hook, "j'"] & \mathcal{N}_i \end{tikzcd}$$

in $\mathrm{Sub}_{/\mathcal{S}}$, where $\mathcal{N}_i$ lives over $\mathcal{S}$ by composing the bundle projection $\mathcal{N}_i \to \mathcal{Z}$ with the map $g_{\mathcal{Z}}\colon \mathcal{Z} \to \mathcal{S}$ and where the maps j and j' are open embeddings of differentiable stacks. The claim thus follows from two instances of Lemma 6.2.7: equivalences

$$\mathrm{Th}_{\mathcal{S}}(\mathcal{X},\mathcal{Z}) \xleftarrow{\sim} \mathrm{Th}_{\mathcal{S}}(\mathcal{U},\mathcal{Z}) \xrightarrow{\sim} \mathrm{Th}_{\mathcal{S}}(\mathcal{N}_i,\mathcal{Z}). \qquad \square$$

Corollary 6.2.9 (Invertibility of Thom spaces)**.** *Let $p\colon \mathcal{X} \to \mathcal{S}$ be a representable submersion and assume p admits a section $i\colon \mathcal{S} \hookrightarrow \mathcal{X}$, which is automatically a closed embedding. Then there is an equivalence*

$$\mathrm{Th}_{\mathcal{S}}(\mathcal{X},\mathcal{S}) \simeq S^{\mathcal{N}_i} \quad \in \quad \mathcal{C}(\mathcal{S}).$$

In particular, the relative Thom space $\mathrm{Th}_{\mathcal{S}}(\mathcal{X},\mathcal{S})$ is an invertible object of $\mathcal{C}(\mathcal{S})$.

Proof. The first statement is immediate from Corollary 6.2.8 and Corollary 6.2.6. The second statement follows from the genuine stability assumption on $\mathcal{C}$. □

Warning 6.2.10. In Corollary 6.2.9, it is crucial that $\mathcal{Z} = \mathcal{S}$: the relative Thom space $\mathrm{Th}_{\mathcal{S}}(\mathcal{X},\mathcal{Z})$ is generally not invertible when $\mathcal{Z} \neq \mathcal{S}$.

Corollary 6.2.11 (Dualizing object is tangent sphere bundle)**.** *Let $f\colon \mathcal{Y} \to \mathcal{X}$ be a representable submersion. Then there is an equivalence $\omega_f \simeq S^{T_f} \in \mathcal{C}(\mathcal{Y})$.*

Proof. By Corollary 6.2.9, there is an equivalence $\omega_f = \mathrm{Th}_{\mathcal{Y}}(\mathcal{Y} \times_{\mathcal{X}} \mathcal{Y}, \mathcal{Y}) \simeq S^{\mathcal{N}_\Delta}$ in $\mathcal{C}(\mathcal{Y})$. Since Proposition 3.5.19 provides an isomorphism of vector bundles $\mathcal{N}_\Delta \cong T_f$ over $\mathcal{Y}$, this finishes the proof. □

Lemma 6.2.12 (Multiplicativity of Thom spaces)**.** *Consider representable submersions $g_i\colon \mathcal{X}_i \to \mathcal{S}$ for $i = 1,2,3$, equipped with sections $s_i\colon \mathcal{S} \to \mathcal{X}_i$. Assume there exists a pullback square*

$$\begin{tikzcd} \mathcal{X}_1 \arrow[r, "a"] \arrow[d, "g_1"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{X}_2 \arrow[d, "b"] \\ \mathcal{S} \arrow[r, "s_3"] & \mathcal{X}_3 \end{tikzcd}$$

over $\mathcal{S}$, where b is a representable submersion and such that $a \circ s_1 = s_2\colon \mathcal{S} \to \mathcal{X}_2$. Then there is an equivalence

$$\mathrm{Th}_{\mathcal{S}}(\mathcal{X}_2) \simeq \mathrm{Th}_{\mathcal{S}}(\mathcal{X}_1) \otimes \mathrm{Th}_{\mathcal{S}}(\mathcal{X}_3) \quad \in \quad \mathcal{C}(\mathcal{S}).$$

In particular, taking $\mathcal{X}_2 = \mathcal{X}_1 \times_{\mathcal{S}} \mathcal{X}_3$ gives $\mathrm{Th}_{\mathcal{S}}(\mathcal{X}_1 \times_{\mathcal{S}} \mathcal{X}_3) \simeq \mathrm{Th}_{\mathcal{S}}(\mathcal{X}_1) \otimes \mathrm{Th}_{\mathcal{S}}(\mathcal{X}_3)$.

Proof. Since $g_2 = g_3 \circ b$ and $s_2 = a \circ s_1$, we have

$$\mathrm{Th}_{\mathcal{S}}(\mathcal{X}_2) = g_{2\sharp}s_{2*} = g_{3\sharp}b_\sharp a_* s_{1*} \xrightarrow{\mathrm{BC}_{\sharp,*}} g_{3\sharp}s_{3*}g_{1\sharp}s_{1*} \simeq \mathrm{Th}_{\mathcal{S}}(\mathcal{X}_1) \otimes \mathrm{Th}_{\mathcal{S}}(\mathcal{X}_3).$$

Here the double Beck-Chevalley map $\mathrm{BC}_{\sharp,*}$ associated to the pullback square is an equivalence by smooth-closed base change and the last equivalence follows from the smooth and closed porjection formulas. □

Remark 6.2.13. One can in fact prove more generally that for closed embeddings $i\colon \mathcal{Z} \to \mathcal{X}$ and $i'\colon \mathcal{Z}' \to \mathcal{X}'$ over $\mathcal{S}$ there is an equivalence

$$\mathrm{Th}_{\mathcal{S}}(\mathcal{X},\mathcal{Z}) \otimes \mathrm{Th}_{\mathcal{S}}(\mathcal{X}',\mathcal{Z}') \simeq \mathrm{Th}_{\mathcal{S}}(\mathcal{X}\times_{\mathcal{S}}\mathcal{X}',\mathcal{Z}\times_{\mathcal{S}}\mathcal{Z}') \quad \in \quad C(\mathcal{S}).$$

We will not need this and leave the verification to the reader.

6.2.2 Pontryagin-Thom construction

Given a closed embedding of smooth manifolds $i\colon Z \hookrightarrow M$, one can define a *Pontryagin-Thom collapse map* $\mathrm{PT}(i)\colon M \to \mathrm{Th}_Z(N_i)$, where N_i denotes the normal bundle of Z in M, and $\mathrm{Th}_Z(N_i)$ is the Thom space of this normal bundle. This map is usually constructed at the point-set level by collapsing the complement of a tubular neighborhood U of Z inside M to a point, and identifying the resulting quotient with the Thom space $\mathrm{Th}_Z(N_i)$. From a more homotopical perspective, we may also think of this map as a composite of the homotopy quotient map $M \to M/(M\setminus Z)$ with the string of identifications

$$M/(M\setminus Z) \simeq U/(U\setminus Z) \simeq N_i/(N_i\setminus Z) \simeq \mathrm{Th}_Z(N_i);$$

here we abuse notation by writing X/A for the *homotopy quotient* of a topological space X by a subspace A, that is, the cofiber of the map $A \hookrightarrow X$ in the ∞-category An of animae. The advantage of the latter formulation is that it admits a simple generalization to the general context of a pullback formalism C over differentiable stacks satisfying the Voevodsky conditions.

Remark 6.2.14. Another approach to a generalization of the Pontryagin-Thom construction to differentiable stacks can be found in [EG11, Section 3].

Construction 6.2.15 (Quotient map)**.** Let $i\colon \mathcal{Z} \hookrightarrow \mathcal{Z}'$ and $i'\colon \mathcal{Z}' \hookrightarrow \mathcal{X}$ be two closed embeddings over $\mathcal{S}$. Then the unit map $\mathbb{1}_{\mathcal{Z}'} \to i_*i^*\mathbb{1}_{\mathcal{Z}'} \simeq i_*\mathbb{1}_{\mathcal{Z}}$ gives rise to a map

$$u_i^*\colon\ \mathrm{Th}_{\mathcal{S}}(\mathcal{X},\mathcal{Z}') = (g_{\mathcal{X}})_\sharp i'_*\mathbb{1}_{\mathcal{Z}'} \to (g_{\mathcal{X}})_\sharp i'_* i_*\mathbb{1}_{\mathcal{Z}} = \mathrm{Th}_{\mathcal{S}}(\mathcal{X},\mathcal{Z}).$$

Construction 6.2.16 (Pontryagin-Thom construction)**.** Let $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be a closed embedding, and let $\mathcal{U}$ be a relative tubular neighborhood of $\mathcal{Z}$ in $\mathcal{X}$ relative to $\mathcal{S}$ in the sense of Definition 3.6.9: a choice of a commutative diagram

$$\begin{tikzcd} & \mathcal{Z} \arrow[dl, hook', bend right, "i"'] \arrow[d, hook] \arrow[dr, hook, bend left, "s_0"] & \\ \mathcal{X} & \mathcal{U} \arrow[l, hook', "j"'] \arrow[r, hook, "j'"] & N_i \end{tikzcd}$$

of stacks over $\mathcal{S}$, where the maps j and j' are open embeddings. We define the *Pontryagin-Thom collapse map* $\mathrm{PT}_{\mathcal{S}}(i)\colon h_{\mathcal{S}}(\mathcal{X}) \to \mathrm{Th}_{\mathcal{S}}(\mathcal{N}_i, \mathcal{Z})$ (with respect to $\mathcal{U}$) as the following composite:

$$\mathrm{PT}_{\mathcal{S}}(i)\colon h_{\mathcal{S}}(\mathcal{X}) \overset{6.2.3}{\simeq} \mathrm{Th}_{\mathcal{S}}(\mathcal{X},\mathcal{X}) \xrightarrow{u_i^*} \mathrm{Th}_{\mathcal{S}}(\mathcal{X},\mathcal{Z}) \overset{6.2.7}{\simeq} \mathrm{Th}_{\mathcal{S}}(\mathcal{U},\mathcal{Z}) \overset{6.2.7}{\simeq} \mathrm{Th}_{\mathcal{S}}(\mathcal{N}_i,\mathcal{Z}).$$

Here the map u_i^* is the quotient map from Construction 6.2.15.

Informally speaking, this map may be thought of as the composite

$$\mathcal{X} \to \mathcal{X}/(\mathcal{X}\setminus\mathcal{Z}) \simeq \mathcal{U}/(\mathcal{U}\setminus\mathcal{Z}) \simeq \mathcal{N}_i/(\mathcal{N}_i\setminus\mathcal{Z}) = \mathrm{Th}(\mathcal{N}_i,\mathcal{Z}),$$

where the first map is the homotopy quotient map $\mathcal{X} \to \mathcal{X}/(\mathcal{X}\setminus\mathcal{Z})$.

Example 6.2.17. Let $\pi\colon \mathcal{E} \to \mathcal{S}$ be a vector bundle and let $s\colon \mathcal{S} \to \mathcal{E}$ be its zero section. Let i be the composite inclusion $\mathcal{S} \hookrightarrow \mathcal{E} \hookrightarrow S^{\mathcal{E}}$, and pick the open neighborhood $\mathcal{E}$ as a tubular neighborhood of $\mathcal{S}$ inside $S^{\mathcal{E}}$. Then by Corollary 6.2.6 there is an equivalence $\mathrm{Th}_{\mathcal{S}}(\mathcal{E},\mathcal{S}) \simeq S^{\mathcal{E}}$, and unwinding that equivalence we see that Pontryagin-Thom map $\mathrm{PT}_{\mathcal{S}}(i)$ fits in a cofiber sequence

$$h_{\mathcal{S}}(\mathcal{S}) \to h_{\mathcal{S}}(S^{\mathcal{E}}) \xrightarrow{\mathrm{PT}_{\mathcal{S}}(i)} \mathrm{Th}_{\mathcal{S}}(\mathcal{E},\mathcal{S}).$$

We will in fact need a slight generalization of the above construction where $\mathcal{X}$ is allowed to be embedded in some larger ambient stack $\mathcal{X}'$:

Construction 6.2.18 (Generalized Pontryagin-Thom construction)**.** Let $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ and $i'\colon \mathcal{X} \hookrightarrow \mathcal{X}'$ be closed embeddings. Choose a tubular neighborhood

$$\begin{tikzcd} & \mathcal{X} \arrow[dl, hook, bend right, "i'"'] \arrow[d, hook] \arrow[dr, hook, bend left, "s_0"] & \\ \mathcal{X}' & \mathcal{U}' \arrow[l, hook', "j'"'] \arrow[r, hook] & \mathcal{N}_{i'} \end{tikzcd}$$

of i' relative to $\mathcal{S}$, and choose another tubular neighborhood

$$\begin{tikzcd} & \mathcal{Z} \arrow[dl, hook, bend right, "j'\circ i"'] \arrow[d, hook] \arrow[dr, hook, bend left, "s_0"] & \\ \mathcal{U}' & \mathcal{U} \arrow[l, hook'] \arrow[r, hook] & \mathcal{N}_{j'\circ i} \cong \mathcal{N}_{i'\circ i} \end{tikzcd}$$

of the composite $j'\circ i\colon \mathcal{Z} \hookrightarrow \mathcal{U}'$ relative to $\mathcal{S}$. We define the map $\mathrm{PT}_{\mathcal{S}}(i,i')$ as the following composite:

$$\mathrm{PT}_{\mathcal{S}}(i,i')\colon \mathrm{Th}_{\mathcal{S}}(\mathcal{N}_{i'},\mathcal{X}) \xrightarrow{u_i^*} \mathrm{Th}_{\mathcal{S}}(\mathcal{N}_{i'},\mathcal{Z}) \simeq \mathrm{Th}_{\mathcal{S}}(\mathcal{U}',\mathcal{Z}) \simeq \mathrm{Th}_{\mathcal{S}}(\mathcal{U},\mathcal{Z}) \simeq \mathrm{Th}_{\mathcal{S}}(\mathcal{N}_{i'\circ i},\mathcal{Z}).$$

Here the last three equivalences are instances of Lemma 6.2.7.

It is clear that Construction 6.2.18 specializes to Construction 6.2.16 when $\mathcal{X}' = \mathcal{X}$ and $i' = \mathrm{id}_{\mathcal{X}}$.

Lemma 6.2.19 (Compatibility with composition)**.** *In the situation of Construction 6.2.18, the following diagram commutes up to preferred homotopy:*

$$\begin{tikzcd}
h_{\mathcal{S}}(\mathcal{X}') \arrow[r, "\mathrm{PT}_{\mathcal{S}}(i')"] \arrow[d, "\mathrm{PT}_{\mathcal{S}}(i' \circ i)"'] & \mathrm{Th}_{\mathcal{S}}(\mathcal{N}_{i'}, \mathcal{X}) \arrow[d, "\mathrm{PT}_{\mathcal{S}}(i, i')"] \\
\mathrm{Th}_{\mathcal{S}}(\mathcal{N}_{i' \circ i}, \mathcal{Z}) \arrow[r, equal] & \mathrm{Th}_{\mathcal{S}}(\mathcal{N}_{i' \circ i}, \mathcal{Z}).
\end{tikzcd}$$

Proof. This follows from the following commutative diagram:

$$\begin{tikzcd}
h_{\mathcal{S}}(\mathcal{X}') \arrow[d, "u_{i'}^*"] \arrow[dr, "\mathrm{PT}_{\mathcal{S}}(i')"] \arrow[drr, bend left, "\mathrm{PT}_{\mathcal{S}}(i' \circ i)"] \arrow[dd, bend right, "u_{i' \circ i}^*"'] & & \\
\mathrm{Th}_{\mathcal{S}}(\mathcal{X}', \mathcal{X}) \arrow[r, "\simeq"'] \arrow[d, "u_i^*"] & \mathrm{Th}_{\mathcal{S}}(\mathcal{N}_{i'}, \mathcal{X}) \arrow[d, "u_i^*"] \arrow[dr, "\mathrm{PT}_{\mathcal{S}}(i, i')"] & \\
\mathrm{Th}_{\mathcal{S}}(\mathcal{X}', \mathcal{Z}) \arrow[r, "\simeq"'] & \mathrm{Th}_{\mathcal{S}}(\mathcal{N}_{i'}, \mathcal{Z}) \arrow[r, "\simeq"'] & \mathrm{Th}_{\mathcal{S}}(\mathcal{N}_{i' \circ i}, \mathcal{Z})
\end{tikzcd}$$

Note that the bottom composite is given by the composite of the equivalences

$$\mathrm{Th}_{\mathcal{S}}(\mathcal{X}', \mathcal{Z}) \xleftarrow{\sim} \mathrm{Th}_{\mathcal{S}}(\mathcal{U}, \mathcal{Z}) \xrightarrow{\sim} \mathrm{Th}_{\mathcal{S}}(\mathcal{N}_{i' \circ i}, \mathcal{Z}).$$

from Lemma 6.2.7. The two small triangles and the outer triangle then commute by definition and the lower left square commutes by naturality. □

Lemma 6.2.20 (Compatibility with base change)**.** *Let $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ be a closed embedding over $\mathcal{S}$. Let $f\colon \mathcal{S}' \to \mathcal{S}$ be a morphism of differentiable stacks and write $f^*\colon \mathrm{Sub}_{\mathcal{S}} \to \mathrm{Sub}_{\mathcal{S}'}$ for the pullback functor.*

(1) If $\mathcal{U}$ is a tubular neighborhood of $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ relative to $\mathcal{S}$, then $f^*\mathcal{U}$ is a tubular neighborhood of $f^*i\colon f^*\mathcal{Z} \hookrightarrow f^*\mathcal{X}$ relative to $\mathcal{S}'$.

(2) There are preferred equivalences

$$h_{\mathcal{S}'}(f^*\mathcal{X}) \simeq f^* h_{\mathcal{S}}(\mathcal{X}) \qquad \textit{and} \qquad \mathrm{Th}_{\mathcal{S}'}(\mathcal{N}_{f^*i}, f^*\mathcal{Z}) \simeq f^* \mathrm{Th}_{\mathcal{S}}(\mathcal{N}_i, \mathcal{Z})$$

(3) With respect to the choice of tubular neighborhood from (1), there is a preferred homotopy making the following diagram commute:

$$\begin{tikzcd}
h_{\mathcal{S}'}(f^*\mathcal{X}) \arrow[r, "\simeq"] \arrow[d, "\mathrm{PT}_{\mathcal{S}'}(f^*(i))"'] & f^* h_{\mathcal{S}}(\mathcal{X}) \arrow[d, "f^* \mathrm{PT}_{\mathcal{S}}(i)"] \\
\mathrm{Th}_{\mathcal{S}'}(\mathcal{N}_{f^*i}, f^*\mathcal{Z}) \arrow[r, "\simeq"] & f^* \mathrm{Th}_{\mathcal{S}}(\mathcal{N}_i, \mathcal{Z}).
\end{tikzcd}$$

Proof. This is immediate from spelling out the definitions and using that by smooth and closed base change the functor f^* commutes with all the constructions that appear. $\square$

Lemma 6.2.21 (Invariance under homotopy)**.** *Let $i\colon \mathcal{Z}\times\mathbb{R} \hookrightarrow \mathcal{X}\times\mathbb{R}$ be closed embedding over $\mathcal{S}\times\mathbb{R}$, and let $\mathcal{Z}\times\mathbb{R} \hookrightarrow \mathcal{U} \hookrightarrow \mathcal{X}\times\mathbb{R}$ be a tubular neighborhood. Then:*

(1) For every $r\in\mathbb{R}$, the restriction $\mathcal{Z}\hookrightarrow \mathcal{U}_r \hookrightarrow \mathcal{X}$ is a tubular neighborhood of the closed embedding $i_r\colon \mathcal{Z}\hookrightarrow\mathcal{X}$;

(2) For every $r\in R$, there is an isomorphism $\alpha\colon \mathcal{N}_{i_r}\cong\mathcal{N}_{i_0}$ of vector bundles over $\mathcal{Z}$ making the following diagram commute up to homotopy:

$$\begin{tikzcd} & h_{\mathcal{S}}(\mathcal{X}) \arrow[dl, "\mathrm{PT}(i_r)"'] \arrow[dr, "\mathrm{PT}(i_0)"] & \\ \mathrm{Th}_{\mathcal{S}}(\mathcal{N}_{i_r},\mathcal{Z}) \arrow[rr, "\alpha", "\simeq"'] & & \mathrm{Th}_{\mathcal{S}}(\mathcal{N}_{i_0},\mathcal{Z}). \end{tikzcd}$$

Proof. Part (1) is an instance of part (1) of Lemma 6.2.20. The first part of (2) is a special case of Lemma 3.5.18. For the convenience of the reader, we shall recall how the isomorphism α is constructed. Considering the normal bundle $\mathcal{N}_i\to\mathcal{Z}\times\mathbb{R}$ of i, it follows from Corollary 2.5.8 that $\mathcal{N}_i$ is of the form $\pi\times\mathbb{R}\colon \mathcal{N}\times\mathbb{R}\to\mathcal{Z}\times\mathbb{R}$ for some vector bundle $\pi\colon\mathcal{N}\to\mathcal{Z}$ over $\mathcal{Z}$. As $\mathcal{N}_{i_r}$ is obtained by pulling back $\mathcal{N}_i$ along the inclusion $\mathcal{Z}\times\{r\}\hookrightarrow\mathcal{Z}\times\mathbb{R}$, we obtain an isomorphism $\mathcal{N}_{i_r}\cong\mathcal{N}$ for every $r\in\mathbb{R}$, and thus we may define α as the composite $\mathcal{N}_{i_r}\cong\mathcal{N}\cong\mathcal{N}_{i_0}$.

Next consider the Pontryagin-Thom collapse map for the embedding i relative to $\mathcal{S}\times\mathbb{R}$, which takes the form

$$\mathrm{PT}_{\mathcal{S}\times\mathbb{R}}(i)\colon h_{\mathcal{S}\times\mathbb{R}}(\mathcal{X}\times\mathbb{R})\to\mathrm{Th}_{\mathcal{S}\times\mathbb{R}}(\mathcal{N}_i,\mathcal{Z}\times\mathbb{R}) \quad \in \quad \mathcal{C}(\mathcal{S}\times\mathbb{R}).$$

Letting $\mathrm{pr}\colon\mathcal{S}\times\mathbb{R}\to\mathcal{S}$ denote the projection, observe that both source and target of this map lie in the image of the fully faithful functor $\mathrm{pr}^*\colon\mathcal{C}(\mathcal{S})\hookrightarrow\mathcal{C}(\mathcal{S}\times\mathbb{R})$, so that the map $\mathrm{PT}_{\mathcal{S}\times\mathbb{R}}(i)$ is of the form $\mathrm{pr}^*(\varphi)$ for some morphism $\varphi\colon h_{\mathcal{S}}(\mathcal{X})\to\mathrm{Th}_{\mathcal{S}}(\mathcal{N},\mathcal{Z})$ in $\mathcal{C}(\mathcal{S})$. Applying Lemma 6.2.20 to the inclusion $\mathcal{S}=\mathcal{S}\times\{r\}\hookrightarrow\mathcal{S}\times\mathbb{R}$ and using that this map is a section of the projection map pr, we obtain for every $r\in\mathbb{R}$ a commutative square as follows:

$$\begin{tikzcd} h_{\mathcal{S}}(\mathcal{X}) \arrow[r, equal] \arrow[d, "\mathrm{PT}_{\mathcal{S}}(i_r)"'] & h_{\mathcal{S}}(\mathcal{X}) \arrow[d, "\varphi"] \\ \mathrm{Th}_{\mathcal{S}}(\mathcal{N}_{i_r},\mathcal{Z}) \arrow[r, "\simeq"] & \mathrm{Th}_{\mathcal{S}}(\mathcal{N},\mathcal{Z}). \end{tikzcd}$$

Combining this square with the analogous square for $r=0$ then proves the claim. $\square$

6.3 Kernel operators

We continue fixing a pullback formalism C satisfying the Voevodsky conditions. In this section, we will study the behavior of functors $C(\mathcal{X}) \to C(\mathcal{Y})$ of a specific form, which we will call *kernel operators*. As in the previous section, we fix a base stack $\mathcal{S}$ over which all other stacks live.

Definition 6.3.1 (Kernel operator)**.** Let $\mathcal{X}, \mathcal{Y} \in \mathrm{Sub}_{/\mathcal{S}}$ and let $D \in C(\mathcal{X} \times_{\mathcal{S}} \mathcal{Y})$. We define the *kernel operator F_D of D (relative to $\mathcal{S}$)* as the functor $F_D \colon C(\mathcal{X}) \to C(\mathcal{Y})$ given as the following composite:

$$F_D : C(\mathcal{X}) \xrightarrow{(\mathrm{pr}_{\mathcal{X}})^*} C(\mathcal{X} \times_{\mathcal{S}} \mathcal{Y}) \xrightarrow{-\otimes D} C(\mathcal{X} \times_{\mathcal{S}} \mathcal{Y}) \xrightarrow{(\mathrm{pr}_{\mathcal{Y}})_\sharp} C(\mathcal{Y}).$$

A functor $F \colon C(\mathcal{X}) \to C(\mathcal{Y})$ is called a *kernel operator (relative to $\mathcal{S}$)* if it comes equipped with an equivalence $F \simeq F_D$ for some $D \in C(\mathcal{X} \times_{\mathcal{S}} \mathcal{Y})$. The object D is called the *kernel* of F.

If $D' \in C(\mathcal{X} \times_{\mathcal{S}} \mathcal{Y})$ is another kernel, then every morphism $\alpha \colon D \to D'$ induces a natural transformation $F_\alpha \colon F_D \to F_{D'}$ on kernel operators.

Warning 6.3.2. It is not necessarily true that the functor F_D uniquely determines the object D. For this reason, the kernel D is required as data.

The notion of a kernel operator is formally analogous to the notion of a *Fourier-Mukai transform* in algebraic geometry, where one would instead consider functors of the form $(\mathrm{pr}_{\mathcal{Y}})_*((\mathrm{pr}_{\mathcal{X}})^*(-) \otimes D) \colon C(\mathcal{X}) \to C(\mathcal{Y})$, using the right adjoint $(\mathrm{pr}_{\mathcal{Y}})_*$ rather than the left adjoint $(\mathrm{pr}_{\mathcal{Y}})_\sharp$. All the results below on kernel operators are straightforward adaptations of well-known results for Fourier-Mukai transforms. A similar discussion appears in [FS21, p.263].

The following are the main examples of kernel operators that we will use.

Example 6.3.3 (Pullback)**.** Let $f \colon \mathcal{X} \to \mathcal{S}$ be a representable submersion and let $\mathcal{Y} = \mathcal{S}$. Then the functor $f^* \colon C(\mathcal{S}) \to C(\mathcal{X})$ is a kernel operator relative to $\mathcal{S}$, with kernel $\mathbb{1}_{\mathcal{X}} \in C(\mathcal{X})$.

Example 6.3.4 (Pushforward)**.** Let $g \colon \mathcal{Y} \to \mathcal{S}$ be a representable submersion and let $\mathcal{X} = \mathcal{S}$. Then the functor $g_\sharp \colon C(\mathcal{Y}) \to C(\mathcal{S})$ is a kernel operator relative to $\mathcal{S}$, with kernel $\mathbb{1}_{\mathcal{Y}} \in C(\mathcal{Y})$.

Example 6.3.5. Let $f: \mathcal{X} \to \mathcal{S}$ be a representable submersion. Then the composite $f^* f_\sharp : C(\mathcal{X}) \to C(\mathcal{X})$ is equivalent to $\mathrm{pr}_{1\sharp}\mathrm{pr}_2^* : C(\mathcal{X}) \to C(\mathcal{X})$ by smooth base change and thus is a kernel operator with kernel $\mathbb{1}_{\mathcal{X}\times_{\mathcal{S}}\mathcal{X}} \in C(\mathcal{X} \times_{\mathcal{S}} \mathcal{X})$. Given the previous two examples, this may also be deduced from Lemma 6.3.9 below.

Example 6.3.6 (Tensor product)**.** Letting $\mathcal{X} = \mathcal{Y} = \mathcal{S}$, every object $D \in C(\mathcal{S})$ gives rise to a kernel operator $F_D = - \otimes D : C(\mathcal{S}) \to C(\mathcal{S})$.

Example 6.3.7 (Suspension by a vector bundle)**.** Let $\pi: \mathcal{E} \to \mathcal{S}$ be a vector bundle and let $s: \mathcal{S} \to \mathcal{E}$ be its zero section. We define the *suspension functor* $\Sigma^{\mathcal{E}}$ as

$$\Sigma^{\mathcal{E}} := \pi_\sharp s_* : C(\mathcal{S}) \to C(\mathcal{S}).$$

Note that $\Sigma^{\mathcal{E}} \mathbb{1}_{\mathcal{S}} = \mathrm{Th}_{\mathcal{S}}(\mathcal{E}, \mathcal{S})$, which by Corollary 6.2.6 is equivalent to the sphere bundle $S^{\mathcal{E}}$. By the smooth and closed projection formulas, it then follows that there is a natural equivalence

$$\Sigma^{\mathcal{E}} \simeq - \otimes S^{\mathcal{E}},$$

so that $\Sigma^{\mathcal{E}}$ is a kernel functor relative to $\mathcal{S}$.

Example 6.3.8 (Twist functor)**.** Let $f: \mathcal{Y} \to \mathcal{S}$ be a representable submersion and consider its dualizing object $\omega_f \in C(\mathcal{Y})$ from Definition 6.1.3. We claim that the twist functor

$$- \otimes \omega_f : C(\mathcal{Y}) \to C(\mathcal{Y})$$

is a kernel operator over $\mathcal{S}$ with kernel $\Delta_* \mathbb{1}_{\mathcal{Y}} \in C(\mathcal{Y} \times_{\mathcal{S}} \mathcal{Y})$. Indeed, by the smooth and closed projection formulas we obtain natural equivalences

$$\begin{aligned} A \otimes \omega_f = A \otimes \mathrm{pr}_{1\sharp}\Delta_* \mathbb{1}_{\mathcal{Y}} &\simeq \mathrm{pr}_{1\sharp}(\mathrm{pr}_1^* A \otimes \Delta_* \mathbb{1}_{\mathcal{Y}}) \\ &\simeq \mathrm{pr}_{1\sharp}\Delta_*(\Delta^* \mathrm{pr}_1^* A \otimes \mathbb{1}_{\mathcal{Y}}) \\ &\simeq \mathrm{pr}_{1\sharp}\Delta_*(\Delta^* \mathrm{pr}_2^* A \otimes \mathbb{1}_{\mathcal{Y}}) \\ &\simeq \mathrm{pr}_{1\sharp}(\mathrm{pr}_2^* A \otimes \Delta_* \mathbb{1}_{\mathcal{Y}}) = F_{\Delta_* \mathbb{1}_{\mathcal{Y}}}(A), \end{aligned}$$

where we use that $\mathrm{pr}_1 \circ \Delta = \mathrm{pr}_2 \circ \Delta$ as both are the identity on $\mathcal{Y}$.

Kernel operators behave well under composition:

Lemma 6.3.9. *Consider objects $\mathcal{X}, \mathcal{Y}, \mathcal{Z} \in \mathrm{Sub}_{/\mathcal{S}}$ and consider kernels $D \in C(\mathcal{X} \times_{\mathcal{S}} \mathcal{Y})$ and $D' \in C(\mathcal{Y} \times_{\mathcal{S}} \mathcal{Z})$. Then the composite $F_{D'} \circ F_D : C(\mathcal{X}) \to C(\mathcal{Z})$ is again a kernel operator, with kernel given by*

$$D' \odot D := (\mathrm{pr}_{13})_\sharp(\mathrm{pr}_{12}^* D \otimes \mathrm{pr}_{23}^* D') \quad \in \quad C(\mathcal{X} \times_{\mathcal{S}} \mathcal{Z}),$$

where $\mathrm{pr}_{12}\colon \mathcal{X}\times_{\mathcal{S}}\mathcal{Y}\times_{\mathcal{S}}\mathcal{Z}\to\mathcal{X}\times_{\mathcal{S}}\mathcal{Y}$ *denotes the projection on the first two factors, and similarly for* pr_{23} *and* pr_{13}.

Proof. This is a straightforward computation using smooth base change and the smooth projection formula, which we will leave to the reader. See Lemma I.3.24 in Part I for an analogous computation. □

Lemma 6.3.10. *Let* $\mathcal{X},\mathcal{Y},\mathcal{Z},\mathcal{W}\in\mathrm{Sub}_{/\mathcal{S}}$ *and consider a kernel* $D\in C(\mathcal{X}\times_{\mathcal{S}}\mathcal{Y})$ *with associated kernel operator* $F_D\colon C(\mathcal{X})\to C(\mathcal{Y})$.

(1) For a representable submersion $f\colon\mathcal{Y}\to\mathcal{Z}$*, the composite* $f_\sharp\circ F_D\colon C(\mathcal{X})\to C(\mathcal{Z})$ *is again a kernel operator with kernel* $(\mathcal{X}\times f)_\sharp D\in C(\mathcal{X}\times_{\mathcal{S}}\mathcal{Z})$.

(2) For a representable submersion $g\colon\mathcal{Z}\to\mathcal{Y}$*, the composite* $g^*\circ F_D\colon C(\mathcal{X})\to C(\mathcal{Z})$ *is again a kernel operator with kernel* $(\mathcal{X}\times g)^*D\in C(\mathcal{X}\times_{\mathcal{S}}\mathcal{Z})$.

(3) For a representable submersion $h\colon\mathcal{W}\to\mathcal{X}$*, the composite* $F_D\circ h_\sharp\colon C(\mathcal{W})\to C(\mathcal{Y})$ *is again a kernel operator with kernel* $(h\times\mathcal{Y})^*D\in C(\mathcal{W}\times_{\mathcal{S}}\mathcal{Y})$.

(4) For a representable submersion $k\colon\mathcal{X}\to\mathcal{W}$*, the composite* $F_D\circ k^*\colon C(\mathcal{W})\to C(\mathcal{Y})$ *is again a kernel operator with kernel* $(k\times\mathcal{Y})_\sharp D\in C(\mathcal{W}\times_{\mathcal{S}}\mathcal{Y})$.

Proof. This is a straightforward computation using smooth base change and the smooth projection formula, which we will leave to the reader. □

Warning 6.3.11. We warn the reader that the identity $\mathrm{id}_{C(\mathcal{X})}\colon C(\mathcal{X})\to C(\mathcal{X})$ is not necessarily a kernel operator relative to $\mathcal{S}$, unlike in the situation for Fourier-Mukai transforms. The problem is that the diagonal $\Delta\colon\mathcal{X}\to\mathcal{X}\times_{\mathcal{S}}\mathcal{X}$ of $\mathcal{X}$ over $\mathcal{S}$ is usually not a representable submersion. If it is, then the identity *is* a kernel operator with kernel given by $\Delta_\sharp\mathbb{1}_{\mathcal{X}}\in C(\mathcal{X}\times_{\mathcal{S}}\mathcal{X})$.

As a consequence, it is not necessarily true that for a representable submersion $f\colon\mathcal{X}\to\mathcal{Y}$ over $\mathcal{S}$, the functors $f_\sharp\colon C(\mathcal{X})\to C(\mathcal{Y})$ and $f^*\colon C(\mathcal{Y})\to C(\mathcal{X})$ are kernel functors relative to $\mathcal{S}$.

6.4 Proof of Main Theorem

In this section, we will prove relative Poicaré duality for separated differentiable stacks, Theorem 6.1.7. Recall the setup: we are given a pullback formalism C satisfying the

Voevodsky conditions and a proper representable submersion $f\colon \mathcal{Y} \to \mathcal{X}$ of separated differentiable stacks. Our goal is to show that the natural transformation

$$\mathfrak{p}_f\colon f_\sharp(-) \Rightarrow f_*(- \otimes \omega_f),$$

constructed in Construction 6.1.5, is an equivalence. The proof will consist of two steps: in Subsection 6.4.1 we prove the theorem in the special case where $\mathcal{Y}$ is a closed substack of a vector bundle over $\mathcal{X}$, and in Section 6.4.2 we argue how to reduce the general case to this special case.

6.4.1 A special case

We will start by proving relative Poincaré duality under the following additional assumption on the proper submersion f:

(A) The stack $\mathcal{Y}$ is a closed substack of a vector bundle over $\mathcal{X}$: there is a commutative diagram

$$\begin{tikzcd} \mathcal{Y} \arrow[r, hook, "i"] \arrow[dr, "f"'] & \mathcal{E} \arrow[d, "\pi"] \\ & \mathcal{X}, \end{tikzcd}$$

where i is a closed embedding and π is a vector bundle.

Our proof strategy in this case will be as follows: we will use the embedding i to construct explicit unit and counit transformations that exhibit the functor $f_\sharp(- \otimes \omega_f^{-1})$ as a right adjoint of f^*, and then show that the resulting equivalence $f_* \simeq f_\sharp(- \otimes \omega_f^{-1})$ is compatible with the Poincaré duality map $\mathfrak{p}_f$.

Let $\pi_i\colon \mathcal{N}_i \to \mathcal{Y}$ denote the normal bundle of the embedding i and let $\pi_\Delta\colon \mathcal{N}_\Delta \to \mathcal{Y}$ denote the normal bundle of the diagonal map $\Delta = \Delta_f\colon \mathcal{Y} \to \mathcal{Y} \times_\mathcal{X} \mathcal{Y}$. Via Example 6.3.7, these bundles give rise to suspension functors $\Sigma^{N_\Delta}, \Sigma^{N_i}\colon C(\mathcal{Y}) \to C(\mathcal{Y})$. The following lemma shows that their composite is given by suspension along the pullback bundle $f^*\pi\colon f^*\mathcal{E} \to \mathcal{Y}$. We thank Marc Hoyois for suggesting an argument which greatly simplified our original proof.

Lemma 6.4.1. *There is a short exact sequence of vector bundles over $\mathcal{Y}$ of the form $\mathcal{N}_\Delta \to f^*\mathcal{E} \to \mathcal{N}_i$. In particular, there is an equivalence $S^{f^*\mathcal{E}} \simeq S^{N_\Delta} \otimes S^{N_i}$ in $C(\mathcal{Y})$, giving rise to natural equivalences of functors*

$$\Sigma^{N_i}\Sigma^{N_\Delta} \simeq \Sigma^{f^*\mathcal{E}} \simeq \Sigma^{N_\Delta}\Sigma^{N_i}\colon C(\mathcal{Y}) \to C(\mathcal{Y}).$$

Proof. Observe that the second statement follows from the first using Lemma 6.2.12 and Corollary 6.2.6, while the final statement then follows from the fact that $\Sigma^{\mathcal{E}}\colon \mathcal{C}(\mathcal{X}) \to \mathcal{C}(\mathcal{X})$ is given by tensoring with $S^{\mathcal{E}} \in \mathcal{C}(\mathcal{X})$ for a vector bundle $\mathcal{E}$ over $\mathcal{X}$, see Example 6.3.7. It thus remains to prove the first statement.

Consider the embedding $i'\colon \mathcal{Y} \hookrightarrow S^{\mathcal{E}}$, given as the composite of $i\colon \mathcal{Y} \hookrightarrow \mathcal{E}$ with the inclusion $\mathcal{E} \hookrightarrow S^{\mathcal{E}}$. Since the map f is proper and $S^{\mathcal{E}} \to \mathcal{X}$ is representable, it follows from Lemma 2.4.10 that the map i' is proper, and thus it is a closed embedding. Consider now the composite

$$(i',\mathrm{id})\colon \mathcal{Y} \overset{\Delta}{\hookrightarrow} \mathcal{Y} \times_{\mathcal{X}} \mathcal{Y} \xhookrightarrow{i' \times_{\mathcal{X}} \mathcal{Y}} S^{\mathcal{E}} \times_{\mathcal{X}} \mathcal{Y}.$$

Observe that the normal bundle of the map $i' \times_{\mathcal{X}} \mathcal{Y}$ is $\mathcal{N}_i \times_{\mathcal{X}} \mathcal{Y}$, and the pullback of this bundle along the diagonal Δ is simply the map $\pi_i\colon \mathcal{N}_i \to \mathcal{Y}$. By Lemma 3.5.16, we thus obtain a short exact sequence

$$\mathcal{N}_\Delta \to \mathcal{N}_{(i',\mathrm{id})} \to \mathcal{N}_i$$

of vector bundles over $\mathcal{Y}$. It thus remains to show that the normal bundle of the map (i',id) is isomorphic to $f^*\mathcal{E}$.

Letting $s_0\colon \mathcal{X} \to \mathcal{E}$ denote the zero-section of $\pi\colon \mathcal{E} \to \mathcal{X}$, observe that the embedding $i\colon \mathcal{Y} \hookrightarrow \mathcal{E}$ is homotopic to the *zero map* $0\colon \mathcal{Y} \xrightarrow{f} \mathcal{X} \xrightarrow{s_0} \mathcal{E} \hookrightarrow S^{\mathcal{E}}$ via the straight-line homotopy $\mathcal{Y} \times \mathbb{R} \to S^{\mathcal{E}}, (y,r) \mapsto r \cdot i(y)$. It follows that the map (i',id) is isotopic to the closed embedding $(0,\mathrm{id})\colon \mathcal{Y} \hookrightarrow S^{\mathcal{E}} \times_{\mathcal{X}} \mathcal{Y}$. Lemma 3.5.18 thus provides an isomorphism of normal bundles

$$\mathcal{N}_{(i',\mathrm{id})} \cong \mathcal{N}_{(0,\mathrm{id})} \quad \in \quad \mathrm{Vect}(\mathcal{Y}).$$

Since $(0,\mathrm{id})$ is the zero-section of the vector bundle $f^*\pi\colon f^*\mathcal{E} \to \mathcal{Y}$, its normal bundle is simply given by the map $f^*\mathcal{E} \to \mathcal{Y}$, finishing the proof. □

We now move to the construction of the aforementioned unit and counit transformations.

Construction 6.4.2. We define a natural transformations $\varepsilon\colon f^*f_\sharp \to \Sigma^{\mathcal{N}_\Delta}$ and $\eta\colon \Sigma^{\mathcal{E}} \to f_\sharp \Sigma^{\mathcal{N}_i} f^*$:

- For ε, recall from Example 6.3.5 and Example 6.3.8 that the functors $f^*f_\sharp$ and $\Sigma^{\mathcal{N}_\Delta} \simeq \mathrm{pr}_{1\sharp}\Delta_*$ are kernel functors relative to $\mathcal{X}$, in the sense of Definition 6.3.1, with kernels given by with kernels given by $\mathbb{1}_{\mathcal{Y}\times_{\mathcal{X}}\mathcal{Y}}$ and $\Delta_*\mathbb{1}_{\mathcal{Y}}$, respectively. We define ε as the transformation induced by the unit map $u_\Delta^*\colon \mathbb{1}_{\mathcal{Y}\times_{\mathcal{X}}\mathcal{Y}} \to \Delta_*\Delta^*\mathbb{1}_{\mathcal{Y}\times_{\mathcal{X}}\mathcal{Y}} = \Delta_*\mathbb{1}_{\mathcal{Y}}$ in $\mathcal{C}(\mathcal{Y} \times_{\mathcal{X}} \mathcal{Y})$. More explicitly, ε is given by the following composite:

$$f^*f_\sharp \simeq \mathrm{pr}_{1\sharp}\mathrm{pr}_2^* \xrightarrow{u_\Delta^*} \mathrm{pr}_{1\sharp}\Delta_*\Delta^*\mathrm{pr}_2^* \simeq \mathrm{pr}_{1\sharp}\Delta_* \simeq \Sigma^{\mathcal{N}_\Delta} \quad \in \quad \mathrm{Fun}(\mathcal{C}(\mathcal{Y}),\mathcal{C}(\mathcal{Y})).$$

- For η, its source and target are also kernel functors relative to $\mathcal{X}$, as they are given by tensoring with the objects $S^{\mathcal{E}}$ and $f_\sharp S^{\mathcal{N}_i}$, respectively. We define η as the transformation induced by the Pontryagin-Thom map $\mathrm{PT}(i')$ of the closed embedding $i'\colon \mathcal{Y} \hookrightarrow S^{\mathcal{E}}$:

$$\mathrm{PT}(i')\colon S^{\mathcal{E}} \to \mathrm{Th}_{\mathcal{X}}(\mathcal{N}_{i'}, \mathcal{Y}) = \mathrm{Th}_{\mathcal{X}}(\mathcal{N}_i, \mathcal{Y}) \simeq f_\sharp S^{\mathcal{N}_i}.$$

The following result is the main ingredient for relative Poincaré duality in the special case.

Proposition 6.4.3 (cf. [Hoy17, Theorem 5.22])**.** *The composites*

$$f^*\Sigma^{\mathcal{E}} \xrightarrow{f^*\eta} f^* f_\sharp \Sigma^{\mathcal{N}_i} f^* \xrightarrow{\varepsilon\Sigma^{\mathcal{N}_i} f^*} \Sigma^{\mathcal{N}_\Delta}\Sigma^{\mathcal{N}_i} f^* \overset{6.4.1}{\simeq} \Sigma^{f^*\mathcal{E}} f^* \simeq f^*\Sigma^{\mathcal{E}} \tag{II.6.1}$$

$$\Sigma^{\mathcal{E}} f_\sharp \xrightarrow{\eta f_\sharp} f_\sharp \Sigma^{\mathcal{N}_i} f^* f_\sharp \xrightarrow{f_\sharp\Sigma^{\mathcal{N}_i}\varepsilon} f_\sharp\Sigma^{\mathcal{N}_i}\Sigma^{\mathcal{N}_\Delta} \overset{6.4.1}{\simeq} f_\sharp \Sigma^{f^*\mathcal{E}} \simeq \Sigma^{\mathcal{E}} f_\sharp \tag{II.6.2}$$

are homotopic to the identity.

Proof. As ε and η are defined as transformations between kernel functors induced by a map on kernels, it follows from Lemma 6.3.9 that also each of the functors appearing in (II.6.1) and (II.6.2) are kernel functors relative to $\mathcal{X}$ and that each of the transformations is induced by a morphism on kernels. It will thus suffice to show that the composites at the level of the kernels are homotopic to the respective identities.

It follows from Lemma 6.3.10 that the transformations $f^*\eta$ and $\eta f_\sharp$ are both induced at the level of kernels by the morphism $f^*\mathrm{PT}(i')\colon f^*S^{\mathcal{E}} \to f^*\mathrm{Th}_{\mathcal{X}}(\mathcal{N}_i, \mathcal{Y})$. Similarly, it follows from Lemma 6.3.9 that the transformation $\varepsilon\Sigma^{\mathcal{N}_i f^*}$ is induced at the level of kernels by the composite

$$f^* f_\sharp S^{\mathcal{N}_i} \simeq \mathrm{pr}_{1\sharp}\mathrm{pr}_2^* S^{\mathcal{N}_i} \xrightarrow{u_\Delta^*} \mathrm{pr}_{1\sharp}\Delta_*\Delta^*\mathrm{pr}_2^* S^{\mathcal{N}_i} \simeq S^{\mathcal{N}_\Delta} \otimes S^{\mathcal{N}_i},$$

and the transformation $f_\sharp\Sigma^{\mathcal{N}_i}\varepsilon$ is induced by the analogous map $f^*f_\sharp S^{\mathcal{N}_i} \to S^{\mathcal{N}_\Delta} \otimes S^{\mathcal{N}_i}$ where the roles of the two projection maps pr_1 and pr_2 are swapped. By symmetry, it will thus suffice to prove the statement for the composite (II.6.1). To this end, consider the following diagram:

$$\begin{tikzcd}
f^*S^{\mathcal{E}} \arrow[r, "f^*\mathrm{PT}_{\mathcal{X}}(i')"] \arrow[d, "\simeq"'] \arrow[dr, phantom, "(1)"] & f^*\mathrm{Th}_{\mathcal{X}}(\mathcal{N}_i,\mathcal{Y}) \arrow[r, "\simeq"] \arrow[d, "\simeq"'] & f^*f_\sharp S^{\mathcal{N}_i} \arrow[r, "\varepsilon"] \arrow[d, phantom, "(2)"] & S^{\mathcal{N}_\Delta}\otimes S^{\mathcal{N}_i} \arrow[d, "\simeq"] \\
S^{\mathcal{E}}\times_{\mathcal{X}}\mathcal{Y} \arrow[r, "\mathrm{PT}_{\mathcal{Y}}(f^*i')"] \arrow[d, equal] & \mathrm{Th}_{\mathcal{Y}}(\mathcal{N}_i\times_{\mathcal{X}}\mathcal{Y},\mathcal{Y}\times_{\mathcal{X}}\mathcal{Y}) \arrow[rr, "\mathrm{PT}(\Delta,f^*i')"] \arrow[d, phantom, "(3)"] & & S^{\mathcal{N}_{(i',\mathrm{id})}} \arrow[d, equal] \\
S^{\mathcal{E}}\times_{\mathcal{X}}\mathcal{Y} \arrow[rrr, "\mathrm{PT}((i',\mathrm{id}))"] \arrow[d, equal] & \arrow[d, phantom, "(4)"] & & S^{\mathcal{N}_{(i',\mathrm{id})}} \arrow[d, "\simeq"] \\
S^{\mathcal{E}}\times_{\mathcal{X}}\mathcal{Y} \arrow[rrr, "\mathrm{PT}((0,\mathrm{id}))"'] & & & S^{f^*\mathcal{E}}.
\end{tikzcd}$$

Note that by Example 6.2.17 the bottom map is the canonical equivalence, as the map $(0,\mathrm{id})\colon \mathcal{Y}\to\mathcal{E}\times_{\mathcal{X}}\mathcal{Y}$ is the zero-section of the vector bundle $f^*\mathcal{E}$ over $\mathcal{Y}$. Also note that the right vertical composite $S^{\mathcal{N}_\Delta}\otimes S^{\mathcal{N}_i}\simeq S^{f^*\mathcal{E}}$ is precisely the equivalence from Lemma 6.4.1. It follows that the composite around the top, right and bottom is the map on kernels inducing the composite (II.6.1). Since the left side of the diagram is the identity, it thus remains to show that the diagram commutes up to homotopy. The square (1) commutes by Lemma 6.2.20. Square (3) commutes by Lemma 6.2.19. Square (4) commutes by Lemma 6.2.21. It thus remains to show that square (2) commutes.

From the description for the map ε obtained before, we see that we may write the square (2) as follows:

$$\begin{tikzcd}
h_{\mathcal{Y}}(\mathcal{Y}\times_{\mathcal{X}}\mathcal{Y})\otimes\mathrm{Th}_{\mathcal{Y}}(\mathcal{N}_i,\mathcal{Y}) \arrow[r, "\mathrm{PT}_{\mathcal{Y}}(\Delta)\otimes\mathrm{id}"] \arrow[d, "\simeq"'] & \mathrm{Th}_{\mathcal{Y}}(\mathcal{N}_\Delta,\mathcal{Y})\otimes\mathrm{Th}_{\mathcal{Y}}(\mathcal{N}_i,\mathcal{Y}) \arrow[r, "\simeq"] \arrow[d, phantom, "(2)"] & S^{\mathcal{N}_\Delta}\otimes S^{\mathcal{N}_i} \arrow[d, "\simeq"] \\
\mathrm{Th}_{\mathcal{Y}}(\mathcal{N}_i\times_{\mathcal{X}}\mathcal{Y},\mathcal{Y}\times_{\mathcal{X}}\mathcal{Y}) \arrow[rr, "\mathrm{PT}_{\mathcal{Y}}(\Delta,f^*i')"] & & S^{\mathcal{N}_{(i,\mathrm{id})}},
\end{tikzcd}$$

where the left equivalence is the equivalence from Lemma 6.2.12 using the identification of $\mathcal{N}_i\times_{\mathcal{X}}\mathcal{Y}$ with $\mathcal{N}_i\times_{\mathcal{Y}}(\mathcal{Y}\times_{\mathcal{X}}\mathcal{Y})$. Both of the Pontryagin-Thom maps $\mathrm{PT}_{\mathcal{Y}}(\Delta)$ and $\mathrm{PT}_{\mathcal{Y}}(\Delta,f^*i')$ start with the quotient map u_Δ^*. On the top, we use a choice of tubular neighborhood $\mathcal{U}_\Delta$ for the diagonal embedding Δ, while on the bottom we use a choice of tubular neighborhood $\mathcal{U}_{(i,\mathrm{id})}$ for the embedding $(i,\mathrm{id})\colon\mathcal{Y}\hookrightarrow\mathcal{E}\times_{\mathcal{X}}\mathcal{Y}$. We thus need to show the commutativity of the following diagram:

$$\begin{tikzcd}
\mathrm{Th}_{\mathcal{Y}}(\mathcal{Y}\times_{\mathcal{X}}\mathcal{Y},\mathcal{Y})\otimes\mathrm{Th}_{\mathcal{Y}}(N_i,\mathcal{Y}) \arrow[d, "(5)"', "\simeq"] & \mathrm{Th}_{\mathcal{Y}}(\mathcal{U}_\Delta,\mathcal{Y})\otimes\mathrm{Th}_{\mathcal{Y}}(N_i,\mathcal{Y}) \arrow[l, "(1)"', "\simeq"] & \mathrm{Th}_{\mathcal{Y}}(N_\Delta,\mathcal{Y})\otimes\mathrm{Th}_{\mathcal{Y}}(N_i,\mathcal{Y}) \arrow[l, "(2)"', "\simeq"] \arrow[d, "\simeq"', "(6)"] \\
\mathrm{Th}_{\mathcal{Y}}(\mathcal{N}_i\times_{\mathcal{X}}\mathcal{Y},\mathcal{Y}) \arrow[r, "(3)", "\simeq"'] & \mathrm{Th}_{\mathcal{Y}}(\mathcal{U}_{(i,\mathrm{id})},\mathcal{Y}) \arrow[r, "(4)", "\simeq"'] & \mathrm{Th}_{\mathcal{Y}}(N_{(i,\mathrm{id})},\mathcal{Y}).
\end{tikzcd}$$

Each of the six equivalences appearing in the above square are specific instances of double Beck-Chevalley maps. The equivalences labeled (1)-(4) are obtained from Lemma 6.2.7, and are thus given by the double Beck-Chevalley maps $\mathrm{BC}_{\sharp,*}$ associated to the following four pullback squares:

$$\begin{tikzcd}
\mathcal{Y} \arrow[r, equal] \arrow[d, hook, "\Delta"'] \arrow[dr, phantom, "(1)"] & \mathcal{Y} \arrow[r, equal] \arrow[d, hook] \arrow[dr, phantom, "(2)"] & \mathcal{Y} \arrow[d, hook, "s_0"] \\
\mathcal{Y} \times_{\mathcal{X}} \mathcal{Y} & \mathcal{U}_{\Delta} \arrow[l, hook'] \arrow[r, hook] & \mathcal{N}_{\Delta}
\end{tikzcd} \quad \text{and} \quad \begin{tikzcd}
\mathcal{Y} \arrow[r, equal] \arrow[d, hook, "{(i',\mathrm{id})}"'] \arrow[dr, phantom, "(3)"] & \mathcal{Y} \arrow[r, equal] \arrow[d, hook] \arrow[dr, phantom, "(4)"] & \mathcal{Y} \arrow[d, hook] \\
\mathcal{N}_i \times_{\mathcal{X}} \mathcal{Y} & \mathcal{U}_{(i',\mathrm{id})} \arrow[l, hook'] \arrow[r, hook] & \mathcal{N}_{(i',\mathrm{id})}.
\end{tikzcd}$$

The two vertical equivalences are both instances of the multiplicativity of Thom spaces, Lemma 6.2.12, which in turn are obtained from the double Beck-Chevalley equivalences $\mathrm{BC}_{\sharp,*}$ associated to the following two pullback squares:

$$\begin{tikzcd}
\mathcal{Y} \times_{\mathcal{X}} \mathcal{Y} \arrow[r, hook, "{(s_0,\mathrm{id})}"] \arrow[d, "\mathrm{pr}_1"'] \arrow[dr, phantom, "(5)"] & \mathcal{N}_i \times_{\mathcal{X}} \mathcal{Y} \arrow[d, "\mathrm{pr}_1"] \\
\mathcal{Y} \arrow[r, hook, "s_0"'] & \mathcal{N}_i
\end{tikzcd} \quad \text{and} \quad \begin{tikzcd}
\mathcal{N}_{\Delta} \arrow[r, hook] \arrow[d, "\pi_{\Delta}"'] \arrow[dr, phantom, "(6)"] & \mathcal{N}_{(i,\mathrm{id})} \arrow[d] \\
\mathcal{Y} \arrow[r, hook, "s_0"'] & \mathcal{N}_i.
\end{tikzcd}$$

The trick now is to observe that the tubular neighborhood $\mathcal{U}_{\Delta}$ of Δ can be chosen as a pullback along $s_0 \colon \mathcal{Y} \hookrightarrow \mathcal{N}_i$ of a tubular neighborhood $\mathcal{U}_{(i,\mathrm{id})}$ of (i,id). To see this, consider the following pullback square:

$$\begin{tikzcd}
\mathcal{Y} \arrow[r, hook, "\Delta"] \arrow[d, "s_0"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{Y} \times_{\mathcal{X}} \mathcal{Y} \arrow[d, "s_0 \times \mathrm{id}"] \\
\mathcal{N}_i \arrow[r, hook, "\Delta'"] & \mathcal{N}_i \times_{\mathcal{X}} \mathcal{Y};
\end{tikzcd}$$

here the map $\Delta' = (\mathrm{id}, \pi_i)$ is the identity in the first component and is the bundle projection $\pi_i \colon \mathcal{N}_i \to \mathcal{Y}$ in the second component. By Lemma 6.2.20, may obtain a tubular neighborhood $\mathcal{U}_{\Delta}$ of Δ relative to $\mathcal{Y}$ by pulling back a tubular neighborhood $\mathcal{U}_{\Delta'}$ of Δ' along the inclusion $s_0 \colon \mathcal{Y} \to \mathcal{N}_i$. We claim that the composite $\mathcal{N}_{\Delta} \to \mathcal{N}_i \xrightarrow{\pi_i} \mathcal{Y}$ is isomorphic to the normal bundle of the map (i,id). Indeed, note that the normal bundle of (i,id) is the same as that of $(s_0,\mathrm{id}) \colon \mathcal{Y} \to \mathcal{N}_i \times_{\mathcal{X}} \mathcal{Y}$, which is the diagonal composite in the previous pullback square. In turn, the normal bundle of (s_0,id) is isomorphic to the normal bundle of the composite $\mathcal{Y} \xrightarrow{s_0} \mathcal{N}_i \xrightarrow{s_0} \mathcal{N}_{\Delta}$, which is the zero-section of the composite bundle $\mathcal{N}_{\Delta} \to \mathcal{N}_i \xrightarrow{\pi_i} \mathcal{Y}$, thus showing the claim. It follows that $\mathcal{U}_{\Delta'}$ also serves as a tubular neighborhood $\mathcal{U}_{(i,\mathrm{id})}$ for (i,id).

All in all, we see that the six pullback squares (1)-(6) above all fit in a large pullback diagram of the following form:

$$
\begin{tikzcd}
\mathcal{Y} \arrow[rr, equal] \arrow[dr, hook, "\Delta"'] & & \mathcal{Y} \arrow[rr, equal] \arrow[dr, hook] & & \mathcal{Y} \arrow[dr, hook, "s_0"] & & \\
& \mathcal{Y}\times_{\mathcal{X}}\mathcal{Y} \arrow[dr, "\mathcal{Y}\times s_0"'] \arrow[dd, "\mathrm{pr}_1"'] & & \mathcal{U}_\Delta \arrow[ll, hook'] \arrow[rr, hook] \arrow[dr, hook] \arrow[dd] & & \mathcal{N}_\Delta \arrow[dr, hook, "s_0"] \arrow[dd, "\pi_\Delta"'] & \\
& & N_i\times_{\mathcal{X}}\mathcal{Y} \arrow[dd, "\mathrm{pr}_1"'] & & \mathcal{U}_{(i,\mathrm{id})} \arrow[ll, hook'] \arrow[rr, hook] \arrow[dd] & & \mathcal{N}_{(i,\mathrm{id})} \arrow[dd] \\
& \mathcal{Y} \arrow[rr, equal] \arrow[dr, hook, "s_0"] & & \mathcal{Y} \arrow[rr, equal] \arrow[dr, hook, "s_0"] & & \mathcal{Y} \arrow[dr, hook, "s_0"] & \\
& & \mathcal{N}_i \arrow[rr, equal] & & \mathcal{N}_i \arrow[rr, equal] & & \mathcal{N}_i.
\end{tikzcd}
$$

The result now follows from the pasting law for double Beck-Chevalley transformations, Lemma F.13. ◻

We are now ready to prove the special case of relative Poincaré duality, Theorem 6.1.7.

Proposition 6.4.4 (Special case of relative Poincaré duality)**.** *Consider a commutative diagram of separated differentiable stacks*

$$
\begin{tikzcd}
\mathcal{Y} \arrow[r, hook, "i"] \arrow[dr, "f"'] & \mathcal{E} \arrow[d, "\pi"] \\
& \mathcal{X},
\end{tikzcd}
$$

where f is a proper representable submersion, i is a closed embedding and π is a vector bundle. Then the Poincaré duality map $\mathfrak{p}_f\colon f_\sharp \to f_*(-\otimes\omega_f)$ is an equivalence.

Proof. We first show that the transformation $\varepsilon\colon f^*f_\sharp \to \Sigma^{\mathcal{N}_\Delta}$ adjoins over to an equivalence $f_\sharp \simeq f_*\circ\Sigma^{\mathcal{N}_\Delta}$. Equivalently, since $\Sigma^{\mathcal{N}_\Delta}$ is invertible, we may show that the induced map $\varepsilon\colon f^*f_\sharp\Sigma^{-\mathcal{N}_\Delta} \to \mathrm{id}$ exhibits the functor $f_\sharp\Sigma^{-\mathcal{N}_\Delta}\colon C(\mathcal{Y}) \to C(\mathcal{X})$ as a right adjoint to $f^*\colon C(\mathcal{X}) \to C(\mathcal{Y})$. Indeed, it is a direct consequence of the triangle identities from Proposition 6.4.3 that a compatible unit transformation $\mathrm{id} \to f_\sharp\Sigma^{-\mathcal{N}_\Delta}f^*$ is given by whiskering the transformation $\eta\colon \Sigma^{\mathcal{E}} \to f_\sharp\Sigma^{\mathcal{N}_i}f^*$ with $\Sigma^{-\mathcal{E}}\colon C(\mathcal{Y}) \to C(\mathcal{Y})$.

It thus remains to show that the transformation $\mathfrak{p}_f$ is obtained under adjunction from the transformation $\varepsilon\colon f^*f_\sharp \to \mathrm{pr}_{1\sharp}\Delta_* \simeq -\otimes\omega_f$. It follows from the explicit description of ε from Construction 6.4.2 that the mate transformation arising in this way is given by the following composite:

$$f_\sharp \xrightarrow{u_f^*} f_* f^* f_\sharp \simeq f_* \mathrm{pr}_{1\sharp} \mathrm{pr}_2^* \xrightarrow{u_\Delta^*} f_* \mathrm{pr}_{1\sharp} \Delta_* \Delta^* \mathrm{pr}_2^* \simeq f_* \mathrm{pr}_{1\sharp} \Delta_* \simeq f_*(- \otimes \omega_f).$$

To unwind the definition of $\mathfrak{p}_f$, recall that the double Beck-Chevalley transformation $\mathrm{BC}_{\sharp,*}$ is given by the composite

$$f_\sharp \mathrm{pr}_{2*} \xrightarrow{u_f^*} f_* f^* f_\sharp \mathrm{pr}_{2*} \simeq f_* \mathrm{pr}_{1\sharp} \mathrm{pr}_2^* \mathrm{pr}_{2*} \xrightarrow{c_{\mathrm{pr}_2}^*} f_* \mathrm{pr}_{1\sharp}.$$

Plugging this into the definition of $\mathfrak{p}_f$, we thus see that it unwinds to the composite

$$f_\sharp \xrightarrow{u_f^*} f_* f^* f_\sharp \simeq f_* \mathrm{pr}_{1\sharp} \mathrm{pr}_2^* \simeq f_* \mathrm{pr}_{1\sharp} \mathrm{pr}_2^* \mathrm{pr}_{2*} \Delta_* \xrightarrow{c_{\mathrm{pr}_2}^*} f_* \mathrm{pr}_{1\sharp} \Delta_* \simeq f_*(- \otimes \omega_f).$$

The claim thus follows from the commutativity of the following square:

$$\begin{tikzcd}
\mathrm{pr}_2^* \arrow[r, "u_\Delta^*"] \arrow[d, "\simeq"'] & \Delta_* \Delta^* \mathrm{pr}_2^* \arrow[d, "\simeq"] \\
\mathrm{pr}_2^* \mathrm{pr}_{2*} \Delta_* \arrow[r, "c_{\mathrm{pr}_2}^*"] & \Delta_*,
\end{tikzcd}$$

which holds by the triangle identity and the fact that the equivalence $\mathrm{id} \simeq (\mathrm{pr}_2 \Delta)_* \simeq \mathrm{pr}_{2*} \Delta_*$ can be taken to be the following composite:

$$\mathrm{id} \xrightarrow{\text{unit}} \mathrm{pr}_{2*} \mathrm{pr}_2^* \xrightarrow{\text{unit}} \mathrm{pr}_{2*} \Delta_* \Delta^* \mathrm{pr}_2^* \simeq \mathrm{pr}_{2*} \Delta_* (\mathrm{pr}_2 \circ \Delta)^* = \mathrm{pr}_{2*} \Delta_*.$$ □

6.4.2 The general case

We will now prove relative Poincaré duality for an arbitrary proper representable submersion $f\colon \mathcal{Y} \to \mathcal{X}$ of separated differentiable stacks. The key ingredient is the fact that the property of the Poincaré map $\mathfrak{p}_f$ to be an equivalence can be tested locally in the base stack $\mathcal{X}$, see Corollary 6.4.6 below, which lets us reduce to the special case treated in the previous subsection. This in turn is a consequence of the following compatibility between the Poincaré duality transformations and pullbacks of stacks:

Lemma 6.4.5 (Poincaré map commutes with base change)**.** *Consider a pullback square of differentiable stacks*

$$\begin{tikzcd}
\mathcal{Y}' \arrow[r, "h"] \arrow[d, "f'"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{Y} \arrow[d, "f"] \\
\mathcal{X}' \arrow[r, "g"] & \mathcal{X}
\end{tikzcd}$$

such that the morphisms f and f' are representable submersions.

(1) There is an equivalence

$$\alpha\colon \omega_{f'} \simeq h^*\omega_f \quad \in \quad \mathcal{C}(\mathcal{Y}').$$

(2) There is a commutative diagram

$$\begin{tikzcd}
f'_\sharp h^* \arrow[r, "\mathfrak{p}_{f'}"] \arrow[dd, "\mathrm{BC}_\sharp"'] & f'_*(h^*(-)\otimes\omega_{f'}) \\
& f'_* h^*(-\otimes\omega_f). \arrow[u, "\alpha"'] \\
g^* f_\sharp \arrow[r, "\mathfrak{p}_f"'] & g^* f_*(-\otimes\omega_f) \arrow[u, "\mathrm{BC}_*"']
\end{tikzcd}$$

Proof. Consider the following commutative diagram, in which all faces are pullback squares:

$$\begin{tikzcd}
& \mathcal{Y}'\times_{\mathcal{X}'}\mathcal{Y}' \arrow[rr, "k"] \arrow[dl, "p'_1"'] \arrow[dd, "p'_2"] & & \mathcal{Y}\times_{\mathcal{X}}\mathcal{Y} \arrow[dl, "p_1"'] \arrow[dd, "p_2"] \\
\mathcal{Y}' \arrow[rr, "h"] \arrow[dd, "f'"'] & & \mathcal{Y} \arrow[dd, "f"] & \\
& \mathcal{Y}' \arrow[rr, "h"] \arrow[dl, "f'"] & & \mathcal{Y} \arrow[dl, "f"] \\
\mathcal{X}' \arrow[rr, "g"'] & & \mathcal{X}. &
\end{tikzcd}$$

Part (1) follows from the following sequence of equivalences:

$$h^*\omega_f = h^*\mathrm{pr}_{1\sharp}\Delta_{f_*}\mathbb{1}_{\mathcal{Y}} \simeq \mathrm{pr}_{1\sharp}k^*\Delta_{f_*}\mathbb{1}_{\mathcal{Y}} \simeq \mathrm{pr}_{1\sharp}\Delta_* h^*\mathbb{1}_{\mathcal{Y}} = \mathrm{pr}_{1\sharp}\Delta_*\mathbb{1}_{\mathcal{Y}'} = \omega_{f'},$$

where we used smooth and closed base change associated to the above pullback squares.

For part (2), we consider the following large diagram:

$$
\begin{tikzcd}
f'_\sharp h^* \arrow[rrr, "\mathfrak{p}_{f'}"] \arrow[dddd, equal] \arrow[dr, "\simeq"] & & & f'_*(h^*(-)\otimes\omega_{p'}) \arrow[dl, "\simeq"'] \\
& f_\sharp p'_{2*}\Delta'_* h^* \arrow[r, "\mathrm{BC}_{\sharp,*}"] & f'_* p'_{1\sharp}\Delta'_* h^* & \\
& f'_\sharp p'_{2*} k^* \Delta_* \arrow[u, "\mathrm{BC}_*"] \arrow[r, "\mathrm{BC}_{\sharp,*}"] & f'_* p'_{1\sharp} k^* \Delta_* \arrow[u, "\mathrm{BC}_*"'] \arrow[d, "\mathrm{BC}_\sharp"'] & \\
\arrow[ur, phantom, "(1)"] & & & \arrow[ul, phantom, "(3)"] \\
f'_\sharp h^* \arrow[r, dashed, "\simeq"] \arrow[dd, "\mathrm{BC}_\sharp"'] & f'_\sharp h^* p_{2*}\Delta_* \arrow[uu, "\mathrm{BC}_*"] \arrow[d, "\mathrm{BC}_\sharp"] \arrow[r, phantom, "(2)"] & f'_* h^* p_{1\sharp}\Delta_* & f'_* h^*(-\otimes\omega_p) \arrow[l, dashed, "\simeq"'] \arrow[uuuu, "\alpha"'] \\
& g^* f_\sharp p_{2*}\Delta_* \arrow[r, "\mathrm{BC}_{\sharp,*}"'] & g^* f_* p_{1\sharp}\Delta_* \arrow[u, "\mathrm{BC}_*"'] & \\
g^* f_\sharp \arrow[rrr, "\mathfrak{p}_f"'] \arrow[ur, "\simeq"] & & & g^* f_*(-\otimes\omega_f). \arrow[ul, "\simeq"'] \arrow[uu, "\mathrm{BC}_*"']
\end{tikzcd}
$$

Apart from the faces labeled (1), (2) and (3), all faces either by naturality or by definition. Commutativity of (1) holds by Lemma F.6. Commutativity of (2) holds by Proposition F.14. Commutativity of (3) is immediate from unwinding the constructions of the equivalences from part (1) and Remark 6.1.4. □

Corollary 6.4.6. *Let $f\colon \mathcal{Y} \to \mathcal{X}$ be a representable submersion between separated differentiable stacks. Then the property that the Poincaré duality transformation $\mathfrak{p}_f\colon f_\sharp \to f_*(-\otimes\omega_f)$ is an equivalence can be checked locally in $\mathcal{X}$.*

Proof. Let $\{j_i\colon \mathcal{U}_i \hookrightarrow \mathcal{X}\}_{i\in I}$ be an open cover of $\mathcal{X}$ and for every $i \in I$ and let $f_i\colon \mathcal{Y}\times_{\mathcal{X}}\mathcal{U}_i \to \mathcal{U}_i$ denote the base change of f along the inclusion $j_i\colon \mathcal{U}_i \hookrightarrow \mathcal{X}$. Assume that for every $i \in I$, the transformation $\mathfrak{p}_{f_i}\colon f_{i\sharp}(-) \Rightarrow f_{i*}(-\otimes\omega_f)$ is an equivalence. Our goal is to show that also the transformation $\mathfrak{p}_f\colon f_\sharp(-) \Rightarrow f_*(-\otimes\omega_f)$ is an equivalence.

Since C is a sheaf of ∞-categories with respect to the open cover topology on SepStk, the collection of pullback functors $j_i^*\colon C(\mathcal{X}) \to C(\mathcal{U}_i)$ is jontly conservative. It will thus suffice to show that the transformation $j_i^*\mathfrak{p}_f\colon j_i^* f_\sharp \Rightarrow j_i^* f_*(-\otimes\omega_f)$ is an equivalence for every $i \in I$. By Lemma 6.4.5, this follows directly from the condition that $\mathfrak{p}_{f_i}$ is an equivalence for every $i \in I$, finishing the proof. □

We may now finish the proof of Theorem 6.1.7: for every proper representable submersion $f\colon \mathcal{Y} \to \mathcal{X}$, the Poincaré duality map $\mathfrak{p}_f\colon f_\sharp(-) \to f_*(-\otimes\omega_f)$ is an equivalence.

Proof of Theorem 6.1.7. By Proposition 3.7.3, we may find an open cover $\{j_\alpha \colon \mathcal{U}_\alpha \hookrightarrow \mathcal{X}\}_{\alpha\in I}$ of $\mathcal{X}$ such that for every $\alpha \in I$ the restriction $f_\alpha \colon \mathcal{X}_{\mathcal{U}_\alpha} \to \mathcal{U}_\alpha$ factors as

$$\mathcal{X}_{\mathcal{U}_\alpha} \xrightarrow{i_\alpha} S^{\mathcal{E}_\alpha} \xrightarrow{\pi_\alpha} \mathcal{U}_\alpha,$$

where $\pi_\alpha \colon \mathcal{E}_\alpha \to \mathcal{U}_\alpha$ is a vector bundle and where i_α is a closed embedding factoring through $\mathcal{E} \hookrightarrow S^{\mathcal{E}}$. By Proposition 6.4.4 we get that the transformation $\mathfrak{p}_{f_\alpha}$ is an equivalence for every α. It then follows from Corollary 6.4.6 that also the transformation $\mathfrak{p}_f$ is an equivalence, finishing the proof. □

6.5 Consequences of relative Poincaré duality

This section will discuss several important consequences of relative Poincaré duality. We continue working in the setting of a pullback formalism C on SepStk satisfying the Voevodsky conditions.

6.5.1 Relative Atiyah duality

Given a proper representable submersion $\mathcal{Y} \to \mathcal{X}$, it follows from relative Poincaré duality that the image of $\mathcal{Y}$ in $C(\mathcal{X})$ is dualizable, with dual given by the Thom object of the inverse tangent sphere bundle:

Proposition 6.5.1 (Relative Atiyah duality)**.** *Let $f \colon \mathcal{Y} \to \mathcal{X}$ be a proper representable submersion of separated differentiable stacks. Then the object $h_{\mathcal{X}}(\mathcal{Y}) \in C(\mathcal{X})$ is dualizable with dual given by $f_\sharp(S^{-T_f})$.*

Proof. The functor $h_{\mathcal{X}}(\mathcal{Y}) \otimes - \colon C(\mathcal{X}) \to C(\mathcal{X})$ is equivalent to $f_\sharp f^*$, regarded as a $C(\mathcal{X})$-linear functor. Its right adjoint is equivalent to $f_* f^*$, which is given by tensoring with $f_* \mathbb{1}_{\mathcal{Y}}$ by the projection formula for f_*. It follows that $h_{\mathcal{X}}(\mathcal{Y})$ is dualizable with dual $f_* \mathbb{1}_{\mathcal{Y}}$. But by relative Poincaré duality there is an equivalence $f_* \mathbb{1}_{\mathcal{Y}} \simeq f_\sharp(S^{-T_f})$ in $C(\mathcal{X})$, finishing the proof. □

6.5.2 Proper base change

As another consequence of relative Poincaré duality, we obtain good properties of the pushforward functors $p_* \colon C(\mathcal{Y}) \to C(\mathcal{X})$ for proper morphisms $p \colon \mathcal{Y} \to \mathcal{X}$ of separated differentiable stacks: proper base change, the proper projection formula and smooth-proper base change.

Proposition 6.5.2 (Proper base change)**.** *Let $p\colon \mathcal{Y} \to \mathcal{X}$ be a proper morphism of separated differentiable stacks. For every pullback square of differentiable stacks*

$$\begin{tikzcd} \mathcal{Y}' \arrow[r, "h"] \arrow[d, "p'"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{Y} \arrow[d, "p"] \\ \mathcal{X}' \arrow[r, "g"] & \mathcal{X}, \end{tikzcd}$$

the right Beck-Chevalley map

$$\mathrm{BC}_*\colon g^* p_* \Rightarrow p'_* h^*$$

is an equivalence in $\mathrm{Fun}(\mathcal{C}(\mathcal{Y}), \mathcal{C}(\mathcal{X}'))$.

Proof. Observe that the statement is local in $\mathcal{X}$, hence we may assume by Proposition 3.7.3 that the map p factors as a closed embedding followed by a proper representable submersion. By Lemma F.6 it thus suffices to prove the claim in these two cases separately. For closed embeddings this is closed base change, see Proposition 5.1.3. So assume that p is a proper representable submersion. Since the functor $- \otimes \omega_p\colon \mathcal{C}(\mathcal{Y}) \to \mathcal{C}(\mathcal{Y})$ is an equivalence by Corollary 6.2.11, it will suffice to show that the transformation $\mathrm{BC}_*\colon g^* p_*(- \otimes \omega_p) \to p'_* h^*(- \otimes \omega_p)$ is an equivalence. This follows from Lemma 6.4.5, since the map $\mathrm{BC}_\sharp$ is an equivalence by smooth base change and the transformations $\mathfrak{p}_f$ and $\mathfrak{p}_{f'}$ are equivalences by relative Poincaré duality, Theorem 6.1.7. This finishes the proof. □

Proposition 6.5.3 (Proper projection formula)**.** *For every proper map $p\colon \mathcal{X} \to \mathcal{Y}$ of separated differentiable stacks and objects $A \in \mathcal{C}(\mathcal{X})$, $B \in \mathcal{C}(\mathcal{Y})$, the exchange map*

$$\mathrm{PF}_*\colon p_*(A) \otimes B \to p_*(A \otimes p^* B)$$

is an equivalence in $\mathcal{C}(\mathcal{Y})$.

Proof. The proof is analogous to Proposition 6.5.2. Using smooth base change and Lemma F.19(2), we see that the claim is local in the target. By Proposition 3.7.3, we may thus assume that the map p factors as a closed embedding followed by a proper representable submersion, and hence by Lemma F.17 it suffices to prove the claim in these two cases. For closed embeddings this is the closed projection formula, see Proposition 5.1.3. So assume that p is a proper representable submersions. By relative Poincaré duality and the smooth projection formula, it suffices to show that the following diagram commutes:

$$\begin{tikzcd} p_\sharp(A) \otimes B \arrow[r, "\mathfrak{p}_p"] & p_*(A \otimes \omega_p) \otimes B \arrow[d, "\mathrm{PF}_*"] \\ p_\sharp(A \otimes p^* B) \arrow[u, "\mathrm{PF}_\sharp"] \arrow[r, "\mathfrak{p}_p"'] & p_*(A \otimes p^* B \otimes \omega_p). \end{tikzcd}$$

This is a consequence of the following large commutative diagram:

$$
\begin{tikzcd}
p_\sharp A \otimes B \arrow[rrr, "\mathfrak{p}_p"] \arrow[dr, "\simeq"] & & & p_*(A \otimes \omega_p) \otimes B \arrow[dd, "\mathrm{PF}_*"] \\
& p_\sharp p_{1*}\Delta_* A \otimes B \arrow[r, "\mathrm{BC}_{\sharp,*}"] & p_* p_{1\sharp}\Delta_* A \otimes B \arrow[ur, "\simeq"] \arrow[d, "\mathrm{PF}_*"] & \\
p_\sharp(A \otimes p^*B) \arrow[uu, "\mathrm{PF}_\sharp"] \arrow[r, dashed, "\simeq"] \arrow[dddd, equal] & p_\sharp(p_{1*}\Delta_* A \otimes p^*B) \arrow[u, "\mathrm{PF}_\sharp"'] \arrow[d, "\mathrm{PF}_*"] \arrow[r, phantom, "(2)"] & p_*(p_{1\sharp}\Delta_* A \otimes p^*B) \arrow[r, dashed, "\simeq"] & p_*(A \otimes \omega_p \otimes p^*B) \arrow[dddd, "\simeq"] \\
& p_\sharp p_{1*}(\Delta_* A \otimes p_1^* p^*B) \arrow[r, "\mathrm{BC}_{\sharp,*}"] \arrow[d, "\mathrm{PF}_*"] & p_* p_{1\sharp}(\Delta_* A \otimes p_1^* p^*B) \arrow[u, "\mathrm{PF}_\sharp"] \arrow[d, "\mathrm{PF}_*"] & \\
(1) & p_\sharp p_{1*}\Delta_*(A \otimes \Delta^* p_1^* p^*B) \arrow[d, "\simeq"] & p_* p_{1\sharp}\Delta_*(A \otimes \Delta^* p_1^* p^*B) \arrow[d, dashed, "\simeq"] & (3) \\
& p_\sharp p_{1*}\Delta_*(A \otimes p^*B) \arrow[r, "\mathrm{BC}_{\sharp,*}"] & p_* p_{1\sharp}\Delta_*(A \otimes p^*B) \arrow[dr, "\simeq"] & \\
p_\sharp(A \otimes p^*B) \arrow[ur, "\simeq"] \arrow[rrr, "\mathfrak{p}_p"] & & & p_*(A \otimes p^*B \otimes \omega_p).
\end{tikzcd}
$$

The unlabeled faces of the diagram commute either by naturality or by definition. Face (1) commutes by Lemma F.17. Face (2) commutes by Lemma F.20. The commutativity of face (3) follows from unwinding the definition of the equivalence from Remark 6.1.4. ◻

Proposition 6.5.4 (Smooth-proper base change)**.** *For every pullback square of separated differentiable stacks*

$$
\begin{tikzcd}
\mathcal{Y}' \arrow[r, "f'"] \arrow[d, "p'"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{Y} \arrow[d, "p"] \\
\mathcal{X}' \arrow[r, "f"] & \mathcal{X}
\end{tikzcd}
$$

where p is a proper morphism and f is a representable submersion, the double Beck-Chevalley map

$$\mathrm{BC}_{\sharp,*} \colon f_\sharp p'_* \Rightarrow p_* f'_\sharp$$

is an equivalence in $\mathrm{Fun}(\mathcal{C}(\mathcal{Y}), \mathcal{C}(\mathcal{X}))$.

Proof. The proof is analogous to Proposition 6.5.2. Using smooth base change and Proposition F.14, we see that the claim is local in the target. By Proposition 3.7.3, we may thus assume the map p factors as a closed embedding followed by a proper representable submersion, and hence by Lemma F.13 it suffices to prove the claim in these two cases. For closed embeddings this holds by smooth-closed base change, see Proposition 5.1.3, so assume that p is a proper representable submersions. Since the functor $-\otimes\omega_{p'}$ is invertible, it will suffice to show that the map $\mathrm{BC}_{\sharp,*}\colon f_\sharp p'_*(-\otimes\omega_{p'}) \to p_* f'_\sharp(-\otimes\omega_{p'})$ is an equivalence. By relative Poincaré duality and smooth base change, this follows from the following large commutative diagram:

$$\begin{tikzcd}
f_\sharp p'_\sharp \arrow[rrr, "\mathfrak{p}_{p'}"] \arrow[dr, "\simeq"] \arrow[dd, "\simeq"'] & & & f_\sharp p'_*(-\otimes\omega_{p'}) \arrow[dd, "\mathrm{BC}_{\sharp,*}"] \\
& f_\sharp p'_\sharp p'_{2*}\Delta'_* \arrow[r, "\mathrm{BC}_{\sharp,*}"] \arrow[d, dashed, "\simeq"] & f_\sharp p'_* p'_{1\sharp}\Delta'_* \arrow[ur, "\simeq"] \arrow[d, "\mathrm{BC}_{\sharp,*}"] & \\
p_\sharp f'_\sharp \arrow[r, dashed, "\simeq"] \arrow[ddd, equal] & p_\sharp f'_\sharp p'_{2*}\Delta'_* \arrow[d, "\mathrm{BC}_{\sharp,*}"'] \arrow[r, phantom, "(2)"] & p_* f'_\sharp p'_{1\sharp}\Delta'_* \arrow[r, dashed, "\simeq"] \arrow[d, dashed, "\simeq"] & p_* f'_\sharp(-\otimes\omega_{p'}) \arrow[d, "\simeq"] \\
& p_\sharp p_{2*} f''_\sharp \Delta'_* \arrow[r, "\mathrm{BC}_{\sharp,*}"] \arrow[d, "\mathrm{BC}_{\sharp,*}"'] \arrow[l, phantom, "(1)"] & p_* p_{1\sharp} f''_\sharp \Delta'_* \arrow[d, "\mathrm{BC}_{\sharp,*}"] \arrow[r, phantom, "(3)"] & p_* f'_\sharp(-\otimes f'^*\omega_p) \arrow[dd, "\mathrm{PF}_\sharp"] \\
& p_\sharp p_{1\sharp}\Delta_* f'_\sharp \arrow[r, "\mathrm{BC}_{\sharp,*}"] & p_* p_{1\sharp}\Delta_* f'_\sharp \arrow[dr, "\simeq"] & \\
p_\sharp f'_\sharp \arrow[ur, "\simeq"] \arrow[rrr, "\mathfrak{p}_p"'] & & & p_*(f'_\sharp(-)\otimes\omega_p).
\end{tikzcd}$$

The unlabeled faces of the diagram commute either by naturality or by definition. Faces (1) and (2) commute by Lemma F.13. The commutativity of face (3) is somewhat involved and we will only give the core idea of the proof. Spelling out the definitions of the equivalences from Remark 6.1.4 and Lemma 6.4.5, the claim will follow from the following commutative diagram:

$$
\begin{tikzcd}
f'_\sharp p'_{1\sharp}\Delta'_* \arrow[r, "\simeq"] \arrow[d, "\simeq"'] & f'_\sharp p'_{1\sharp}\Delta'_*(f'^*\mathbb{1}\otimes\Delta'^* {p'_1}^*(-)) & f'_\sharp p'_{1\sharp}(\Delta'_* f'^*\mathbb{1}\otimes {p'_1}^*(-)) \arrow[l, "\mathrm{PF}_*"'] \arrow[r, "\mathrm{PF}_\sharp"] \arrow[dd, bend left, "\mathrm{BC}_*"] & f'_\sharp(p'_{1\sharp}\Delta'_* f'^*\mathbb{1}\otimes -) \arrow[dd, "\mathrm{BC}_*"] \\
p_{1\sharp}f''_\sharp\Delta'_* \arrow[r, phantom, "\simeq"] \arrow[dd, "\mathrm{BC}_{\sharp,*}"'] & p_{1\sharp}f''_\sharp\Delta'_*(f'^*\mathbb{1}\otimes\Delta'^* {p'_1}^*(-)) \arrow[u, "\simeq"] \arrow[d, "\mathrm{BC}_{\sharp,*}"'] & p_{1\sharp}f''_\sharp(\Delta'_* f'^*\mathbb{1}\otimes {p'_1}^*(-)) \arrow[l, "\mathrm{PF}_*"'] \arrow[u, "\simeq"] \arrow[dd, bend right, "\mathrm{BC}_*"'] & \\
 & p_{1\sharp}\Delta_* f'_\sharp(f'^*\mathbb{1}\otimes\Delta'^* {p'_1}^*(-)) \arrow[d, "\mathrm{PF}_\sharp"'] & f'_\sharp p'_{1\sharp}(f''^*\Delta_*\mathbb{1}\otimes {p'_1}^*(-)) \arrow[r, "\mathrm{PF}_\sharp"] & f'_\sharp(p'_{1\sharp}f''^*\Delta_*\mathbb{1}\otimes -) \arrow[dd, "\mathrm{BC}_\sharp"] \\
p_{1\sharp}\Delta_* f'_\sharp \arrow[r, dashed, "\simeq"] \arrow[dd, equal] & p_{1\sharp}\Delta_*(\mathbb{1}\otimes f'_\sharp\Delta'^* {p'_1}^*(-)) \arrow[d, "\mathrm{BC}_\sharp"'] \arrow[r, phantom, "(2)"] & p_{1\sharp}f''_\sharp(f''^*\Delta_*\mathbb{1}\otimes {p'_1}^*(-)) \arrow[u, "\simeq"'] \arrow[d, "\mathrm{PF}_\sharp"] \arrow[dr, phantom, "(3)"] & \\
(1) & p_{1\sharp}\Delta_*(\mathbb{1}\otimes\Delta^* f''_\sharp {p'_1}^*(-)) \arrow[d, "\mathrm{BC}_\sharp"'] & p_{1\sharp}(\Delta_*\mathbb{1}\otimes f''_\sharp {p'_1}^*(-)) \arrow[l, "\mathrm{PF}_*"'] \arrow[d, "\mathrm{BC}_\sharp"] & f'_\sharp(f'^* p_{1\sharp}\Delta_*\mathbb{1}\otimes -) \arrow[d, "\mathrm{PF}_\sharp"] \\
p_{1\sharp}\Delta_* f'_\sharp \arrow[r, "\simeq"'] & p_{1\sharp}\Delta_*(\mathbb{1}\otimes\Delta^* p_1^* f'_\sharp(-)) & p_{1\sharp}(\Delta_*\mathbb{1}\otimes p_1^* f'_\sharp(-)) \arrow[l, "\mathrm{PF}_*"] \arrow[r, "\mathrm{PF}_\sharp"'] & p_{1\sharp}\Delta_*\mathbb{1}\otimes f'_\sharp(-).
\end{tikzcd}
$$

Except for the labeled faces, all faces commute by naturality. Face (1) commutes by Lemma F.6. Faces (2) and (3) are formal properties of mate transformations, whose verification we will leave to the reader. □

Proposition 6.5.5 (Proper exceptional pullback)**.** *Let $p\colon \mathcal{Y}\to\mathcal{X}$ be a proper map of separated differentiable stacks. Then the functor $p_*\colon \mathcal{C}(\mathcal{Y})\to\mathcal{C}(\mathcal{X})$ admits a right adjoint $p^!\colon \mathcal{C}(\mathcal{X})\to\mathcal{C}(\mathcal{Y})$.*

Proof. Combining smooth base change with the fact that a limit of a diagram of left adjoint functors is again left adjoint, we see that the claim is local in $\mathcal{X}$. Using Proposition 3.7.3 we may thus assume that the map p factors as a closed embedding followed by a proper representable submersion. If p is a closed embedding, the claim holds by Corollary 5.1.5. If p is a proper representable submersion, there is an equivalence $p_*(-)\simeq p_\sharp(-\otimes\omega_f^{-1})$ and the latter functor admits a right adjoint $\omega_f\otimes f^*(-)$. □

6.5.3 Relative Poincaré duality for stacks on SepStk

Thus far, our discussion of the categorical properties for the pullback functors $f^*\colon \mathcal{C}(\mathcal{X})\to\mathcal{C}(\mathcal{Y})$ has been restricted to morphisms $f\colon\mathcal{Y}\to\mathcal{X}$ between differentiable stacks. As we will see now, these properties formally extend to more general morphisms in $\mathrm{Shv}(\mathrm{SepStk})$.

This generalization will be used in Section III.4 to deduce relative Poincaré duality for proper genuine sheaves of spectra.

Recall that, since $\mathcal{C}$ is a sheaf of ∞-categories on SepStk, it admits a unique extension to a limit-preserving functor $\mathcal{C}\colon \mathrm{Shv}(\mathrm{SepStk})^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$. We will refer to objects $\mathfrak{X} \in \mathrm{Shv}(\mathrm{SepStk})$ as *stacks on* SepStk. We will identify a differentiable stack $\mathcal{X}$ with its image under the Yoneda embedding.

Definition 6.5.6 (Representable submersions and proper maps of stacks on SepStk)**.** Let $f\colon \mathfrak{Y} \to \mathfrak{X}$ be a morphism of stacks on SepStk.

(1) We say that f is a *representable submersion* if for every separated differentiable stack $\mathcal{X}$ and every pullback square

$$\begin{tikzcd} \mathcal{Y} \arrow[r] \arrow[d, "f'"'] \arrow[dr, phantom, "\lrcorner"] & \mathfrak{Y} \arrow[d, "f"] \\ \mathcal{X} \arrow[r] & \mathfrak{X} \end{tikzcd}$$

in $\mathrm{Shv}(\mathrm{SepStk})$, the object $\mathcal{Y}$ is also a separated differentiable stack and the base change morphism $f'\colon \mathcal{Y} \to \mathcal{X}$ is a representable submersion;

(2) We say that f is an *open embedding* if it is a representable submersion and the base change morphism $f'\colon \mathcal{Y} \to \mathcal{X}$ in part (1) is even an open embedding for all $\mathcal{X} \to \mathfrak{X}$;

(3) We say that f is *representable* if for every separated differentiable stack $\mathcal{X}$ and every pullback square

$$\begin{tikzcd} \mathcal{Y} \arrow[r] \arrow[d, "f'"'] \arrow[dr, phantom, "\lrcorner"] & \mathfrak{Y} \arrow[d, "f"] \\ \mathcal{X} \arrow[r] & \mathfrak{X} \end{tikzcd}$$

in $\mathrm{Shv}(\mathrm{SepStk})$ in which the bottom map $\mathcal{X} \to \mathfrak{X}$ is a representable submersion, the object $\mathcal{Y}$ is also a separated differentiable stack;

(4) We say that f is *proper* if it is representable and the base change morphism $f'\colon \mathcal{Y} \to \mathcal{X}$ in the previous point is proper for every $\mathcal{X} \to \mathfrak{X}$.

Remark 6.5.7. By definition, the condition that the base change morphism $f'\colon \mathcal{Y} \to \mathcal{X}$ is a representable submersion or a proper morphism can be tested by further pulling back along a map $M \to \mathcal{X}$ from a smooth manifold M, hence the same could be done in Definition 6.5.6. We have chosen this formulation to enhance the analogy with Definition 2.1.4.

Proposition 6.5.8. *Let* $\mathcal{C}\colon \mathrm{Shv}(\mathrm{SepStk})^{\mathrm{op}} \to \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$ *be a pullback formalism on* SepStk *satisfying the Voevodsky conditions. Then the following hold:*

(1) *For every representable submersion $f\colon \mathfrak{Y} \to \mathfrak{X}$ of stacks on* SepStk*, the pullback functor $f^*\colon C(\mathfrak{X}) \to C(\mathfrak{Y})$ admits a left adjoint $f_\sharp\colon C(\mathfrak{Y}) \to C(\mathfrak{X})$. Furthermore, these left adjoints satisfy smooth base change and the smooth projection formula.*

(2) *For every proper morphism $p\colon \mathfrak{Y} \to \mathfrak{X}$ of stacks on* SepStk*, the pullback functor $p^*\colon C(\mathfrak{X}) \to C(\mathfrak{Y})$ admits a right adjoint $p_*\colon C(\mathfrak{Y}) \to C(\mathfrak{X})$. Furthermore, these right adjoints satisfy proper base change and the proper projection formula.*

(3) *The left adjoints $f_\sharp$ and right adjoints p_* from parts (1) and (2) satisfy smooth-proper base change.*

(4) *For every proper representable submersion $f\colon \mathfrak{Y} \to \mathfrak{X}$ of stacks on* SepStk *the Poincaré duality map $\mathfrak{p}_f\colon f_\sharp(-) \to f_*(- \otimes \omega_f)$ is an equivalence of functor $C(\mathfrak{Y}) \to C(\mathfrak{X})$.*

Proof. Each of these properties are local in the morphism $f\colon \mathfrak{Y} \to \mathfrak{X}$, in the sense that they may be tested after pulling back along an effective epimorphism $\mathfrak{X}' \twoheadrightarrow \mathfrak{X}$ in $\mathrm{Shv}(\mathrm{SepStk})$. They then follow immediately from the analogous properties for morphisms of differentiable stacks:

(1) Smooth base change and the smooth projection formula hold by assumption on C;

(2) Proper base change and the proper projection formula hold by Proposition 6.5.2 and Proposition 6.5.3, respectively;

(3) Smooth-proper base change holds by Proposition 6.5.4;

(4) Relative Poincaré duality holds by Theorem 6.1.7.

This finishes the proof. $\square$

Appendix

C Smooth manifolds

We collect some basic results on smooth manifolds. Let Diff denote the category of smooth manifolds and smooth maps. It will be convenient for us to allow *impure* smooth manifolds, meaning that different path components might have distinct dimensions.

Definition C.1. Let $f\colon M \to N$ be a smooth map between smooth manifolds.

- The map f is called a *submersion* if for every $x \in M$ the map $T_x f\colon T_x M \to T_{f(x)} N$ on tangent spaces is surjective.
- The map f is called an *immersion* if for every $x \in M$ the map $T_x f\colon T_x M \to T_{f(x)} N$ on tangent spaces is injective.
- The map f is called an *embedding* if it is an injective immersion such that f restricts to a homeomorphism $M \xrightarrow{\cong} f(M)$.
- The map f is called *proper* if the preimage $f^{-1}(K) \subseteq M$ of a compact subspace $K \subseteq N$ is compact.

Definition C.2 (Embedded submanifold)**.** A subset $N \subseteq M$ of an n-dimensional smooth manifold M is called an *embedded k-submanifold* for $k \leq n$ if around every point $x \in N$ there exists a chart $\varphi\colon U \xrightarrow{\cong} V \subseteq \mathbb{R}^n$ such that

$$\varphi(U \cap N) = \{(x_1, \ldots, x_k, x_{k+1}, \ldots, x_n) \in V \cap x_{k+1} = \cdots = x_n = 0\}.$$

This definition is local in M and hence gives a notion of an *embedded submanifold* also for impure smooth manifolds.

Proposition C.3 ([Lee02, Proposition 5.5])**.** *Let $N \subseteq M$ be an embedded submanifold, equipped with the subspace topology. Then N admits a unique structure of a smooth manifold such that the inclusion $N \hookrightarrow M$ is a smooth embedding.* □

Proposition C.4 ([Mic08, Theorem 1.15]). *The category* Diff *is idempotent complete: given a smooth manifold M and a smooth endomorphism $f\colon M \to M$ satisfying $f \circ f = f$, the image $f(M) \subseteq M$ is an embedded submanifold of M.* □

Corollary C.5. *The image of the Yoneda embedding $y\colon \mathrm{Diff} \hookrightarrow \mathrm{PSh}(\mathrm{Diff})$ is closed under retracts.* □

Smooth manifolds are closed under closed equivalence relations.

Proposition C.6 (Godement, [Ser92, Section III.12, Theorem 2], [Bou67, Section 5.9.5]). *Let M be a smooth manifold and let $R \subseteq M \times M$ be a closed smooth submanifold defining an equivalence relation on M. Assume that the first projection $\mathrm{pr}_1\colon R \to M$ is a smooth submersion. Then the quotient space M/R admits a unique structure of a smooth manifold making the quotient map $M \to M/R$ a smooth submersion.* □

The following lemma discusses various cancellation properties of smooth maps between smooth manifolds:

Lemma C.7. *Consider a commutative triangle of smooth maps between smooth manifolds:*

$$\begin{tikzcd} & M \arrow[dl, "f"'] \arrow[dr, "gf"] & \\ N \arrow[rr, "g"] & & O. \end{tikzcd}$$

(1) If gf is an embedding, then f is an embedding;

(2) If gf is proper, then f is proper;

(3) If gf is a closed embedding, then f is a closed embedding.

(4) If f is a surjective submersion, then gf is a surjective submersion if and only if g is a surjective submersion.

Proof. For (1), assume that gf is an embedding. As gf is injective, also f is injective. As the composite

$$T_xM \xrightarrow{T_xf} T_{f(x)}N \xrightarrow{T_{f(x)}g} T_{gf(x)}O$$

is injective, so is T_xf. So f is an injective immersion. If $h\colon gf(M) \to M$ is a continuous inverse to the homeomorphism $gf\colon M \to gf(M)$, then the composite

$$f(M) \xrightarrow{g} gf(M) \xrightarrow{h} M$$

will be a continuous inverse to $f\colon M \to f(M)$.

For (2), assume that gf is proper. Let $K \subseteq N$ be a compact subspace. Since gf is proper, the subspace $(gf)^{-1}(g(K)) \subseteq N$ is compact. The preimage $f^{-1}(K)$ is a closed subspace of $(gf)^{-1}(g(K))$, hence it is also compact.

For (3), it suffices to observe that a smooth map is a closed embedding if and only if it is a proper embedding.

For (4), it is clear that if f and g are surjective submersions then so is gf, so assume that f and gf are surjective submersions. For any $y \in N$, take $x \in M$ such that $f(x) = y$. Then both $T_x f\colon T_x M \to T_{f(x)}N$ as well as the composite

$$T_x M \xrightarrow{T_x f} T_y N \xrightarrow{T_y g} T_{gy} O$$

are surjective, and thus so is $T_y g$ as desired. □

Every smooth submersion locally admits a simple 'normal form':

Theorem C.8 (Local submersion theorem)**.** *Let $f\colon M \to N$ be a smooth submersion between smooth manifolds. Then for every point $x \in M$ there exists an open neighborhood $x \in U \subseteq M$ equipped with an isomorphism $U \cong f(U) \times \mathbb{R}^n$ for some natural number n such that the following diagram commutes:*

$$\begin{tikzcd} f(U) \times \mathbb{R}^n \arrow[r, "\cong"] \arrow[dr, "\mathrm{pr}_{f(U)}"'] & U \arrow[r, hook] \arrow[d, "f|_U"] & M \arrow[d, "f"] \\ & f(U) \arrow[r, hook] & N. \end{tikzcd} \tag{C.1}$$

Proof. This is a special case of the Constant Rank Theorem of [Lee02, Theorem 5.13]: as f has constant rank $k = \dim(N)$, one may choose local coordinates on M and N such that f is of the form $f(x_1, \ldots, x_k, x_{k+1}, \ldots, x_{k+n}) = (x_1, \ldots, x_k)$. Choosing small open balls around x and $f(x)$ thus shows that f is locally given by a projection $\mathbb{R}^{k+n} \to \mathbb{R}^k$ onto the first k coordinates. □

It is a folklore theorem that base changes of smooth submersions exist in Diff and are again smooth immersions. More precisely:

Proposition C.9. *Let M, N and N' be smooth manifolds, let $f\colon M \to N$ be a smooth submersion, and let $g\colon N' \to N$ be a smooth map. Then the pullback space $M \times_N N'$ admits a unique smooth structure such that a map $h\colon X \to M \times_N N'$ is smooth if and only if both composites $\mathrm{pr}_M \circ h\colon X \to M$ and $\mathrm{pr}_{M'} \circ h\colon X \to N'$ are smooth. Furthermore, the projection $M \times_N N' \to N'$ is a smooth submersion.*

Proof. This is a special case of the fact that pullbacks in Diff of transversal maps exist. Although this result is widely known, we were not able to find a proof in the literature, hence we will provide one for the special case f is a smooth submersion.

Observe that the statement is local in M, in the sense that if $\{U_\alpha\}$ is an open cover of M such that the statement holds for the composite $f_\alpha = U_\alpha \hookrightarrow M \xrightarrow{f} N$, then it also holds for f. By Theorem C.8, it will thus suffice to prove the claim when f is either an open embedding, where the statement is clear, or a projection $N \times \mathbb{R}^n \to N$, where the pullback is the projection $N' \times \mathbb{R}^n \to N'$. $\square$

Lemma C.10. *Let $f\colon M \to N$ be a submersion between smooth manifolds. Then it is surjective if and only if it admits local sections.*

Proof. If f admits local sections, it is clear that f is surjective. Conversely, consider a point $y \in N$. We have to find an open neighborhood around y on which f admits a section. By surjectivity of f, we may pick $x \in f^{-1}(\{y\})$. Using Theorem C.8, we may find an open neighborhood $x \in U \subseteq M$ equipped with an isomorphism $U \cong f(U) \times \mathbb{R}^n$ for some natural number n such that the diagram (C.1) commutes. Since the left diagonal map $f(U) \times \mathbb{R}^n \to f(U)$ admits a section $y' \mapsto (y', 0)$, it follows that f admits a partial section defined on the open neighborhood $f(U)$ around y, proving that f admits local sections. $\square$

We finish the chapter by discussing relative tangent bundles and normal bundles.

Definition C.11. Let $f\colon M \to N$ be a smooth map between smooth manifolds. We define its *relative tangent bundle* T_f as the kernel of the derivative df of f:

$$T_f := \ker(df\colon TM \to f^*TN) \quad \in \quad \mathrm{Vect}(M).$$

We define its *normal bundle* as the cokernel of df:

$$N_f := \mathrm{coker}(df\colon TM \to f^*TN) \quad \in \quad \mathrm{Vect}(M).$$

Relative tangent bundles and normal bundles behave well under pullback:

Lemma C.12. *Consider a pullback square of smooth manifolds*

$$\begin{tikzcd} M' \arrow[r, "h"] \arrow[d, "f'"'] \arrow[dr, phantom, "\lrcorner"] & M \arrow[d, "f"] \\ N' \arrow[r, "g"] & N \end{tikzcd}$$

such that g is a smooth submersion. Then the induced square on tangent bundles

$$\begin{tikzcd} TM' \arrow[r, "dh"] \arrow[d, "df'"'] & TM \arrow[d, "df"] \\ TN' \arrow[r, "dg"] & TN \end{tikzcd}$$

is again a pullback square.

Proof. The statement is clear when g is an open embedding. Theorem C.8 then allows us to reduce to the case where $N' = N \times \mathbb{R}^n$ and $g = \mathrm{pr}_N : N \times \mathbb{R}^n \to N$ is the projection. In this case we have $M' = M \times \mathbb{R}^n$, and the statement now follows as $T(N \times \mathbb{R}^n) \cong TN \times \mathbb{R}^{n+n}$ and similarly for $T(M \times \mathbb{R}^n)$. □

Corollary C.13. *For a pullback square of smooth manifolds as in Lemma C.12, the following square is also a pullback square:*

$$\begin{tikzcd} T_{f'} \arrow[r, "dh"] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & T_f \arrow[d] \\ M' \arrow[r, "h"] & M. \end{tikzcd}$$

If f and (thus) f' are immersions, than also the following square is a pullback square:

$$\begin{tikzcd} N_{f'} \arrow[r, "dg"] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & N_f \arrow[d] \\ M' \arrow[r, "h"] & M. \end{tikzcd}$$

Proof. For every $x \in M'$, the induced square of vector bundles

$$\begin{tikzcd} T_x M' \arrow[r, "dh"] \arrow[d, "df'"'] \arrow[dr, phantom, "\lrcorner"] & T_{h(x)} M \arrow[d, "df"] \\ T_{f'(x)} N' \arrow[r, "dg"] & T_{f(h(x))} N \end{tikzcd}$$

is a pullback square by Lemma C.12. Passing to vertical kernels shows that the map $dh : T_{f'} \to T_f$ induces isomorphisms on fibers, proving that the first square is a pullback square. If f and f' are immersions, then df and df' are injections. Passing to vertical cokernels in the above square shows that the map $dg : N_{f'} \to N_f$ induces isomorphisms on fibers, proving that the second square is a pullback square. □

D Lie groupoids

In this appendix, we recall some background material on Lie groupoids freely used in the body of the text.

D.1 Definition and examples

We recall the definition and some basic examples of Lie groupoids.

Definition D.1 (Lie groupoid)**.** A *Lie groupoid* $\mathcal{G}$ is a groupoid ($\mathcal{G}_1 \rightrightarrows \mathcal{G}_0$) in which $\mathcal{G}_1$ and $\mathcal{G}_0$ are smooth manifolds, the structure maps $s, t\colon \mathcal{G}_1 \to \mathcal{G}_0$, $m\colon \mathcal{G}_1 \times_{s,\mathcal{G}_0,t} \mathcal{G}_1 \to \mathcal{G}_1$ and $u\colon \mathcal{G}_0 \to \mathcal{G}_1$ are all smooth, and the source and target maps s, t are smooth submersions. A *morphism of Lie groupoids* $f\colon \mathcal{G} \to \mathcal{H}$ is a smooth functor, i.e. a pair of smooth maps $f_0\colon \mathcal{G}_0 \to \mathcal{H}_0$ and $f_1\colon \mathcal{G}_1 \to \mathcal{H}_1$ commuting with the structuer maps. We denote by LieGrpd the category of Lie groupoids and morphisms of Lie groupoids.

Given a smooth manifold M, we say that $\mathcal{G}$ is a *Lie groupoid over* M if $\mathcal{G}_0 = M$.

Warning D.2. While one sometimes reads that 'Lie groupoids are groupoid objects internal to the category Diff of smooth manifolds', this statement needs to be taken with care, as the category Diff does not admit all pullbacks. The assumption that source and target maps are smooth submersions is needed to get a well-behaved theory.

Example D.3 (Classifying groupoid)**.** Let G be a Lie group. We let $G \ltimes \mathrm{pt}$ be the Lie groupoid given by $(G \ltimes \mathrm{pt})_0 = \mathrm{pt}$ and $(G \ltimes \mathrm{pt})_1 = G$, with composition and inversion given by multiplication and inversion in G. We call $G \ltimes \mathrm{pt}$ the *classifying groupoid* of G.

Example D.4 (Action groupoid)**.** More generally, let M be a smooth G-manifold, i.e., a smooth manifold equipped with a smooth G-action. We let $G \ltimes M$ be the Lie groupoid given by $(G \ltimes M)_0 = M$ and $(G \ltimes M)_1 = G \times M$. The structure maps are given by

$$s(g,x) = x, \qquad t(g,x) = g \cdot x, \qquad e(x) = (e,x),$$
$$i(g,x) = (g^{-1}, gx) \qquad m((g', gx),(g,x)) = (g'g, x).$$

We call $G \ltimes M$ the *action groupoid* of M. It defines a functor $G \ltimes -\colon \mathrm{Diff}_G \to \mathrm{LieGrpd}$, where Diff_G denotes the category of smooth G-manifolds.

Example D.5 (Čech groupoid)**.** Let $p\colon M \to N$ be a smooth submersion between smooth manifolds. Then its *Čech groupoid* $\check{C}(p)$ is defined as the Lie groupoid with $\check{C}(p)_0 = M$, $\check{C}(p)_1 = M \times_N M$, and structure maps

$$s(x,x') = x, \qquad t(x,x') = x', \qquad e(x) = (x,x),$$
$$i(x,x') = (x',x), \qquad m((x,x'),(x',x'')) = (x,x'').$$

When $N = \mathrm{pt}$ is a point, the resulting Lie groupoid $M \times M \rightrightarrows M$ is known as the *pair groupoid* of M.

Example D.6 (Gauge groupoid)**.** Let $P \to M$ be a principal G-bundle. The *gauge groupoid* is the Lie groupoid $(P \times P)/G \rightrightarrows M$, where G acts diagonally on $P \times P$. The composition is defined as in the pair groupoid.

Example D.7 (Pullback groupoid)**.** Let $\mathcal{G}$ be a Lie groupoid over M, let N be a smooth manifold and let $f : N \to M$ be a smooth submersion. We define the *pullback groupoid* $f^*\mathcal{G}$ by letting $(f^*\mathcal{G})_0 = N$ and defining

$$(f^*\mathcal{G})_1 = N \times_M \mathcal{G}_1 \times_M N = \{(x,g,y) \in N \times \mathcal{G}_1 \times N \mid s(g) = f(x), t(g) = f(y)\},$$

with structure maps inherited from $\mathcal{G}$. As a special case, we get for every open subspace $U \subseteq M$ a groupoid $\mathcal{G}|_{\mathcal{U}}$, called the *restriction of $\mathcal{G}$ to U*:

$$\mathcal{G}|_U = \left(\{g \in \mathcal{G}_1 \mid s(g), t(g) \in U\} \overset{s}{\underset{t}{\rightrightarrows}} U \right).$$

More generally, we may define a *topological* groupoid $\mathcal{G}|_X$ for any subspace $X \subseteq M$. However, it is not always true that $\mathcal{G}|_X$ is a Lie groupoid.

We recall the definition of the orbits and isotropy groups of a Lie groupoid.

Definition D.8 (Orbits)**.** Let $\mathcal{G}$ be a Lie groupoid over M. Two points $x, y \in M$ are said to *lie in the same orbit of $\mathcal{G}$* if there exists an arrow $g\colon x \to y$, i.e., a point $g \in \mathcal{G}_1$ with $s(g) = x$ and $t(g) = y$. This gives rise to a partition of M into the *orbits of $\mathcal{G}$*: every point $x \in M$ is contained in a unique orbit:

$$\mathcal{O}_x := \{t(g) \in M \mid g \in \mathcal{G}_1, s(g) = x\} \subseteq M.$$

Definition D.9 (Isotropy group)**.** Let $\mathcal{G}$ be a Lie groupoid over M and let $x, y \in M$. We define

$$
\begin{aligned}
\mathcal{G}(x,-) &:= s^{-1}(x) = \{g \in \mathcal{G}_1 \mid s(g) = x\} \\
\mathcal{G}(-,y) &:= t^{-1}(y) = \{g \in \mathcal{G}_1 \mid t(g) = y\} \\
\mathcal{G}(x,y) &:= \mathcal{G}(x,-) \cap \mathcal{G}(-,y) = \{g \in \mathcal{G}_1 \mid s(g) = x, t(g) = y\} \\
G_x &:= \mathcal{G}(x,x).
\end{aligned}
$$

The composition in $\mathcal{G}$ restricts to a group structure on G_x, and we call G_x the *isotropy group of* $\mathcal{G}$ *at* x.

Example D.10. The isotropy group of the action groupoid $G \ltimes M$ at a point $x \in M$ is the usual isotropy group $G_x = \{g \in G \mid gx = x\}$ of the G-action on M.

As s and t are smooth submersions, the fibers $\mathcal{G}(x,-)$ and $\mathcal{G}(-,y)$ are closed submanifolds of $\mathcal{G}_1$. One can prove that also the subspaces $\mathcal{G}(x,y) \subseteq \mathcal{G}_1$ and $\mathcal{O}_x \subseteq M$ are smooth manifolds, implying in particular that the isotropy group G_x is a Lie group:

Proposition D.11 (Moerdijk and Mrčun [MM03, Theorem 5.4])**.** *Let $\mathcal{G}$ be a Lie groupoid over M and let $x, y \in M$.*

(1) The subspace $\mathcal{G}(x,y)$ is a closed submanifold of $\mathcal{G}_1$;

(2) The isotropy group G_x is a Lie group;

(3) The orbit $\mathcal{G} \cdot x$ is an immersed submanifold of M;

(4) The target map $t\colon \mathcal{G}(x,-) \to \mathcal{O}_x$ is a principal G_x-bundle.

Definition D.12. Let $\mathcal{G}$ be a Lie groupoid over M and let $S \subseteq M$ be a subspace. We say that X is *saturated* if it contains all orbits that intersect S: if $x \in S$, then also $gx \in S$ for all $g \in \mathcal{G}_1$ with $s(g) = x$. For an arbitrary subspace $S \subseteq M$, we define the $\mathcal{G}$*-saturation* $\mathcal{G} \cdot S$ *of* S as the union in M of all orbits of $\mathcal{G}$ intersecting S non-trivially:

$$
\mathcal{G} \cdot S := \bigcup_{x \in S} \mathcal{O}_x = \{gx \in M \mid g \in \mathcal{G}_1, s(g) = x \in X\}.
$$

Definition D.13 (Proper Lie groupoids)**.** A Lie groupoid $\mathcal{G}$ is called *proper* if the map $(s,t)\colon \mathcal{G}_1 \to \mathcal{G}_0 \times \mathcal{G}_0$ is a proper map.

An action groupoid $G \ltimes M$ is proper if and only if the action of the Lie group G on the smooth manifold M is proper. Note that the isotropy groups G_x of a proper Lie groups are compact, being given as the fibers of the map $(s,t)\colon \mathcal{G}_1 \to \mathcal{G}_0 \times \mathcal{G}_0$ at (x,x).

D.2 Principal bundles

We recall the notions left actions of Lie groupoids and of smooth principal bundles for Lie groupoids.

Definition D.14 (Left $\mathcal{G}$-action)**.** Let $\mathcal{G}$ be a Lie groupoid, let M be a smooth manifold and let $M \to \mathcal{G}_0$ be a smooth submersion. A *left $\mathcal{G}$-action on $M \to \mathcal{G}_0$* consists of a smooth map $a\colon \mathcal{G}_1 \times_{s,\mathcal{G}_0} M \to M$ over $\mathcal{G}_0$ which is associative and unital in the sense that the diagrams

$$\begin{tikzcd} \mathcal{G}_1 \times_{\mathcal{G}_0} \mathcal{G}_1 \times_{\mathcal{G}_0} M \arrow[r, "m\times 1"] \arrow[d, "1\times a"] & \mathcal{G}_1 \times_{\mathcal{G}_0} M \arrow[d, "a"] \\ \mathcal{G}_1 \times_{\mathcal{G}_0} M \arrow[r, "a"] & M \end{tikzcd} \quad \text{and} \quad \begin{tikzcd} \mathcal{G}_0 \times_{\mathcal{G}_0} M \arrow[r, "i\times 1"] \arrow[d, "\cong"] & \mathcal{G}_1 \times_{\mathcal{G}_0} M \arrow[d, "a"] \\ M \arrow[r, equal] & M \end{tikzcd}$$

commute. A *smooth $\mathcal{G}$-manifold* is a smooth manifold $M \to \mathcal{G}_0$ over $\mathcal{G}_0$ equipped with a left $\mathcal{G}$-action.

Given another map $M' \to \mathcal{G}_0$ equipped with a left $\mathcal{G}$-action, a *morphism of smooth $\mathcal{G}$-manifolds $\varphi\colon M \to M'$* is a map $\varphi\colon M \to M'$ which commutes with the action maps $a\colon \mathcal{G}_1 \times_{\mathcal{G}_0} M \to M$ and $a'\colon \mathcal{G}_1 \times_{\mathcal{G}_0} M' \to M'$. We denote the resulting category of smooth $\mathcal{G}$-manifolds by $\mathrm{Diff}_{\mathcal{G}}$.

Note that for a Lie group G, a smooth $(G \ltimes \mathrm{pt})$-manifold is simply a smooth G-manifold, i.e. a smooth manifold M equipped with a smooth G-action. We let Diff_G denote the category of smooth G-manifolds.

Definition D.15 (Smooth principal $\mathcal{G}$-bundle)**.** Let $\mathcal{G}$ be a Lie groupoid and let N be a smooth manifold. A *smooth principal $\mathcal{G}$-bundle* is a smooth manifold P equipped with smooth maps $P \to \mathcal{G}_0$ and $p\colon P \to N$ and equipped with a left $\mathcal{G}$-action $a\colon \mathcal{G}_1 \times_{\mathcal{G}_0} P \to P$ on P satisfying the following conditions:

(1) The map $p\colon P \twoheadrightarrow N$ is a surjective smooth submersion.

(2) The action lives over N, in that the following diagram commutes:

$$\begin{tikzcd} \mathcal{G}_1 \times_{\mathcal{G}_0} P \arrow[rr, "a"] \arrow[dr, "p\circ \mathrm{pr}_2"'] & & P \arrow[dl, "p"] \\ & N. & \end{tikzcd}$$

(3) The map

$$\mathcal{G}_1 \times_{\mathcal{G}_0} P \xrightarrow{(a,\mathrm{pr}_2)} P \times_N P$$

is a diffeomorphism.

If $P' \to N$ is another smooth principal $\mathcal{G}$-bundle over N, a *morphism $P \to P'$ of smooth principal $\mathcal{G}$-bundles over N* consists of a map $\varphi\colon P \to P'$ of smooth $\mathcal{G}$-manifolds which lies over N. We let $\mathrm{PrnBdl}_{\mathcal{G}}(N)$ denote the resulting category of smooth principal $\mathcal{G}$-bundles over N.

Construction D.16. Let $p\colon P \twoheadrightarrow N$ be a smooth principal $\mathcal{G}$-bundle and consider a point $x \in N$. As p is a surjective smooth submersion, it admits local sections, and thus there exists an open neighborhood $x \in U \subseteq N$ and a smooth map $s\colon U \hookrightarrow P$ such that the following diagram commutes:

$$\begin{tikzcd} & P \arrow[d, "p"] \\ U \arrow[ur, hook, "s"] \arrow[r, hook] & N. \end{tikzcd}$$

Now, pulling back the diffeomorphism

$$\mathcal{G}_1 \times_{\mathcal{G}_0} P \xrightarrow{(a,\mathrm{pr}_2)} P \times_N P$$

along the map $s\colon U \hookrightarrow P$, we obtain a diffeomorphism

$$\mathcal{G}_1 \times_{\mathcal{G}_0} U \xrightarrow{\cong} P|_U$$

over U, where the map $U \to \mathcal{G}_0$ is the composite of $s\colon U \hookrightarrow P$ and the structure map $P \to \mathcal{G}_0$. In particular, for a point $x \in U \subseteq N$ the fiber P_x of p over x is given by $\mathcal{G}(x', -)$, where x' is the image of x under the map $U \to \mathcal{G}_0$.

We may think of the diffeomorphisms $\mathcal{G}_1 \times_{\mathcal{G}_0} U \xrightarrow{\cong} P|_U$ as analogue to the classical local triviality condition on principal bundles.

Corollary D.17. *Every morphism $\varphi\colon P \to P'$ of smooth principal $\mathcal{G}$-bundles over N is invertible. In other words, the category $\mathrm{PrnBdl}_{\mathcal{G}}(N)$ is a groupoid.*

Proof. It suffices to check this locally on N. Choosing a point $x \in N$ and proceeding as in Construction D.16, we find an open neighborhood $x \in U \subseteq N$ and a commutative diagram

$$\begin{tikzcd} & \mathcal{G}_1 \times_{\mathcal{G}_0} U \arrow[dl, "\cong"'] \arrow[dr, "\cong"] & \\ P|_U \arrow[rr, "\varphi|_U"] & & P'|_U, \end{tikzcd}$$

proving that the restriction of φ to U is a diffeomorphism. The claim follows. □

D.3 Morita equivalence

We recall the notion of Morita equivalence of Lie groupoids. The relevance of this notion of equivalence is that it corresponds to an equivalence between the underlying differentiable stacks of the two Lie groupoids, see Proposition II.2.3.14.

Definition D.18 (Morita equivalence)**.** Let $\mathcal{G}$ be a Lie groupoid over M and let $\mathcal{H}$ be a Lie groupoid over N. A *Morita equivalence* between $\mathcal{G}$ and $\mathcal{H}$ consists of:

(1) Surjective smooth submersions $\alpha\colon P \to M$ and $\beta\colon P \to N$;

(2) A left $\mathcal{G}$-action $a_{\mathcal{G}}\colon \mathcal{G}_1 \times_M P \to P$ on P over M making $\beta\colon P \to N$ into a principal $\mathcal{G}$-bundle;

(3) A left $\mathcal{H}$-action $a_{\mathcal{H}}\colon \mathcal{H}_1 \times_N P \to P$ on P over N making $\alpha\colon P \to M$ into a principal $\mathcal{H}$-bundle;

(4) We require that the actions $a_{\mathcal{G}}$ and $a_{\mathcal{H}}$ commute with each other.

We say that $\mathcal{G}$ and $\mathcal{H}$ are *Morita equivalent* if there exists a Morita equivalence between $\mathcal{G}$ and $\mathcal{H}$.

We mention some standard examples of Morita equivalences, whose proofs we leave to the reader.

Example D.19. Let G and H be Lie groups. Then the Lie groupoids $G \ltimes \mathrm{pt}$ and $H \ltimes \mathrm{pt}$ are Morita equivalent if and only if G and H are isomorphic.

Example D.20. Let $\mathcal{G}$ be a Lie groupoid over M with a single orbit. Then there is a Morita equivalence between $\mathcal{G}$ and $G_x \ltimes \mathrm{pt}$ for any $x \in M$.

Example D.21. Let $p\colon M \to N$ be a smooth submersion. Then the Čech groupoid $\check{C}(p)$ from Example D.5 is Morita equivalent to the submanifold $p(M) \subseteq N$, regarded as a Lie groupoid with only identity arrows.

Example D.22. Given a smooth, free and proper action of a Lie group G on a smooth manifold M, the action groupoid $G \ltimes M$ is equivalent to the quotient $N = M/G$.

Example D.23. Let $\mathcal{G}$ be a Lie groupoid over M and let $U \subseteq M$ be an open subset. Let $U' := \mathcal{G} \cdot U \subseteq M$ be the $\mathcal{G}$-saturation of U. Then the restriction $\mathcal{G}|_U$ of $\mathcal{G}$ to U is Morita equivalent to the restriction $\mathcal{G}|_{U'}$ of $\mathcal{G}$ to U'.

E Recollections on ∞-topoi

Throughout this dissertation, we make extensive use of the theory of ∞-topoi, due to Rezk, Toën-Vezzosi and Lurie. For the convenience of the reader, we will provide in this appendix a thorough treatment of this theory, covering all the foundations that we will need in this dissertation.

The appendix is organized as follows. We start in Appendix E.1 by recalling the definition and most important characterizations of ∞-topoi. The correspondence between groupoid objects and effective epimorphisms is discussed in Appendix E.2. In Appendix E.3 we define groupoid actions and principal bundles in ∞-topoi and show how they can be classified using the classifying stack of the groupoid, following the treatment of [NSS15] and [SS21, Section 3.2] for *group* actions. We discuss sheaf topoi in Appendix E.4 and hypercompleteness and hyperdescent in Appendix E.5.

E.1 Definition and characterizations of ∞-topoi

Recall the definition of an ∞-topos and of a geometric morphism between ∞-topoi.

Definition E.1 (∞-topos)**.** An ∞-category $\mathcal{B}$ is called an ∞-*topos* if it is equivalent to an accessible left exact Bousfield localization of a presheaf category $\mathrm{PSh}(C)$, for some small ∞-category C.

Definition E.2 (Geometric morphism)**.** A functor $\psi_* \colon \mathcal{B} \to \mathcal{B}'$ between ∞-topoi is called a *geometric morphism* if it admits a left exact left adjoint $\psi^* \colon \mathcal{B}' \to \mathcal{B}$.

The following theorem provides criteria for testing whether an ∞-category is an ∞-topos:

Theorem E.3 (Lurie [Lur09, Proposition 6.1.3.9, Proposition 6.1.0.6])**.** *Let $\mathcal{B}$ be an ∞-category. The following conditions are equivalent:*

(1) The ∞-category $\mathcal{B}$ is an ∞-topos;

(2) The ∞-category $\mathcal{B}$ satisfies the following ∞-categorical analogues of the Giraud's axioms:

(i) The ∞-category $\mathcal{B}$ is presentable;

(ii) Colimits in $\mathcal{B}$ are universal, see Definition E.4;

(iii) Coproducts in $\mathcal{B}$ are disjoint, see Definition E.6;

(iv) Every groupoid object of $\mathcal{B}$ is effective, see Definition E.7.

(3) The ∞-category $\mathcal{B}$ is presentable and the cartesian fibration $t\colon \operatorname{Ar}(\mathcal{B}) \to \mathcal{B}$ is classified by a limit preserving functor

$$\mathcal{B}_{/-}\colon \mathcal{B}^{\mathrm{op}} \to \mathrm{Pr}^{\mathrm{L}},\ X \mapsto \mathcal{B}_{/X}.$$

Condition (3) is often referred to as *descent for ∞-topoi*.

We will now explain the meaning of some of the conditions in the above theorem.

Universal colimits

Let $f: T \to S$ be a morphism in a presentable ∞-category $\mathcal{B}$. Since $\mathcal{B}$ has all limits, the functor $f_!\colon \mathcal{B}_{/T} \to \mathcal{B}_{/S}$ given by composition with f admits a right adjoint

$$\begin{aligned} f^*\colon \mathcal{B}_{/S} &\to \mathcal{B}_{/T} \\ (X \to S) &\mapsto (X \times_S T \to T). \end{aligned}$$

Definition E.4. Let $\mathcal{B}$ be a presentable ∞-category. We say that *colimits in $\mathcal{B}$ are universal* if, for any morphism $f: T \to S$ in $\mathcal{B}$, the associated pullback functor

$$f^*\colon \mathcal{B}_{/S} \to \mathcal{B}_{/T}$$

preserves colimits.

By the adjoint functor theorem, f^* preserves all colimits if and only if it admits a right adjoint $f_*\colon \mathcal{B}_{/T} \to \mathcal{B}_{/S}$.

Proposition E.5 (Lurie [Lur09, Proposition 6.1.1.4])**.** *Let $\mathcal{B}$ be an ∞-category which admits finite limits. The following conditions are equivalent:*

(1) The ∞-category $\mathcal{B}$ is presentable, and colimits in $\mathcal{B}$ are universal;

(2) The cartesian fibration $t\colon \operatorname{Ar}(\mathcal{B}) \to \mathcal{B}$ is classified by a functor $\mathcal{B}_{/-}\colon \mathcal{B}^{\mathrm{op}} \to \mathrm{Pr}^{\mathrm{L}}$ into the ∞-category of presentable ∞-categories and colimit preserving functors.

Disjoint coproducts

Definition E.6. If C is an ∞-category that admits finite coproducts, then we will say that *coproducts in C are disjoint* if every cocartesian diagram

$$\begin{tikzcd} \emptyset \arrow[r] \arrow[d] & Y \arrow[d] \\ X \arrow[r] & X \sqcup Y \end{tikzcd}$$

is also cartesian, provided that $\emptyset$ is an initial object of C. More informally this says that the intersection of X and Y inside the union $X \sqcup Y$ is empty.

Effective groupoid objects

We refer to Appendix E.2 for the definition of a groupoid object in an ∞-category and for the definition of the Čech nerve of a morphism in an ∞-category.

Definition E.7. Let $\mathcal{G}\colon \Delta^{\mathrm{op}} \to C$ be a groupoid object in a presentable ∞-category C. We say that $\mathcal{G}$ is *effective* if it is equivalent to the Čech nerve of the canonical map $\mathcal{G}_0 \to |\mathcal{G}| := \mathrm{colim}_{[n]\in\Delta}\, \mathcal{G}_n$.

E.2 Groupoids and effective epimorphisms

We recall from [Lur09, Section 6.2.3] the correspondence between groupoid objects and effective epimorphisms in an ∞-topos $\mathcal{B}$. Throughout this section, $\mathcal{B}$ will stay fixed.

Definition E.8 (Groupoid object, [Lur09, Definition 6.1.2.7])**.** Let C be an ∞-category. A simplicial object $\mathcal{G} : \Delta^{\mathrm{op}} \to C$ is called a *groupoid object* if for every $n \in \mathbb{N}$ and every (not necessarily order-preserving) partition

$$[n] \simeq \{i_0, \ldots, i_k\} \sqcup_{\{i_k\}} \{i_k, \ldots, i_n\},$$

the induced diagram

$$\begin{tikzcd} \mathcal{G}_n \arrow[r, "(i_0,\ldots,i_k)^*"] \arrow[d, "(i_k,\ldots,i_n)^*"'] \arrow[dr, phantom, "\lrcorner"] & \mathcal{G}_k \arrow[d, "i_k^*"] \\ \mathcal{G}_{n-k} \arrow[r, "i_k^*"] & \mathcal{G}_0 \end{tikzcd}$$

is a pullback in C. We let $\mathrm{Grpd}(C) \subseteq \mathrm{Fun}(\Delta^{\mathrm{op}}, C)$ denote the full subcategory spanned by the groupoid objects in C.

Definition E.9. Let C be an ∞-category admitting simplicial colimits and let $U\colon \Delta^{\mathrm{op}} \to C$ be a simplicial object. We define is *geometric realization* $|U|$ of U as its colimit in C:

$$|U| := \operatorname{colim}_{[n]\in\Delta^{\mathrm{op}}} U_n.$$

It comes equipped with a structure map $U_0 \to |U|$.

Construction E.10 (Čech nerve)**.** Let C be a presentable ∞-topos and let $f\colon X \to Y$ be a morphism C. We will construct a groupoid object $\check{C}(f)$ called the *Čech nerve* of f, informally given by

$$\check{C}(f)_n := X \times_Y X \times_Y \cdots \times_Y X,$$

the n-fold fiber product of X over Y. More formally, we let Δ_+ denote the augmented simplex category and let $\Delta_+^{\leq 0} \subseteq \Delta_+$ the full subcategory on the objects $[-1]$ and $[0]$. As $\Delta_+^{\leq 0}$ has a single non-identity morphism $[-1] \to [0]$, we may regard f as as a functor $f\colon (\Delta_+^{\leq 0})^{\mathrm{op}} \to C$, sending $[0]$ to X, $[-1]$ to Y and the morphism $[-1] \to [0]$ to $f\colon X \to Y$. Let $\iota_*(f)\colon \Delta_+^{\mathrm{op}} \to C$ denote the right Kan extension of f along the inclusion $\iota\colon \Delta_+^{\leq 0} \hookrightarrow \Delta_+$. Then the *Čech nerve* $\check{C}(f)$ of f is defined as the restriction of $\iota_*(f)$ to Δ^{op}.

Lemma E.11. *For every morphism $f\colon X \to Y$ in a presentable ∞-category C, the Čech nerve $\check{C}(f)$ is a groupoid object in C.*

Proof. This follows immediately from [Lur09, Proposition 6.1.2.11]. $\square$

Definition E.12 (Effective epimorphism, cf. [Lur09, Corollary 6.2.3.5])**.** Consider a morphism $f\colon X \to Y$ in $\mathcal{B}$. The diagram $\iota_*(f)\colon \Delta_+^{\mathrm{op}} \to \mathcal{B}$ from Construction E.10 defines a cocone from the Čech nerve $\check{C}(f)$ to the object Y, giving rise to a comparison map

$$|\check{C}(f)| = \operatorname{colim}_{[n]\in\Delta^{\mathrm{op}}} X^{\times_Y^n} \to Y.$$

We call f an *effective epimorphism* if this comparison map is an equivalence. In this case, we will also refer to f as exhibiting X as an *atlas* for Y. We write

$$\mathrm{Atl}(\mathcal{B}) \subseteq \mathrm{Fun}(\Delta^1, \mathcal{B})$$

for the full subcategory spanned by the effective epimorphisms $f\colon X \twoheadrightarrow Y$ in $\mathcal{B}$.

The above constructions of geometric realization and Čech nerve are inverse to each other:

Proposition E.13 (cf. [Lur09, Discussion below Corollary 6.2.3.5])**.** *Let $\mathcal{B}$ be a ∞-topos. Then the Čech nerve construction restricts to an equivalence of ∞-categories*

$$\check{C}\colon \operatorname{Atl}(\mathcal{B}) \xrightarrow{\sim} \operatorname{Grpd}(\mathcal{B}), \quad f \mapsto \check{C}(f).$$

The inverse is given by sending a groupoid object $\mathcal{G}$ to the map $\mathcal{G}_0 \to |\mathcal{G}|$.

Proof. By definition, the effective epimorphisms are precisely those morphisms $f\colon X \to Y$ for which the canonical comparison map

$$\begin{tikzcd} \check{C}(f)_0 \arrow[r, equal] \arrow[d] & X \arrow[d, "f"] \\ |\check{C}(f)| \arrow[r] & Y \end{tikzcd}$$

in $\operatorname{Fun}(\Delta^1, \mathcal{B})$ is an equivalence. Conversely, given a groupoid object $\mathcal{G}$ in $\mathcal{B}$, the colimit cone defining $|\mathcal{G}|$ corresponds to an extension of $\mathcal{G}$ to Δ_+^{op}, which in turn gives rise to a comparison map $\mathcal{G} \to \check{C}(\mathcal{G}_0 \to |\mathcal{G}|)$ of groupoid objects. A groupoid object $\mathcal{G}$ is called *effective* if this comparison map is an equivalence. But in an ∞-topos, every groupoid object is effective, see [Lur09, Theorem 6.1.0.6(3)]. It follows in particular that the morphism $\mathcal{G}_0 \to |\mathcal{G}|$ is an effective epimorphism. □

Definition E.14. If $\mathcal{G}$ is a groupoid object in an ∞-topos, we denote its geometric realization alternatively by $\mathbb{B}\mathcal{G}$ and refer to it as the *classifying stack* of $\mathcal{G}$:

$$\mathbb{B}\mathcal{G} := |\mathcal{G}| = \operatorname{colim}_{[n]\in\Delta^{\mathrm{op}}} \mathcal{G}_n.$$

Corollary E.15. *The structure map $\mathcal{G}_0 \to \mathbb{B}\mathcal{G}$ of the colimit is an effective epimorphism.*

Proof. This is immediate from Proposition E.13. □

We finish this subsection by showing that the equivalence $\operatorname{Atl}(\mathcal{B}) \simeq \operatorname{Grpd}(\mathcal{B})$ of Proposition E.13 sends cartesian squares on the left to cartesian natural transformations on the right, and vice versa.

Lemma E.16. *Let $\alpha\colon \mathcal{G} \to \mathcal{H}$ be a morphism of groupoid objects in $\mathcal{B}$, and let*

$$\begin{tikzcd} \mathcal{G}_0 \arrow[r, "\alpha_0"] \arrow[d, two heads] & \mathcal{H}_0 \arrow[d, two heads] \\ |\mathcal{G}| \arrow[r, "|\alpha|"] & |\mathcal{H}| \end{tikzcd} \tag{E.1}$$

be the map in $\operatorname{Atl}(\mathcal{B})$ *induced by the equivalence of Proposition E.13. Then α is cartesian as a natural transformation of simplicial objects in $\mathcal{B}$ if and only if the diagram* (E.1) *is a cartesian square.*

Proof. If α is cartesian, then the square (E.1) is cartesian by [Lur09, Theorem 6.1.3.9(4)]. For the converse, it suffices by the pasting law of pullback squares to show that the diagram

$$\begin{tikzcd}
\mathcal{G}_n \arrow[r, "\alpha_n"] \arrow[d] & \mathcal{H}_n \arrow[d] \\
\mathcal{G}_0 \arrow[r, "\alpha_0"] & \mathcal{H}_0
\end{tikzcd}$$

is a pullback square for each $n \geq 0$. For $n = 0$ this is clear. For $n = 1$, this follows by applying the pasting law of pullback squares to the following diagram:

$$\begin{tikzcd}
& \mathcal{G}_1 \arrow[rr, "\alpha_1"] \arrow[dl] \arrow[dd] & & \mathcal{H}_1 \arrow[dl] \arrow[dd] \\
\mathcal{G}_0 \arrow[rr, "\alpha_0", near end] \arrow[dd] & & \mathcal{H}_0 \arrow[dd] & \\
& \mathcal{G}_0 \arrow[rr, "\alpha_0", near start] \arrow[dl] & & \mathcal{H}_0 \arrow[dl] \\
|\mathcal{G}| \arrow[rr, "|\alpha|"] & & |\mathcal{H}|. &
\end{tikzcd}$$

The general case then follows by induction by applying the pasting law of pullback squares to the diagram

$$\begin{tikzcd}
& \mathcal{G}_n \arrow[rr, "\alpha_n"] \arrow[dl] \arrow[dd] & & \mathcal{H}_n \arrow[dl] \arrow[dd] \\
\mathcal{G}_{n-1} \arrow[rr, "\alpha_{n-1}", near end] \arrow[dd] & & \mathcal{H}_{n-1} \arrow[dd] & \\
& \mathcal{G}_1 \arrow[rr, "\alpha_1", near start] \arrow[dl] & & \mathcal{H}_1 \arrow[dl] \\
\mathcal{G}_0 \arrow[rr, "\alpha_0"] & & \mathcal{H}_0. &
\end{tikzcd}$$

This finishes the proof. ◻

E.3 Groupoid actions and principal bundles

In this subsection we discuss $\mathcal{G}$-actions for a groupoid object $\mathcal{G}$ in an ∞-topos $\mathcal{B}$ and prove that the ∞-category of $\mathcal{G}$-actions is equivalent to the slice of $\mathcal{B}$ over the classifying stack $\mathbb{B}\mathcal{G}$ of $\mathcal{G}$. We further recall the notion of a principal $\mathcal{G}$-bundle over an object $B \in \mathcal{B}$, and show that these are classified by morphisms $B \to \mathbb{B}\mathcal{G}$. Our treatment is based on [NSS15] and [SS21, Section 3.2], where the analogous situation for *group* actions is discussed.

Actions by groupoid objects

We introduce the notion of an action of a groupoid object in an ∞-topos $\mathcal{B}$, generalizing the treatment of Sati and Schreiber [SS21] of group actions in ∞-topoi.

Definition E.17. Let $\mathcal{G}$ be a groupoid object in an ∞-topos $\mathcal{B}$, and consider an object $X \in \mathcal{B}$. Then an *action* of $\mathcal{G}$ on X consists of the following data:

- a groupoid object $\mathcal{G} \ltimes X$ in $\mathcal{B}$;
- an equivalence $(\mathcal{G} \ltimes X)_0 \simeq X$;
- a *cartesian* natural transformation $c\colon \mathcal{G} \ltimes X \to \mathcal{G}$ of simplicial objects in $\mathcal{B}$.

We let $\mathrm{Act}_{\mathcal{G}}(\mathcal{B}) \subseteq \mathrm{Grpd}(\mathcal{B})_{/\mathcal{G}}$ denote the full subcategory of $\mathcal{G}$-actions in $\mathcal{B}$.

Remark E.18. Let $\mathcal{G} \ltimes X$ be an action of $\mathcal{G}$ on X. Since $(\mathcal{G} \ltimes X)_0 \simeq X$, the map $c\colon \mathcal{G} \ltimes X \to \mathcal{G}$ induces a structure map $c_0\colon X \to \mathcal{G}_0$. Since c is a cartesian natural transformation, there are cartesian squares

$$\begin{tikzcd}
\cdots \arrow[r] & (\mathcal{G} \ltimes X)_2 \arrow[r, "d_0"] \arrow[d, "c_2"] \arrow[dr, phantom, "\lrcorner", very near start] & (\mathcal{G} \ltimes X)_1 \arrow[r, "d_0"] \arrow[d, "c_1"] \arrow[dr, phantom, "\lrcorner", very near start] & X \arrow[d, "c_0"] \\
\cdots \arrow[r] & \mathcal{G}_2 \arrow[r, "d_0"] & \mathcal{G}_1 \arrow[r, "t=d_0"] & \mathcal{G}_0,
\end{tikzcd}$$

and in particular we obtain equivalences

$$\begin{aligned}
(\mathcal{G} \ltimes X)_1 &\xrightarrow{\sim} \mathcal{G}_1 \times_{\mathcal{G}_0} X; \\
(\mathcal{G} \ltimes X)_2 &\xrightarrow{\sim} \mathcal{G}_2 \times_{\mathcal{G}_0} X \simeq \mathcal{G}_1 \times_{\mathcal{G}_0} \mathcal{G}_1 \times_{\mathcal{G}_0} X; \\
&\vdots
\end{aligned}$$

Under these equivalences, the map $s = d_1\colon (\mathcal{G} \ltimes X)_1 \to (\mathcal{G} \ltimes X)_0$ corresponds to a map $a\colon \mathcal{G}_1 \times_{\mathcal{G}_0} X \to X$, thought of as the *action map*. The rest of the simplicial diagram $\mathcal{G} \ltimes X$ should be thought of the data witnessing that this action is unital and associative up to coherent homotopy.

Definition E.19. If $\mathcal{G} \ltimes X$ is an action of a groupoid object $\mathcal{G}$ on an object $X \in \mathcal{B}$, we define the *quotient* $X//\mathcal{G}$ as the geometric realization of $\mathcal{G} \ltimes X$:

$$X//\mathcal{G} := |\mathcal{G} \ltimes X| \simeq \mathrm{colim}_{[n] \in \Delta^{\mathrm{op}}} \mathcal{G}_n \times_{\mathcal{G}_0} X.$$

The map $\mathcal{G} \ltimes X \to \mathcal{G}$ of groupoids induces a map $X//\mathcal{G} = |\mathcal{G} \ltimes X| \to |\mathcal{G}| = \mathbb{B}\mathcal{G}$, and this defines a functor $-//\mathcal{G}\colon \mathrm{Act}_{\mathcal{G}}(\mathcal{B}) \to \mathcal{B}_{/\mathbb{B}\mathcal{G}}$.

Proposition E.20 (Classification of groupoid-actions, cf. [SS21, Proposition 3.2.63])**.** *The quotient functor $-//\mathcal{G}$ is an equivalence of ∞-categories:*

$$-//\mathcal{G}\colon \mathrm{Act}_{\mathcal{G}}(\mathcal{B}) \xrightarrow{\sim} \mathcal{B}_{/\mathbb{B}\mathcal{G}}.$$

Proof. By definition, $\mathrm{Act}_{\mathcal{G}}(\mathcal{B})$ is a full subcategory of the slice $\mathrm{Grpd}(\mathcal{B})_{/\mathcal{G}}$ spanned by those morphisms into $\mathcal{G}$ which are cartesian. Similarly, the slice $\mathcal{B}_{/\mathbb{B}\mathcal{G}}$ embeds as a full subcategory of the slice $\mathrm{Atl}(\mathcal{B})_{/(\mathcal{G}_0\to\mathbb{B}\mathcal{G})}$ by sending a morphism $B\to\mathbb{B}\mathcal{G}$ to the square

$$\begin{tikzcd}
B\times_{\mathbb{B}\mathcal{G}}\mathcal{G}_0 \arrow[r] \arrow[d, two heads] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{G}_0 \arrow[d, two heads] \\
B \arrow[r] & \mathbb{B}\mathcal{G}.
\end{tikzcd}$$

The essential image of this embedding $\mathcal{B}_{/\mathbb{B}\mathcal{G}} \hookrightarrow \mathrm{Atl}(\mathcal{B})_{/(\mathcal{G}_0\to\mathbb{B}\mathcal{G})}$ precisely consists of the pullback squares. Under these identifications, the functor $-//\mathcal{G}\colon \mathrm{Act}_{\mathcal{G}}(\mathcal{B})\to\mathcal{B}_{/\mathbb{B}\mathcal{G}}$ is given by the restriction of the equivalence $\mathrm{Grpd}(\mathcal{G})_{/\mathcal{G}}\simeq\mathrm{Atl}(\mathcal{B})_{/(\mathcal{G}_0\to\mathbb{B}\mathcal{G})}$ of Proposition E.13. By Lemma E.16, the subcategory $\mathrm{Act}_{\mathcal{G}}(\mathcal{B})$ on the left precisely corresponds to the subcategory $\mathcal{B}_{/\mathbb{B}\mathcal{G}}$ on the right, finishing the proof □

Free and transitive actions

Given a $\mathcal{G}$-action in $\mathcal{B}$, one can define when the action is *free* or *transitive*. Actions which are both free and transitive will correspond to principal $\mathcal{G}$-bundles over the terminal object of $\mathcal{B}$.

Definition E.21 (Higher shear maps, cf. [SS21, Definition 3.2.73])**.** Let $\mathcal{G}\ltimes X$ be an action of a groupoid object $\mathcal{G}$ on an object X in the fixed ∞-topos $\mathcal{B}$. We define the *shear map* of $\mathcal{G}\ltimes X$ as the map

$$\mathrm{shear}_1 = (a,\mathrm{pr}_X)\colon \mathcal{G}_1\times_{\mathcal{G}_0}X\to X\times X,$$

where $a\colon \mathcal{G}_1\times_{\mathcal{G}_0}X\to X$ is the action map and pr_X is the projection onto X. More generally, we define the *higher shear maps* of $\mathcal{G}\ltimes X$ as follows. Consider the following commutative square in $\mathcal{B}$:

$$\begin{tikzcd}
X \arrow[r, equal] \arrow[d, two heads] & X \arrow[d] \\
X//\mathcal{G} \arrow[r] & \mathrm{pt}.
\end{tikzcd}$$

Regarding this as a morphism from $X\twoheadrightarrow X//\mathcal{G}$ to $X\to\mathrm{pt}$ in $\mathrm{Fun}(\Delta^1,\mathcal{B})$ and passing to Čech nerves, we obtain a morphism of groupoid objects

$$\mathrm{shear}\colon (\mathcal{G}\ltimes X)\to \check{C}(X\to\mathrm{pt}).$$

Under the identifications of Remark E.18, it may be displayed as follows:

$$\begin{tikzcd}
\mathcal{G} \ltimes X \arrow[d, "\mathrm{shear}"] & = & \cdots \arrow[r, shift left=3] \arrow[r, shift left=1] \arrow[r, shift right=1] \arrow[r, shift right=3] & \mathcal{G}_2 \times_{\mathcal{G}_0} X \arrow[r, shift left=2] \arrow[r] \arrow[r, shift right=2] \arrow[d, "\mathrm{shear}_2"] & \mathcal{G}_1 \times_{\mathcal{G}_0} X \arrow[r, shift left, "a"] \arrow[r, shift right, "\mathrm{pr}_X"'] \arrow[d, "\mathrm{shear}_1"] & X \arrow[d, equal, "\mathrm{shear}_0"] \\
\check{C}(X \to \mathrm{pt}) & = & \cdots \arrow[r, shift left=3] \arrow[r, shift left=1] \arrow[r, shift right=1] \arrow[r, shift right=3] & X \times X \times X \arrow[r, shift left=2] \arrow[r] \arrow[r, shift right=2] & X \times X \arrow[r, shift left, "\mathrm{pr}_1"] \arrow[r, shift right, "\mathrm{pr}_2"'] & X.
\end{tikzcd}$$

Definition E.22 (Free, transitive and regular ∞-actions, cf. [SS21, Definition 3.2.75]). Let $\mathcal{G} \ltimes X$ be an action of a groupoid object $\mathcal{G}$ on an object $X \in \mathcal{B}$. We say that this action is

- *free* if its shear map shear_1 is a monomorphism:
$$\mathcal{G} \ltimes X \text{ is free} \quad \Leftrightarrow \quad \mathcal{G}_1 \times_{\mathcal{G}_0} X \xhookrightarrow{\mathrm{shear}_1} X \times X\,;$$
- *transitive* if its shear map shear_1 is an effective epimorphism:
$$\mathcal{G} \ltimes X \text{ is transitive} \quad \Leftrightarrow \quad \mathcal{G}_1 \times_{\mathcal{G}_0} X \xrightarrow{\mathrm{shear}_1} \mathrel{\mkern-14mu}\rightarrow X \times X\,;$$
- *regular* if its shear map shear_1 is an equivalence:
$$\mathcal{G} \ltimes X \text{ is regular} \quad \Leftrightarrow \quad \mathcal{G}_1 \times_{\mathcal{G}_0} X \xrightarrow[\sim]{\mathrm{shear}_1} X \times X\,.$$

Note that $\mathcal{G} \ltimes X$ is regular if and only if it is free and transitive.

In case the morphism $X \to \mathrm{pt}$ is an effective epimorphism in $\mathcal{B}$, regularity of $\mathcal{G} \ltimes X$ can be expressed as the triviality of the quotient $X//\mathcal{G}$:

Lemma E.23 (cf. [SS21, Proposition 3.2.77]). *Let $X \in \mathcal{B}$ be an object such that the terminal morphism $X \to \mathrm{pt}$ is an effective epimorphism in $\mathcal{B}$. Let $\mathcal{G} \ltimes X$ be an action of a groupoid object on X. Then the action $\mathcal{G} \ltimes X$ is regular if and only if the quotient $X//\mathcal{G}$ is the terminal object of $\mathcal{B}$.*

Proof. First assume that the map $X//\mathcal{G} \to \mathrm{pt}$ is an equivalence. Then the map $X \twoheadrightarrow X//\mathcal{G}$ is equivalent to the map $X \to \mathrm{pt}$. It follows that the shear map $\mathrm{shear}\colon (\mathcal{G} \ltimes X) \to \check{C}(X \to \mathrm{pt})$ is an equivalence, being defined by applying Čech nerves to these two morphisms.

Conversely, assume that $\mathcal{G} \times X$ is regular and that $X \twoheadrightarrow \mathrm{pt}$ is an effective epimorphism. Then the map $X//G \to \mathrm{pt}$ is obtained from the higher shear map $\mathrm{shear}\colon (\mathcal{G} \ltimes X) \to \check{C}(X \to \mathrm{pt})$ by passing to colimits. It will thus suffice to show that each of the higher shear maps $\mathrm{shear}_n\colon \mathcal{G}_n \times_{\mathcal{G}_0} X \to X^n$ is an equivalence. We proceed by induction. The case $n = 1$ is assumed, so assume that shear_{n-1} is an equivalence. Consider the following diagram:

$$\begin{tikzcd}
& \mathcal{G}_n \times_{\mathcal{G}_0} X \arrow[rr, "\mathrm{shear}_n"] \arrow[dl] \arrow[dd] & & X^{n+1} \arrow[dl] \arrow[dd] \\
\mathcal{G}_{n-1} \times_{\mathcal{G}_0} X \arrow[rr, "\mathrm{shear}_{n-1}", "\simeq"'] \arrow[dd] & & X^n \arrow[dd] & \\
& \mathcal{G}_1 \times_{\mathcal{G}_0} X \arrow[rr, "\mathrm{shear}_1", "\simeq"'] \arrow[dl] & & X^2 \arrow[dl] \\
X \arrow[rr, equal, "\mathrm{shear}_0"] & & X &
\end{tikzcd}$$

Since the left and right squares are pullback squares, it follows that shear_n is an equivalence as well, finishing the proof. □

Principal bundles

Given a groupoid object $\mathcal{G}$ in $\mathcal{B}$ and an object $B \in \mathcal{B}$, we obtain a notion of *principal $\mathcal{G}$-bundles over B* by applying the above definition of regular actions to the slice topos $\mathcal{B}_{/B}$. Observe that $\mathcal{G}$ gives rise to a groupoid $\mathcal{G} \times B$ in the slice $\mathcal{B}_{/B}$ by applying the pullback-preserving functor $B \times -\colon \mathcal{B} \to \mathcal{B}_{/B}$. For every object $X \in \mathcal{B}_{/B}$, there is an equivalence

$$(\mathcal{G}_n \times B) \times_{\mathcal{G}_0 \times B} X \simeq \mathcal{G}_n \times_{\mathcal{G}_0} X \quad \in \quad \mathcal{B}_{/B},$$

and thus a $(\mathcal{G} \times B)$-action on X is nothing but a $\mathcal{G}$-action $\mathcal{G} \ltimes X \colon \Delta^{\mathrm{op}} \to \mathcal{B}$ of $\mathcal{G}$ on the underlying object of X together with a lift of $\mathcal{G} \ltimes X$ along the forgetful functor $\mathcal{B}_{/B} \to \mathcal{B}$. For this reason, we will refer to a $(\mathcal{G} \times B)$-action in the slice $\mathcal{B}_{/B}$ simply as a *$\mathcal{G}$-action over B*.

Definition E.24 (Principal bundles, cf. [SS21, Definition 3.2.79])**.** Consider an object $B \in \mathcal{B}$ and let $\mathcal{G} \in \mathrm{Grpd}(\mathcal{B})$ be a groupoid in $\mathcal{B}$.

(1) Let $p\colon P \to B$ be an object of $\mathcal{B}_{/B}$ equipped with a $\mathcal{G}$-action over B. We say that the map p is a *formally principal $\mathcal{G}$-bundle over B* if this action corresponds to a regular $(\mathcal{G} \times B)$-action in the slice $\mathcal{B}_{/B}$:

$$\mathcal{G}_1 \times_{\mathcal{G}_0} P \xrightarrow[\sim]{\mathrm{shear}_1} P \times_B P\,.$$

(2) A formally principal $\mathcal{B}$-bundle $p\colon P \to B$ is called a *principal $\mathcal{G}$-bundle* if it is an effective epimorphism.

(3) We write

$$\mathrm{PrnBdl}_{\mathcal{G}}(\mathcal{B})_B \hookrightarrow \mathrm{Act}_{\mathcal{G}}(\mathcal{B}_{/B})$$

for the full subcategory of principal $\mathcal{G}$-bundles over B.

Remark E.25. One observes that for a morphism $f\colon B' \to B$ in $\mathcal{B}$, the pullback functor $f^*\colon \mathcal{B}_{/B} \to \mathcal{B}_{/B'}$ sends principal $\mathcal{G}$-bundles to principal $\mathcal{G}$-bundles.

We end this section with a classification of principal $\mathcal{G}$-bundles over B in terms of morphisms $B \to \mathbb{B}\mathcal{G}$ in $\mathcal{B}$.

Lemma E.26. *Let $p\colon P \twoheadrightarrow B$ be an effective epimorphism in $\mathcal{B}$ which comes equipped with a $\mathcal{G}$-action over B. Let $P//\mathcal{G} \in \mathcal{B}_{/B}$ denote the quotient of the $\mathcal{G}$-action. Then the map $p\colon P \to B$ is a principal $\mathcal{G}$-bundle if and only if the canonical map $P//\mathcal{G} \to B$ is an equivalence.*

Proof. This is an instance of Lemma E.23. □

It follows from Lemma E.26 that the forgetful functor from principal $\mathcal{G}$-bundles over B to $\mathcal{G}$-actions in $\mathcal{B}$ lands in the fiber over $B \in \mathcal{B}$ of the quotient functor $-//\mathcal{G}\colon \mathrm{Act}_{\mathcal{G}}(\mathcal{B}) \to \mathcal{B}$.

Proposition E.27. *The functor*

$$\mathrm{PrnBdl}_{\mathcal{G}}(\mathcal{B})_B \to \mathrm{Act}_{\mathcal{G}}(\mathcal{B}) \times_{\mathcal{B}} \{B\}$$

is an equivalence of ∞-categories.

Proof. We will construct an explicit inverse of this functor. Consider a $\mathcal{G}$-action $\mathcal{G} \ltimes P$ in $\mathcal{B}$ equipped with an equivalence $B \simeq P//\mathcal{G}$. The cocone $(\Delta^{\mathrm{op}})^{\triangleright} \to \mathcal{B}$ which exhibits $B \simeq P//\mathcal{G}$ as a colimit of $\mathcal{G} \ltimes P$ then adjoints over to provide a lift of the simplicial object $\mathcal{G} \ltimes P\colon \Delta^{\mathrm{op}} \to \mathcal{B}$ along the forgetful functor $\mathcal{B}_{/B} \to \mathcal{B}$; we will abuse notation and again denote this lift by $\mathcal{G} \ltimes P$. By adjunction, this lift comes equipped with a groupoid map to $\mathcal{G} \times B$, and since the forgetful functor $\mathcal{B}_{/B} \to B$ preserves pullbacks this is a cartesian morphism of groupoids, thus defining a $(\mathcal{G} \times B)$-action in $\mathcal{G}_{/B}$. This construction thus defines a functor

$$\mathrm{Act}_{\mathcal{G}}(\mathcal{B}) \times_{\mathcal{B}} \{B\} \to \mathrm{Act}_{\mathcal{G}}(\mathcal{B}_{/B}).$$

It follows directly from Lemma E.26 that this functor factors through the subcategory $\mathrm{PrnBdl}_{\mathcal{G}}(\mathcal{B})_B$ of principal $\mathcal{G}$-bundles over B. It is clear that the two functors we have constructed are inverse to each other, finishing the proof. □

Theorem E.28 (Classification of principal $\mathcal{G}$-bundles, cf. [SS21, Theorem 3.2.82])**.** *Given a groupoid object $\mathcal{G} \in \mathrm{Grpd}(\mathcal{B})$ and an object $B \in \mathcal{B}$, there is an equivalence of ∞-categories*

$$\mathrm{Hom}_{\mathcal{B}}(B, \mathbb{B}\mathcal{G}) \xrightarrow{\sim} \mathrm{PrnBdl}_{\mathcal{G}}(\mathcal{B})_B,$$

given on objects by sending a morphism $c\colon B \to \mathbb{B}\mathcal{G}$ to the principal $\mathcal{G}$-bundle $P \to B$ defined by the pullback square

$$\begin{tikzcd} P \arrow[r] \arrow[d, two heads] \arrow[dr, phantom, "\lrcorner"] & \mathcal{G}_0 \arrow[d, two heads] \\ B \arrow[r, "c"] & \mathbb{B}\mathcal{G}. \end{tikzcd}$$

Proof. Recall from Proposition E.20 the equivalence $-//\mathcal{G}\colon \mathrm{Act}_{\mathcal{G}}(\mathcal{B}) \xrightarrow{\sim} \mathcal{B}_{/\mathbb{B}\mathcal{G}}$, which induces an equivalence on fibers over $\mathcal{B}$:

$$\mathrm{Act}_{\mathcal{G}}(\mathcal{B}) \times_{\mathcal{B}} \{B\} \simeq \mathcal{B}_{/\mathbb{B}\mathcal{G}} \times_{\mathcal{B}} \{B\} \simeq \mathrm{Hom}_{\mathcal{B}}(B, \mathbb{B}\mathcal{G}).$$

Its inverse sends a morphism $c\colon B \to \mathbb{B}\mathcal{G}$ to the base change $P = B \times_{\mathbb{B}\mathcal{G}} \mathcal{G}_0 \twoheadrightarrow B$ along c of the map $\mathcal{G}_0 \twoheadrightarrow \mathbb{B}\mathcal{G}$. Combining this equivalence with the equivalence from Proposition E.27 then finishes the proof. □

Corollary E.29. *For every groupoid object $\mathcal{G}$ in $\mathcal{B}$ and every object $B \in \mathcal{B}$, the ∞-category* $\mathrm{PrnBdl}_{\mathcal{G}}(\mathcal{B})_B$ *is an ∞-groupoid.* □

E.4 Sheaf topoi

We recall the definition of a Grothendieck topology τ on an ∞-category C and the definition of the associated ∞-topos $\mathrm{Shv}_\tau(C)$ of τ-sheaves on C.

Definition E.30 (Sieve, [Lur09, Definition 6.2.2.1])**.** Let C be an ∞-category. A *sieve on* C is a full subcategory $C^{(0)} \subseteq C$ having the property that if $f\colon X \to Y$ is a morphism in C and Y belongs to $C^{(0)}$, then X also belongs to $C^{(0)}$. If $X \in C$ is an object, then a *sieve on* X is a sieve on the ∞-category $C_{/X}$. Given a morphism $f\colon X \to Y$ and a sieve $C^{(0)}_{/Y}$ on Y, we let $f^*C^{(0)}_{/Y} \subseteq C_{/X}$ denote subcategory spanned by those morphisms $(g\colon Z \to X) \in C_{/X}$ such that $f \circ g \in C^{(0)}_{/Y}$.

Remark E.31. By [Lur09, Lemma 6.2.2.5], the data of a sieve $C(0)$ on C is equivalenc to the data of a subterminal object R of $\mathrm{PSh}(C)$: $C^{(0)}$ defines the presheaf

$$R(X) = \begin{cases} \mathrm{pt} & X \in C^{(0)}, \\ \emptyset & \text{otherwise.} \end{cases}$$

Because of the equivalence $\mathrm{PSh}(C_{/X}) \simeq \mathrm{PSh}(C)_{/y(X)}$, it follows that the data of a sieve on X is the same as the data of a monomorphism $R \hookrightarrow y(X)$ of presheaves on C, where $y(X) := \mathrm{Hom}_C(-, X)\colon C^{\mathrm{op}} \to \mathrm{Spc}$ is the presheaf represented by X.

Definition E.32 (Grothendieck topology, [Lur09, Definition 6.2.2.1])**.** Let C be an ∞-category. A *Grothendieck topology* on C consists of a specification, for each object X of C, of a collection of sieves on C which we will refer to as *covering sieves*. The collections of covering sieves are required to satisfy the following properties:

(1) For every object X of C, the sieve $C_{/X} \subseteq C_{/X}$ on X is a covering sieve;

(2) For every morphism $f\colon X \to Y$ in C and every covering sieve $C^{(0)}_{/Y}$ on Y, the sieve $f^*C^{(0)}_{/Y}$ is a covering sieve on X;

(3) Let X be an object of C, $C^{(0)}_{/X}$ a covering sieve on X, and $C^{(1)}_{/X}$ an arbitrary sieve on X. Suppose that, for every morphism $f\colon Y \to X$ belonging to the sieve $C^{(0)}_{/X}$, the pullback sieve $f^*C^{(1)}_{/X}$ is a covering sieve on Y. Then $C^{(1)}_{/X}$ is a covering sieve on X.

A *site* is an ∞-category C equipped with a Grothendieck topology.

Definition E.33 (Sheaf category, [Lur09, Definition 6.2.2.6])**.** Let C be an ∞-category equipped with a Grothendieck topology τ and let S denote the collections of monomorphisms $\{R \hookrightarrow y(X)\}$ which corresponds to covering sieves on X. An object $\mathcal{F} \in \mathrm{PSh}(C)$ is called a *τ-sheaf*, or simply a *sheaf* if τ is clear from context, if it is S-local: for every covering sieve $R \hookrightarrow y(X)$, the induced map of spaces

$$\mathcal{F}(X) = \mathrm{Hom}_{\mathrm{PSh}(C)}(y(X), \mathcal{F}) \to \mathrm{Hom}_{\mathrm{PSh}(C)}(R, \mathcal{F})$$

is an equivalence. We let $\mathrm{Shv}_\tau(C) \subseteq \mathrm{PSh}(C)$ denote the full subcategory of sheaves with respect to τ.

Proposition E.34 ([Lur09, Proposition 6.2.2.7])**.** *The inclusion* $\mathrm{Shv}_\tau(C) \subseteq \mathrm{PSh}(C)$ *admits a left exact accessible left adjoint* $L_\tau\colon \mathrm{PSh}(C) \to \mathrm{Shv}_\tau(C)$*. In particular,* $\mathrm{Shv}_\tau(C)$ *is an* ∞*-topos.* □

The functor L_τ is known as the *sheafification functor*.

An equivalent description of the ∞-category $\mathrm{Shv}_\tau(C)$ is in terms of *Čech descent*.

Definition E.35 (Čech descent)**.** Let C be an ∞-category and let $\mathcal{U} = \{U_i \to X\}_{i \in I}$ be a collection of morphisms. We will denote by $\check{C}(\mathcal{U})$ the Čech nerve of the morphism

$$\bigsqcup_{i \in I} y(U_i) \to y(X)$$

in $\mathrm{PSh}(C)$. A presheaf $\mathcal{F} \in \mathrm{PSh}(C)$ is said to *satisfy Čech descent with respect to* $\mathcal{U}$ if the map

$$\mathcal{F}(X) = \mathrm{Hom}_{\mathrm{PSh}(C)}(y(X), \mathcal{F}) \to \lim_{[n]\in\Delta^{\mathrm{op}}} \mathrm{Hom}_{\mathrm{PSh}(C)}(\check{C}(\mathcal{U})_n, \mathcal{F})$$

is an equivalence. More concretely, $\mathcal{F}$ satisfies Čech descent with respect to $\mathcal{U}$ if the diagram

$$\mathcal{F}(X) \longrightarrow \prod_{i\in I} \mathcal{F}(U_i) \rightrightarrows \prod_{i,j\in I} \mathcal{F}(U_i \times_X U_j) \mathrel{\substack{\textstyle\rightarrow\\[-0.6ex]\textstyle\rightarrow\\[-0.6ex]\textstyle\rightarrow}} \cdots$$

is a limit diagram.

Proposition E.36 (cf. [Lur09, Lemma 6.2.3.18])**.** *Let $\mathcal{U} = \{U_i \to X\}_{i\in I}$ be a collection of morphisms in an ∞-category C and let $R \hookrightarrow X$ be the covering sieve generated by $\mathcal{U}$. Then a presheaf $\mathcal{F}$ on C satisfies Čech descent with respect to $\mathcal{U}$ if and only if it is local with respect to $R \hookrightarrow X$.*

Proof. The morphism $\bigsqcup_{i\in I} y(U_i) \to y(X)$ factors as an effective epimorphism followed by a monomorphism:

$$\bigsqcup_{i\in I} y(U_i) \twoheadrightarrow R \hookrightarrow y(X).$$

By [Lur09, Lemma 6.2.3.18], the map $R \hookrightarrow y(X)$ is precisely the monomorphism corresponding to the covering sheaf generated by $\mathcal{U}$. It follows that R is equivalent to the colimit of the Čech nerve $\check{C}(\mathcal{U})$ of $\mathcal{U}$, and we obtain for every $\mathcal{F} \in \mathrm{PSh}(C)$ an equivalence

$$\mathrm{Hom}_{\mathrm{PSh}(C)}(R, \mathcal{F}) \simeq \mathrm{Hom}_{\mathrm{PSh}(C)}(\mathrm{colim}_{[n]\in\Delta} \check{C}(\mathcal{U})_n, \mathcal{F}) \simeq \lim_{[n]\in\Delta^{\mathrm{op}}} \mathrm{Hom}_{\mathrm{PSh}(C)}(\check{C}(\mathcal{U})_n, \mathcal{F}).$$

As this map is compatible with the map from $\mathcal{F}(X)$, the claim follows. ◻

The effective epimorphisms in a sheaf topos $\mathrm{Shv}_\tau(C)$ can be characterized as those morphisms which *admit local sections*, in the following sense:

Definition E.37. Let C be an ∞-category equipped with a Grothendieck topology τ. Let $f \colon X \to Y$ be a morphism in $\mathrm{Shv}_\tau(C)$ and assume that $Y = y(C)$ lies in the image of the sheafified Yoneda functor $y \colon C \to \mathrm{Shv}_\tau(C)$. We say that f *admits local sections* if there exists a covering family $\{U_i \to C\}_{i\in I}$ of C such that the base change $f' \colon X \times_Y y(U_i) \to y(U_i)$ of f along the map $y(U_i) \to Y$ admits a section for every $i \in I$.

If $f \colon X \to Y$ is an arbitrary morphism in $\mathrm{Shv}_\tau(C)$, we say that f *admits local sections* if its base change along any map $y(C) \to Y$ from a representable admits local sections.

Lemma E.38. *A morphism $f \colon X \to Y$ in $\mathrm{Shv}_\tau(C)$ is an effective epimorphism if and only if it admits local sections.*

Proof. By [Lur09, Proposition 6.2.3.15], a morphism $f\colon X \to Y$ is an effective epimorphism if and only if its base change along any map $y(C) \to Y$ is an effective epimorphism. We may thus assume that $Y = y(C)$ is a representable object.

Consider the sieve $C_{/C}^{(f)} \subseteq C_{/C}$ on C generated by the map $f\colon X \to C$:

$$C_{/C}^{(f)} := \{g\colon C' \to C \mid y(g)\colon y(C') \to y(C) \text{ factors through } f\colon X \to y(C)\}.$$

Notice that a morphism $g\colon C' \to C$ factors through f if and only if the base change $y(C') \times_{y(C)} X \to y(C')$ admits a section. Hence we have to prove that f is an effective epimorphism if and only if $C_{/C}^{(f)}$ contains an covering family of C, i.e., if $C_{/C}^{(f)}$ is a covering sieve. Under the equivalence of sieves over C and monomorphisms $Z \hookrightarrow y(C)$ from Remark E.31, the sieve $C_{/C}^{(f)}$ corresponds to a monomorphism $\tau_{\leq -1}(f) \hookrightarrow y(C)$ which by (a slight generalization of) [Lur09, Lemma 6.2.3.18] is equivalent to the (-1)-truncation of f inside the slice $\mathrm{PSh}(C)_{/C}$. By [Lur09, Lemma 6.2.2.16] the sieve $C_{/C}^{(f)}$ is a covering sieve if and only if the sheafification of this map $\tau_{\leq -1}(f) \hookrightarrow C$ is an equivalence in $\mathrm{Shv}_\tau(C)$. By [Lur09, Proposition 6.2.3.4], the (-1)-truncation $\tau_{\leq -1}(f)$ of f is computed by the colimit of its Čech nerve inside $\mathrm{PSh}(C)$. Since colimits in $\mathrm{Shv}_\tau(C)$ are computed by sheafifying the colimit in $\mathrm{PSh}(C)$, it follows that $C_{/C}^{(f)}$ is a covering sieve if and only if the colimit of the Čech nerve $\check{C}(f)$ inside $\mathrm{Shv}_\tau(C)$ is equivalent to c, which is by definition what it means for $f\colon X \to y(C)$ to be an effective epimorphism. This finishes the proof. □

Morphisms of sites

We discuss continuous and cocontinuous functors between Grothendieck sites.

Definition E.39 (Continuous functor between sites)**.** Let (C, τ) and $(\mathcal{D}, \tau')$ be ∞-categories equipped with Grothendieck topologies. A functor $u\colon C \to \mathcal{D}$ is called *continuous* if the functor $\mathrm{PSh}(\mathcal{D}) \to \mathrm{PSh}(C)$ given by precomposition with u restricts to a functor

$$u_*\colon \mathrm{Shv}_{\tau'}(\mathcal{D}) \to \mathrm{Shv}_\tau(C),$$

i.e., sends τ'-sheaves on $\mathcal{D}$ to τ-sheaves on C. In this case, u_* admits a left adjoint

$$u^*\colon \mathrm{Shv}_\tau(C) \to \mathrm{Shv}_{\tau'}(\mathcal{D})$$

given by the composite $\mathrm{Shv}_\tau(C) \hookrightarrow \mathrm{PSh}(C) \xrightarrow{\mathrm{LKE}_u} \mathrm{PSh}(\mathcal{D}) \xrightarrow{L_{\tau'}} \mathrm{Shv}_{\tau'}(\mathcal{D})$, where LKE_u denotes left Kan extension along u.

Remark E.40. Assume that C and $\mathcal{D}$ admit pullbacks and that u preserves pullbacks. Then $u\colon C \to \mathcal{D}$ is continuous if and only if u sends τ-covering sieves to τ'-covering sieves.

Definition E.41 (Morphism of sites)**.** Let $(\mathcal{C}, \tau)$ and $(\mathcal{D}, \tau')$ be ∞-categories equipped with Grothendieck topologies. A functor $u \colon \mathcal{C} \to \mathcal{D}$ is called a *morphism of sites* $\mathcal{D} \to \mathcal{C}$ if it is a continuous functor and the pullback functor

$$u^* \colon \operatorname{Shv}_\tau(\mathcal{C}) \to \operatorname{Shv}_{\tau'}(\mathcal{D})$$

preserves finite limits. In other words, the functor $u_* \colon \operatorname{Shv}_{\tau'}(\mathcal{D}) \to \operatorname{Shv}_\tau(\mathcal{C})$ is a geometric morphism of ∞-topoi.

Example E.42. Let $u \colon \mathcal{C} \to \mathcal{D}$ be a continuous functor and assume that u admits a left adjoint $u' \colon \mathcal{D} \to \mathcal{C}$. Then the functor $\mathrm{LKE}_u \colon \mathrm{PSh}(\mathcal{C}) \to \mathrm{PSh}(\mathcal{D})$ given by left Kan extension along u is equivalent to the restriction functor along u'. In particular, the functor u^* is a composite of three left exact functors and thus is itself left exact. It follows that u is a morphism of sites.

Definition E.43 (Cocontinuous functor between sites)**.** Let $(\mathcal{C}, \tau)$ and $(\mathcal{D}, \tau')$ be ∞-categories equipped with Grothendieck topologies. A functor $v \colon \mathcal{C} \to \mathcal{D}$ is called *cocontinuous* if for every object $X \in \mathcal{C}$ and every covering sieve $\mathcal{D}^{(0)}_{/v(X)} \subseteq \mathcal{D}_{/v(X)}$ on $v(X) \in \mathcal{D}$, the pullback sieve

$$v^*\mathcal{D}^{(0)}_{/X} := \{(\psi \colon Y \to X) \in \mathcal{C}_{/X} \mid (v(\psi) \colon v(Y) \to v(X)) \in \mathcal{D}^{(0)}_{/v(X)}\} \subseteq \mathcal{C}_{/X}$$

is a covering sieve on X.

Lemma E.44. *Let $v \colon (\mathcal{C}, \tau) \to (\mathcal{D}, \tau')$ be a functor between sites. Then v is cocontinuous if and only if the right Kan extension functor* $\mathrm{PSh}(\mathcal{C}) \to \mathrm{PSh}(\mathcal{D})$ *restricts to a functor*

$$v_* \colon \operatorname{Shv}_\tau(\mathcal{C}) \to \operatorname{Shv}_{\tau'}(\mathcal{D}),$$

i.e., sends τ-sheaves on $\mathcal{C}$ to τ'-sheaves on $\mathcal{D}$. In this case, v_* admits a left exact left adjoint v^*, so that v_* is a geometric morphism of ∞-topoi.

Proof. The right Kan extension functor sends τ-sheaves on $\mathcal{C}$ to τ'-sheaves on $\mathcal{D}$ if and only if the composite

$$\mathrm{PSh}(\mathcal{D}) \xrightarrow{- \circ v} \mathrm{PSh}(\mathcal{C}) \xrightarrow{L_\tau} \operatorname{Shv}_\tau(\mathcal{C})$$

sends every covering sieve $R \hookrightarrow y(X)$ to an equivalence. Since this composite preserves finite limits, it preserves monomorphisms, and thus the image of $R \hookrightarrow y(X)$ is again a monomorphism. It follows that the map $R \hookrightarrow y(X)$ is sent to an equivalence in $\operatorname{Shv}_\tau(\mathcal{C})$ if and only if it is sent to an effective epimorphism. By Lemma E.38, this is in turn equivalent to the statement that this map admits local sections. Spelling out the definition, this is precisely the condition that f is cocontinuous.

The last statement holds because the left adjoint v^* of v_* is given by the composite

$$\mathrm{Shv}_{\tau'}(\mathcal{D}) \hookrightarrow \mathrm{PSh}(\mathcal{D}) \xrightarrow{-\circ v} \mathrm{PSh}(C) \xrightarrow{L_\tau} \mathrm{Shv}_\tau(C),$$

and each of these three functors is left-exact. □

Corollary E.45. *Let $v\colon (C,\tau) \to (\mathcal{D},\tau')$ be a cocontinuous functor between sites. Then the following diagram commutes:*

$$\begin{tikzcd} \mathrm{PSh}(\mathcal{D}) \arrow[r, "-\circ v"] \arrow[d, "L_{\tau'}"'] & \mathrm{PSh}(C) \arrow[d, "L_\tau"] \\ \mathrm{Shv}_{\tau'}(\mathcal{D}) \arrow[r, "v^*"] & \mathrm{Shv}_\tau(C). \end{tikzcd}$$

Proof. This diagram is obtained by passing to left adjoints from the fact that right Kan extension along v preserves sheaves by Lemma E.44. □

Warning E.46. Let $u\colon (C,\tau) \hookrightarrow (\mathcal{D},\tau')$ is a functor between sites which is both continuous and cocontinuous, then the notation u_* is overloaded: it could both stand for restriction along u as well as for right Kan extension along u. In such cases, we will explicitly state what the intended meaning of u_* is.

Corollary E.47. *Let $v\colon (C,\tau) \hookrightarrow (\mathcal{D},\tau')$ be a functor between sites. Assume that v is fully faithful and cocontinuous. Then the geometric morphism*

$$v_*\colon \mathrm{Shv}_\tau(C) \hookrightarrow \mathrm{Shv}_{\tau'}(\mathcal{D})$$

of Lemma E.44 is fully faithful.

Proof. This is immediate from the fact that the right Kan extension functor along a fully faithful functor is again fully faithful. □

Corollary E.48. *Let $v\colon (C,\tau) \hookrightarrow (\mathcal{D},\tau')$ be a functor between sites and assume that v admits a right adjoint $u\colon \mathcal{D} \to C$. Then v is cocontinuous if and only if u is a morphism of sites. In this case, we have $v_* \simeq u_*$ and $v^* \simeq u^*$.*

Proof. Observe that right Kan extension along v is equivalent to restriction along u, and thus the former preserves sheaves if and only if the latter does. This shows that v is cocontinuous if and only if u is continuous, and that in this case we have $v_* = u_*$ and thus $v^* = u^*$. It follows from Example E.42 that u is continuous if and only if it is a morphism of sites. This finishes the proof. □

E.5 Hypercompleteness

We discuss the notion of hypercomplete ∞-topoi. Throughout, we work in a fixed ∞-topos $\mathcal{B}$.

Definition E.49 (n-truncated morphisms)**.** A morphism $f\colon X \to Y$ in $\mathcal{B}$ is called (-2)-*truncated* if it is an equivalence. It is called *n-truncated*, for $n \geq -1$, if its diagonal $\Delta_f\colon X \to X \times_Y X$ is $(n-1)$-truncated. An object X of $\mathcal{B}$ is *n-truncated* if the map $X \to \mathrm{pt}$ to the terminal object is n-truncated.

Definition E.50 (n-connective morphisms, cf. [Lur09, Proposition 6.5.1.18])**.** A morphism $f\colon X \to Y$ is called 0-*connective* if it is an effective epimorphism. It is called *n-connective*, for $n \geq 1$, if it is an effective epimorphism and its diagonal $\Delta_f\colon X \to X \times_Y X$ is $(n-1)$-connective.

Remark E.51. Let $\psi_*\colon \mathcal{B} \to \mathcal{B}'$ be a geometric morphism of ∞-topoi. Then both the functor ψ_* as well as its left adjoint ψ^* are left-exact and thus preserve n-truncated morphisms for every n. The left adjoint ψ^* further preserves effective epimorphisms by [Lur09, Remark 6.2.3.6], and thus also preserves n-connected morphisms for every n.

Definition E.52 (Hypercomplete ∞-topoi, [Lur09, Section 6.5.2])**.** A morphism $f\colon X \to Y$ in an ∞-topos $\mathcal{B}$ is called *∞-connected* if it is n-connective for every $n \geq 0$. The ∞-topos $\mathcal{B}$ is called *hypercomplete* if every ∞-connected morphism in $\mathcal{B}$ is an equivalence.

Example E.53 ([Lur09, Theorem 7.2.3.6, Corollary 7.2.1.12])**.** Let X be a paracompact topological space of covering dimension $\leq n$. Then the ∞-topos $\mathrm{Shv}(X)$ of sheaves on X is hypercomplete.

Example E.54. Let M be a smooth manifold. Then M has finite covering dimension, and thus by the previous example the ∞-topos $\mathrm{Shv}(M)$ of sheaves on M is hypercomplete.

Hypercovers

Given an effective epimorphism $f\colon X \twoheadrightarrow Y$ in an ∞-topos $\mathcal{B}$, we may think of X as a *cover* of Y. The Čech nerve $\check{C}(f)$ f encodes all the iterated self-intersections $X^{\times^n_Y}$ of X over Y, and since f is an effective epimorphism this expresses Y as a colimit of these iterated self-intersections.

In certain situations, it is convenient to work with a more general notion of cover, known as a *hypercover*. A hypercover of Y also comes with an effective epimorphism $X \twoheadrightarrow Y$, but rather than simply using the iterated self-intersections $X^{\times_Y^n}$ one is allowed to further refine this: there is some object X_1 covering the intersection $X \times_Y X$, some object X_2 covering the double intersections of X_1 and X, and so forth. The colimit of a hypercover of Y admits a map to Y which one can show is always ∞-connected. In particular, if the ∞-topos is hypercomplete then the hypercover expresses Y as a colimit of the objects X_i. We will now recall the precise definitions and statements.

Notation E.55. For each natural number $n \geq 0$, let $\Delta^{\leq n}$ denote the full subcategory of Δ spanned by the set of objects $\{[0], [1], \ldots, [n]\}$. If C is a presentable ∞-category, the restriction functor

$$\mathrm{sk}_n \colon \mathrm{Fun}(\Delta^{\mathrm{op}}, C) \to \mathrm{Fun}((\Delta^{\leq n})^{\mathrm{op}}, C)$$

admits a right adjoint given by right Kan extension along the inclusion functor $(\Delta^{\leq n})^{\mathrm{op}} \hookrightarrow \Delta^{\mathrm{op}}$. We let

$$\mathrm{cosk}_n \colon \mathrm{Fun}(\Delta^{\mathrm{op}}, C) \to \mathrm{Fun}(\Delta^{\mathrm{op}}, C)$$

denote the composition of sk_n with its right adjoint and refer to cosk_n as the *n-coskeleton functor*.

Definition E.56 (Hypercover, [Lur09, Definition 6.5.3.2])**.** Let $\mathcal{B}$ be an ∞-topos. A simplicial object $U_\bullet \in \mathrm{Fun}(\Delta^{\mathrm{op}}, \mathcal{B})$ is called a *hypercover* of $\mathcal{B}$ if, for each $n \geq 0$, the unit map

$$U_n \to (\mathrm{cosk}_{n-1} U_\bullet)_n$$

is an effective epimorphism. We will say that $U_\bullet$ is an *effective hypercover* of $\mathcal{B}$ if the colimit of $U_\bullet$ is a terminal object of $\mathcal{B}$. For an object $B \in \mathcal{B}$, a *hypercover of B* is a hypercover of the slice topos $\mathcal{B}_{/B}$.

For $n = 0$, the map $U_n \to (\mathrm{cosk}_{n-1} U_\bullet)_n$ is simply the map $U_0 \to 1_{\mathcal{B}}$ to the terminal object of $\mathcal{B}$. For $n = 1$, it is the map $(d_1, d_0) \colon U_1 \to U_0 \times U_0$. For general n, the object $(\mathrm{cosk}_{n-1} U_\bullet)_n$ is known as the *n-th matching object*.

As mentioned above, one can show that every hypercover is effective in a hypercomplete ∞-topos. In fact, this property completely characterizes the hypercomplete ∞-topoi:

Theorem E.57 ([Lur09, Theorem 6.5.3.12])**.** *Let $\mathcal{B}$ be an ∞-topos. Then the following conditions are equivalent:*

(1) The ∞-topos $\mathcal{B}$ is hypercomplete;

(2) For every object $B \in \mathcal{B}$, every hypercover $U_\bullet$ of B is effective. □

Complete covers

An important example of a family of hypercovers comes from the notion of a *complete cover* of a topological space. We follow the discussion of [DI04, Subsection 4.4].

Definition E.58 ([DI04, Definition 4.5])**.** Let X be a topological space and let $\mathcal{U} = \{U_i\}_{i\in I}$ be an open cover of X. We say that $\mathcal{U}$ is a *complete cover* if for every finite collection $\sigma = \{i_1, \ldots, i_n\}$ of indices, the intersection

$$U_\sigma := U_{i_1} \cap \cdots \cap U_{i_n}$$

is covered by elements of $\mathcal{U}$. It is called a *Čech cover* if each of the intersections U_σ is again in $\mathcal{U}$.

In what follows, we identify an open cover of X with the associated subposet of the poset $\mathrm{Open}(X)$ of open subsets of X.

Construction E.59 (Dugger-Isaksen)**.** Given a cover $\mathcal{U}$ of a topological space X, we define a simplicial topological space $\Omega(\mathcal{U})_\bullet$. For any $n \geq 0$, let P_n denote the poset of nonempty subsets of $\{0, \ldots, n\}$ and inclusions. The assignment $[n] \mapsto P_n$ defines a cosimplicial poset in the obvious way. We then define $\Omega(\mathcal{U})_\bullet$ to be the simplicial topological space

$$[n] \mapsto \coprod_{F\colon P_n^{\mathrm{op}} \to \mathcal{U}} F(\{0, \ldots, n\}),$$

where the coproduct runs over all maps of posets $F\colon P_n^{\mathrm{op}} \to \mathcal{U}$. The face and degeneracy maps are induced by those in $P_\bullet$ in the expected way.

To illustrate the definition, observe that a point in $\Omega(\mathcal{U})_3$ is given by the following data:

(1) A sequence of open subsets $U_0, \ldots, U_3$ of X which are in $\mathcal{U}$;

(2) Six open subsets $U_{01}, U_{02}, \ldots, U_{23}$ in $\mathcal{U}$ such that $U_{ij} \subseteq U_i \cap U_j$ for all $i < j$;

(3) Four open subsets $U_{012}, \ldots, U_{123}$ in $\mathcal{U}$ such that $U_{ijk} \subseteq U_{ij} \cap U_{jk} \cap U_{ik}$ for all $i < j < k$;

(4) An open subset U_{0123} in $\mathcal{U}$ which is contained in all the U_{ijk};

(5) A point in U_{0123}.

It is usually helpful to think of these open sets as indexed by the faces of a 3-simplex.

Proposition E.60 ([DI04, Proposition 4.6])**.** *Let $\mathcal{U}$ be a complete cover of a topological space X. Then the simplicial topological space $\Omega(\mathcal{U})_\bullet$ is a hypercover of X, in the sense that for every $n \geq 0$ the matching morphism $\Omega(\mathcal{U})_n \to (\mathrm{cosk}_{n-1}\, \Omega(\mathcal{U})_\bullet)_n$ is an open cover.*

List of Symbols

$\hookrightarrow$ A monomorphism or an embedding

$\twoheadrightarrow$ An effective epimorphism in an ∞-topos

$\odot$ The composition product, Definition I.3.23

$\emptyset$ The empty space or empty stack

$\mathbb{1}$ The monoidal unit of a symmetric monoidal ∞-category C

$[1]$ The poset $\{0 < 1\}$, regarded either as a category or as an object of Δ

$[n]$ The poset $\{0 < 1 < \cdots < n\}$, regarded either as a category or as an object of Δ

$[\alpha]$ The poset of ordinals $\beta \leq \alpha$ for an ordinal α

$\mathrm{Act}_{\mathcal{G}}(\mathcal{B})$ The ∞-category of objects of an ∞-category $\mathcal{B}$ equipped with the action of a groupoid object $\mathcal{G}$ in $\mathcal{B}$, Definition E.17

An The ∞-category of animae/∞-groupoids

An_G The ∞-category of genuine G-animae/G-spaces

$\mathrm{Atl}(\mathcal{B})$ The ∞-category of effective epimorphisms in an ∞-topos $\mathcal{B}$, Definition E.12

$\mathrm{Atl}^{\mathrm{rep}}(\mathrm{Shv}(\mathrm{Diff}))$ The ∞-category of representable atlases $M \twoheadrightarrow \mathcal{X}$, Definition II.2.3.6

$\mathcal{B}$ A generic ∞-topos, Definition E.1

$\mathbb{B}G$ Classifying stack of a Lie group G, Example II.2.3.4

$\mathbb{B}\mathcal{G}$ Classifying stack of a Lie groupoid $\mathcal{G}$, Definition II.2.3.2

$\mathrm{BC}_\sharp$ Left Beck-Chevalley transformation, Definition F.2

BC_* Right Beck-Chevalley transformation, Definition F.2

$\mathrm{BC}_{\sharp,*}$ Double Beck-Chevalley transformation, Definition F.12

$\mathrm{CAlg}(C)$ The ∞-category of commutative algebras in a symm. mon. ∞-category C

$\mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})_{\mathrm{aug}}$ The (very large) ∞-category of augmented presentably symmetric monoidal ∞-categories, Definition I.2.41

Cat_∞ The (very large) ∞-category of large ∞-categories

$\mathrm{Cat}(\mathcal{B})$ The (very large) ∞-category of large $\mathcal{B}$-categories, Definition I.2.2

$\mathrm{Cat}_{\mathcal{B}}$ The (very large) $\mathcal{B}$-category of large $\mathcal{B}$-categories, see page 23

C^{op} The opposite of an ∞-category C

$C^{\simeq}$ The core, or underlying ∞-groupoid, of an ∞-category C

C^A The C-linear $\mathcal{B}$-category given by $C^A(B) = C(A \times B)$, Definition I.2.23

$C[A]$ The free C-linear $\mathcal{B}$-category on $A \in \mathcal{B}$, Definition I.2.23

$C[S^{-1}]$ The formal inversion of objects S in a presentably symmetric monoidal $\mathcal{B}$-category C, Definition I.2.35

$\check{C}(f)$ The Čech nerve of a morphism $f\colon X \to Y$ in an ∞-category, Construction E.10

coev A coevaluation map

$\mathrm{cofib}(f)$ The cofiber of a map f

coind_H^G Coinduction, right adjoint to res_H^G

const A constant functor

cosk_n The n-coskeleton of a simplicial object

D_A The dualizing object of an object $A \in \mathcal{B}$ (with respect to a presentably symmetric monoidal $\mathcal{B}$-category C), Definition I.3.1

D_f The relative dualizing object of morphism $f\colon A \to B$ in $\mathcal{B}$, Definition I.3.5

$D_B^{CW}(X)$ The weak Costenoble-Waner dual of an object $X \in C(B)$, Definition I.3.32

Δ The simplex category

$\Delta^{\leq n}$ The full subcategory of Δ spanned by the set of objects $\{[0], [1], \dots, [n]\}$

Δ^n_{alg} The algebraic n-simplex, Construction II.4.2.2

Δ_f The diagonal of a morphism f

$\Delta^k(f)$ The k-fold iterated diagonal of a morphism f, Definition I.3.15

Diff The site of smooth manifolds and open covers, Definition II.2.1.1

DiffStk The (2,1)-category of differentiable stacks, Definition II.2.2.1

ev An evaluation map

$\mathcal{E}$ A generic vector bundle over a stack

f^* The pullback functor along a morphism f

f_* The pushforward functor, right adjoint to f^*

$f_!$ In Part I: the left adjoint to f^*. In Part III: the exceptional pushforward functor in a six-functor formalism

$f_\sharp$ In parts II and III: the left adjoint to f^*

fgt A forgetful functor

$\mathrm{fib}(f)$ The fiber of a map f

F_D The $\mathcal{B}$-functor $\mathcal{C}^A \to \mathcal{C}^B$ classified by an object $D \in \mathcal{C}(A \times B)$, Definition I.2.33

$\mathrm{Fun}(\mathcal{C},\mathcal{D})$ The ∞-category of functors $\mathcal{C} \to \mathcal{D}$

$\mathrm{Fun}^{\mathrm{L}}(\mathcal{C},\mathcal{D})$ The ∞-category of colimit-presering functors $\mathcal{C} \to \mathcal{D}$

$\mathrm{Fun}^{\mathrm{R}}(\mathcal{C},\mathcal{D})$ The ∞-category of limit-presering functors $\mathcal{C} \to \mathcal{D}$

$\underline{\mathrm{Fun}}_{\mathcal{B}}(\mathcal{C},\mathcal{D})$ The parametrized functor category, Example I.2.7

$\mathcal{F}$ A generic sheaf

$\Gamma(\mathcal{C})$ The underlying ∞-category of a parametrized ∞-category $\mathcal{C}$, Definition I.2.2

$\Gamma(\mathcal{F})$ The global sections of a sheaf $\mathcal{F}$

γ^* Comparison functor between genuine and ordinary sheaves on a differentiable stack $\mathcal{X}$, Construction II.4.2.25

$g_{\mathcal{X}}$ The structure map $g_{\mathcal{X}}\colon \mathcal{X} \to \mathcal{S}$ of a stack $\mathcal{X}$ over a base stack $\mathcal{S}$

G_x The isotropy group of a differentiable stack at a point x, Definition II.3.4.1

$G \ltimes M$ The action groupoid of a smooth G-manifold M, Example D.4

Glo The global indexing category, Definition I.4.12

Grpd The ordinary category of groupoids

$\mathrm{Grpd}(C)$ The ∞-category of groupoid objects in an ∞-category C, Definition E.8

$\mathcal{G}$ A generic Lie groupoid, Definition D.1, or a generic groupoid object in an ∞-category, Definition E.8

$\mathcal{G}_n$ The manifold of n-tuples of composable morphisms in a Lie groupoid $\mathcal{G}$, Definition 2.3.1, or the n-th level of a groupoid object in an ∞-category

$\mathcal{G}_U$ The restriction of a Lie groupoid, Example D.7

$h_{\mathcal{X}}(\mathcal{Y})$ The image of $\mathcal{Y} \in \mathrm{Sub}_{/\mathcal{X}}$ in $C(\mathcal{X})$, Notation II.4.5.13

$h^{\mathcal{Z}}_{\mathcal{X}}(\mathcal{Y}, t)$ The presheaf of $\mathcal{Z}$-trivialized morphisms, Construction II.5.2.9

$\mathrm{H}(\mathcal{X})$ The ∞-category of genuine sheaves on a differentiable stack $\mathcal{X}$, Definition II.4.2.6

$\mathrm{Hom}_C(-,-)$ The space of morphisms in an ∞-category C

$\mathrm{Hom}^{\Delta}_C(-,-)$ The simplicial hom-set of a simplicially enriched category C

$\underline{\mathrm{Hom}}_C$ The internal hom object in a symmetric monoidal ∞-category C

i A generic closed embedding of differentiable stacks

id An identity map

ind^G_H Induction from H-equivariant objects to G-equivariant objects, left adjoint to res^G_H

j A generic open embedding of differentiable stacks

l.b.c. Abbreviation of ‘left base change, Definition I.2.9

l.p.f. Abbreviation of ‘left projection formula, see (2) on page 24

$\mathrm{L}_{\mathrm{open}}$ The sheafification functor $L_{\mathrm{open}} \colon \mathrm{PSh}(\mathrm{Diff}) \to \mathrm{Shv}(\mathrm{Diff})$ with respect to the open cover topology

$\mathrm{L}_{\mathbb{R}}$ The homotopy localization functor at the level of presheaves inverting $\mathbb{R} \to \mathrm{pt}$, Construction II.4.2.2

$\mathrm{L}_{\mathrm{htp}}$ The homotopy localization functor at the level of sheaves inverting $\mathbb{R} \to \mathrm{pt}$, Definition II.4.2.6

L^{α} The α-th iteration of the functor $L^1 = L_{\mathrm{open}} \circ L_{\mathbb{R}}$ for an ordinal α, Construction II.4.2.11

$\mathrm{LConst}(\mathcal{E})$ The locally constant $\mathcal{B}$-category associated to an ∞-category $\mathcal{E}$, Example I.2.5

LieGrpd The ordinary category of Lie groupoids

$\mathrm{LMod}_R(\mathcal{C})$ The ∞-category of left modules over an associative algebra R in a monoidal ∞-category $\mathcal{C}$

$\mathcal{L}$ The formal inversion functor, Lemma I.2.42

M//G The quotient stack of a smooth G-manifold M, Example II.2.3.5

$\mathrm{Mod}_R(\mathcal{C})$ The ∞-category of modules over a commutative algebra R in a symmetric monoidal ∞-category $\mathcal{C}$

$\mathrm{Nat}(F, G)$ The space of natural transformations $F \Rightarrow G$ between two functors $F, G \colon \mathcal{C} \to \mathcal{D}$

$\mathrm{Nat}_{\mathcal{C}}(F, G)$ The space of $\mathcal{C}$-linear natural transformations $F \Rightarrow G$ between two $\mathcal{C}$-linear functors $F, G \colon \mathcal{D} \to \mathcal{E}$,

$\mathrm{N}^{\mathrm{Diff}}$ The continuous diffeology functor, Definition II.3.1.1

Nm_A The twisted norm map of an object $A \in \mathcal{B}$, Definition I.3.3

$\widetilde{\mathrm{Nm}}_A$ The adjoint twisted norm map of an object $A \in \mathcal{B}$, Definition I.3.3

Nm_f The twisted norm map of a morphism f in $\mathcal{B}$, Definition I.3.5

$\mathcal{N}_i$ The normal bundle of an embedding of stacks i, Definition II.3.5.14

$\mathrm{Open}(\mathcal{X})$ The poset of open substacks of a stack $\mathcal{X}$

Orb The global orbit category, the wide subcategory of Glo spanned by the injective group homomorphisms, Definition I.4.13

Orb_G The orbit category of a Lie group G

$\mathrm{Orb}_G^{\mathrm{pr}}$ The proper orbit category of a Lie group G, Definition I.4.22

$\underline{\mathrm{OrbSp}}$ The orbicategory of orbispectra, Definition I.4.16

$\underline{\mathrm{OrbSpc}}$ The orbicategory of orbispaces, Definition I.4.15

ω_f The dualizing object of a morphism of stacks f, Definition II.6.1.3

$\Omega_{\mathcal{B}}$ The $\mathcal{B}$-category of $\mathcal{B}$-groupoids, Definition I.2.11

PB(SepStk) The (very large) ∞-category of pullback formalisms on SepStk, Definition 4.5.5

$\mathrm{PF}_\sharp$ The left projection formula map, Definition F.15

PF_* The right projection formula map, Definition F.15

$\mathfrak{p}_f$ The Poincaré duality map $f_\sharp \to f_*(- \otimes \omega_f)$, Construction II.6.1.5

pr A projection map

Pr^{L} The (very large) ∞-category of presentable ∞-categories and colimit-preserving functors

Pr^{R} The (very large) ∞-category of presentable ∞-categories and accessible limit-preserving functors

$\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ The ∞-category of stable presentable ∞-categories and colimit-preserving functors

$\mathrm{Pr}^{\mathrm{L}}(\mathcal{B})$ The ∞-category of presentable $\mathcal{B}$-categories and colimit-preserving $\mathcal{B}$-functors, Definition I.2.12

$\mathrm{Pr}^{\mathrm{L}}_{\mathcal{B}}$ The $\mathcal{B}$-category of presentable $\mathcal{B}$-categories and colimit-preserving $\mathcal{B}$-functors, see page 23

$\mathrm{Pr}^{\mathrm{L}}_G$ The ∞-category $\mathrm{Pr}^{\mathrm{L}}(\mathrm{Spc}_G)$ of presentable G-categories

$\mathrm{PrnBdl}_{\mathcal{G}}(\mathcal{B})_B$ The ∞-category of principal $\mathcal{G}$-bundles over an object B in an ∞-topos $\mathcal{B}$, Definition E.24

$\mathrm{PrnBdl}_{\mathcal{G}}(N)$ The ∞-category of smooth principal $\mathcal{G}$-bundles over a smooth manifold N, Definition D.15

$\mathrm{PSh}(C)$ The ∞-category of presheaves on a small ∞-category C

$\mathrm{PSh}^{\mathrm{htp}}(-)$ The ∞-category of homotopy invariant presheaves, Definition II.4.2.1

pt The one-point space

$\mathrm{PT}_{\mathcal{S}}(i)$ The Pontryagin-Thom collapse map of a closed embedding $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ over $\mathcal{S}$, Construction II.6.2.16

$\pi_B^* C$ The pullback $\mathcal{B}_{/B}$-category of a $\mathcal{B}$-category C, Example I.2.6

$(\pi_B)_* C$ The pushforward $\mathcal{B}$-category of a $\mathcal{B}_{/B}$-category C, Example I.2.6

$\Pi_\infty(X)$ The fundamental ∞-groupoid of a topological space X

$\Pi_G(M)$ The G-equivariant homotopy type of a smooth G-manifold M, Definition II.4.4.8

$\Pi_{\mathrm{Glo}}(\mathcal{X})$ The global homotopy type of a differentiable stack, Section III.2.1

QtStk The (2,1)-category of global quotient stacks $M//G$, Example II.2.3.5

Rep_G The category of G-representations

res^G_H Restriction from G-equivariant objects to H-equivariant objects

$\mathrm{RMod}_R(C)$ The ∞-category of right modules over an associative algebra R in a monoidal ∞-category C

$S^{\mathcal{E}}$ Sphere bundle of a vector bundle $\mathcal{E} \to \mathcal{X}$ of stacks, Definition II.2.5.2, Notation II.4.5.25

$\mathbb{S}_G$ The G-equivariant sphere spectrum

SepStk The (2,1)-category of separated differentiable stacks, Definition II.3.3.1

$\mathrm{SH}(\mathcal{X})$ The ∞-category of genuine sheaves of spectra on a separated differentiable stack $\mathcal{X}$, Definition II.4.3.14

shear The shear map, Definition E.21

$\mathrm{Shv}(C)$ The ∞-category of sheaves on a Grothendieck site C, Definition E.33

$\mathrm{Shv}(\mathcal{X})$ The ∞-category of ordinary sheaves on a differentiable stack $\mathcal{X}$, Definition II.4.1.17

sk_n The n-skeleton of a simplicial object

$\mathrm{Span}(\mathcal{B}, E)$ The span category of a geometric setup $(\mathcal{B}, E)$, Definition III.3.1

Sp The ∞-category of spectra

Spc The ∞-category of spaces/∞-groupoids

Sp^G The ∞-category of genuine G-spectra for a compact Lie group G, Definition I.4.1

Spc^G The ∞-category of G-spaces for a compact Lie group G

$\mathrm{Spc}^G_{\mathrm{pr}}$ The ∞-category of proper G-spaces for a Lie group G, Definition I.4.22

$\underline{\mathrm{Sp}}_G$ The G-category of genuine G-spectra, Definition I.4.2

$\underline{\mathrm{Spc}}_G$ The G-category of G-spaces, Definition I.4.2

$\mathrm{Sub}_{/\mathcal{X}}$ The category of representable submersions $\mathcal{Y} \to \mathcal{X}$, Definition II.4.1.1

$\mathrm{Th}_{\mathcal{S}}(\mathcal{X}, \mathcal{Z})$ The relative Thom space of a closed embedding $i\colon \mathcal{Z} \hookrightarrow \mathcal{X}$ relative to a base stack $\mathcal{S}$, Definition II.6.2.1

$T\mathcal{G}$ The tangent groupoid of a Lie groupoid $\mathcal{G}$, Definition II.3.5.1

T_f The relative tangent bundle of a morphism f of stacks, Definition II.3.5.5

Top The ordinary category of topological spaces

$\mathrm{TopGrpd}_1$ The ordinary category of topological groupoids, Definition I.4.11

TopGrpd The ∞-category of topological groupoids, Definition I.4.11

U_A Unit for the composition product, Definition I.3.23

$\mathcal{U}$ A generic open substack of some stack

$\mathrm{Vect}(\mathcal{X})$ The ordinary category of vector bundles over a differentiable stack $\mathcal{X}$, Definition II.2.5.3

Vect_k The ordinary category of vector spaces over a field k

$\mathcal{X} \setminus \mathcal{Z}$ The open complement of a closed substack $\mathcal{Z} \subseteq \mathcal{X}$, Definition II.3.2.2

$|\mathcal{X}|_{\mathrm{mod}}$ The coarse moduli space of a stack $\mathcal{X}$, Definition II.3.1.1

$\mathfrak{X}$ A generic sheaf on SepStk

y The Yoneda embedding $y\colon \mathcal{C} \hookrightarrow \mathrm{PSh}(\mathcal{C})$

$\mathcal{Z}$ A generic closed substack of some stack

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